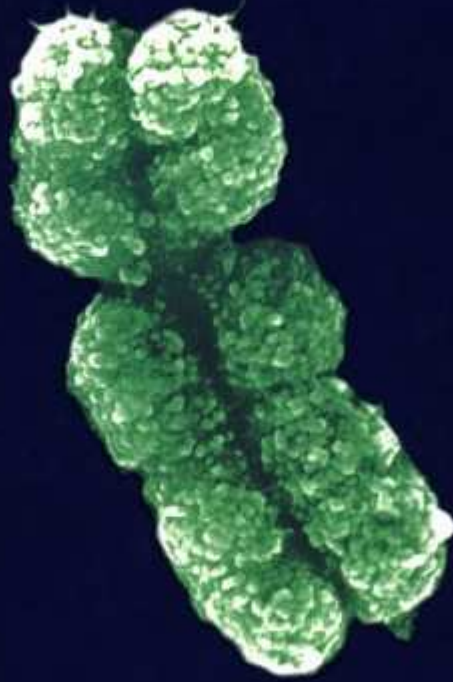


GENETICS

Principles and Analysis

Fourth Edition



Daniel L. Hartl
Elizabeth W. Jones

Genetics: Principles and Analysis

Fourth Edition

Daniel L. Hartl
Harvard University

Elizabeth W. Jones
Carnegie Mellon University



Jones and Bartlett Publishers
Sudbury, Massachusetts

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TO THE BEST TEACHERS WE EVER HAD—OUR PARENTS AND OUR STUDENTS

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ABOUT THE COVER

A human chromosome.

ABOUT THE PUBLISHER

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Jones and Bartlett Publishers
40 Tall Pine Drive
Sudbury, MA 01776
978-443-5000
info@jbpub.com
<http://www.jbpub.com>

Jones and Bartlett Publishers Canada
P.O. Box 19020
Toronto, ON M5S 1X1
CANADA

Jones and Bartlett Publishers International
Barb House, Barb Mews
London W6 7PA
UK

ABOUT THE BOOK

Chief Executive Officer: Clayton Jones
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* * *

Development Editor: Richard Morel
Book and Cover Design: J/B Woolsey Associates
Art Development and Rendering: J/B Woolsey Associates
Composition and Book Layout: Thompson Steele, Inc.
Prepress: Westwords, Inc.
Cover Manufacture: Coral Graphic Services, Inc.
Book Manufacture: World Color Book Services

Library of Congress Cataloging-in-Publication Data

Hartl, Daniel L.

Genetics: Principles and analysis / Daniel L. Hartl, Elizabeth W. Jones.—4th ed.
p. cm.

Includes bibliographical references (p.).

ISBN 0-7637-0489-X

1. Genetics. I. Jones, Elizabeth W. II. Title.

QH430.H3733 1998

576.5—dc21

97-40566

CIP

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Printed in the United States

03 02 01 00 99 98 9 8 7 6 5 4 3 2

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Preface

This book is titled *Genetics: Principles and Analysis*, Fourth Edition, because it embodies our belief that a good course in genetics should maintain the right balance between two important aspects of the science. The first aspect is that genetics is a body of knowledge pertaining to genetic transmission, function, and mutation. This constitutes the *Principles*. The second aspect is that genetics is an experimental approach, or a kit of "tools," for the study of biological processes such as development or behavior. This is *Analysis*.

The overall aim of *Genetics: Principles and Analysis*, Fourth Edition, is to provide a clear, comprehensive, rigorous, and balanced introduction to genetics at the college level. It is a guide to learning a critically important and sometimes difficult subject. The rationale of the book is that any student claiming a knowledge of genetics must:

- Understand the basic processes of gene transmission, mutation, expression, and regulation;
- Be able to think like a geneticist at the elementary level of being able to formulate genetic hypotheses, work out their consequences, and test the results against observed data;
- Be able to solve problems of several types, including problems that ask the student to verbalize genetic principles in his or her own words, single-concept exercises that require application of definitions or the basic principles of genetics, genetic analysis in which several concepts must be applied in logical order, and quantitative problems that call for some numerical calculation;
- Gain some sense of the social and historical context in which genetics has developed and is continuing to develop; and
- Have some familiarity with the genetic resources and information that are available through the Internet.

Genetics: Principles and Analysis, Fourth Edition, incorporates many special features to help students achieve these learning goals. The text is clearly and concisely written in a somewhat relaxed prose style without being chummy or excessively familiar. Each chapter is headed by a list of **Principles** that are related at numerous points to the larger whole. Each chapter contains two or three **Connections** in which the text material is connected to excerpts of classic papers that report key experiments in genetics or that raise important social, ethical, or legal issues in genetics. Each Connection has a brief introduction of its own, explaining the importance of the experiment and the historical context in which it was carried out. At the end of each chapter is a complete **Summary**, **Key Terms**, **GeNETics on the web** exercises that guide students in the use of Internet resources in genetics, and several different types and levels of **Problems**. These features are discussed individually below.

In recent decades, both the amount of genetic knowledge and its rate of growth have exploded. Many of the new discoveries have personal and social relevance through applications of genetics to human affairs in prenatal diagnosis, testing for carriers, and identification of genetic risk factors for complex traits, such as breast cancer and heart disease. There are also ethical controversies: Should genetic manipulation be used on patients for the treatment of disease? Should human fetuses be used in research? Should human beings be cloned? There are also social controversies—for example, when insurance companies exclude coverage of people because of their inherited risks of certain diseases.

Inspired in part by the controversies and the publicity, many of today's students come to a course in

genetics with great enthusiasm. The challenges for the teacher are:

- To sustain this enthusiasm;
- To help motivate a desire to understand the principles of genetics in a comprehensive and rigorous way;
- To guide students in gaining an understanding that genetics is not only a set of principles but also an experimental approach to solving a wide range of biological problems; and
- To help students learn to think about genetic problems and about the wider social and ethical issues arising from genetics.

While addressing these challenges in *Principles and Analysis*, we have also tried to show the beauty, logical clarity, and unity of the subject. Endlessly fascinating, genetics is the material basis of the continuity of life.

Chapter Organization

In order to help the student keep track of the main issues and avoid being distracted by details, each chapter begins with a list of the Principles that provide the main focus of the chapter. There is also an **Outline**, showing step by step the path along the way. An opening paragraph gives an overview of the chapter, illustrates the subject with some specific examples, and shows how the material is connected to genetics as a whole. The text makes liberal use of numbered lists and "bullets" in order to help students organize their learning, as well as summary statements set off in special type in order to emphasize important principles. Each chapter ends with a **Summary** and list of **Key Terms** as well as the **Problems**. There is a

Concise Dictionary of Genetics at the end of the book for students to check their understanding of the Key Terms or look up any technical terms they may have forgotten. The Dictionary includes not only the Key Terms but also genetic terms that students are likely to encounter in exploring the Internet or in their further reading. The Dictionary also includes page references for terms defined in the text.

Contents

The organization of the chapters is that favored by the majority of instructors who teach genetics. It is the organization we use in our own courses. An important feature is the presence of an introductory chapter providing a broad overview of genes—what they are, how they function, how they change by mutation, and how they evolve through time. Today, most students learn about DNA in grade school or high school; in our teaching, we have found it rather strange to pretend that DNA does not exist until the middle of the term. The introductory chapter serves to connect the more advanced concepts that students are about to learn with what they already know. It also serves to provide each student with a solid framework for integrating the material that comes later.

Throughout each chapter, there is a balance between observation and theory, between principle and concrete example, and between challenge and motivation. Molecular, classical, and evolutionary genetics are integrated throughout. A number of points related to organization and coverage should be noted:

- **Chapter 1** is an overview of genetics designed to bring students with disparate backgrounds to a common level of understanding. This chapter enables classical, molecular, and evolutionary genetics to be integrated in the rest of the book. Included in Chapter 1 are the basic concepts of genetics: trait, gene, genotype, phenotype, gene interaction, and so forth. Chapter 1 also includes a discussion of the experimental evidence that DNA is the genetic material, as well as a description of DNA structure and how DNA codes for proteins.

- **Chapters 2 through 4** are the core of Mendelian genetics, including segregation and independent assortment, the chromosome theory of heredity, mitosis and meiosis, linkage and chromosome mapping, and tetrad analysis in fungi. Also included is the basic probability framework of Mendelian genetics and the testing of genetic models by means of the chisquare test.

An important principle of genetics, too often ignored or given inadequate treatment, is that of the complementation test and how complementation differs from segregation or other genetic principles. Chapter 2 includes a clear and concise description of complementation, with examples, showing how complementation is used in genetic analysis to group mutations into categories corresponding to genes. This chapter also introduces the use of molecular markers, especially with reference to human genetic analysis, because these are the principal types of genetic markers often used in modern genetics.

- **Chapters 5 and 6** deal with the molecular structure and replication of DNA and with the molecular organization of chromosomes. A novel feature is a description of how basic research that revealed the molecular mechanisms of DNA replication ultimately led to such important practical applications as DNA sequencing and the polymerase chain reaction. This example illustrates the value of basic research in leading, often quite unpredictably, to practical applications. The chapter on chromosome structure also includes a discussion of repetitive DNA sequences in eukaryotic genomes, including transposable elements.

- **Chapter 7** covers the principles of cytogenetics, including variation in chromosome number and the chromosome mechanics of deletions, duplications, inversions, and translocations. Also included is the subject of the human genome with special reference to human chromosome number and structure and the types of aberrations that are found in human chromosomes.

- **Chapter 8** deals with the principles of genetics in prokaryotes with special emphasis on *E. coli* and temperate and virulent bacteriophages. There is an extensive discussion of mechanisms of genetic recombination in microbes, including transformation, conjugation, transduction, and the horizontal transfer of genes present in plasmids, such as F' plasmids.

- **Chapter 9** focuses on recombinant DNA and genome analysis. Included are the use of restriction enzymes and vectors in recombinant DNA, cloning strategies, site-directed mutagenesis, "reverse genetics" (the production of genetically defined, transgenic animals and plants), and applications of genetic engineering. Also discussed are methods used in the analysis of complex genomes, such as the human genome, in which a gene that has been localized by genetic mapping to a region of tens of millions of base pairs must be isolated in cloned form and identified.

- **Chapters 10 through 12** deal with molecular genetics in the strict sense. These chapters include the principles of gene expression, gene regulation, and the genetic control of development. The chapter on development focuses especially on genetic analysis of development in nematodes (*Caenorhabditis elegans*) and *Drosophila*, and there is a thorough examination of the exciting new work on the genetic basis of floral development in *Arabidopsis thaliana*.

- **Chapter 13** covers the molecular details of mutation and the effects of mutagens, including new information on the genetic effects of the Chernobyl nuclear accident. It also covers the rapidly growing field of DNA repair mechanisms, as well as the molecular mechanisms of recombination.
- **Chapter 14** covers organelle genetics.
- **Chapters 15 and 16** deal with population and evolutionary genetics. The discussion of population genetics includes DNA typing in criminal investigations and paternity testing. The material on quantitative genetics includes a discussion of methods by which particular genes influencing quantitative traits (QTLs, or quantitative-trait loci) may be identified and mapped by linkage analysis. QTL mapping is presently one of the most important approaches for identifying the genetic basis of human disease.
- **Chapter 17**, entitled the *Genetics of Biorhythms and Behavior*, illustrates the genetic analysis of behavior with experimental models, including chemotaxis in bacteria, mating behavior in *Drosophila*, and learning in laboratory rodents. This chapter also includes a section on mad cow disease and its relation to the molecular basis of biological rhythms. There is also a section on the genetic determinants of human behavior with examples of the approach using "candidate" genes that led to the identification of the "natural Prozac" polymorphism in the human serotonin transporter gene.

Integrated throughout the book are frequent references to **human genetics**, including sections on the fragile-X syndrome, imprinting, the genetic basis of cancer, expansion of unstable repeats in diseases such as Huntington disease, the relationship of DNA repair enzymes to hereditary colon cancer, the controversial mitochondrial "Eve," genetic diseases associated with defects in biorhythms, and many other special topics, including the human genome project.

Connections

A unique special feature of this book is found in boxes called **Connections**. Each chapter has two or three of these boxes. They are our way of connecting genetics to the world outside the classroom. All of the Connections include short excerpts from the original literature of genetics, usually papers, each introduced with a short explanatory passage. Many of the Connections are excerpts from classic papers, such as Mendel's paper, but by no means all of the "classic" papers are old papers. More than a quarter were published more recently than 1980, including the paper in which the cloning of the sheep *Dolly* was reported.

The pieces are called Connections because each connects the material in the text to something that broadens or enriches its implications. Some of the Connections raise issues of ethics in the application of genetic knowledge, social issues that need to be addressed, or issues related to the proper care of laboratory animals. They illustrate other things as well. Because each Connection names the place where the research was carried out, the student will learn that great science is done in many universities and research institutions throughout the world. Some of the pieces were published originally in French, others in German. These appear in English translation. In papers that use outmoded or unfamiliar terminology, or that use archaic gene symbols, we have substituted the modern equivalent because the use of a consistent terminology in the text and in the Connections makes the material more accessible to the student.

Genetics on the Internet

More than in most fields of biology, genetic resources and genetic information are abundant on the Internet. The most useful sites are not always easy to find. A recent search of Internet sites using the Alta Vista search engine and the keyword *genetics* yielded about 500,000 hits. Most of these are of limited usefulness, but quite a few are invaluable to the student and to the practicing geneticist. The

problem is how to find the really useful ones among the 500,000 sites.

To make the genetic information explosion on the Internet available to the student, we have developed **Internet Exercises**, called **GeNETics on the web**, which make use of Internet resources. One reason for developing these exercises is that genetics is a dynamic science, and most of the key Internet resources are kept up to date. Continually updated, the Internet exercises introduce the newest discoveries as soon as they appear, and this keeps the textbook up to date as well.

The addresses of the relevant genetic sites are not printed in the book. Instead, the sites are accessed through the use of key words that are highlighted in each exercise. The key words are maintained as hot links at the publisher's web site (<http://www.jbpub.com/genetics>) and are kept constantly up to date, tracking the address of each site if it should change. The use of key words also allows an innovation: one exercise in each chapter makes use of a **mutable site** that changes frequently in both the site accessed and the exercise. Students should look at the Internet Exercises. The instructor may wish to make short assignments from some of them, or use them for extra credit or as short term papers. We have included a suggested assignment for each of the exercises, but many instructors may wish to develop their own. We would be pleased to receive suggestions for new web exercises at the Jones and Bartlett home page: <http://www.jbpub.com>.

Problems

Each chapter provides numerous problems for solution, graded in difficulty, for the students to test their understanding. The problems are of three different types:

Review the Basics problems ask for genetic principles to be restated in the student's own words; some are matters of definition or call for the application of elementary principles.

Analysis and Applications problems are more traditional types of genetic problems in which several concepts must be applied in logical order and often require some numerical calculation. The level of mathematics is that of arithmetic and elementary probability as it pertains to genetics. None of the problems uses mathematics beyond elementary algebra.

Challenge Problems are similar to those in Analysis and Applications, but they are a degree more challenging, often because they require a more extensive analysis of data before the question can be answered.

Supplementary Problems, in a special section at the end of the book, consist of over 300 additional problems. These include representatives of all three types of problems found at the ends of the chapters, and they are graded in difficulty. The Supplementary Problems may be used for additional assignments, more practice, or even as examination questions. The problems were generously contributed by geneticist Elena R. Lozovskaya of Harvard University, and they were selected and edited by the authors. Unlike the other problems, the solutions to the Supplementary Problems are not included in the answer section at the end of the book. Solutions are available for the instructor in the *Test Bank and Solutions Manual*.

Guide to Problem Solving

Each chapter contains a **Guide to Problem Solving** that demonstrates problems worked in full. The concepts needed to solve the problem, and the reasoning behind the answer, are explained in detail. The Guide to Problem Solving serves as another level of review of the important concepts used in working problems. It also highlights some of the most common mistakes made by beginning students and gives pointers on how the student can avoid falling into these conceptual traps.

Solutions

All Analysis and Applications Problems and all Challenge Problems are answered in full, with complete methods and explanations, in the answer section at the end of the book. The rationale for giving all the answers is that problems are valuable opportunities to learn. Problems that the student cannot solve are usually more important than the ones that can be solved, because the sticklers usually identify trouble spots, areas of confusion, or gaps in understanding. As often as not, the conceptual difficulties are resolved when the problem is worked in full and the correct approach explained, and the student seldom stumbles over the same type of problem again.

Further Reading

Each chapter also includes recommendations for **Further Reading** for the student who either wants more information or who needs an alternative explanation for the material presented in the book. Some additional "classic" papers and historical perspectives are included. Complete author lists are also given for a few Connections that had too many authors to cite individually in the text.

Illustrations

The art program is spectacular, thanks to the creative efforts of J/B Woolsey Associates, with special thanks to John Woolsey and Patrick Lane. Every chapter is richly illustrated with beautiful graphics in which color is used functionally to enhance the value of each illustration as a learning aid. The illustrations are also heavily annotated with "process labels" explaining step-by-step what is happening at each level of the illustration. These labels make the art inviting as well as informative. They also allow the illustrations to stand relatively independently of the text, enabling the student to review material without rereading the whole chapter.

The art program is used not only for its visual appeal but also to increase the pedagogical value of the book:

- Characteristic colors and shapes have been used consistently throughout the book to indicate different types of molecules—DNA, mRNA, tRNA, and so forth. For example, DNA is illustrated in any one of a number of ways, depending on the level of resolution necessary for the illustration, and each time a particular level of resolution is depicted, the DNA is shown in the same way. It avoids a great deal of potential confusion that DNA, RNA, and proteins are represented in the same manner in Chapter 17 as they are in Chapter 1.
- There are numerous full-color photographs of molecular models in three dimensions; these give a strong visual reinforcement of the concept of macromolecules as physical entities with defined three-dimensional shapes and charge distributions that serve as the basis of interaction with other macromolecules.
- The page design is clean, crisp, and uncluttered. As a result, the book is pleasant to look at and easy to read.

Flexibility

There is no necessary reason to start at the beginning and proceed straight to the end. Each chapter is a self-contained unit that stands on its own. This feature gives the book the flexibility to be used in a variety of course formats. Throughout the book, we have integrated classical and molecular principles, so you can begin a course with almost any of the chapters. Most teachers will prefer starting with the overview in Chapter 1, possibly as suggested reading, because it brings every student to the same basic level of understanding. Teachers preferring the Mendeleary format should continue with Chapter 2; those preferring to teach the details of DNA early should continue with Chapter 5. Some teachers are partial to a chromosomes-early format, which would suggest continuing with Chapter 3, followed by Chapters 2 and 4. A novel approach would be a genomes-first format, which could be implemented by continuing with Chapter 9. Some teachers like to discuss mechanisms of mutation early in the course, and Chapter 13 can easily be assigned early. The writing and illustration program was designed to accommodate a variety of formats, and we encourage teachers to take advantage of this flexibility in order to meet their own special needs.

Supplements

An unprecedented offering of traditional and interactive multimedia supplements is available to assist instructors and aid students in mastering genetics. Additional information and review copies of any of the following items are available through your Jones and Bartlett Sales Consultant.

For the Instructor

- **Test Bank and Solutions Manual.** The Test Bank, authored by Michael Draper of Tufts University with contributions from Patrick McDermot of Tufts University, contains 850 test items, with 50 questions per chapter. There is a mix of factual, descriptive, and quantitative question types. A typical chapter file contains 20 multiple-choice objective questions, 15 fillins, and 15 quantitative problems. A Solutions Manual containing worked solutions of all the supplemental problems in the main text is bound together with the Test Bank. Both the problems and solutions were authored by Elena R. Lozovskaya of Harvard University.
- **Electronic Test Bank.** An electronic version of the Test Bank for preparing customized tests is included in the Instructor's ToolKit. It is available for Macintosh or Windows operating systems.
- **Instructor's ToolKit CD-ROM.** This easy-to-use multimedia tool contains an Image Bank of over 450 figures from the text specially enhanced for classroom presentation. You select the images you need by chapter, topic, and figure number. This lecture aid readily interfaces with other presentation tools, including a complete set of PowerPoint lecture outlines. It also contains key simulated web sites that allow you to bring the Internet into the classroom without the need for a live Internet connection.
- **Visual Genetics Plus: Tutorial and Laboratory Simulations. Faculty Version.** This Mac/IBM CDROM, created by Alan W. Day and Robert L. Dean of the University of Western Ontario and Harry Roy of Rensselaer Polytechnic Institute, is already in use at over 200 institutions worldwide. Visual Genetics 3.0 continues to provide a unique, dynamic presentation tool for viewing key genetic and molecular processes in the classroom. With this new, greatly expanded version of the Virtual Genetics Lab 2.0, instructors can now assign 17 comprehensive lab simulations. You can also bring the lab into the classroom, as the program allows you to perform on-screen tasks such as the selection of mutant colonies, using a pipette to make a dilution series, inoculating mutants to petri dishes to test for response to growth factors, and then to analyze and interpret the data. Through the testing feature and presentation capabilities, you can offer a complete lab environment. Site Licenses and Instructor Copies are available.

- **Video Resource Library.** A full complement of quality videos is available to qualified adopters. Genetics-related topics include: Origin and Evolution of Life, Human Gene Therapy, Biotechnology, the Human Genome Project, Oncogenes, and Science and Ethics.

For the Student

- ***The Gist of Genetics: Guide to Learning and Review.*** Written by Rowland H. Davis and Stephen G. Weller of the University of California, Irvine, this study aid uses illustrations, tables, and text outlines to review all of the fundamentals of genetics. It includes extensive practice problems and review questions with solutions for self-check. The Gist helps students formulate appropriate questions and generate hypotheses that can be tested with classical principles and modern genetic techniques.
- ***GeNETics on the web.*** Corresponding to the end-of-chapter GeNETics on the web exercises, this World Wide Web site offers genetics-related links, articles and monthly updates to other genetics sites on the Web. Material for this site is carefully selected and updated by the authors. Jones and Bartlett Publishers ensures that links for the site are regularly maintained. Visit the *GeNETics on the web* site at <http://www.jbpub.com/genetics>.
- ***An Electronic Companion to Genetics™.*** This Mac/IBM CD-ROM, by Philip Anderson and Barry Ganetzky of the University of Wisconsin, Madison, reviews

important genetics concepts covered in class using state-of-the-art interactive multimedia. It consists of hundreds of animations, diagrams, and videos that dynamically explain difficult concepts to students. In addition, it contains over 400 interactive multiple-choice, "drag and drop," true/false, and fill-in problems. These resources will prove invaluable to students in a self-study environment and to instructors as a lecture-enhancement tool. This CD-ROM is available for packaging exclusively with Jones and Bartlett Publishers texts.

• **Visual Genetics Plus: Tutorial and Laboratory Simulations. Student Version.** This Mac/IBM CD-ROM, created by Alan W. Day and Robert L. Dean of the University of Western Ontario and Harry Roy of Rensselaer Polytechnic Institute, is already in use at over 200 institutions worldwide. Visual Genetics 3.0 affords a dynamic multimedia review of key genetic and molecular processes, including a greatly expanded version of the Virtual Genetics Lab 2.0, with which students can work on 17 comprehensive lab simulations. The lab allows students to perform tasks on screen—such as selecting mutant colonies, making a dilution series, inoculating mutants into petri dishes to test for response to growth factors—and then guides them in analyzing and interpreting the data. The Student Version is available for purchase and can be packaged with our text.

Acknowledgments

We are indebted to the many colleagues whose advice and thoughts were immensely helpful throughout the preparation of this book. These colleagues range from specialists in various aspects of genetics who checked for accuracy or suggested improvement to instructors who evaluated the material for suitability in teaching or sent us comments on the text as they used it in their courses.

Jeremy C. Ahouse, Brandeis University

John C. Bauer, Stratagene, Inc., La Jolla, CA

Mary K. B. Berlyn, Yale University

Pierre Carol, Université Joseph Fourier, Grenoble, France

John W. Drake, National Institute of Environmental Health Sciences, Research Triangle Park, NC

Jeffrey C. Hall, Brandeis University

Steven Henikoff, Fred Hutchinson Cancer Research Center, Seattle, WA

Joyce Katich, Monsanto, Inc., St. Louis, MO

Jeane M. Kennedy, Monsanto, Inc., St. Louis, MO

Jeffrey King, University of Berne, Switzerland

K. Brooks Low, Yale University

Gustavo Maroni, University of North Carolina

Jeffrey Mitton, University of Colorado, Boulder

Gisela Mosig, Vanderbilt University

Robert K. Mortimer, University of California, Berkeley

Ronald L. Phillips, University of Minnesota

Robert Pruitt, Harvard University

Pamela Reinagel, California Institute of Technology, Pasadena

Kenneth E. Rudd, National Library of Medicine

Leslie Smith, National Institute of Environmental Health Sciences, Research Triangle Park, NC

Johan H. Stuy, Florida State University

Irwin Tessman, Purdue University

Kenneth E. Weber, University of Southern Maine

We would also like to thank the reviewers, listed below, who reviewed one or more chapters and who, in several cases, reviewed the complete fourth edition manuscript. Their comments and recommendations helped improve the content, organization, and presentation of the material. We offer special thanks to Dick Morel, who carefully reviewed and commented on all of the illustrations as well as the text.

Laura Adamkewicz, George Mason University

Peter D. Ayling, University of Hull (UK)

Anna W. Berkovitz, Purdue University

John Celenza, Boston University

Stephen J. D'Surney, University of Mississippi

Kathleen Dunn, Boston College

David W. Francis, University of Delaware

Mark L. Hammond, Campbell University

Richard Imberski, University of Maryland

Sally A. MacKenzie, Purdue University

Kevin O'Hare, Imperial College (UK)

Peggy Redshaw, Austin College

Thomas F. Savage, Oregon State University

David Shepard, University of Delaware

Charles Staben, University of Kentucky

David T. Sullivan, Syracuse University

James H. Thomas, University of Washington

We also wish to acknowledge the superb art, production, and editorial staff who helped make this

book possible: Mary Hill, Patrick Lane, Andrea Fincke, Judy Hauck, Bonnie Van Slyke, Sally Steele, John Woolsey, Brian McKean, Kathryn Twombly, Rich Pirozzi, Mike Campbell, and Tom Walker. Much of the credit for the attractiveness and readability of the book should go to them. Thanks also to Jones and Bartlett, the publishers, for the high quality of the book production. We are also grateful to the many people, acknowledged in the legends of the illustrations, who contributed photographs, drawings, and micrographs from their own research and publications, especially those who provided color photographs for this edition. Every effort has been made to obtain permission to use copyrighted material and to make full disclosure of its source. We are grateful to the authors, journal editors, and publishers for their cooperation. Any errors or omissions are wholly inadvertant and will be corrected at the first opportunity.

Introduction: For the Student

In signing up for a genetics course, our students often wonder how much work is going to be required, how much time it will take to do the reading and written assignments, how hard the examinations will be, and what is their likelihood of getting a good grade. These are perfectly legitimate issues, and you should not feel guilty if they are foremost in your mind.

You may also be wondering what you are going to learn by taking a course in genetics. Will the material be interesting? Is there any reason to study genetics other than to satisfy an academic requirement? At the end of the course, will you be glad that you took it? Will there be any practical value to what you will learn? This introduction is designed to reassure you that the answer to each question is yes. The study of genetics is relevant not only to biologists but to all members of our modern, complex, technological society. Understanding the principles of genetics will help you to make informed decisions about numerous matters of political, scientific, and personal concern.

At least 4000 years ago in the Caucasus, the Middle East, Egypt, South America, and other parts of the world, farmers recognized that they could improve their crops and their animals by selective breeding. Their knowledge was based on experience and was very incomplete, but they did recognize that many features of plants and animals were passed from generation to generation. They discovered that desirable traits—such as size, speed, and weight of animals—could sometimes be combined by controlled mating and that, in plants, crop yield and resistance to arid conditions could be combined by cross-pollination. The ancient breeding programs were not based on much solid information because nothing was known about genes or any of the principles of heredity. In a few instances, the pattern of hereditary transmission of a human trait came to be recognized. One example is *hemophilia*, or failure of the blood to clot, which results in life-threatening bleeding from small cuts and bruises. By the second century of the present era, rules governing exemptions from circumcision had been incorporated into the Talmud, indicating that several key features of the mode of inheritance of hemophilia were understood. The Talmud's exemptions apply in the case of a mother who lost two sons from excessive bleeding following circumcision: Subsequent boys born to the same mother, and all boys born to her sisters, were exempt. However, the paternal half brothers of a boy who had died from excessive bleeding were not exempt. (Paternal half brothers have the same father but a different mother.) These rules of exemption from circumcision make very good sense when judged in light of our modern understanding of the inheritance of hemophilia, as you will learn in Chapter 3.

The scientific study of heredity is called **genetics**. The modern approach to genetics can be traced to the mid-nineteenth century with Gregor Mendel's careful analyses of inheritance in peas. Mendel's experiments were simple and direct and brought forth the most significant principles that determine how traits are passed from one generation to the next. In Chapter 2, you will learn the rules followed by genes and chromosomes as they pass from generation to generation, and you will be able to calculate in many instances the probabilities by which organisms with particular traits will be produced. Mendel's kind of experiments, which occupied most of genetic research until the middle of the twentieth century, is called **transmission genetics**. Some people have called it formal genetics, because the subject can be understood and the rules clearly seen without any reference to the biochemical nature of genes or gene products.

Beginning about 1900, geneticists began to wonder about a subject we now call **molecular genetics**. Is the gene a known kind of molecule? How can genetic information be encoded in a molecule? How is the genetic information transmitted from one generation to the next? In what way is the genetic information changed in a mutant organism? At that time, there was no logical starting point

for such an investigation, no experimental "handle." In the 1940s, critical observations were made that implicated the molecule deoxyribonucleic acid (DNA), first discovered in 1869. You will learn about these experiments in Chapter 1. With the discovery of the structure of DNA in 1953 by Watson and Crick, genetics entered the DNA age. Within a decade, there came an understanding of the chemical nature of genes and how genetic information is stored, released to a cell, and transmitted from one generation to the next. During the first three decades after the discovery of DNA structure, the body of genetic knowledge grew with a two-year doubling time. These were exciting times, and you will be presented with a distillation of these findings in the chapters of this book that deal with **molecular genetics**.

Since the early 1970s, genetics has undergone yet another revolution: the development of recombinant DNA technology. This technology is a collection of methods that enable genes to be transferred, at the will of the molecular geneticist, from one organism to another. This branch of genetics is known as **genetic engineering**. Genetic engineering has had an enormous impact in genetic research, particularly in our ability to understand

Connection Balancing Act

Thomas Hunt Morgan 1913

Columbia University, New York,
New York

Genetics and cell biology have both advanced with surprising rapidity in recent years. Hardly a week goes by without a new discovery of notable importance being reported in the pages of Science or Nature or some other major research journal. Nontechnical accounts of new discoveries are regularly reported in the popular press and on television. We are in the midst of a knowledge explosion—doubtless you remember being told this before. We are so often reminded that we live in a fast-paced world and should be proud to be speeding along. But hit the brakes, and pause for a moment, to reread the first sentence of this book is an almost direct quotation of the words that Thomas Hunt Morgan wrote to introduce his first book on genetics. This was in 1913. Morgan was one of the pioneers of modern genetics, and genetics in 1913 was poised for truly spectacular advances. He could scarcely have imagined what modern genetics would be like—how much we would know about some things, how little we would know about others; how powerful the methods would be in some ways, how limited they would be in others. Morgan did see one thing clearly. It was that the key to understanding biology is to maintain the right balance among different approaches of studying organisms—through genetics, cell biology, molecular biology, biochemistry, biophysics, developmental biology, neurobiology, evolutionary biology, and ecology. Maintaining the right balance for today's students has been our primary goal in writing this book.

Two lines of research have developed with surprising rapidity in recent years. Their development has been independent, but at many stages in their progress they have looked to each other for help. The study of the cell has furnished some fundamental facts connected with problems of heredity. The modern study of heredity has proven itself to be an instrument even more subtle in the analysis of the materials of the cells than actual observations on the germ cells themselves. The time has come, we think, when a failure to recognize the close bond between these two modern lines of advance can no longer be interpreted as wise or cautious skepticism. An anarchistic spirit in science does not always mean greater profundity; it is our attitude toward science more correct because we are unduly skeptical toward every advance. To maintain the right balance is the hardest task we have to meet. What we most fear is that in attempting to formulate some of the difficult problems of present-day interest we may appear to make at times unqualified statements in a dogmatic spirit. All conclusions in science are relative and subject to change for change in science does not mean so much that what has gone before is wrong, as the discovery of a better strategic position than the one last held.

Source: *Heredity and Sex*. NY: Columbia University Press.

gene expression and its regulation in plants and animals. Topics previously unapproachable suddenly became amenable to experimental investigation. Currently, genetic engineering is providing us with new tools of great economic importance and of value in medical practice. Current projects of great interest include the genetic modification of plants and domesticated animals and the production of clinically important substances.

Beginning in the 1980s came the new emphasis on **genomics**, the application of recombinant DNA strategies to the study of whole **genomes** (the totality of genetic information in an organism) rather than single genes. The complete set of DNA instructions has been determined by direct DNA sequencing in a number of viruses, cellular organelles such as

mitochondria, several bacteria, and the yeast *Saccharomyces cerevisiae*. Programs are also underway to determine the complete DNA sequence of other **model organisms**. (In genetics, a **model organism** is a species that is studied as an example to learn basic principles that we hope will be applicable to other organisms.) Just on the horizon is the capability of determining the complete DNA sequence in the human genome. The availability of genomic sequences opens up new approaches for genetics because it turns the subject on its head. Instead of starting with a mutant organism that has some physical abnormality, attempting to identify the gene responsible, and determining the DNA sequence, one can now start with a DNA sequence that has already been determined and try to learn what the gene does.

By far the greatest practical influence of genetics has been in the fields of medicine and agriculture. There have been many important contributions to modern clinical practice, and progress is accelerating because of the increased emphasis on genomic analysis. Genetic experiments have revealed thousands of new genes and markers in the human genome and have given us new methods for the detection of mutant genes—not only in affected individuals but also in their relatives and in members of the population at large. These methods have given genetic counseling new meaning. Human beings are at risk for

any of several thousand different inherited diseases. Married couples can be informed of the possibility of their producing an affected offspring and can now make choices between childbearing and adoption. Consider the relief of a woman and man who learn that they do not carry a particular defective gene and can produce a child without worry. Even when an offspring might be affected with a genetic disorder, techniques are available to determine if a fetus does, in fact, carry a mutant gene.

In agriculture, studies of the genetic composition of economically important plants have enabled plant breeders to institute rational programs for developing new varieties. Among the more important plants that have been developed are high-yielding strains of corn and dwarf wheat, disease-resistant rice, corn with an altered and more nutritious amino acid composition (high-lysine corn), and wheat that grows faster, allowing crops to be grown in short-season regions such as Canada and Sweden. You will be introduced to the techniques for developing some of these strains in this book. Often new plant varieties have shortcomings, such as a requirement for increased amounts of fertilizer or a decreased resistance to certain pests. How to overcome these shortcomings is a problem for the modern geneticist, who has the job of manipulating the inherited traits. Genetic engineering is also providing new procedures for such manipulations, and quite recently there have been dramatic successes.

A few words about the book. Each chapter contains two or three **Connections** set off in special boxes. Each connects the material in the text to the real world of genetics outside the classroom. Some of the Connections are excerpts from classic papers, including Mendel's paper. Others are very recent, such as the paper that reports the cloning of an adult sheep. Some of the Connections raise issues of ethics in the application of genetic knowledge, social issues that need to be addressed, issues related to the proper care of laboratory animals, or other matters. We have included a Connection in this Introduction to give you a taste. For an appreciation of genetics in a broad historical context related at many points to contemporary research and social and ethical issues, we urge you to connect with the Connections. There is a complete listing, chapter by chapter, of all the Connections in the Table of Contents. Following the Table of Contents is a complete list of all the material excerpted, shown in chronological order.

Each chapter comes with a set of **Internet Exercises**, called **GeNETics on the web**, which will introduce you to the genetic resources and information that can be accessed through the Internet. These are important because genetics is more richly represented on the Internet than any other field of biology. Each exercise uses a key word in describing an issue or a problem. The key words are maintained as hot links at the publisher's web site (<http://www.jbpub.com/genetics>) and are kept constantly up to date. Each exercise comes with a short written component that your instructor may wish to assign. We urge you to go through the web exercises even if they are not assigned, as they will help you to become familiar with some of the extraordinary resources that are out there. We should mention two special types of exercises. One is the **mutable site** in which the site and the exercise are changed frequently. You can check back on a mutable site that you have explored before, and there will be a good chance that it will have changed in the meantime. The other special site is the **PIC site**, which connects you to a genetics site chosen for its visual appeal.

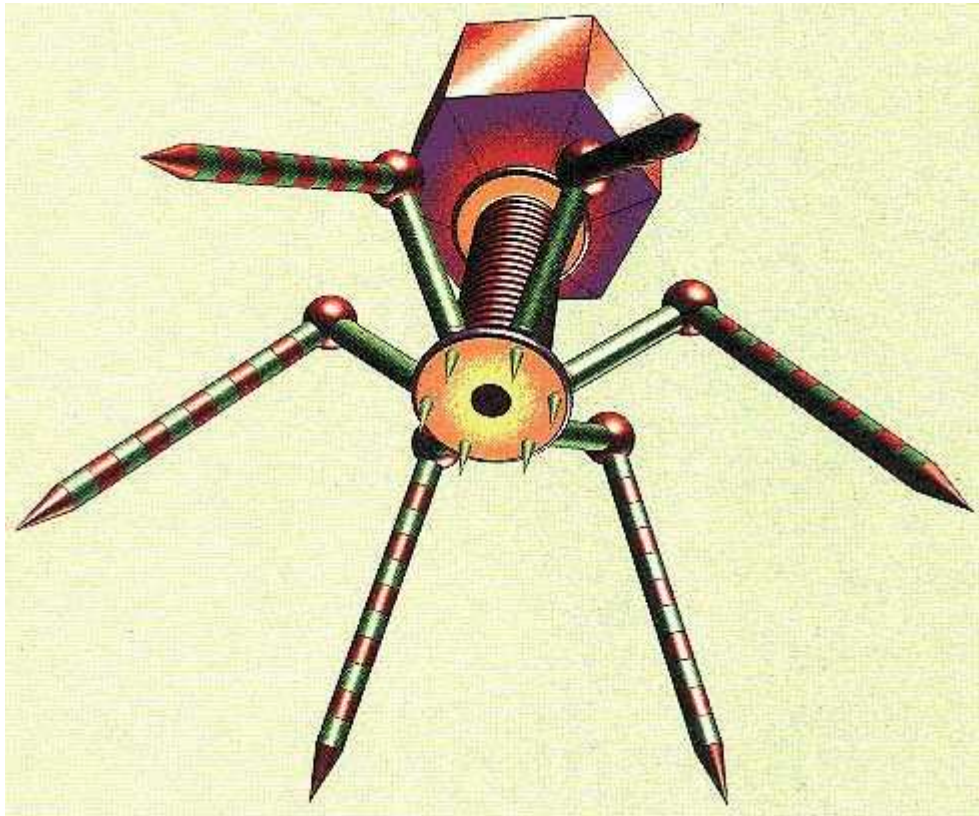
As a pedagogical aid, important terms are printed in **boldface** in the text. These terms are collected at the end of each chapter in a section entitled **Key Terms**. You should know their meanings because they form the basic vocabulary of genetics. If necessary, you can look them up in the **Concise Dictionary of Genetics** at the back of the book. Each chapter also includes a **Summary** at the end of the text. Sample problems are worked in the section titled **Guide to Problem Solving**. Each chapter ends with a fairly large collection of problems. These are of three types:

Review the Basics problems ask you to restate genetic principles or definitions in your own words or to apply elementary principles.

Analysis and Applications problems require you to apply several concepts in logical order and usually to do some numerical calculation. (The calculations use only simple arithmetic, so there is no reason to be intimidated even if higher mathematics is not a comfortable part of your repertoire.)

Challenge Problems are similar in nature but a little more difficult because you may need to analyze some data to solve the problem.

It is essential that you work as many of the problems as you can, because experience has shown that practice with problems is a good way to learn genetics and to identify particular points or concepts that have been misunderstood. Sometimes it is not even necessary to solve a problem completely but only to read the problem and decide whether you could solve it if asked to do so. The **Answers** to all of the problems, and full explanations, are given at the back of the book. A problem will be more useful to you if you take a fair shot at it before turning to the answer. The back of the book also includes a large set of **Supplementary Problems**, without answers, for still more practice. There is nothing better than solving problems not only to test your knowledge but to make it part of your long-term memory.



A stylized version of a bacteriophage that very much resembles the phage T2 used in the Hershey-Chase experiments.
[Courtesy of Paul Dowrick, © Phage *et al* Ltd.]

Chapter 1— The Molecular Basis of Heredity and Variation

CHAPTER OUTLINE

1-1 DNA: The Genetic Material

Experimental Proof of the Genetic Function of DNA

Genetic Role of DNA in Bacteriophage

1-2 DNA Structure: The Double Helix

1-3 An Overview of DNA Replication

1-4 Genes and Proteins

Transcription of DNA Makes RNA

Translation of RNA Makes Protein

1-5 Mutation

1-6 How Genes Determine Traits

Pleiotropy: One Gene Can Affect More Than One Trait

Epistasis: One Trait Can Be Affected by More Than One Gene

Effects of the Environment

1-7 Evolution

The Molecular Continuity of Life

Adaptation and Diversity

The Role of Chance in Evolution

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Further Reading

GeNETics on the web

PRINCIPLES

- Genes control biologically inherited traits; a trait that is genetically determined can also be influenced by environmental factors.
- Genes are composed of the chemical deoxyribonucleic acid (DNA).
- DNA replicates to form (usually identical) copies of itself.
- DNA contains a code specifying what types of enzymes and other proteins are made in cells.
- DNA occasionally mutates, and the mutant forms specify altered proteins.
- Genes interact with one another in sometimes complex ways.
- Organisms change genetically through generations in the process of biological evolution.

CONNECTIONS

CONNECTION: It's the DNA!

Oswald T. Avery, Colin M. MacLeod, and Maclyn McCarty 1944

Studies on the chemical nature of the substance inducing transformation of pneumococcal types

CONNECTION: Shear Madness

Alfred D. Hershey and Martha Chase 1952

Independent functions of viral protein and nucleic acid in growth of bacteriophage

The members of any biological species are similar in some characteristics but different in others. For example, all human beings share a set of observable characteristics, or **traits**, that define us as a species. We have a backbone and a spinal cord; these traits are among those that define us as a type of vertebrate. We are warm blooded and feed our young with milk from mammary glands; these traits are among those that define us as a type of mammal. We are, in finer detail, a type of primate that habitually stands upright and has long legs, relatively little body hair, a large brain, a flat face with a prominent nose, jutting chin, distinct lips, and small teeth. These traits set us apart from other primates, such as chimpanzees and gorillas. The biological characteristics that define us as a species are inherited, but they do not differ from one person to the next.

Within the human species, however, there is also much variation. Traits such as hair color, eye color, skin color, height, weight, and personality characteristics are tremendously variable from one person to the next. There is also variation in health-related traits, such as predisposition to high blood pressure, diabetes, chemical dependence, mental depression, and the Alzheimer disease. Some of these traits are inherited biologically, others are inherited culturally. Eye color results from biological inheritance; the native language we speak results from cultural inheritance. Many traits are influenced jointly by biological inheritance and environmental factors. For example, weight is determined in part by inheritance but also in part by eating habits and level of physical activity.

The study of biologically inherited traits is **genetics**. Among the traits studied in genetics are those that are influenced in part by the environment. The fundamental concept of genetics is

Inherited traits are determined by elements of heredity, called **genes**, that are transmitted from parents to offspring in reproduction.

The elements of heredity and some basic rules governing their transmission from generation to generation were discovered by Gregor Mendel in experiments with garden peas. His results were published in 1866. Mendel's experiments are among the most beautifully designed, carefully executed, and elegantly interpreted in the history of experimental science. Mendel interpreted his data in terms of a few abstract rules by which hereditary elements are transmitted from parents to offspring. Three years later, in 1869, Friedrich Miescher discovered a new type of weakly acid substance, abundant in the nuclei of salmon sperm and white blood cells. At the time he had no way of knowing that it would turn out to be the chemical substance of which genes are made. Miescher's weak acid, the chemical substance of the gene, is now called **deoxyribonucleic acid (DNA)**. However, the connection between DNA and heredity was not demonstrated until about the middle of the twentieth century. How was this connection established?

1.1—

DNA: The Genetic Material

The importance of the cell nucleus in inheritance became apparent in the 1870s with the observation that the nuclei of male and female reproductive cells fuse in the process of fertilization. This observation suggested that there was something inside the sperm and egg nucleus that was responsible for inherited characteristics. The next major advance was the discovery of thread-like objects inside the nucleus that become visible in the light microscope when stained with certain dyes; these threads were called **chromosomes**. As we shall see in Chapter 3, chromosomes have a characteristic "splitting" behavior in cell division, which ensures that each daughter cell receives an identical complement of chromosomes. By 1900 it had become clear that the number of chromosomes is constant within each species but differs among species. The characteristics of chromosomes made it seem likely that they were the carriers of the genes.

By the 1920s, more and more evidence suggested a close relationship between DNA and the genetic material. Studies using special stains showed that DNA, in addition to certain proteins, is present in chromosomes. Furthermore, investigations

disclosed that almost all cells of a given species contain a constant amount of DNA, whereas the amount and kinds of proteins and other molecules differ greatly in different cell types. The indirect evidence that genes are DNA was rejected because crude chemical analyses of DNA had suggested (incorrectly) that it lacks the chemical diversity needed for a genetic substance. In contrast, proteins were known to be an exceedingly diverse collection of molecules. And so, on the basis of incorrect data, it became widely accepted that proteins were the genetic material and that DNA merely provided the structural framework of chromosomes. Against the prevailing opinion that genes are proteins, experiments purporting to demonstrate that DNA is the genetic material had also to demonstrate that proteins are *not* the genetic material. Two of the experiments regarded as decisive are described in this section.

Experimental Proof of the Genetic Function of DNA

The first evidence that genes are DNA came from studies of bacteria that cause pneumonia. Bacterial pneumonia in mammals is caused by strains of *Streptococcus pneumoniae* that are able to synthesize a slimy "capsule" around each cell. Strains that lack a capsule do not cause pneumonia. The capsule is composed of a complex carbohydrate (polysaccharide) that protects the bacterium from the immune response of the infected animal and enables the bacterium to cause the disease. When a bacterial cell is grown on solid medium, it undergoes repeated cell divisions to form a visible clump of cells called a **colony**. The enveloping capsule gives the colony a glistening or smooth (S) appearance. Some strains of *S. pneumoniae* are not able to synthesize a capsule. As a result, they form colonies that have a rough (R) surface (Figure 1.1). The R strains do not cause pneumonia, because without their capsules, the bacteria are attacked by the immune system of the host. Both types of bacteria "breed true" in the sense that the progeny formed by cell division have the capsular type of the parent, either S or R.

When mice are injected either with living R cells or with dead S cells killed by heat, they remain healthy. However, it was discovered in 1928 that mice often died of pneumonia when injected with a mixture containing a small number of living R cells and a large number of dead S cells. Bacteria isolated from blood samples of the mice infected with the mixture produced S cultures with a capsule typical of the injected S cells, even though the injected S cells had been killed by heat. Therefore, the material containing the dead S cells that was injected must have included a substance that could convert, or transform, otherwise harmless cells of the R bacterial strain into S strain cells with the ability to resist the immunological system of the mouse, multiply, and cause pneumonia. In other words, there was a genetic **transformation** of an R cell into an S cell. Furthermore, the new genetic characteristics were inherited by descendants of the transformed bacteria.

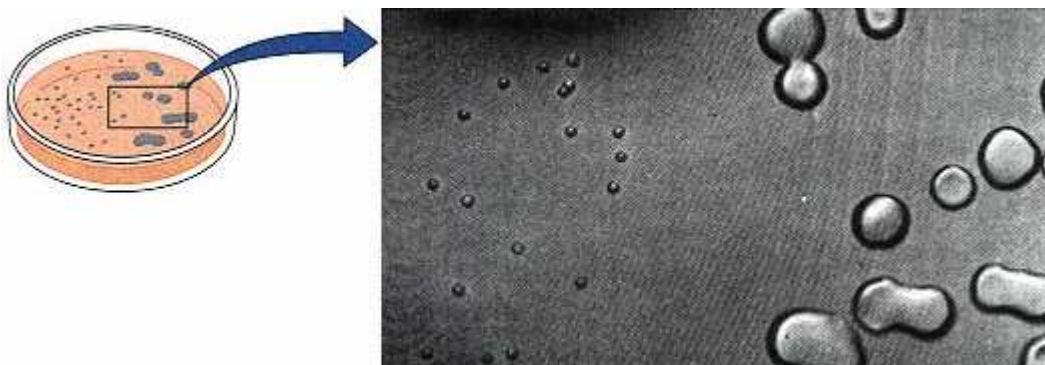


Figure 1.1

Colonies of rough (R, the small colonies) and smooth (S, the large colonies) strains of *Streptococcus pneumoniae*. The S colonies are larger because of the capsule on the S cells.

[Photograph from O.T. Avery, C.M. MacLeod, and M. McCarty. 1944. *J. Exp. Med.* 79: 137.]

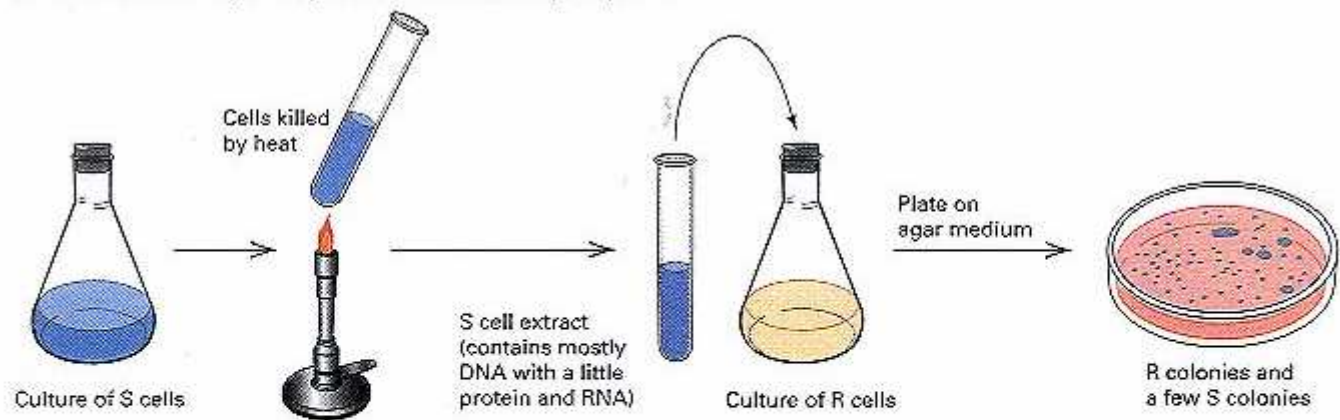
Connection It's the DNA!

Oswald T. Avery, Colin M. MacLeod, and Maclyn McCarty 1944
The Rockefeller University, New York, New York <i>Studies on the Chemical Nature of the Substance Inducing Transformation of Pneumococcal Types</i>
<i>This paper is one of the milestones of molecular biology. Genetics and biochemistry were at last united through the finding that DNA was the chemical substance of the genetic material. There is very little biology in the paper beyond the use of Streptococcus (then called Pneumococcus) to ascertain whether a particular batch of extract, or an extract treated in some manner, contained the active substance able to transform type R into type S cells. The thrust of the paper is biochemistry: purifying the substance, showing that no known macromolecules other than DNA could be found in the extract, and demonstrating that the transforming activity could be destroyed by enzymes that attack DNA but not by protease or RNase enzymes.</i>
Biologists have long attempted by chemical means to induce in higher organisms predictable and specific changes which thereafter could be transmitted as hereditary characters. Among microorganisms the most striking example of inheritable and specific alterations in cell structure and function that can be experimentally induced is the transformation of specific types of <i>Pneumococcus</i>. This phenomenon was first described by Griffith, who succeeded in transforming an attenuated [nonvirulent] and nonencapsulated (R) variant into fully encapsulated and virulent (S) cells. . . . The present paper is concerned
<i>Within the limit of the analytical methods, the active fraction contains no demonstrable protein, lipid, or polysaccharide and consists principally, if not solely, of a highly polymerized form of deoxyribonucleic acid.</i>
with a more detailed analysis of the phenomenon of transformation of specific types of <i>Pneumococcus</i>. The major interest has centered in attempts to isolate the active principle from crude extracts and to identify its chemical nature, or at least to characterize it sufficiently to place it in a general group of known chemical substances. . . . A biologically active fraction has been isolated in highly purified form which in exceedingly minute amounts is capable under appropriate cultural conditions of inducing the transformation of unencapsulated R variants into fully encapsulated forms of the same specific type as that of the heat-killed microorganisms from which the inducing material was recovered. . . . Within the limit of the analytical methods, the active fraction contains no demonstrable protein, lipid, or polysaccharide and consists principally, if not solely, of a highly polymerized form of deoxyribonucleic acid. . . . Various enzymes have been tested for their capacity to destroy the transforming activity. Extracts to which were added crystalline trypsin and chymotrypsin [proteases], or combinations of both, suffered no loss in activity. . . . Prolonged treatment with crystalline ribonuclease under optimal conditions caused no demonstrable decrease in transforming activity. . . . The blood serum of several mammalian species contains an enzyme which causes the depolymerization of deoxyribonucleic acid; fresh dog and rabbit serum are capable of completely destroying transforming activity. . . . The evidence presented supports the belief that a nucleic acid of the deoxyribose type is the fundamental unit of the transforming principle.
Source: <i>Journal of Experimental Medicine</i> 79: 137-158

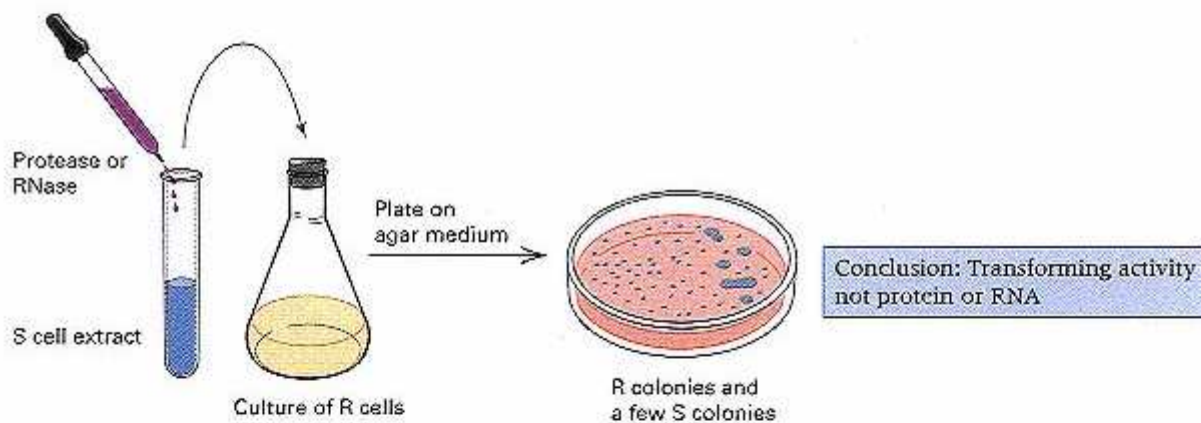
What substance was present in the dead S cells that made transformation possible? In the early 1940s, components of dead S cells were extracted and added to R cell cultures. The key experiment was one in which DNA was extracted from dead S cells and added to growing cultures of R cells and the resulting mixture spread onto an agar surface (Figure 1.2A). Among the R colonies, a few of type S appeared! Although the DNA preparations may still have contained traces of protein and RNA, the addition of an enzyme that destroys proteins (a *protease* enzyme) or one that destroys RNA (an *RNase* enzyme) did not eliminate the transforming activity (Figure 1.2B). On the other hand, the addition of an enzyme that destroys DNA completely eliminated the transforming activity (Figure 1.2C). These experiments were carried out by Oswald Avery, Colin MacLeod, and Maclyn McCarty at the Rockefeller University. They concluded their landmark report by noting that "the evidence presented supports the belief that a

nucleic acid of the deoxyribose type is fundamental

(A) The transforming activity in S cells is not destroyed by heat



(B) The transforming activity is not destroyed by either protease or RNase



(C) The transforming activity is destroyed by DNase

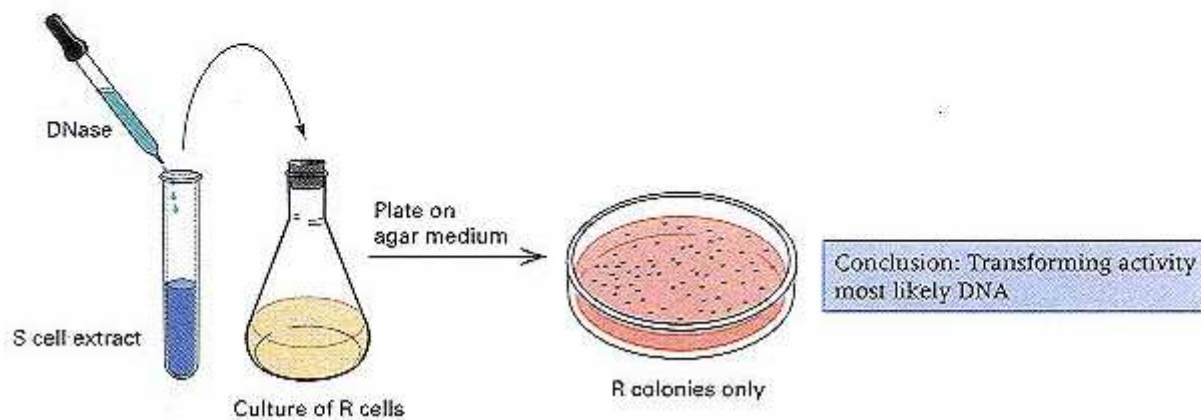


Figure 1.2

A diagram of the experiment that demonstrated that DNA is the active material in bacterial transformation. (A) Purified DNA extracted from heat-killed S cells can convert some living R cells into S cells, but the material may still contain undetectable traces of protein and/or RNA. (B) The transforming activity is not destroyed by either protease or RNase. (C) The transforming activity is destroyed by DNase and so probably consists of DNA.

unit of the transforming principle." In other words, DNA seems to be the genetic material.

Genetic Role of DNA in Bacteriophage

A second important finding concerned a type of virus that infects bacterial cells. The virus, T2 by name, is known as a **bacteriophage**, or **phage** for short, because it infects bacterial cells. Bacteriophage means "bacteria-eater." T2 infects cells of the intestinal bacterium *Escherichia coli*. A T2 particle is illustrated in Figure 1.3. It is exceedingly small, yet it has a complex structure composed of head (which contains the phage DNA), collar, tail, and tail fibers. (For comparison, consider that the head of a human sperm is about 30 to 50 times larger in both length and width than the T2 head.) T2 infection begins with attachment of a phage particle by the tip of its tail to the bacterial cell wall, entry of phage material into the cell, multiplication of this material to form a hundred or more progeny phage, and release of progeny by disruption of the bacterial host cell.

Because DNA contains phosphorus but no sulfur, and proteins usually contain some sulfur but no phosphorus, the DNA and proteins in a phage particle can be labeled differentially by the use of radioactive isotopes of the two elements. This difference was put to use by Alfred Hershey and Martha Chase in 1952, working at the Cold Spring Harbor Laboratories. By that time it was already known that T2 particles are composed of DNA and protein in approximately equal amounts. Hershey and Chase produced particles with radioactive DNA by infecting *E. coli* cells that had been grown for several generations in a medium containing ^{32}P (a radioactive isotope of phosphorus) and then collecting the phage progeny. Other particles with labeled proteins were obtained in the same way, using a medium that contained ^{35}S (a radioactive isotope of sulfur).

The experiments are summarized in Figure 1.4. Nonradioactive *E. coli* cells were infected with phage labeled with *either* ^{32}P (Figure 1.4A) or ^{35}S (Figure 1.4B) in order to follow the proteins and DNA separately. Infected cells were concentrated by centrifugation, resuspended in fresh medium, and then agitated in a kitchen blender to shear attached phage material from the cell surfaces. The blending was found to have no effect on the subsequent course of the infection, which implies that the genetic material must enter the infected cells very soon after phage attachment. When intact bacteria were separated from the material removed by blending, most of the radio

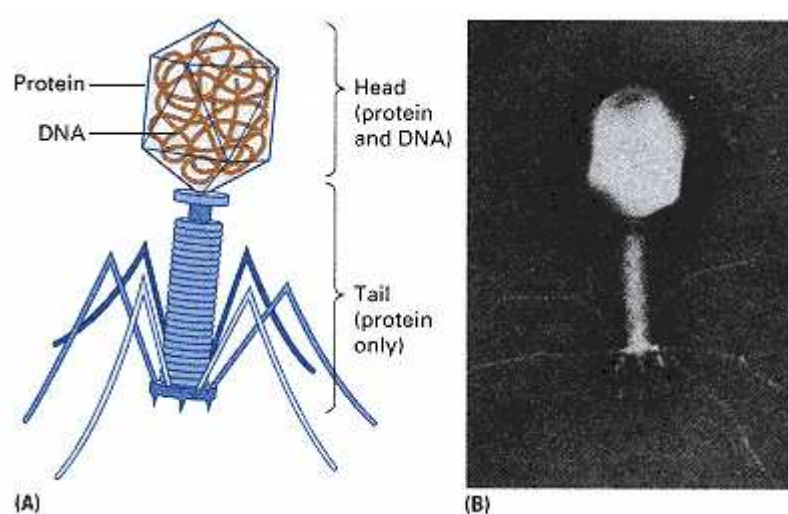


Figure 1.3

(A) Drawing of *E. coli* phage T2, showing various components. The DNA is confined to the interior of the head. (B) An electron micrograph of phage T4, a closely related phage.

[Electron micrograph courtesy of Robley Williams.]

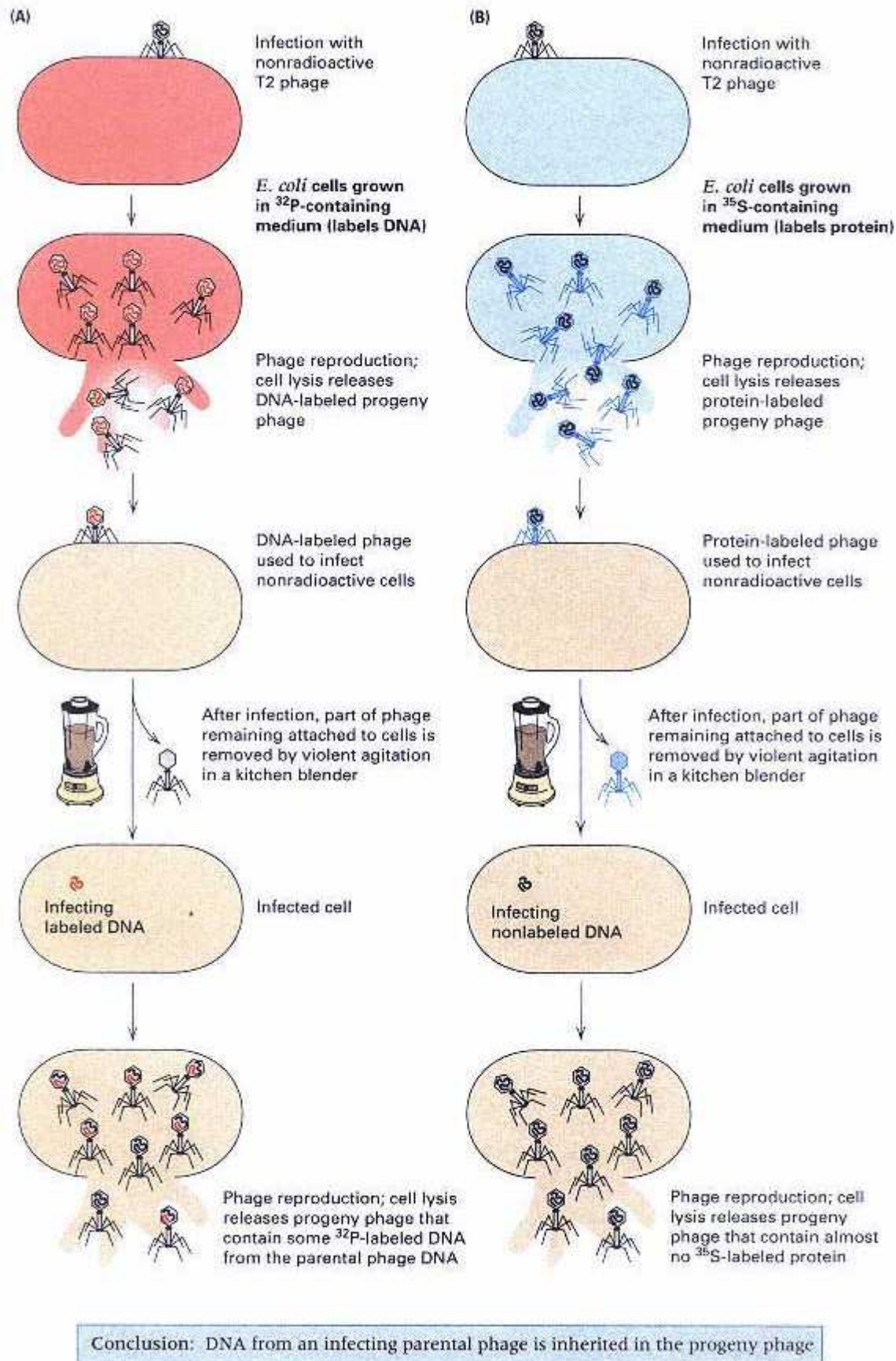


Figure 1.4

The Hershey-Chase ("blender") experiment, which demonstrated that DNA, not protein, is responsible for directing the reproduction of phage T2 in infected *E. coli* cells. (A) Radioactive DNA is transmitted to progeny phage in substantial amounts. (B) Radioactive protein is transmitted to progeny phage in negligible amounts.

Connection Shear Madness

Alfred D. Hershey and Martha Chase 1952
Cold Spring Harbor Laboratories, Cold Spring Harbor, New York <i>Independent Functions of Viral Protein and Nucleic Acid in Growth of Bacteriophage</i>
<i>Published a full eight years after the paper of Avery, MacLeod and McCarty, the experiments of Hershey and Chase get equal billing. Why? Some historians of science suggest that the Avery et al. experiments were "ahead of their time." Others suggest that Hershey had special standing because he was a member of the "in group" of phage molecular geneticists. Max Delbrück was the acknowledged leader of this group, with Salvador Luria close behind. (Delbrück, Luria and Hershey shared a 1969 Nobel Prize.) Another possible reason is that whereas the experiments of Avery et al. were feats of strength in biochemistry, those of Hershey and Chase were quintessentially genetic. Which macromolecule gets into the hereditary action, and which does not? Buried in the middle of this paper, and retained in the excerpt, is a sentence admitting that an earlier publication by the researchers was a misinterpretation of their preliminary results. This shows that even first-rate scientists, then and now, are sometimes misled by their preliminary data. Hershey later explained, "We tried various grinding arrangements, with results that weren't very encouraging. When Margaret McDonald loaned us her kitchen blender the experiment promptly succeeded."</i>
The work [of others] has shown that bacteriophages T2, T3, and T4 multiply in the bacterial cell in a non-infective [immature] form. Little else is known about the vegetative [growth] phase of these viruses. The experiments reported in this paper show that one of the first steps in the growth of T2 is the release from its protein coat of the
<i>Our experiments show clearly that a physical separation of the phage T2 into genetic and nongenetic parts is possible.</i>
nucleic acid of the virus particle, after which the bulk of the sulfur-containing protein has no further function. . . . Anderson has obtained electron micrographs indicating that phage T2 attaches to bacteria by its tail. . . . It ought to be a simple matter to break the empty phage coats off the infected bacteria, leaving the phage DNA inside the cells. . . . When a suspension of cells with ³⁵S- or ³²P-labeled phage was spun in a blender at 10,000 revolutions per minute, . . . 75 to 80 percent of the phage sulfur can be stripped from the infected cells. . . . These facts show that the bulk of the phage sulfur remains at the cell surface during infection. . . . Little or no ³⁵S is contained in the mature phage progeny. . . . Identical experiments starting with phage labeled with ³²P show that phosphorus is transferred from parental to progeny phage at yields of about 30 phage per infected bacterium. . . . [Incomplete separation of phage heads] explains a mistaken preliminary report of the transfer of ³⁵S from parental to progeny phage. . . . The following questions remain unanswered. (1) Does any sulfur-free phage material other than DNA enter the cell? (2) If so, is it transferred to the phage progeny? (3) Is the transfer of phosphorus to progeny direct or indirect? . . . Our experiments show clearly that a physical separation of the phage T2 into genetic and nongenetic parts is possible. The chemical identification of the genetic part must wait until some of the questions above have been answered. . . . The sulfur-containing protein of resting phage particles is confined to a protective coat that is responsible for the adsorption to bacteria, and functions as an instrument for the injection of the phage DNA into the cell. This protein probably has no function in the growth of the intracellular phage. The DNA has some function. Further chemical inferences should not be drawn from the experiments presented.
Source: <i>Journal of General Physiology</i> 36: 39–56

activity from ³²P-labeled phage was found to be associated with the bacteria; however, when the infecting phage was labeled with ³⁵S, only about 20 percent of the radioactivity was associated with the bacterial cells. From these results, it was apparent that a T2 phage transfers most of its DNA, but not much of its protein, to the cell it infects.

The critical finding (Figure 1.4) was that about 50 percent of the transferred ^{32}P -labeled DNA, but less than 1 percent of the transferred ^{35}S -labeled protein, was inherited by the *progeny* phage particles. Because some protein was transferred to infected cells and transmitted to the progeny phage, the Hershey-Chase experiment was not nearly so rigorous as the transformation experiments in implicating DNA as the genetic material. Nevertheless, owing to its consistency with the DNA hypothesis, the experiment was very influential.

The transformation experiment and the Hershey-Chase experiment are regarded as classics in the demonstration that genes consist of DNA. At the present time, many research laboratories throughout the world carry out the equivalent of the transformation experiment on a daily basis, generally

using bacteria, yeast, or animal or plant cells grown in culture. These experiments indicate that DNA is the genetic material in these organisms as well as phage T2.

There are no known exceptions to the generalization that DNA is the genetic material in all cellular organisms.

It is worth noting, however, that in a few types of viruses, the genetic material consists of another type of nucleic acid called RNA.

1.2—

DNA Structure: The Double Helix

Even with the knowledge that genes are DNA, many questions still remained. How does the DNA in a gene duplicate when a cell divides? How does the DNA in a gene control a hereditary trait? What happens to the DNA when a mutation (a change in the DNA) takes place in a gene? In the early 1950s, a number of researchers began to try to understand the detailed molecular structure of DNA in hopes that the structure alone would suggest answers to these questions. The first essentially correct three-dimensional structure of the DNA molecule was proposed in 1953 by James Watson and Francis Crick at Cambridge University. The structure was dazzling in its elegance and revolutionary in suggesting how DNA duplicates itself, controls hereditary traits, and undergoes mutation. Even while the tin sheet and wire model of the DNA molecule was still incomplete, Crick could be heard boasting in his favorite pub that "we have discovered the secret of life."

In the Watson-Crick structure, DNA consists of two long chains of subunits twisted around one another to form a double-stranded helix. The double helix is right-handed, which means that as one looks along the barrel, each chain follows a clockwise path as it progresses. You can see the right-handed coiling in Figure 1.5A if you imagine yourself looking up into the structure from the bottom: The "backbone" of each individual strand coils in a clockwise direction. The subunits of each strand are **nucleotides**, each of which contains any one of four chemical constituents called **bases**. The four bases in DNA are

- **Adenine (A)**
- **Thymine (T)**
- **Guanine (G)**
- **Cytosine (C)**

The chemical structures of the nucleotides and bases are included in Chapter 5. A key point for present purposes is that the bases in the double helix are paired as shown in Figure 1.5B.

At any position on the paired strands of a DNA molecule, if one strand has an A, the partner strand has a T; and if one strand has a G, the partner strand has a C.

The pairing between A—T and G—C is said to be **complementary**: The complement

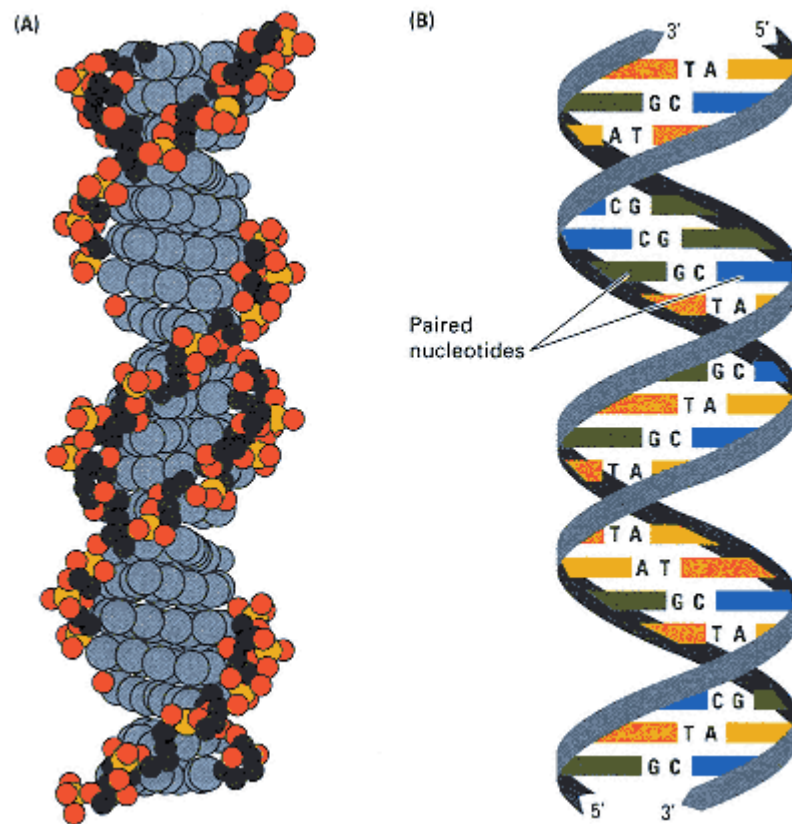


Figure 1.5
Molecular structure of a DNA double helix. (A) A "space-filling" model, in which each atom is depicted as a sphere. (B) A diagram highlighting the helical strands around the outside of the molecule and the A—T and G—C base pairs inside.

of A is T, and the complement of G is C. The complementary pairing in the duplex molecule means that each base along one strand of the DNA is matched with a base in the opposite position on the other strand. Furthermore,

Nothing restricts the sequence of bases in a single strand, so any sequence could be present along one strand.

This principle explains how only four bases in DNA can code for the huge amount of information needed to make an organism. It is the *sequence* of bases along the DNA that encodes the genetic information, and the sequence is completely unrestricted.

The complementary pairing is also called **Watson-Crick pairing**. In the three-dimensional structure (Figure 1.5A), the base pairs are represented by the spheres filling the interior of the double helix. The base pairs lie almost flat, stacked on top of one another perpendicular to the long axis of the double helix, like pennies in a roll. When discussing a DNA molecule, biologists frequently refer to the individual strands as **single-stranded DNA** and to the double helix as **double-stranded DNA** or **duplex DNA**.

Each DNA strand has a *polarity*, or directionality, like a chain of circus elephants linked trunk to tail. In this analogy, each elephant corresponds to one nucleotide along the DNA strand. The polarity is determined by the direction in which the nucleotides are pointing. The "trunk" end of the strand is called the 5' *end* of the strand, and the "tail" end is called the 3' *end*. In double-stranded DNA, the paired strands are oriented in opposite directions, the 5' end of one strand aligned with the 3' end of the other. The molecular basis of the polarity, and the reason for the opposite orientation of the strands in duplex DNA, is explained in Chapter 5.

Beyond the most optimistic hopes, knowledge of the structure of DNA immediately gave clues to its function:

1. The sequence of bases in DNA could be copied by using each of the separate "partner" strands as a pattern for the creation of a new partner strand with a complementary sequence of bases.
2. The DNA could contain genetic information in coded form in the sequence of bases, analogous to letters printed on a strip of paper.
3. Changes in genetic information (mutations) could result from errors in copying in which the base sequence of the DNA became altered.

In the remainder of this chapter, some of the implications of these clues are discussed.

1.3—

An Overview of DNA Replication

In their first paper on the structure of DNA, Watson and Crick remarked that "it has not escaped our notice that the specific base pairing we have postulated immediately suggests a copying mechanism for the genetic material." The copying mechanism they had in mind is illustrated in Figure 1.6; the process is now called **replication**. In replication, the strands of the original (parent) duplex separate, and each individual strand serves as a pattern, or **template**, for the synthesis of a new strand (replica). The replica strands are synthesized by the addition of successive nucleotides in such a way that each base in the replica is complementary (in the Watson-Crick pairing sense) to the base across the way in the template strand. Although the model in Figure 1.6 is simple in principle, it is a complex process with chemical and geometrical problems that require a large number of enzymes and other proteins to resolve. The details are discussed in Chapter 5. For purposes of this overview, the important point is that the replication of a duplex molecule results in two duplex daughter molecules, each with a sequence of nucleotides identical to the parental strand.

In Figure 1.6A, the backbones in the parental DNA strands and those in the newly synthesized strand are shown in contrasting colors. In the process on the left, the top strand is the template present in the parental molecule, and the bottom strand is the newly synthesized partner. In

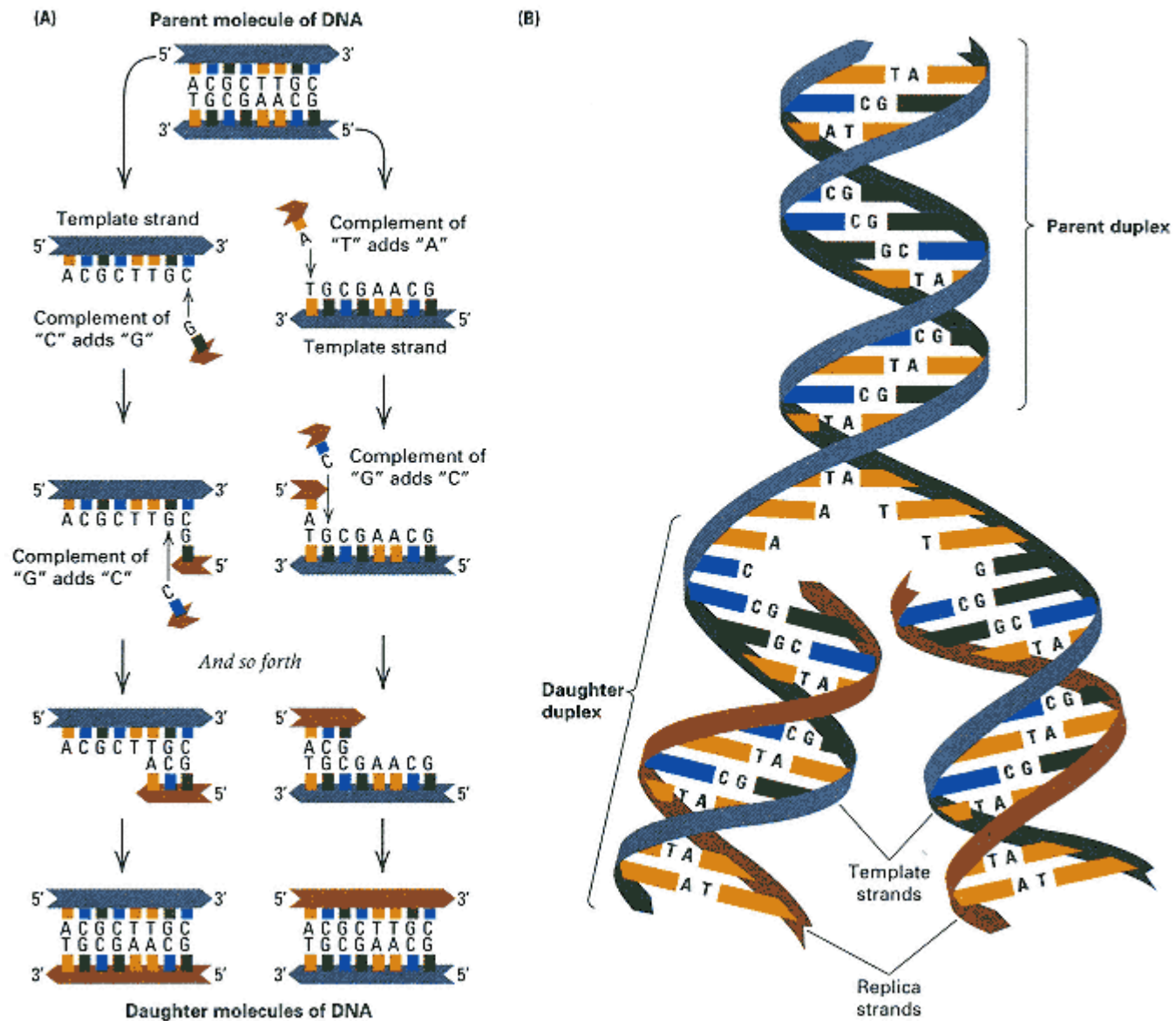


Figure 1.6

Replication of DNA. (A) Each of the parental strands serves as a template for the production of a complementary daughter strand, which grows in length by the successive addition of single nucleotides. (B) Replication in a long DNA duplex as originally proposed by Watson and Crick. As the parental strands separate, each parental strand serves as a template for the formation of a new daughter strand by means of A—T and G—C base pairing.

the process on the right, the bottom strand is the template from the parental molecule, and the top strand is the newly synthesized partner. How the process of replication occurs in a long duplex molecule is shown in Figure 1.6B. The separation of the parental strands and the synthesis of the daughter strands take place simultaneously in different parts of the molecule. In each successive region along the parental duplex, as the parental strands come apart, each of the separated parental strand serves as a template for the synthesis of a new daughter strand.

1.4—

Genes and Proteins

By the beginning of the twentieth century, it had already become clear that proteins were responsible for most of the metabolic activities of cells. Proteins were known to be essential for the breakdown of organic molecules to generate the chemical energy needed for cellular activities. They were also known to be required for the assembly of small molecules into more complex molecules and cellular structures. In 1878, the term **enzyme** was introduced to refer to the biological catalysts that accelerate biochemical reactions in cells. By 1900, owing largely to the genius of the German biochemist Emil Fischer, enzymes had been shown to be proteins. Other proteins are key components of cells; for example, structural proteins give the cell form and mobility, other proteins form pores in the cell membrane and control the traffic of small molecules into and out of the cell, and still other proteins regulate cellular activities in response to molecular signals from the external environment or from other cells.

In 1908, the British physician Archibald Garrod had an important insight into the relationship between enzymes and disease:

Any hereditary disease in which cellular metabolism is abnormal results from an inherited defect in an enzyme.

Such hereditary diseases became known as **inborn errors of metabolism**, a term still in use today. Although the full implications of Garrod's suggestion could not be explored experimentally until many years afterward, the coupling of the concepts of inheritance (gene) with enzyme (protein) was a brilliant simplification of the problem of biochemical genetics because it put the emphasis on the question "How do genes control the structure of proteins?" How biologists pursued this question is summarized in the following sections.

Transcription of DNA Makes RNA

Watson and Crick were quite right in suggesting that the genetic information in DNA is contained in the sequence of bases; it is encoded in a manner analogous to letters (the bases) printed on a strip of paper. However, learning the details of the genetic code and the manner in which it is deciphered took about 20 years of additional work. The long series of investigations showed that in a region of DNA that directs the synthesis of a protein, the genetic code for the protein is contained in a DNA strand. The coded genetic information in this strand is decoded in a linear order in which each successive "word" in the DNA strand specifies the next chemical subunit to be added to the protein as it is being made. The protein subunits are called **amino acids**. Each "word" in the genetic code consists of three adjacent bases.

For example, the base sequence ATG in a DNA strand specifies the amino acid methionine (Met), TTT specifies phenylalanine (Phe), GGA specifies glycine (Gly), and GTG specifies valine (Val). How the genetic information is transferred from the base sequence of a DNA strand into the amino acid sequence of the corresponding protein is shown in Figure 1.7. This scheme, in which DNA codes for RNA and RNA codes for proteins, is known as the **central dogma** of molecular genetics. (The term *dogma* means a set of beliefs. The term dates from the time when the idea was first advanced as a theory; since then, the "dogma" has been confirmed experimentally, but the term persists.)

The main concept in the central dogma is that DNA does not code for protein directly but acts through an intermediary molecule called **ribonucleic acid (RNA)**. The structure of RNA is similar, but not identical, to that of DNA. The sugar is ribose rather than deoxyribose. RNA is usually single-stranded (not a duplex), and RNA contains a base, **uracil (U)**, that takes the place of thymine (T) in DNA (Chapter 5). In the synthesis of proteins, there are actually three types of RNA that participate and that play different roles:

- A **messenger RNA (mRNA)**, which carries the genetic information from DNA and is used as a template for protein synthesis.
- The **ribosomal RNA (rRNA)**, which is a major constituent of the cellular parti-

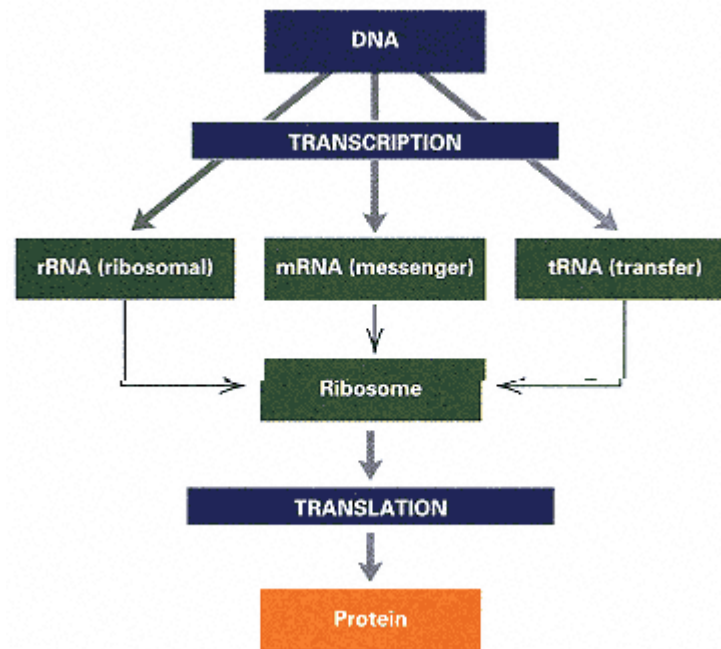


Figure 1.7
The "central dogma" of molecular genetics: DNA codes for RNA, and RNA codes for protein. The DNA $\rightarrow$ RNA step is transcription and the RNA $\rightarrow$ protein step is translation.

cles called **ribosomes** on which protein synthesis actually takes place.

- A set of **transfer RNA (tRNA)** molecules, each of which incorporates a particular amino acid subunit into the growing protein when it recognizes a specific group of three adjacent bases in the mRNA.

Why on Earth should a process as functionally simple as DNA coding for protein have the additional complexity of RNA intermediaries? Certain biochemical features of RNA suggest a hypothesis: that RNA played a central role in the earliest forms of life and that it became locked into the processes of information transfer and protein synthesis. So it remains today: The participation of RNA in protein synthesis is a relic of the earliest stages of evolution—a "molecular fossil." The hypothesis that the first forms of life used RNA both for carrying information (in the base sequence) and as catalysts (accelerating chemical reactions) is supported by a variety of observations. Two examples: (1) DNA replication requires an RNA molecule in order to get started (Chapter 5), and (2) some RNA molecules act to catalyze biochemical reactions important in protein synthesis (Chapter 10). In the later evolution of the early life forms, additional complexity could have been added. The function of information storage and replication could have been transferred from RNA to DNA, and the function of RNA catalysis in metabolism could have been transferred from RNA to protein by the evolution of RNA-directed protein synthesis.

The manner in which genetic information is transferred from DNA to RNA is straightforward (Figure 1.8). The DNA opens up, and one of the strands is used as a template for the synthesis of a complementary strand of RNA. (How the template strand is chosen is discussed in Chapter 10.) The process of making an RNA strand from a DNA template is **transcription**, and the RNA molecule that is made is the **transcript**. The base sequence in the RNA is complementary (in the Watson-Crick pairing sense) to that in the DNA template, except that U (which pairs with A) is present in the RNA in place of T. The base-pairing rules between DNA and RNA are summarized in Figure 1.9. Like DNA, an RNA strand also has a polarity, exhibiting a 5' end and a 3' end determined by the orientation of the nucleotides. The 5' end of the RNA transcript is synthesized first and, in the RNA—DNA duplex formed in transcription, the polarity of the RNA strand is opposite to that of the DNA strand.

from the previous tRNA and attached to the amino acid of the next in line, again increasing the length of the protein chain by one amino acid. A protein is therefore synthesized in a stepwise manner, one amino acid at a time. By way of analogy, the process of protein synthesis by the addition of consecutive amino acids is like the construction of a chain of pop-together plastic beads by the addition of consecutive beads.

The role of tRNA in translation is illustrated in Figure 1.10 and can be described as follows:

The mRNA is read codon by codon. Each codon specifying an amino acid matches with a complementary group of three adjacent bases in a single tRNA molecule, which brings the correct amino acid into line.

The tRNA molecules used in translation do not line up along the mRNA simultaneously as shown in Figure 1.10. The process of translation takes place on a ribosome, which combines with a single mRNA molecule and moves along it from one end (the 5' end) to the other (the 3' end) in steps of three adjacent nucleotides (codon by codon). As each new codon comes into place, the correct tRNA attaches to the ribosome, and the growing chain of amino

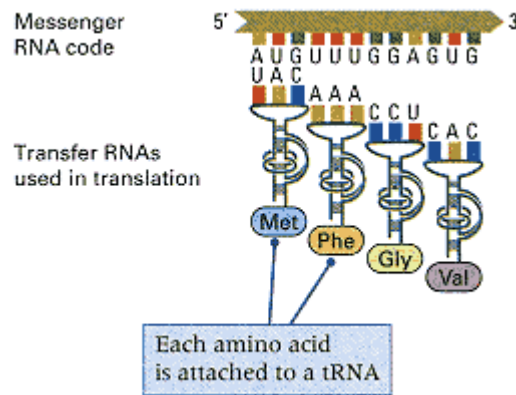


Figure 1.10

The role of transfer RNA in the synthesis of proteins. The sequence of bases in the messenger RNA determines the order in which transfer RNA molecules are lined up. Each group of three adjacent bases in the messenger RNA attracts a transfer RNA containing a complementary sequence of three bases. Each transfer RNA molecule carries a particular amino acid, and the amino acids in the protein join together in the same order in which the transfer RNA molecules line up along the messenger RNA. Because the transfer RNA molecules are aligned in this manner, the sequence of bases in the messenger RNA determines the sequence of amino acids in the protein. Polypeptide chains are synthesized by the sequential addition of amino acids, one at a time. As each transfer RNA molecule is brought into line, the incomplete polypeptide chain grows one amino acid longer by becoming attached to the amino acid linked to the transfer RNA.

acids becomes attached to the amino acid on the tRNA. As the ribosome moves along the mRNA, successive amino acids are added to the growing chain until any one of three particular codons specifying "stop" is encountered. At this point, synthesis of the chain of amino acids is finished, and the protein is released from the ribosome. (Chapter 10 gives a more detailed treatment of translation.)

Technically speaking, the chain of amino acids produced in translation is a **polypeptide**. The distinction between a polypeptide and a protein is that a protein can consist of several polypeptide chains that come together after translation. Some proteins are composed of two or more identical polypeptide chains (encoded in the same gene); others are composed of two or more different polypeptide chains (encoded in different genes). For example, the protein **hemoglobin**, which is the oxygen-carrying protein in red blood cells, is composed of four polypeptide chains encoded in two different genes: Two of the chains are β polypeptide chains translated from the β -globin gene, and the other two are α polypeptide chains translated from the α -globin gene.

1.5— Mutation

Mutation means any heritable change in a gene. The Watson-Crick structure of DNA also suggested that, chemically speaking, a mutation is a change in the sequence of bases along the DNA. The change may be simple, such as the substitution of one pair of bases in a duplex molecule for a different pair of bases. For example, an A—T pair in a duplex molecule may mutate to either T—A, C—G, or G—C. The change in base sequence may also be more complex, such as the deletion or addition of base pairs. These and other types of mutations are discussed in Chapter 13.

One possible consequence of a mutation is illustrated in Figure 1.11. Part A shows a region of duplex DNA and the mRNA transcribed from the bottom strand. The tRNA molecules used in translation result in the amino acid sequence

Met—**Phe**—Gly—Val

What happens if the T—A base pair marked with the "sunburst" mutates to become a C—G base pair? The result is shown in part B. The second codon in the mRNA is now CUU, which codes for leucine (Leu), instead of the codon UUU, which codes for phenylalanine (Phe). In translation, the CUU codon in the mRNA combines with the leucine-bearing tRNA, and the result is the mutant amino acid sequence

Met—**Leu**—Gly—Val

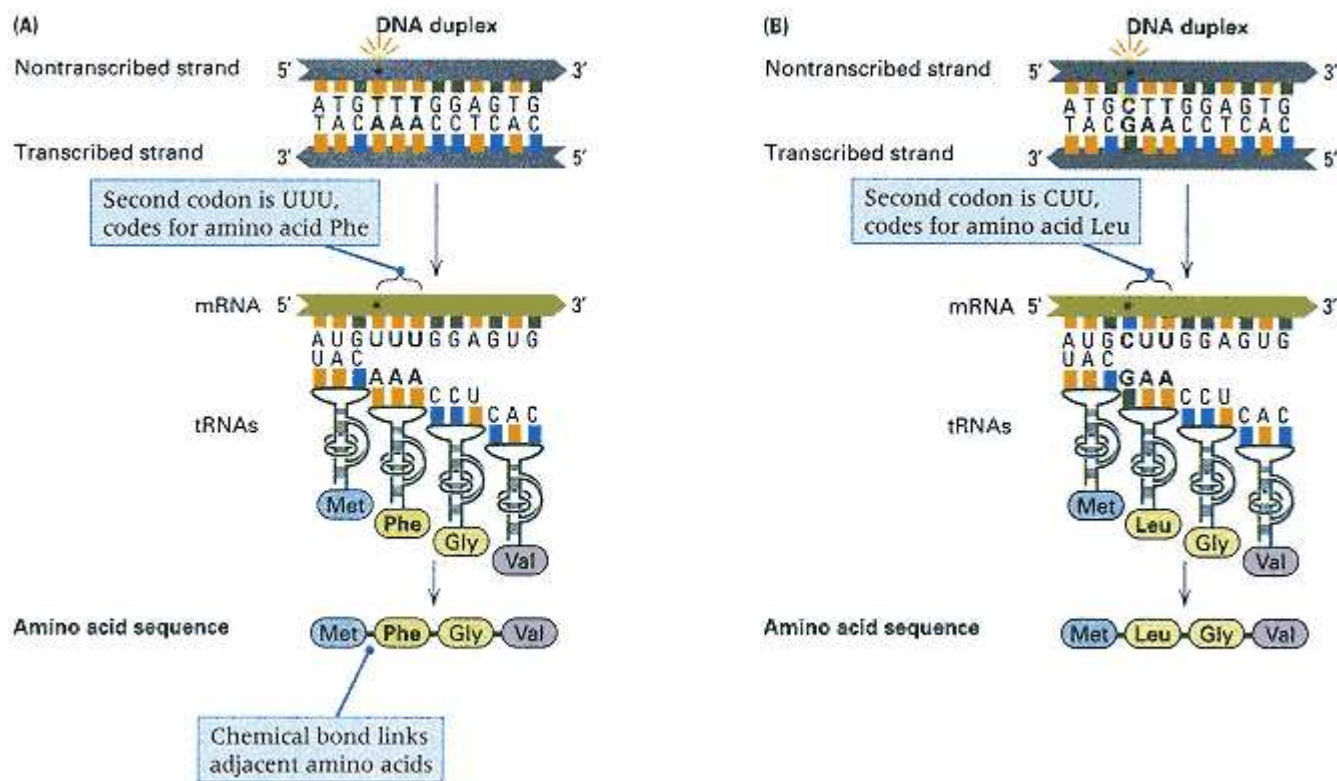


Figure 1.11

A mutation is a change in base sequence in the DNA. Any mutation that causes the insertion of an incorrect amino acid in a protein can impair the function of the protein. (A) The DNA molecule is transcribed into a messenger RNA that codes for the sequence of amino acids Met—Phe—Gly—Val. In the DNA molecule, the marked T—A base pair results in the initial U in the messenger RNA codon UUU for Phe (phenylalanine). (B) Substitution of a C—G base pair for the normal T—A base pair results in a messenger RNA containing the codon CUU instead of UUU. The CUU codon codes for Leu (leucine), which therefore replaces Phe in the mutant polypeptide chain.

1.6— How Genes Determine Traits

We have seen that the key principle of molecular genetics is the central dogma:

The sequence of nucleotides in a gene specifies the sequence of amino acids in a protein using messenger RNA as the intermediary molecule in the coding process.

It is one of the ironies of genetics (and a consequence of the biochemical complexity of organisms) that whereas the connection between genes and proteins is conceptually simple, the connection between genes and traits is definitely not simple. Most visible traits of organisms are the net result of many genes acting together in combination with environmental factors. Therefore, the relationship between genes and traits is often complex for one or more of the following reasons:

1. One gene can affect more than one trait.
2. One trait can be affected by more than one gene.
3. Many traits are affected by environmental factors as well as by genes.

Now let us examine each of these principles, with examples.

Pleiotropy: One Gene Can Affect More Than One Trait

A mutant gene may affect a number of seemingly unrelated traits. The mutation is then said to show **pleiotropy**, and the various manifestations of the mutation are known as **pleiotropic effects**. An example of a mutation in human beings with manifold pleiotropic effects is **sickle-cell anemia**, which affects the major oxygen-carrying protein of the red blood cells. The major organs and organ systems affected by the pleiotropic effects of the mutation are shown in Figure 1.12.

The underlying mutation in sickle-cell anemia is in the gene for β -globin, which codes for the β polypeptide chains present in the oxygen-carrying protein of red blood cells. The molecular basis of the disease is shown in Figure 1.13. Figure 1.13A shows the region of the β -globin gene that codes for amino acids 5 through 8 (the complete polypeptide chain is 146 amino acids in length). The sickle-cell mutation changes the base pair marked with the sunburst. As shown in part B, the mutant form of the gene contains a T—A base pair instead of the normal A—T base pair. As a result of the mutation, the mRNA contains a GUG codon. Because GUG codes for valine (Val), this amino acid is incorporated into the polypeptide in place of the normal glutamic acid (Glu) at position number 6.

The defective β polypeptide chain gives the hemoglobin protein a tendency to form long, needle-like polymers. Red blood cells in which polymerization happens become deformed into crescent, sickle-like shapes. Some of the deformed red blood cells are destroyed immediately (reducing the oxygen-carrying capacity of the blood and causing the anemia), whereas others may clump together and clog the blood circulation in the capillaries.

The consequences of the Glu $\rightarrow$ Val replacement are a profound set of pleiotropic effects. All of these effects are related to the breakdown of red blood cells, to the decreased oxygen-carrying capacity of the blood, or to physiological adjustments the body makes to try to compensate for the disease (such as enlargement of the spleen). Patients with sickle-cell anemia

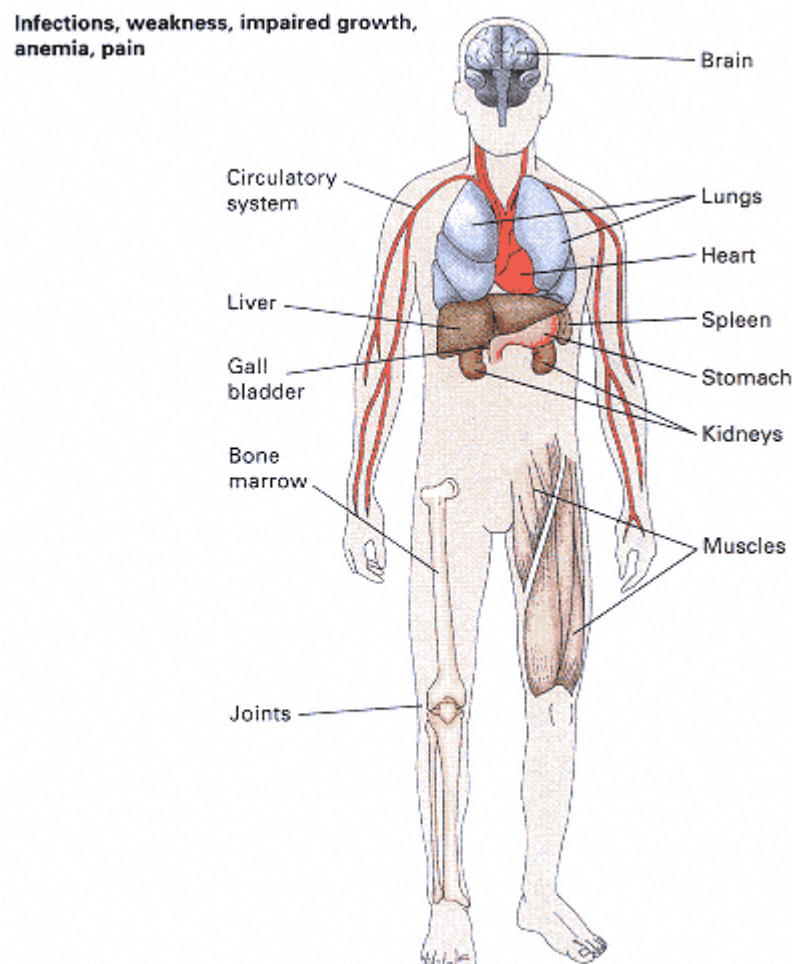


Figure 1.12

Sickle-cell anemia has multiple, seemingly unrelated symptoms known as pleiotropic effects. The primary defect is a mutant form of hemoglobin in the blood. The resulting destruction of red blood cells and the impaired

ability of the blood to carry oxygen affect the circulatory system, bone marrow, muscles, brain, and virtually all major internal organs. The symptoms of the disease are anemia, recurrent pain, weakness, susceptibility to infections, and slowed growth.

suffer bouts of severe pain. The anemia causes impaired growth, weakness, and jaundice. Affected people are so generally weakened that they are susceptible to bacterial infections, which are the most common cause of death in children with the disease.

Although sickle-cell anemia is a severe genetic disease that often results in premature death, it is relatively frequent in areas of Africa and the Middle East in which a type of malaria caused by the protozoan parasite *Plasmodium falciparum* is widespread. The association between sickle-cell anemia and malaria is not coincidental. The

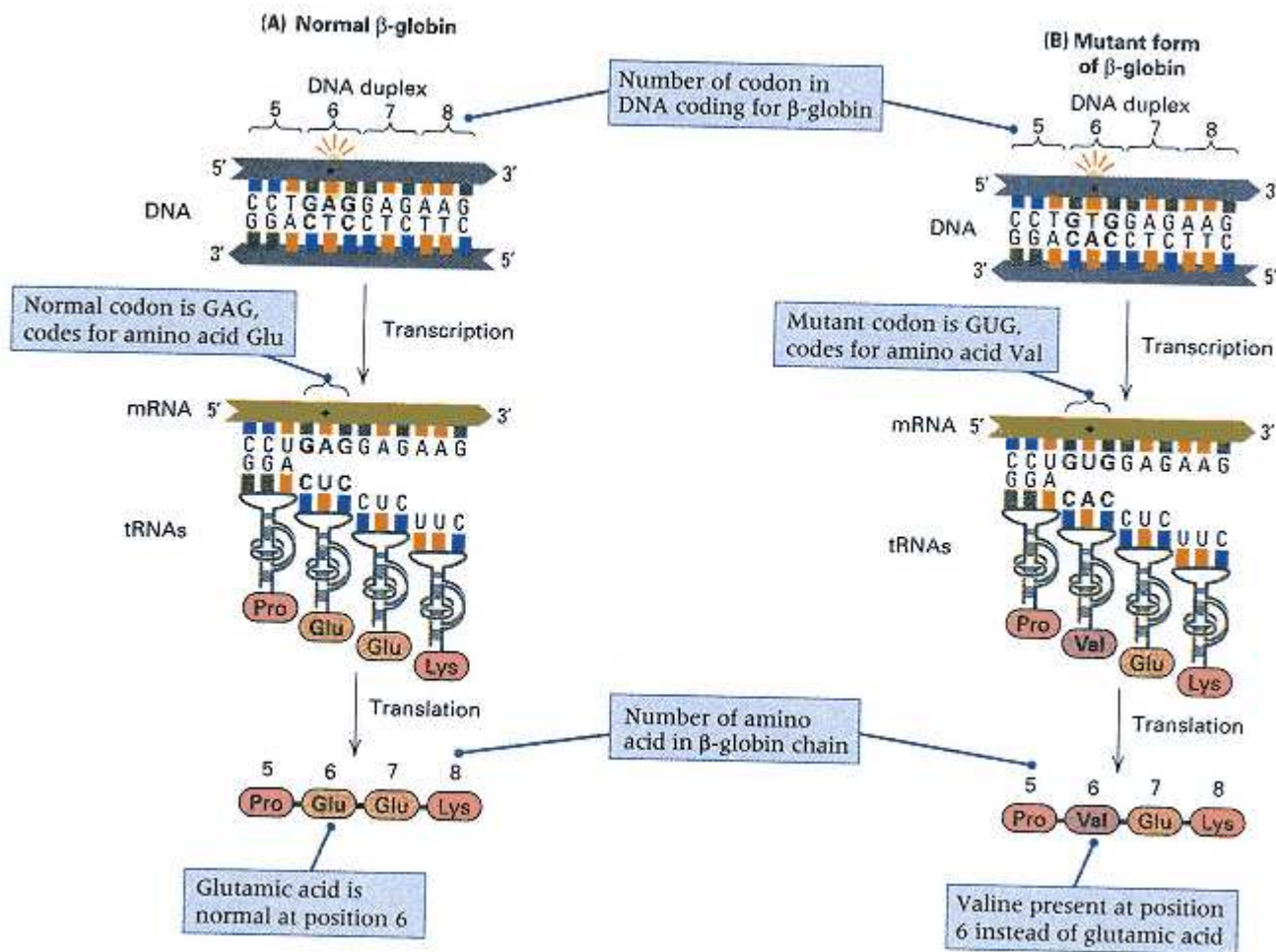


Figure 1.13

Genetic basis of sickle-cell anemia: (A) Part of the DNA in the normal β -globin gene is transcribed into a messenger RNA coding for the amino acid sequence Pro—Glu—Glu—Lys. The T in the marked A—T base pair is transcribed as the A in the GAG codon for Glu (glutamic acid). (B) Mutation of the normal A—T base pair to a T—A base pair results in the codon GUG instead of GAG. The codon GUG codes for Val (valine), so the polypeptide sequence in this part of the molecule is Pro—Val—Glu—Lys. The resulting hemoglobin is defective and tends to polymerize at low oxygen concentration.

association results from the ability of the mutant β -hemoglobin to afford some protection against malarial infection. In the life cycle of the parasite, it passes from a mosquito to a human through the mosquito's bite. The initial stages of infection take place in cells in the liver, where specialized forms of the parasite are produced that are able to infect and multiply in red blood cells. Widespread infection of red blood cells impairs the ability of the blood to carry oxygen, causing the weakness, anemia, and jaundice characteristic of malaria. In people with the mutant β -hemoglobin, however, it is thought that the infected blood cells undergo sickling and are rapidly removed from circulation. The proliferation of the parasite among the red blood cells is thereby checked, and the severity of the malarial infection is reduced. There is consequently a genetic balancing act between the prevalence of the genetic disease sickle-cell anemia and that of the parasitic disease malaria. If the mutant β -hemoglobin becomes too frequent, more lives are lost from sickle-cell anemia than are gained by the protection against malaria; on the other hand, if the mutant β -hemoglobin becomes too rare, fewer lives are lost from sickle-cell anemia but the gain is offset by more deaths from malaria. The end result of this kind of genetic balancing act is discussed in quantitative terms in Chapter 15.

Epistasis: One Trait Can Be Affected by More Than One Gene

Every trait requires numerous genes for its proper development, metabolism, and physiology. Consequently, one trait can be affected by more than one gene. An example of this principle is illustrated in Figure 1.14, which shows the effects of two genes that function in eye pigmentation in *Drosophila*. The genes are *vermilion* (*v*) and *cinnabar* (*cn*). These genes encode enzymes, denoted V and Cn, respectively, that are used in the **biochemical pathway** that converts the amino acid tryptophan into the brown eye pigment *xanthommatin* through a series of intermediate substances I1, 12, and so forth (Figure 1.14A). Each step in the pathway is catalyzed by a different enzyme encoded by a different gene. The nonmutant, or **wildtype**, eye color of *Drosophila* is a brick-like red because the

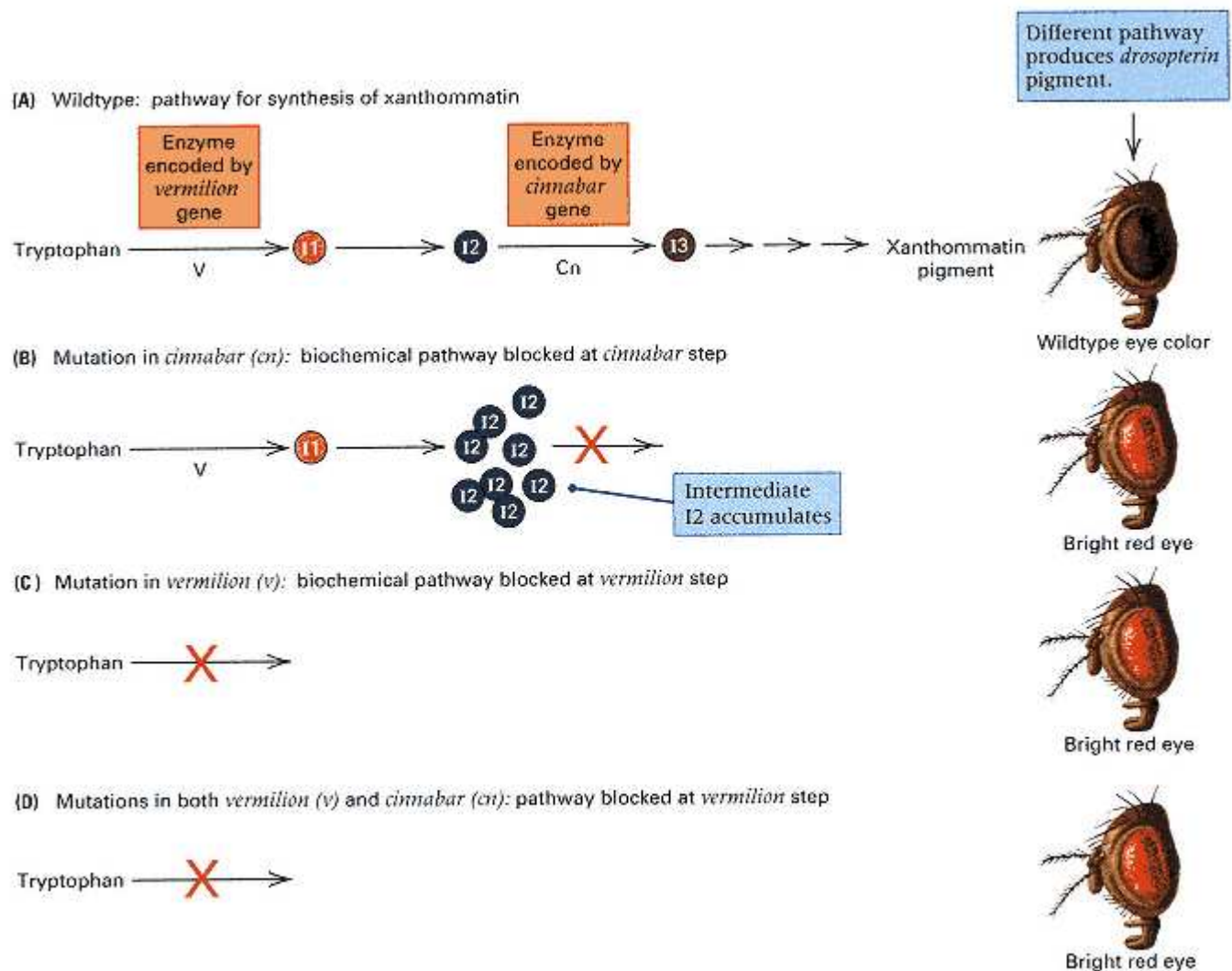


Figure 1.14

The *Drosophila* mutants *vermilion* and *cinnabar* exemplify epistasis between mutant genes affecting eye color. (A) Metabolic pathway for the production of the brown pigment xanthommatin. The intermediate substances are denoted I1, 12, and so forth, and each single arrow represents one step in the pathway. (The multiple arrows at the end represent an unspecified number of steps.) (B) The *cn* gene codes for an enzyme, Cn, that converts 12 to 13. In flies mutant for *cn*, the pathway is blocked at this step. (C) The *v* gene codes for a different enzyme, V, that catalyzes the conversion of tryptophan into intermediate I1. In flies mutant for *v*, the pathway is blocked at this step. (D) In *v cn* double mutants, the pathway is blocked at the earlier step, in this case the conversion of tryptophan to I1.

pigment cells contain not only xanthommatin but also a bright red pigment called *drosopterin* synthesized by a different bio-chemical pathway.

As indicated in Figure 1.14B, flies that are mutant for *cn* lack xanthommatin. They have bright red eyes because of the drosopterin. Flies mutant for *cn* have a nonfunctional Cn enzyme, so the pathway is blocked at the step at which Cn should function. Because there is no functional Cn enzyme to convert intermediate 12 into the next intermediate along the way, 12 accumulates in *cn* flies. Mutant *v* flies also lack xanthommatin but for a different reason (Figure 1.14C). In these flies the pathway is blocked because there is no functional V enzyme. It does not matter whether the Cn enzyme is present, because without the V enzyme, there is no 12 for Cn to work on.

The pathway in flies with a mutation in both *v* and *cn* is illustrated in Figure 1.14D. The situation is identical to that in flies with a *v* mutation only because, lacking functional V enzyme, the pathway is blocked at this step. The general term for gene interaction is **epistasis**. Freely translated from the Greek, epistasis means "standing over." Epistasis means that the presence of one mutation "stands over," or conceals, the effects of a different mutation. In the example in Figure 1.14, we would say that *v* is *epistatic* to *cn*, because in flies with a *v* mutation, it is impossible to determine from the status of the xanthommatin pathway whether the *cn* gene is mutant or wildtype. The converse is not true: In flies with a *cn* mutation, the presence or absence of intermediate 12 shows whether the *v* gene is mutant or wildtype. If 12 accumulates, the V enzyme must be present (and the *v* gene wildtype); whereas if 12 is absent, the V enzyme must be non-functional (and the *v* gene mutant.)

The example in Figure 1.14 also illustrates an important feature of genetic terminology. Although both *vermilion* and *cinnabar* are needed for the synthesis of the *brown* pigment, the names of the genes are shades of *bright red*. At first this seems illogical, but *mutant genes are named for their effects on the organism*. Because mutations in either *vermilion* or *cinnabar* result in bright red eyes, the gene names make sense even though the products of both genes function in the brown-pigment pathway.

Effects of the Environment

Genes and environment also interact. To appreciate the interaction between genes and environment, consider the trait "anemia," which refers to a generalized weakness resulting from an insufficient number of red blood cells or from an inadequate volume of blood. There are many different types of anemia. Some forms of anemia are genetically determined, such as sickle-cell anemia (Figure 1.13). Other forms of anemia are caused by the environment; an example is anemia resulting from chronic deficiency of dietary iron or from infection with malaria. Still other forms of anemia are caused by genetic and environmental factors acting together. For example, people with a mutant form of the enzyme **glucose-6-phosphate dehydrogenase (G6PD)**, an enzyme important in maintaining the integrity of the membrane of red blood cells, become severely anemic when they eat fava beans, because a substance in the beans triggers destruction of red cells. Because of its association with fava beans, the disease is called **favism**, but a more common name is **G6PD deficiency**. Red-cell destruction in people with G6PD deficiency can also be triggered by various chemicals such as naphthalene (used in mothballs) as well as by certain antibiotics and other drugs. G6PD deficiency, which affects primarily males, has a relatively high frequency in populations in coastal regions around the Mediterranean Sea. It is thought that the defect in the red blood cells may increase resistance to malaria.

With these examples as background, consider this question: Is anemia caused by heredity or environment? There is no simple answer. As we have seen, a complex trait such as anemia has many possible causes. Some types are genetically determined, some environmental in origin, and some require both genes and environment for their expression. The genes-versus-environment issue is exceptionally clear in the example of anemia only because various forms of the disorder have already been sorted out and assigned causes, whether they be genetic or environmental or both. However, before the various forms were distinguished, anemia was regarded as a tremendously complex condition, and

all varieties were lumped together. Without separating the disorder into categories, all that one could conclude was that family history seemed to be important in some cases, but not all, and that the environment certainly played a role as well.

Most complex traits are analogous to anemia in consisting of different conditions lumped together because of their overall similarity. A familiar example is heart disease. It is well known that inherited risk factors in heart disease are related to the metabolism of saturated fats and cholesterol. Some rare forms of the disease with a strong genetic component have already been identified. There are also environmental risk factors in heart disease—cigarette smoking, being overweight, lack of exercise, high dietary intake of saturated fats and cholesterol, and so forth. In the population as a whole, the overall risk of heart disease is determined by both genetic and environmental factors, and some of the factors act synergistically, which means that the risk from two factors together is greater than would be predicted from the risk of each factor considered by itself.

The example of heart disease also illustrates that genetic and environmental effects can be offsetting. For example, a person with a family history of heart disease can considerably mitigate the risk by careful diet, exercise, abstention from smoking, and other behaviors. Taking drugs to control high blood pressure is also an example of an environmental intervention that reduces the overall risk of heart disease.

Heart disease is a typical example of a complex trait influenced by multiple genes as well as by many environmental factors. Most of the variation found in human beings falls into this category, including personality and other behavioral characteristics. Some traits are more strongly influenced by genetic factors than others, and it is extremely difficult to sort out the forms of a trait that might share a single cause. An illustration of complex genetic and environmental causation is shown in Figure 1.15. The boxes labeled mild, moderate, and so forth represent various different severities in which a trait can be expressed; these are analogous to the different forms of anemia. Across the top are three genes and three environmental factors that influence the trait. The heavy lines represent major effects, the thin lines minor effects. If the four types of expression of the trait

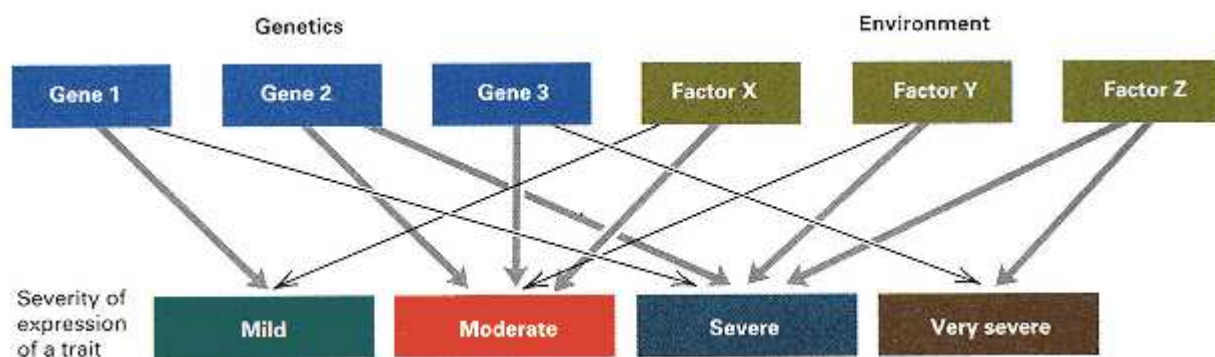


Figure 1.15

Most complex traits are affected by multiple genetic and environmental factors, not all of them equal in influence.

In this example, the severity of expression of a complex disease is affected by three genes (1, 2, 3) and three environmental factors (X, Y, Z) that, in various combinations, determine the particular manner in which the disease will be expressed. Heavy arrows depict major influences, light arrows minor influences. For example, the mild expression of the disease is determined primarily by gene 1 with a minor influence of environmental factors X. The moderate expression of the disease is determined by two genes (2 and 3) and two environmental factors (X having a major effect and Y a minor effect). In a complex trait, therefore, some forms of expression of the disease (mild in this example) may have a relatively simple form of genetic causation, whereas other forms of the same disease (moderate in this example) may have a more complex causation that even includes different genes.

The genetic basis of such diseases is difficult to determine unless the different forms of the disease can be distinguished.

were regarded as a single entity without being distinguished, then the genetic and environmental causation could be characterized only as "three genes and three environmental factors, each with major effects." However, when the different levels of severity of the trait are considered separately, the situation can be clarified. For example, the mild form is determined by one major genetic factor and one minor environmental factor, and the very severe form is determined by one minor genetic factor and one major environmental factor. Real complexity remains in the moderate and severe forms, however: The moderate form is determined by two major genes together with one major and one minor environmental factor, and the severe form is determined by two major environmental factors together with one major and one minor genetic factor. Figure 1.15 also illustrates the more general point that traits do not present themselves already classified in the most informative manner. Progress in genetics has often resulted from the proper subdivision of a complex trait into distinct types that differ in their genetic or environmental causation.

1.7— Evolution

One of the remarkable discoveries of molecular genetics is that organisms that seem very different (for example, plants and animals) share many common features in their genetics and biochemistry. These similarities indicate a fundamental "unity of life":

All creatures on Earth share many features of the genetic apparatus, including genetic information encoded in the sequence of bases in DNA, transcription into RNA, and translation into protein on ribosomes via transfer RNAs. All creatures also share certain characteristics in their biochemistry, including many enzymes and other proteins that are similar in amino acid sequence.

The Molecular Continuity of Life

The molecular unity of life comes about because all creatures share a common origin through **evolution**, the process by which populations of organisms that are descended from a common ancestor gradually become more adapted to their environment and sometimes split into separate species. In the evolutionary perspective, the unity of fundamental molecular processes is derived by inheritance from a distant common ancestor in which many mechanisms were already in place.

Not only the unity of life but also many other features of living organisms become comprehensible from an evolutionary perspective. The importance of the evolutionary perspective in understanding aspects of biology that seem pointless or needlessly complex is summed up in a famous aphorism of the evolutionary biologist Theodosius Dobzhansky: "Nothing in biology makes sense except in the light of evolution."

One indication of the common ancestry among Earth's creatures is illustrated in Figure 1.16. The tree of relationships was inferred from similarities in nucleotide sequence in a type of ribosomal RNA molecule common to all these organisms. Three major kingdoms of organisms are distinguished:

- 1. Bacteria** This group includes most bacteria and cyanobacteria (formerly called blue-green algae). Cells of these organisms lack a membrane-bounded nucleus and mitochondria, are surrounded by a cell wall, and divide by binary fission.
- 2. Archaea** This group was initially discovered among microorganisms that produce methane gas or that live in extreme environments, such as hot springs or high salt concentrations; they are widely distributed in more normal environments as well. Like Bacteria, the cells of Archaea lack internal membranes. DNA sequence analysis indicates that the machinery for DNA replication and transcription resembles that of Eukarya whereas metabolism in Archaea strongly resembles that of Bacteria. About half of their genes are unique to Archaea, however.
- 3. Eukarya** This group includes all organisms whose cells contain an elaborate network of internal membranes, a membrane-bounded nucleus, and mito-

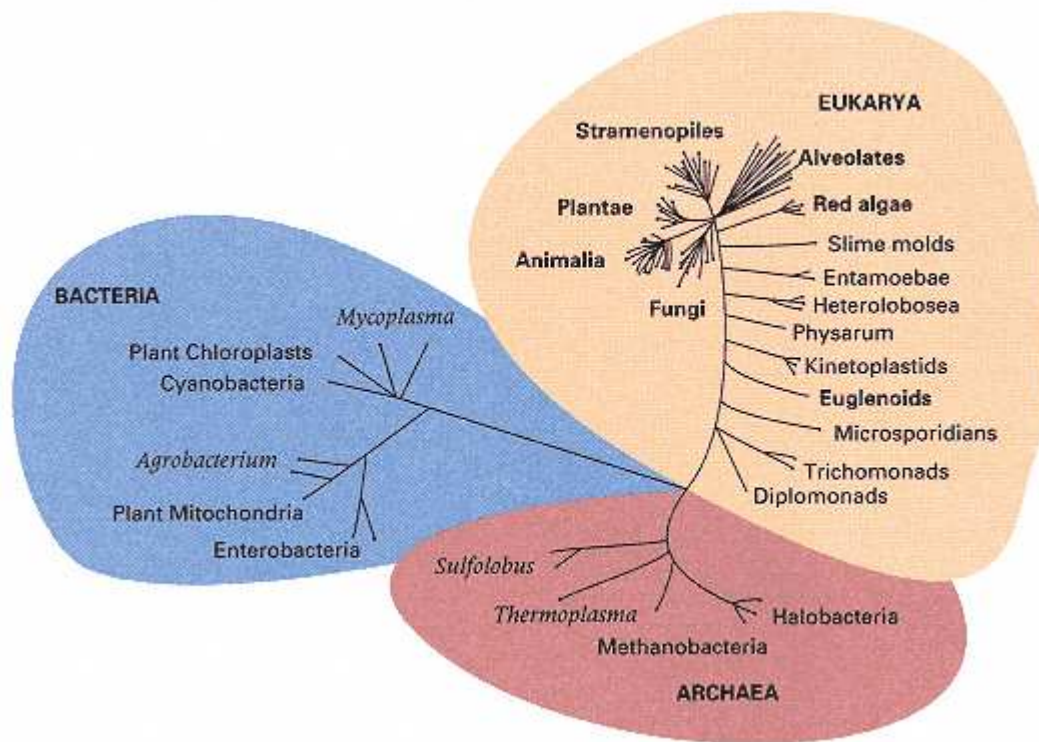


Figure 1.16

Evolutionary relationships among the major life forms as inferred from similarities in nucleotide sequence in an RNA molecule found in the small subunit of the ribosome. The three major kingdoms of Bacteria, Archaea, and Eukarya are apparent. Plants, animals, and fungi are more closely related to each other than to members of either of the other kingdoms. Note the diverse groups of undifferentiated, relatively simple organisms that diverged very early in the eukaryote lineage.

[Courtesy of Mitchell L. Sogin.]

chondria. Their DNA is organized into true chromosomes, and cell division takes place by means of mitosis (discussed in Chapter 3). The eukaryotes include plants and animals as well as fungi and many single-celled organisms, such as amoebae and ciliated protozoa.

The Bacteria and Archaea are often grouped together into a larger assemblage called **prokaryotes**, which literally means "before [the evolution of] the nucleus." This terminology is convenient for designating prokaryotes as a group in contrast with **eukaryotes**, which literally means "good [well-formed] nucleus."

Adaptation and Diversity

Figure 1.16 illustrates the unity of life, but it also illustrates the diversity. Frogs are different from fungi, and beetles are different from bacteria. As a human being, it is sobering to consider that complex, multicellular organisms came relatively late onto the evolutionary scene of life on Earth. Animals came later still and primates very late indeed. What about human evolution? In the time scale of Earth history, human evolution is a matter of a few million years—barely a snap of the fingers.

If common ancestry is the source of the unity of life, what is the source of diversity? Because differences among species are inherited, the original source of the differences must be mutation. However, mutations alone are not sufficient to explain why organisms are adapted to living in their environments—why ocean mammals have special adaptations that make swimming and diving possible, or why desert mammals have special adaptations that enable them to survive on minimal amounts of water. Mutations are chance events not directed toward any particular

adaptive goal, like longer fur among mammals living in the Arctic. The process that accounts for adaptation was outlined by Charles Darwin in his 1859 book *On the Origin of Species*. Darwin proposed that adaptation is the result of **natural selection**, the process in which individual organisms that carry particular mutations or combinations of mutations that equip them to survive or reproduce more effectively in the prevailing environment will leave more offspring than other organisms and so contribute their favorable genes disproportionately to future generations. If this process is repeated throughout the course of many generations, the entire species becomes genetically transformed because a gradually increasing proportion of the population inherits the favorable mutations. The genetic basis of natural selection is discussed in Chapter 15.

The Role of Chance in Evolution

Natural selection is undoubtedly the key process in bringing about the genetic adaptation of organisms to their environments. Hence there is a great deal of appeal in being able to explain why particular traits are adaptive. Unfortunately, the ingenuity of the human imagination makes it all too easy to make up an adaptive story for any trait whatsoever. One example is the adaptive argument that the reason why blood is red is that seeing it scares one's enemies when one is injured. This explanation sounds almost plausible, but the truth is that blood is red for the same reason as rust; it contains oxidized iron. Each hemoglobin chain carries an atom of iron, and oxidized iron is red as a matter of physics, not biological evolution.

Made-up adaptive stories not supported by hard evidence are called **just-so stories** after the title (*Just So Stories*) of a 1902 book by Rudyard Kipling. The stories tell how animal traits came to be: how the elephant got its trunk (a crocodile caught a baby elephant by his nose and "pulled and pulled and pulled it out into a really truly trunk same as all elephants have today"); how the whale got his throat (because he swallowed a sailor who wedged a grate at the back of his throat that prevented him from eating anything except "very, very small fish—and this is the reason why whales nowadays never eat men or boys or little girls"); how the camel got his hump (a wizard cursed it on him for not doing his work); and so forth.

The rationale for inventing evolutionary just-so stories is the assumption that all traits are adaptive by necessity and one needs only to find a reason why. But this is not necessarily so. For example, some traits exist not because they are selectively advantageous in themselves but because they are pleiotropic effects of genes selected for other reasons. Chance may also play a large role in some major events in the history of life. Many evolutionary biologists now believe that a mass extinction was precipitated 65.3 million years ago when an asteroid smashed into the Pacific Ocean off the Yucatan Peninsula and spewed so much debris into the air that Earth went dark for years. The mass extinction triggered by this event was not by any means the largest in Earth history, but it led to the extinction of all dinosaur species and about 90 percent of other species. Until then, dinosaurs were a wonderfully diverse and well-adapted group of organisms. The demise of the dinosaurs made way for the evolutionary diversification and success of mammals, so one could argue that a chance asteroid impact explains in part why human beings are here.

Chapter Summary

Organisms of the same species have some traits (characteristics) in common, but they may differ from one another in innumerable other traits. Many of the differences between individual organisms result from genetic differences, the effects of the environment, or both. Genetics is the study of inherited traits, including those influenced in part by the environment. The elements of heredity consist of genes, which are transmitted from parents to offspring in reproduction. Although the sorting of genes in successive generations was first put into numerical form by Mendel, the chemical basis of genes was discovered by Miescher in the form of a weak acid now called deoxyribonucleic acid (DNA). However, experimental proof that DNA is the genetic material did not come until about the middle of the twentieth century.

The first convincing evidence of the role of DNA in heredity came from experiments of Avery, MacLeod, and McCarty, who showed that genetic characteristics in bacteria could be altered from one type to another by treatment with purified DNA. In studies of *Streptococcus pneumoniae*, they transformed mutant cells unable to cause pneumonia into cells that could by treatment with pure DNA from disease-causing forms. A second important line of evidence was the Hershey-Chase experiment, which showed that the T2 bacterial virus injects primarily DNA into the host bacterium (*Escherichia coli*) and that a much higher proportion of parental DNA, as compared with parental protein, is found among the progeny phage.

The three-dimensional structure of DNA, proposed in 1953 by Watson and Crick, gave many clues about the manner in which DNA functions as the genetic material. A molecule of DNA consists of two long chains of nucleotide subunits twisted around one another to form a right-handed helix. Each nucleotide subunit contains any one of four bases: A (adenine), T (thymine), G (guanine), or C (cytosine). The bases are paired in the two strands of a DNA molecule. Wherever one strand has an A, the partner strand has a T, and wherever one strand has a G, the partner strand has a C. The base pairing means that the two paired strands in a DNA duplex molecule have complementary base sequences along their lengths. The structure of the DNA molecule suggested that genetic information could be coded in DNA in the sequence of bases. Mutations (changes in the genetic material) could result from changes in the sequence of bases, such as by the substitution of one nucleotide for another or by the insertion or deletion of one or more nucleotides. The structure of DNA also suggested a mode of replication in which the two strands of the parental DNA molecule separate and each individual strand serves as a template for the synthesis of a new complementary strand.

Most genes code for proteins. More precisely stated, most genes specify the sequence of amino acids in a polypeptide chain. The transfer of genetic information from DNA into protein is a multistep process that includes several types of RNA (ribonucleic acid). Structurally, an RNA strand is similar to a DNA strand except that the "backbone" contains a different sugar (ribose instead of deoxyribose) and RNA contains the base uracil (U) instead of thymine (T). Also, RNA is usually present in cells in the form of single, unpaired strands. The initial step in gene expression is transcription, in which a molecule of RNA is synthesized that is complementary in base sequence to whichever DNA strand is being transcribed. In polypeptide synthesis, which takes place on a ribosome, the base sequence in the RNA transcript is translated in groups of three adjacent bases (codons). The codons are recognized by different types of transfer RNA (tRNA) through base pairing. Each type of tRNA is attached to a particular amino acid, and when a tRNA base-pairs with the proper codon on the ribosome, the growing end of the polypeptide chain is attached to the amino acid on the tRNA. There are special codons that specify the "start" and "stop" of polypeptide synthesis. The most probable reason why various types of RNA are an intimate part of transcription and translation is that the earliest forms of life used RNA for both genetic information and enzyme catalysis.

A mutation that alters one or more codons in a gene may change the amino acid sequence of the resulting protein synthesized in the cell. Often the altered protein is functionally defective, so an inborn error of metabolism results. The particular manner in which an inborn error of metabolism is expressed can be very complex, because metabolism consists of an intricate branching network of biochemical pathways. Most visible traits of organisms result from many genes acting together in combination with environmental factors. The relationship between genes and traits is often complex because (1) every gene potentially affects many traits (hence a gene may show pleiotropy), (2) every trait is potentially affected by many genes (hence two different genes may interact, or show epistasis), and (3) many traits are significantly affected by environmental factors as well as by genes. Many complex traits include unrecognized subtypes that differ in their genetic or environmental causation. Progress in genetics has often resulted from finding ways to distinguish the subtypes.

All living creatures are united by sharing many features of the genetic apparatus (for example, transcription and translation) and many metabolic features. The unity of life results from common ancestry and is one of the evidences for evolution. There is also great diversity among living creatures. The three major kingdoms of organisms are the bacteria (which lack a membrane-bounded nucleus), the archaea (which share features with both eukarya and bacteria but form a distinct group), and eukarya (all "higher" organisms, whose cells have a membrane-bounded nucleus containing DNA organized into discrete chromosomes). The bacteria and archaea collectively are often called prokaryotes.

The ultimate source of diversity among organisms is mutation. However, natural selection is the process by which mutations that are favorable for survival and reproduction are retained and mutations that are harmful are eliminated. Natural selection, first proposed by Charles Darwin, is therefore the primary mechanism by which organisms become progressively more adapted to their environments. It is not always easy to determine how (or

whether) a particular trait is adaptive.

Key Terms

adenine (A)	base	colony
amino acid	biochemical pathway	complementary
Archaea	central dogma	cytosine (C)
Bacteria	chromosome	deoxyribonucleic acid (DNA)
bacteriophage	codon	double-stranded DNA

duplex DNA	messenger RNA	sickle-cell anemia
enzyme	mRNA	single-stranded DNA
epistasis	mutation	template
Eukarya	natural selection	thymine (T)
eukaryote	nucleotide	trait
evolution	phage	transcript
favism	pleiotropic effect	transcription
genes	pleiotropy	transfer RNA
genetics	polypeptide	transformation
glucose-6-phosphate dehydrogenase	prokaryote	translation
G6PD	replication	tRNA
G6PD deficiency	ribonucleic acid	uracil (U)
guanine (G)	ribosomal RNA	Watson-Crick pairing
hemoglobin	ribosome	wildtype
inborn error of metabolism	RNA	
just-so story	rRNA	

Review the Basics

- What is a trait? Give five examples of human traits. How could you determine whether each of these traits was genetically transmitted?
- How is it possible for a trait to be determined by *both* heredity and environment? Give an example of such a trait.
- How did understanding the molecular structure of DNA give clues to its ability to replicate, to code for proteins, and to undergo mutations?
- Why is pairing of complementary bases a key feature of DNA replication? What is the process of transcription and in what ways does it differ from DNA replication?
- How is the sequence of amino acids in a protein encoded in the sequence of nucleotides in a messenger RNA?
- What is "an inborn error of metabolism"? How did this concept serve as a bridge between genetics and biochemistry?

- What does it mean to say that any gene potentially affects more than one trait? What does it mean to say that one trait is potentially affected by more than one gene?
- If a species A is more closely related evolutionarily to species B than it is to species C, would you expect the DNA sequence of B to be closer to that of A or that of C? Why?

Guide to Problem Solving

Problem 1: A double-stranded DNA molecule has the sequence

5' -ATGCTTCATTTTCAGCTCGAATTTTGCC - 3'
3' -TACGAAGTAAAGTCGAGCTTAAAACGG - 5'

When this molecule is replicated, what is the base sequence of the new partner strand that is synthesized to pair with the upper strand? What is the base sequence of the new partner strand that pairs with the lower strand?

Answer: Both new strands have a nucleotide sequence that follows the Watson-Crick base-pairing rules of A with T and G with C. Therefore, the newly synthesized partner of the upper strand has a base sequence identical to that of the old lower strand, including the same 3' → 5' polarity. Similarly, the newly synthesized partner of the lower strand has a base sequence identical to that of the old upper strand.

Problem 2: Certain enzymes isolated from bacteria can recognize specific, short DNA sequences in duplex DNA and cleave both strands. The enzyme *AluI* is an example. It recognizes the sequence 5'-AGCT-3' in double-stranded DNA and cleaves both strands at the chemical bond connecting the G and C nucleotides. If the DNA duplex in Worked Problem 1 were cleaved with *AluI*, what DNA fragments would result?

Answer: There is only one 5'-AGCT-3' site in the duplex, so both strands would be cleaved once at the position between the G and the C in this sequence. Each cleavage generates a new 5' end and a new 3' end, which must maintain the polarity of the strand cleaved. Therefore, the resulting double-stranded DNA fragments are

5' -ATGCTTCATTTTCAG - 3' 5' -CTCGAATTTTGCC - 3'
3' -TACGAAGTAAAGTC - 5' 3' -GAGCTTAAAACGG - 5'

Problem 3: Suppose that one strand of the DNA duplex in Worked Problem 1 is transcribed from left to right as the molecule is illustrated. Which strand is the one transcribed? What is the sequence of the resulting transcript? (Hint: The 5' end of the RNA transcript is synthesized first.)

Chapter 1 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://www.jbpub.com/genetics>

Select the link to **Genetics: Principles and Analysis** and then choose the link to **GeNETics on the web**. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. James D. Watson once said that he and Francis Crick had no doubt that their proposed DNA structure was essentially correct, because the structure was so beautiful it had to be true. At an internet site accessed by the keyword **DNA**, you can view a large collection of different types of models of DNA structure. Some models highlight the sugar-phosphate backbones, others the A—T and G—C base pairs, still others the helical structure of double-stranded DNA. If assigned to do so, pick one of the models that appeals to you. Make a sketch of the model (or, alternatively, print the model), label the major components, and write a paragraph explaining why you find this representation appealing.

2. One of the first inborn errors of metabolism studied by Archibald Garrod (1902) was a condition called **alkaptonuria**. Use this keyword to learn about the symptoms of this condition and its molecular basis. What enzyme is defective in alkaptonuria? What substance is present in the urine of patients that causes it to turn dark upon standing? If assigned to do so, write a 200-word summary of what you have learned.

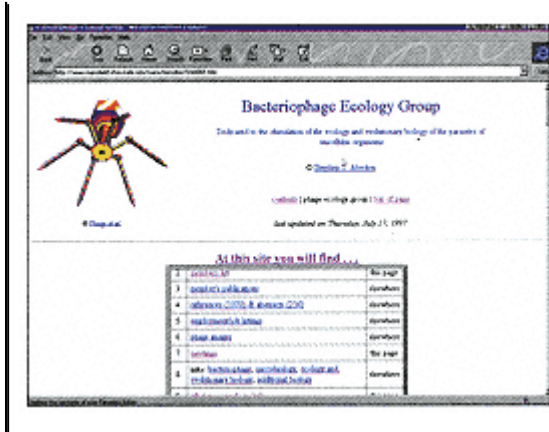
3. Perhaps surprisingly, the history of the bacteriophage T2 that figures so prominently in the experiments of Hershey and Chase is clouded in mystery. Use the keyword **T2** to learn what is known about its origin and the sleuthing required to find it out. If assigned, prepare a timeline (chronology) of T2 phage from the time of its first isolation (under a different name) and its passage from researcher to researcher until it received its "final" name, phage T2.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 1, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 1.



Answer: Because the 5' end of the RNA transcript is synthesized first, and transcription proceeds from left to right as the template molecule is drawn, the transcribed strand must be the lower strand. This is necessary so that the template DNA strand and the RNA transcript will have opposite polarities. In addition, the base uracil (U) in RNA replaces the base thymine (T) in DNA. Therefore, the RNA transcript (shown paired with the DNA template to illustrate the polarity relation) is RNA

RNA 5' - AUGCUUCAUUUCAGCUCGAAUUUUUGCC - 3'
 DNA 3' - TACGAAGTAAAGTCGAGCTTAAACGG - 5'

Problem 4: RNA transcripts can be translated *in vitro* using ribosomes, transfer RNAs, and other necessary constituents extracted from cells, but the first codon can be any sequence of three nucleotides (instead of AUG, which is used *in vivo*). A synthetic mRNA consists of the repeating tetranucleotide 5'-AUGC-3' and hence has the sequence

5' - AUGCAUGCAUGCAUGCAUGCAUGC . . . - 3'.

When this molecule is translated *in vitro*, the resulting polypeptide has the repeating sequence

Met—His—Ala—Cys—Met—His—Ala—Cys . . .

What does this result tell you about the number of nucleotides in a codon? Using the fact that the only methionine codon is 5'-AUG-3', deduce a codon for histidine (His), alanine (Ala), and cysteine (Cys). Would the result differ if

the mRNA were translated from the 3' end to the 5' end instead of in the actual direction from the 5' and to the 3' end?

Answer: The result means that each codon consists of three nucleotides and that they are translated in nonoverlapping groups of three. A repeating sequence of four nucleotides repeats four codons; in this case, the sequence

5' -AUGCAUGCAUGCAUGCAUGCAUGCAUGCAUGC -3'

is translated by grouping into the codons

5' -AUG CAU GCA UGC AUG CAU GCA UGC AUG CAU GC -3'

(You should verify that it does not matter at which nucleotide in the mRNA the translation begins, because all three possible reading frames yield the same set of repeating codons.) Because 5'-AUG-3' codes for Met, it follows that 5'-CAU-3' codes for His (the next amino acid after Met), 5'-GCA-3' codes for Ala (the next in line), and 5'-UGC-3' codes for Cys. Translation of the RNA in the 3' → 5' direction is precluded, because in this direction, there is no AUG codon, so the resulting polypeptide could not contain methionine.

Analysis and Applications

1.1 Considering that favism is brought on by eating broad beans, would you consider this a "genetic" trait or a trait caused by the environment? Why?

1.2 What is the end result of replication of a duplex DNA molecule?

1.3 What is the role of the messenger RNA in translation? What is the role of the ribosome? What is the role of transfer RNA? Is there more than one type of ribosome? Is there more than one type of transfer RNA?

1.4 What important observation about S and R strains of *Streptococcus pneumoniae* prompted Avery, MacLeod, and McCarty to study this organism?

1.5 In the transformation experiments of Avery, MacLeod, and McCarty, what was the strongest evidence that the substance responsible for the transformation was DNA rather than protein?

1.6 What feature of the physical organization of bacteriophage T2 made it suitable for use in the Hershey-Chase experiments?

1.7 Although the Hershey-Chase experiments were widely accepted as proof that DNA is the genetic material, the results were not completely conclusive. Why not?

1.8 The DNA extracted from a bacteriophage contains 28 percent A, 28 percent T, 22 percent G, and 22 percent C. What can you conclude about the structure of this DNA molecule?

1.9 The DNA extracted from a bacteriophage consists of 24 percent A, 30 percent T, 20 percent G, and 26 percent C. What is unusual about this DNA? What can you conclude about its structure?

1.10 A double-stranded DNA molecule is separated into its constituent strands, and the strands are separated in an ultracentrifuge. In one of the strands the base composition is 24 percent A, 28 percent T, 22 percent G, and 26 percent C. What is the base composition of the other strand?

1.11 While studying sewage, you discover a new type of bacteriophage that infects *E. coli*. Chemical analysis reveals protein and RNA but no DNA. Is this possible?

1.12 One strand of a DNA duplex has the base sequence 5'-ATCGTATGCACTTTACCCGG-3'. What is the base sequence of the complementary strand?

1.13 A region along one strand of a double-stranded DNA molecule consists of tandem repeats of the trinucleotide 5'-TCG-3', so the sequence in this strand is

5' -TCGTCGTCGTCGTCG . . . -3'

What is the sequence in the other strand?

1.14 A duplex DNA molecule contains a random sequence of the four nucleotides with equal proportions of each. What is the average spacing between consecutive occurrences of the sequence 5'-GGCC-3'? Between consecutive occurrences of the sequence 5'-GAATTC-3'?

1.15 A region along a DNA strand that is transcribed contains no A. What base will be missing in the corresponding region of the RNA?

1.16 The duplex nucleic acid molecule shown here consists of a strand of DNA paired with a complementary strand of RNA. Is the RNA the top or the bottom strand? One of the base pairs is mismatched. Which pairs is it?

5' -AUCGGUUACAUUCCGACUGA-3'
3' -TAGCCAATGTAAGGGTGACT-5'

1.17 The sequence of an RNA transcript that is initially synthesized is 5'-UAGCUAC-3', and successive nucleotides are added to the 3' end. This transcript is produced from a DNA strand with the sequence

3' -AAGTCGCATATCGATGCTAGCGCAACCT-5'

What is the sequence of the RNA transcript when synthesis is complete?

1.18 An RNA molecule folds back upon itself to form a "hairpin" structure held together by a region of base pairing. One segment of the molecule in the paired region has the base sequence 5'-AUACGAUA-3'. What is the base sequence with which this segment is paired?

1.19 A synthetic mRNA molecule consists of the repeating base sequence

5' -UUUUUUUUUUUU . . . -3'

When this molecule is translated *in vitro* using ribosomes, transfer RNAs, and other necessary constituents from *E. coli*, the result is a polypeptide chain consisting of the repeating amino acid Phe—Phe—Phe—Phe If you assume that the genetic code is a triplet code, what does this result imply about the codon for phenylalanine (Phe)?

1.20 A synthetic mRNA molecule consisting of the repeating base sequence

5' -UUUUUUUUUUUU . . . -3'

is terminated by the addition, to the right-hand end, of a single nucleotide bearing A. When translated *in vitro*, the resulting polypeptide consists of a repeating sequence of phenylalanines terminated by a single leucine. What does this result imply about the codon for leucine?

1.21 With *in vitro* translation of an RNA into a polypeptide chain, the translation can begin anywhere along the RNA molecule. A synthetic RNA molecule has the sequence

5' -CGCUUACCACAUGUCGCGAACUCG-3'

How many reading frames are possible if this molecule is translated *in vitro*? How many reading frames are possible if this molecule is translated *in vivo*, in which translation starts with the codon AUG?

1.22 You have sequenced both strands of a double-stranded DNA molecule. To inspect the potential amino acid coding content of this molecule, you conceptually transcribe it into RNA and then conceptually translate the RNA into a polypeptide chain. How many reading frames will you have to examine?

1.23 A synthetic mRNA molecule consists of the repeating base sequence 5'-UCUCUCUCUCUCUCUC . . . -3'. When this molecule is translated *in vitro*, the result is a polypeptide chain consisting of the alternating amino acids Ser—Leu—Ser—Leu—Ser—Leu Why do the amino acids alternate? What does this result imply about the codons for serine (Ser) and leucine (Leu)?

1.24 A synthetic mRNA molecule consists of the repeating base sequence 5'-AUCAUCAUCAUCAUC . . . -3'. When this molecule is translated *in vitro*, the result is a mixture of three different polypeptide chains. One consists of repeating isoleucines (Ile—Ile—Ile—Ile . . .), another of repeating serines (Ser—Ser—Ser—Ser . . .), and the third of repeating histidines (His—His—His—His . . .). What does this result imply about the manner in which an mRNA is translated?

1.25 How is it possible for a gene with a mutation in the coding region to encode a polypeptide with the same amino acid sequence as the nonmutant gene?

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In this small garden plot adjacent to the monastery of St. Thomas, Gregor Mendel grew more than 33,500 pea plants in the years 1856–1863, including more than 6,400 plants in one year alone. He received some help from two fellow monks who assisted in the experiments. Inside the monastery wall on the right is the Mendel museum (called the Mendelianum). The flowers are maintained as a memorial to Mendel's experiments.

Chapter 2— Principles of Genetic Transmission

CHAPTER OUTLINE

2-1 The Monohybrid Crosses

Traits Present in the Progeny of the Hybrids

Mendel's Genetic Hypothesis and Its Experimental Tests

The Principle of Segregation

Important Genetic Terminology

Verification of Mendelian Segregation by the Testcross

2-2 Segregation of Two or More Genes

The Principle of Independent Assortment

Dihybrid Testcrosses

The Big Experiment

2-3 Mendelian Inheritance and Probability

Mutually Exclusive Events: The Addition Rule

Independent Events: The Multiplication Rule

2-4 Segregation in Human Pedigrees

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Multiple Alleles

2-6 Modified Dihybrid Ratios Caused by Epistasis

2-7 Complications in the Concept of Dominance

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Codominance and the Human ABO Blood Groups

Incomplete Penetrance and Variable Expressivity

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

- Inherited traits are determined by the genes present in the reproductive cells united in fertilization.
- Genes are usually inherited in pairs—one from the mother and one from the father.
- The genes in a pair may differ in DNA sequence and in their effect on the expression of a particular inherited trait.
- The maternally and paternally inherited genes are not changed by being together in the same organism.
- In the formation of reproductive cells, the paired genes separate again into different cells.
- Random combinations of reproductive cells containing different genes result in Mendel's ratios of traits appearing among the progeny.
- The ratios actually observed for any traits are determined by the types of dominance and gene interaction.
- In genetic analysis, the complementation test is used to determine whether two recessive mutations that cause a similar phenotype are alleles of the same gene. The mutant parents are crossed, and the phenotype of the progeny is examined. If the progeny phenotype is nonmutant (complementation occurs), the mutations are in different genes; if the progeny phenotype is mutant (lack of complementation), the mutations are in the same gene.

CONNECTIONS

CONNECTION: What Did Gregor Mendel Think He Discovered?

Gregor Mendel 1866

Experiments on plant hybrids

CONNECTION: This Land Is Your Land, This Land Is My Land

The Huntington's Disease Collaborative Research Group 1993

A novel gene containing a trinucleotide repeat that is expanded and unstable on Huntington's disease chromosomes

Gregor Mendel's story is one of the inspiring legends in the history of modern science. Living as a monk at the distinguished monastery of St. Thomas in the town of Brno (Brünn), in what is now the Czech Republic, Mendel taught science at a local trade school and also carried out biological experiments. The most important experiments were crosses of sweet peas carried out from 1856 to 1863 in a small garden plot nestled in a corner of the monastery grounds. He reported his experiments to a local natural history society, published the results and his interpretation in its scientific journal in 1866, and began exchanging letters with Carl Nägeli in Munich, one of the leading botanists of the time. However, no one understood the significance of Mendel's work. By 1868, Mendel had been elected abbot of the monastery, and his scientific work effectively came to an end. Shortly before his death in 1884, Mendel is said to have remarked to one of the younger monks, "My scientific work has brought me a great deal of satisfaction, and I am convinced that it will be appreciated before long by the whole world." The prophecy was fulfilled 16 years later when Hugo de Vries, Carl Correns, and Erich von Tschermak, each working independently and in a different European country, published results of experiments similar to Mendel's, drew attention to Mendel's paper, and attributed priority of discovery to him.

Although modern historians of science disagree over Mendel's intentions in carrying out his work, everyone concedes that Mendel was a first-rate experimenter who performed careful and exceptionally well-documented experiments. His paper contains the first clear exposition of *transmission genetics*, or the statistical rules governing the transmission of hereditary elements from generation to generation. The elegance of Mendel's experiments explains why they were embraced as the foundation of genetics, and the rules of hereditary transmission inferred from his results are often referred to as **Mendelian genetics**. Mendel's breakthrough experiments and concepts are the subject of this chapter.

2.1—

The Monohybrid Crosses

The principal difference between Mendel's approach and that of other plant hybridizers of his era is that Mendel thought in quantitative terms about traits that could be classified into two contrasting categories, such as round seeds versus wrinkled seeds. He proceeded by carrying out quite simple crossing experiments and then looked for statistical regularities that might suggest general rules. In his own words, he wanted to "determine the number of different forms in which hybrid progeny appear" and to "ascertain their numerical interrelationships."

Mendel selected peas for his experiments for two reasons. First, he had access to varieties that differed in observable alternative characteristics, such as round versus wrinkled seeds and yellow versus green seeds. Second, his earlier studies had indicated that peas usually reproduce by self-pollination, in which pollen produced in a flower is used to fertilize the eggs in the same flower (Figure 2.1). To produce hybrids by cross-pollination, he needed only to open the keel petal (enclosing the reproductive structures), remove the immature anthers (the pollen-producing structures) before they shed pollen, and dust the stigma (the female structure) with pollen taken from a flower on another plant.

Mendel recognized the need to study inherited characteristics that were uniform within any given variety of peas but different between varieties (for example, round seeds always observed in one variety and wrinkled seeds always observed in another). For this reason, at the beginning of his experiments, he established **true-breeding** varieties in which the plants produced only progeny like themselves when allowed to self-pollinate normally. These different varieties, which bred true for seed shape, seed color, flower color, pod shape, or any of the other well-defined characters that Mendel had selected for investigation (Figure 2.2), provided the parents for subsequent hybridization. A **hybrid** is the off-

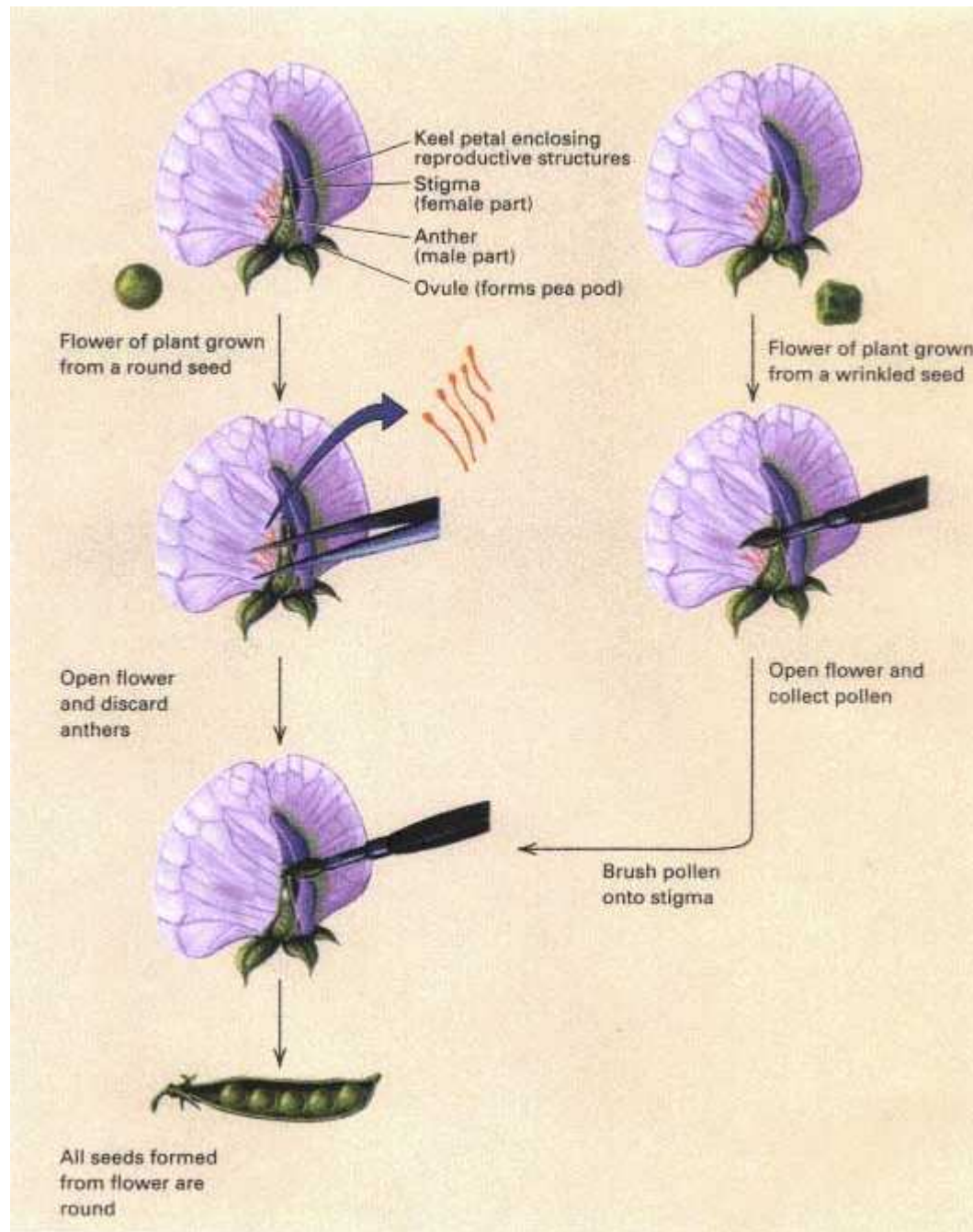


Figure 2.1

Crossing pea plants requires some minor surgery in which the anthers of a flower are removed before they produce pollen. The stigma, or female part of the flower, is not removed. It is fertilized by brushing with mature pollen grains taken from another plant.

spring of a cross between parents that differ in one or more traits. A **monohybrid** is a hybrid in which the parents differ in only one trait of interest. (They may differ in other traits as well, but the other differences are ignored for the purposes of the experiment.)

It is worthwhile to examine a few of Mendel's original experiments to learn what his methods were and how he interpreted his results. One pair of characters that he studied was round versus wrinkled seeds. When pollen from a variety of plants with wrinkled seeds was used to cross-pollinate plants from a variety with round seeds, all of the resulting hybrid seeds were round. Geneticists call the hybrid seeds or plants the **F₁ generation** to distinguish

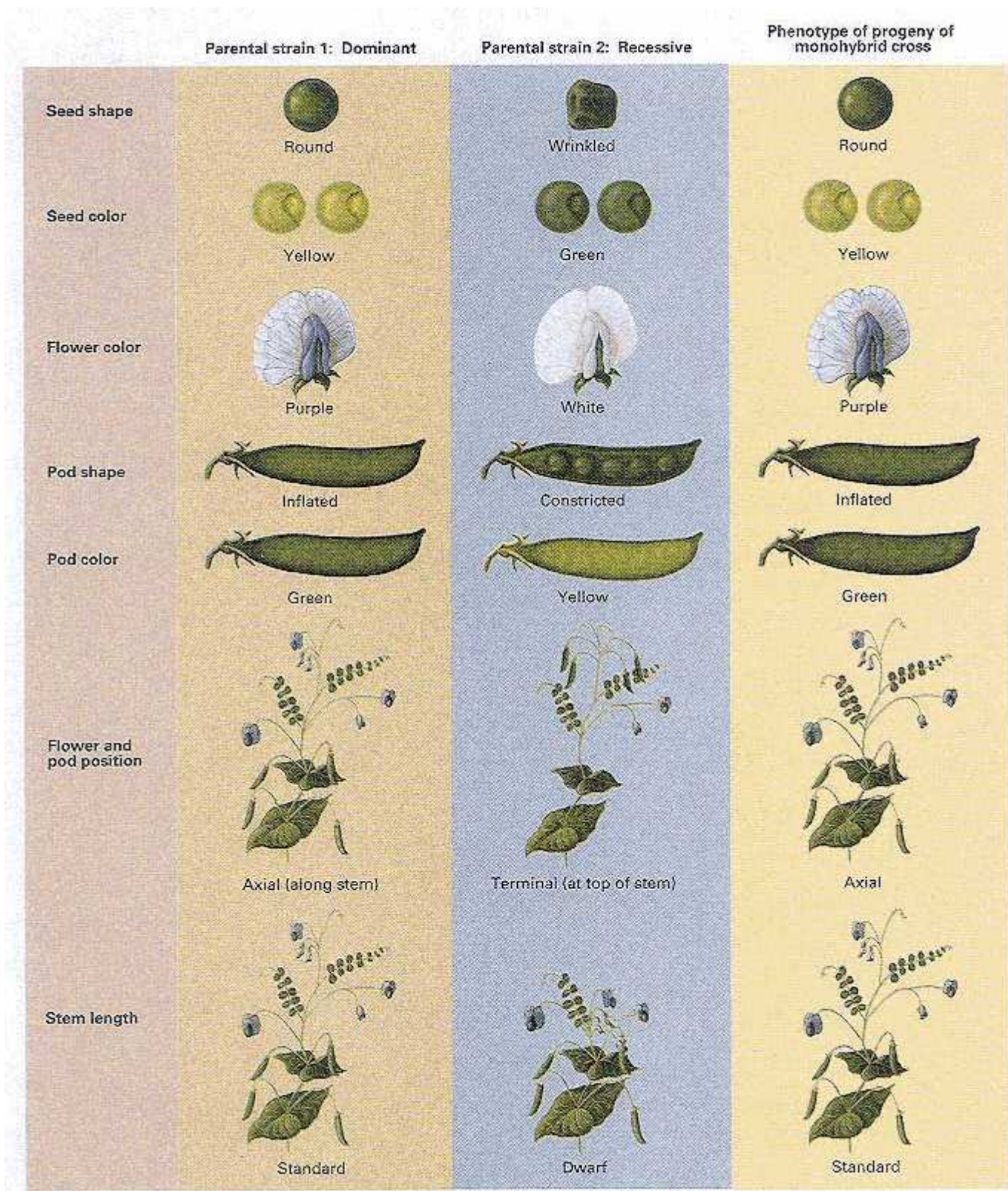


Figure 2.2

The seven character differences in peas studied by Mendel. The characteristic shown at the far right is the dominant trait that appears in the hybrid produced by crossing.

them from the pure-breeding parents, the **P₁ generation**. Mendel also performed the **reciprocal cross**, in which plants from the variety with round seeds were used as the pollen parents and those from the variety with wrinkled seeds as the female parents. As before, all of the F₁ seeds were round (Figure 2.3). The principle illustrated by the equal result of reciprocal crosses is that, with a few important exceptions that will be discussed in later chapters,

The outcome of a genetic cross does not depend on which trait is present in the male and which is present in the female; reciprocal crosses yield the same result.

Similar results were obtained when Mendel made crosses between plants that differed in any of the pairs of alternative characteristics. In each case, all of the F₁ progeny exhibited only one of the parental traits, and the other trait was absent. The trait expressed in the F₁ generation in each of the monohybrid crosses is shown at the right in Figure 2.2. The trait expressed in the hybrids Mendel called the **dominant** trait; the trait not expressed in the hybrids he called **recessive**.

Traits Present in the Progeny of the Hybrids

Although the recessive trait is not expressed in the hybrid progeny of a monohybrid cross, it reappears in the next generation when the hybrid progeny are allowed to undergo self-fertilization. For example, when the round hybrid seeds from the round × wrinkled cross were grown into plants and allowed to undergo self-fertilization, some of the resulting seeds were round and others wrinkled. The two types were observed in definite numerical

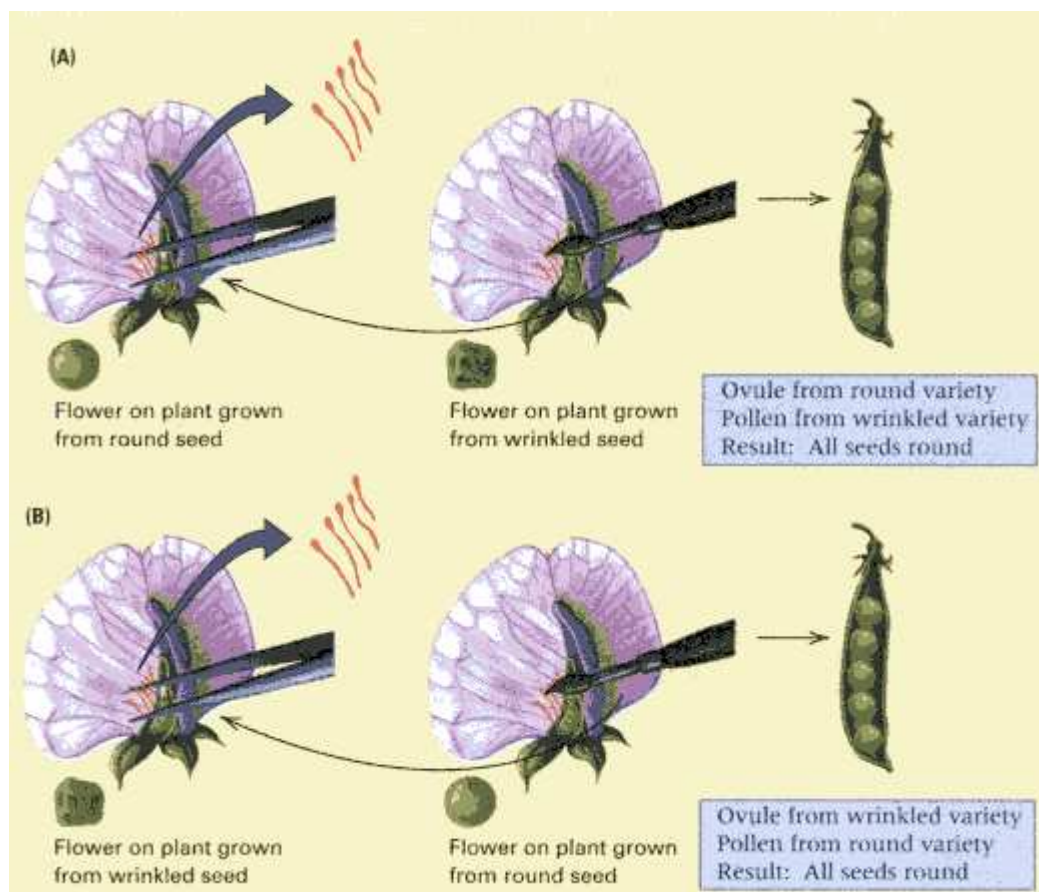
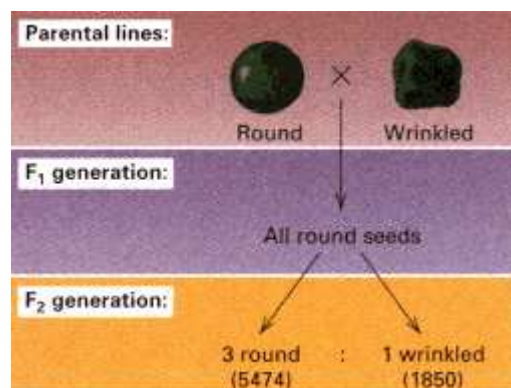


Figure 2.3

Mendel was the first to show that the characteristics of the progeny produced by a cross do not depend on which parent is the male and which the female. In this example, the seeds of the hybrid offspring are round whether the egg came from the round variety and the pollen from the wrinkled variety (A) or the other way around (B).

proportions. Mendel counted 5474 seeds that were round and 1850 that were wrinkled. He noted that this ratio was approximately 3 : 1.

The progeny seeds produced by self-fertilization of the F_1 generation constitute the **F_2 generation**. Mendel found that the dominant and recessive traits appear in the F_2 progeny in the proportions 3 round : 1 wrinkled. The results of crossing the round and wrinkled varieties are summarized in the following diagram.



Similar results were obtained in the F_2 generation of crosses between plants that differed in any of the pairs of alternative characteristics (Table 2.1). Note that the

Table 2.1 Results of Mendel's monohybrid experiments

Parental traits	F_1 trait	Number of F_2 progeny	F_2 ratio
round × wrinkled (Seeds)	round	5474 round, 1850 wrinkled	2.96 : 1
yellow × green (seeds)	yellow	6022 yellow, 2001 green	3.01 : 1
purple × white (flowers)	purple	705 purple, 224 white	3.15 : 1
inflated × constricted (pods)	inflated	882 inflated, 299 constricted	2.95 : 1
green × yellow (unripe pods)	green	428 green, 152 yellow	2.82 : 1
axial × terminal (flower position)	axial	651 axial, 207 terminal	3.14 : 1
long × short (stems)	long	787 long, 277 short	2.84 : 1

first two traits (round versus wrinkled seeds and yellow versus green seeds) have many more observations than any of the others; the reason is that these traits can be classified directly in the seeds, whereas the others can be classified only in the mature plants. The principal observations from the data in Table 2.1 were

- The F_1 hybrids express only the dominant trait.

- In the F_2 generation, plants with either

the dominant or the recessive trait are present.

- In the F_2 generation, there are approximately three times as many plants with the dominant trait as plants with the recessive trait. In other words, the F_2 ratio of dominant : recessive equals approximately 3 : 1.

In the remainder of this section, we will see how Mendel followed up these basic observations and performed experiments that led to his concept of discrete genetic units and to the principles governing their inheritance.

Mendel's Genetic Hypothesis and Its Experimental Tests

In Mendel's monohybrid crosses, the recessive trait that was not expressed in the F_1 hybrids reappeared in unchanged form in the F_2 generation, differing in no discernible way from the trait present in the original P_1 recessive plants. In a letter describing this finding, Mendel noted that in the F_2 generation, "the two parental traits appear, separated and unchanged, and there is nothing to indicate that one of them has either inherited or taken over anything from the other." From this finding, Mendel concluded that the hereditary determinants for the traits in the parental lines were transmitted as two different elements that retain their purity in the hybrids. In other words, the hereditary determinants do not "mix" or "contaminate" each other. Hence, a plant with the dominant trait might carry, in unchanged form, the hereditary determinant for the recessive trait.

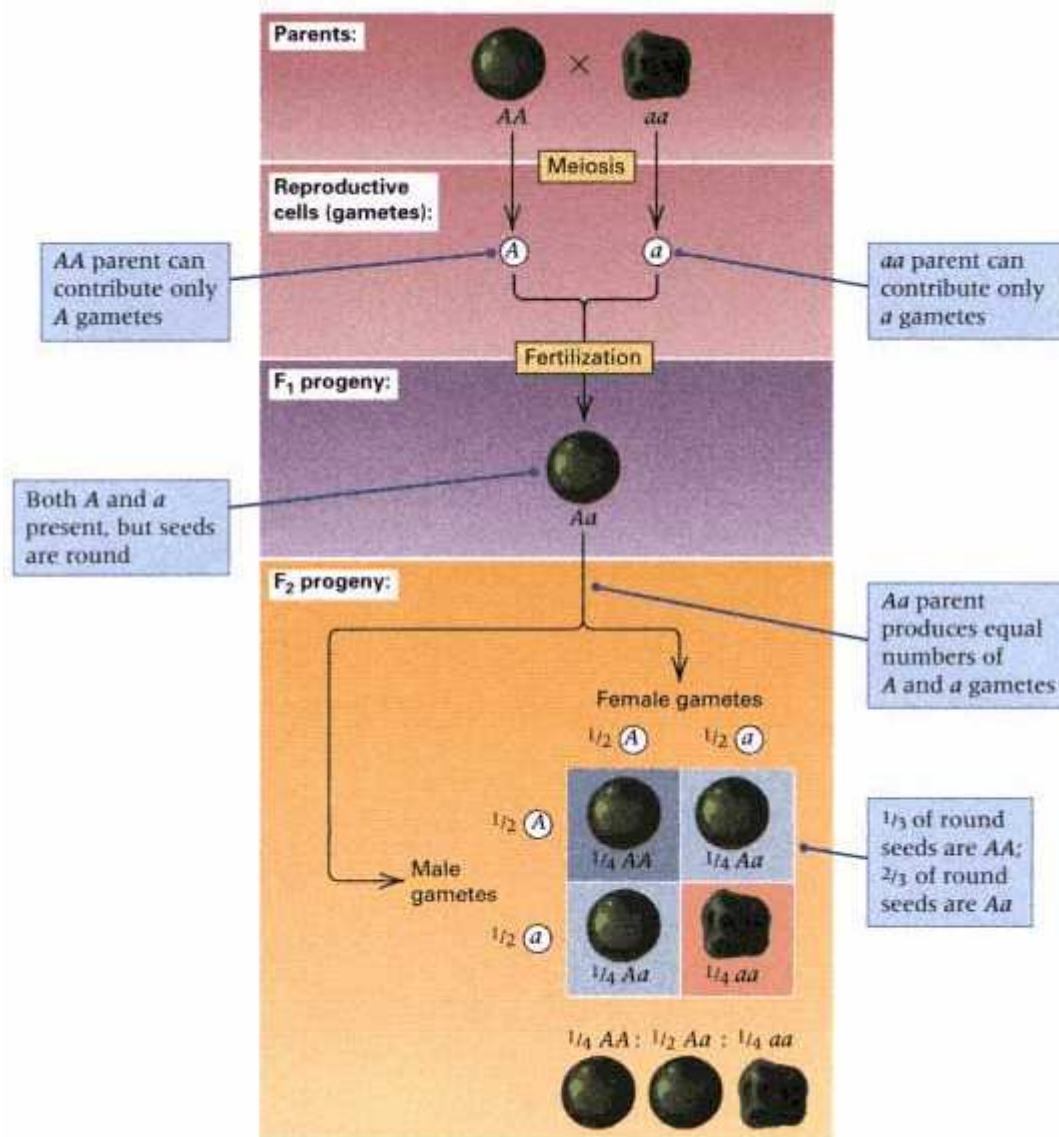


Figure 2.4

A diagrammatic explanation of Mendel's genetic hypothesis to explain the 3 : 1 ratio of dominant: recessive phenotypes observed in the F_2 generation of a monohybrid cross.

Note that the ratio of $AA : Aa : aa$ genetic types in the F_2 generation is 1 : 2 : 1.

To explain his results, Mendel developed a genetic hypothesis that can be understood with reference to Figure 2.4. He assumed that each reproductive cell, or *gamete*, contains one representative of each kind of hereditary determinant in the plant. The hereditary determinant for round seeds he called A ; that for wrinkled seeds he called a . Mendel proposed that in the true-breeding variety with round seeds, all of the reproductive cells would contain A ; in the true-breeding variety with wrinkled seeds, all of the reproductive cells would contain a . When the varieties are crossed, the F_1 hybrid should receive one of each of A and a and so should have the genetic constitution Aa (Figure 2.4). If A is dominant to a , the F_1 seeds should be round.

When an F_1 plant is self-fertilized, the A and a determinants would separate from one another and be included in the gametes in equal numbers. Hence, as shown in Figure 2.4, random combinations of the gametes should result in an F_2 generation with the genetic composition $1/4 AA$, $1/2 Aa$, and $1/4 aa$. The AA and Aa types should have round seeds, and the aa types should have wrinkled seeds, and so the predicted ratio of round : wrinkled seeds would be 3 : 1. (The genetic types AA , Aa , and aa can also be written with slashes as A/A , A/a and a/a , respectively; the two types of symbolism are equivalent.)

The genetic hypothesis in Figure 2.4 also illustrates another of Mendel's important

Connection What Did Gregor Mendel Think He Discovered?

Gregor Mendel 1866

Monastery of St. Thomas, Brno
[then Brünn], Czech Republic
Experiments on Plant Hybrids
(original in German)

Mendel's paper is remarkable for its precision and clarity. It is worth reading in its entirety for this reason alone. Although the most important discovery attributed to Mendel is segregation, he never uses this term. His description of segregation is found in the first passage in italics in the excerpt. (All of the italics are reproduced from the original.) In his description of the process, he takes us carefully through the separation of A and a in gametes and their coming together again at random in fertilization. One flaw in the description is Mendel's occasional confusion between genotype and phenotype, which is illustrated by his writing A instead of AA and a instead of aa in the display toward the end of the passage. Most early geneticists made no consistent distinction between genotype and phenotype until 1909, when the terms themselves were first coined.

Artificial fertilization undertaken on ornamental plants to obtain new color variants initiated the experiments reported here. The striking regularity with which the same hybrid forms always reappeared whenever fertilization between like species took place suggested further experiments whose task it was to follow that development of hybrids in their progeny. . . . This paper discusses the attempt at such a detailed experiment. . . . Whether the plan by which the individual experiments were set up and carried out was adequate to the assigned task should be decided by a benevolent judgment. . . . [Here the experimental results are described in detail.] Thus experimentation also justifies the assumption *that pea hybrids form germinal and pollen cells that in their composition correspond in equal numbers to all the constant forms resulting from the combination of traits united through fertilization.* The difference of forms among the progeny of hybrids, as well as the ratios in which they are observed, find an adequate explanation in the principle [of segregation] just deduced. The simplest case is given by the series for *one pair of differing traits*. It is shown that this series is described by the expression: $A + 2Aa + a$, in which A and a signify the forms with constant differing traits, and Aa the form hybrid for both. The series contains four individuals in three different terms. In their production, pollen and germinal cells of form A and a participate, on the average, equally in fertilization; therefore each form manifests itself twice, since four individuals are produced.

Participating in fertilization are thus:

Pollen cells $A + A + a + a$

Germinal cells $A + A + a + a$

Whether the plan by which the individual experiments were set up and carried out was adequate to the assigned task should be decided by a benevolent judgment

It is entirely a matter of chance which of the two kinds of pollen combines with each single germinal cell. However according to the laws of probability, in an average of many cases it will always happen that every pollen form A and a will unite equally often with every germinal-cell form A and a; therefore, in fertilization, one of the two pollen cells A will meet a germinal cell A, the other a germinal cell a, and equally, one pollen cell a will become associated with a germinal cell A, and the other a.

Pollen cells	A	A	a	a
Germinal cells	A	A	a	a

The result of fertilization can be visualized by writing the designations for associated germinal and pollen cells in the form of fractions, pollen cells above the line, germinal cells below. In the case under discussion one obtains

$$\frac{A}{A} + \frac{A}{a} + \frac{a}{A} + \frac{a}{a}$$

In the first and fourth terms germinal and pollen cells are alike; therefore the products of their association must be constant, namely A and a ; in the second and third, however, a union of the two differing parental traits takes place again, therefore the forms arising from such fertilizations are absolutely identical with the hybrid from which they derive. *Thus, repeated hybridization takes place.* The striking phenomenon, that hybrids are able to produce, in addition to the two parental types, progeny that resemble themselves is thus explained: Aa and aA both give the same association, Aa , since, as mentioned earlier, it makes no difference to the consequence of fertilization which of the two traits belongs to the pollen and which to the germinal cell. Therefore

$$\frac{A}{A} + \frac{A}{a} + \frac{a}{A} + \frac{a}{a} = A + 2Aa + a$$

This represents the *average* course of self-fertilization of hybrids when two differing traits are associated in them. In individual flowers and individual plants, however, the ratio in which the members of the series are formed may be subject to not insignificant deviations. . . . Thus it was proven experimentally that, in ***Pisum***, hybrids form *different kinds* of germinal and pollen cells and that this is the reason for the variability of their offspring.

Source: *Verhandlungen des naturforschenden den Vereines in Brünn* 4: 3–47

deductions: Two plants with the same outward appearance, such as round seeds, might nevertheless differ in their hereditary makeup as revealed by the types of progeny observed when they are crossed. For example, in the true-breeding round variety, the genetic composition of the seeds is AA , whereas in the F_1 hybrid seeds of the round $\times$ wrinkled cross, the genetic composition of the seeds is Aa .

But how could the genetic hypothesis be tested? Mendel realized that a key prediction of his hypothesis concerned the genetic composition of the round seeds in the F_2 generation. If the hypothesis is correct, then one-third of the round seeds should have the genetic composition AA and two-thirds of the round seeds should have the genetic composition Aa . This principle is shown in Figure 2.5. The ratio of $AA : Aa : aa$ in the F_2 generation is $1 : 2 : 1$, but if we disregard the recessives, then the ratio of $AA : Aa$ is $1 : 2$; in other words, $1/3$ of the round seeds are AA and $2/3$ are Aa . Upon self-fertilization, plants grown from the AA types should be true breeding for round seeds, whereas those from the Aa types should yield round and wrinkled seeds in the ratio $3 : 1$. Furthermore, among the wrinkled seeds in the F_2 generation, all should have the genetic composition aa , and so, upon self-fertilization, they should be true breeding for wrinkled seeds.

For several of his traits, Mendel carried out self-fertilization of the F_2 plants in order to test these predictions. His results for round versus wrinkled seeds are summarized in the diagram below:

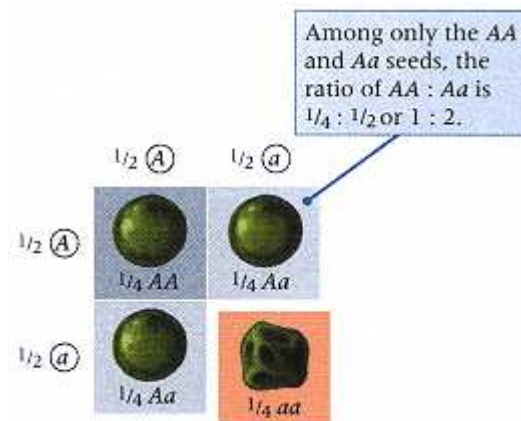
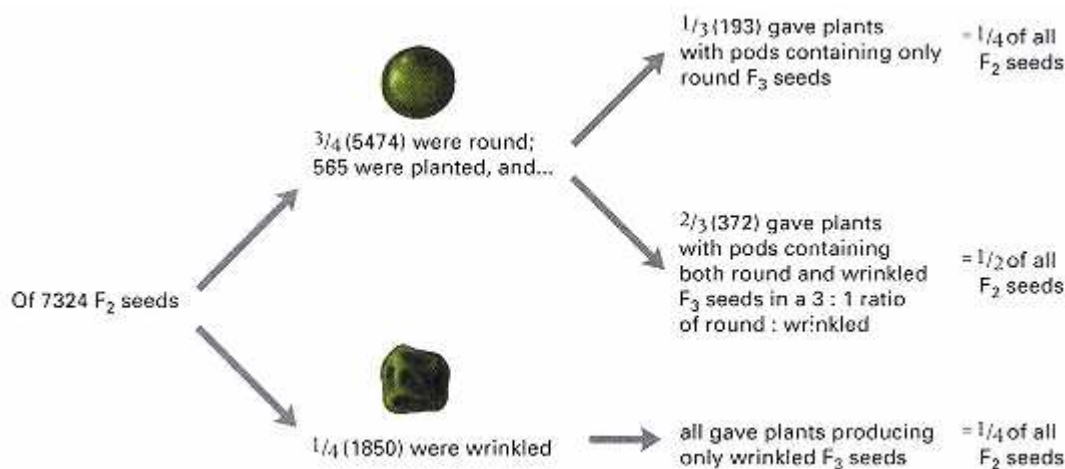


Figure 2.5

In the F_2 generation, the ratio of $AA : Aa$ is $1 : 2$.
Therefore, among those seeds that are round, $1/3$ should be AA and $2/3$ should be Aa .

As predicted from Mendel's genetic hypothesis, the plants grown from F_2 wrinkled seeds were true breeding for wrinkled seeds. They produced only wrinkled seeds in the F_3 generation. Moreover, among 565 plants grown from F_2 round seeds, 193 were true breeding, producing only round seeds in the F_3 generation, whereas the other 372 plants produced both round and wrinkled seeds in a proportion very close to $3 : 1$. The ratio $193 : 372$ equals $1 : 1.93$, which is very close to the ratio $1 : 2$ of $AA : Aa$ types predicted theoretically from the genetic hypothesis in Figure 2.4. Overall, taking all of the F_2 plants into account, the ratio of genetic types observed was very close to the predicted $1 : 2 : 1$ of $AA : Aa : aa$ expected from Figure 2.4.



The Principle of Segregation

The diagram in Figure 2.4 is the heart of Mendelian genetics. You should master it and be able to use it to deduce the progeny types produced in crosses. Be sure you thoroughly understand the meaning, and the biological basis, of the ratios 3 : 1 and 1 : 2 and 1 : 2 : 1. The following list highlights Mendel's key assumptions in formulating his model of inheritance.

1. For each of the traits that Mendel studied, a pea plant contains two hereditary determinants.
2. For each pair of hereditary determinants present in a plant, the members may be identical (for example, *AA*) or different (for example, *Aa*).
3. Each reproductive cell (gamete) produced by a plant contains only one of each pair of hereditary determinants (that is, either *A* or *a*).
4. In the formation of gametes, any particular gamete is equally likely to include either hereditary determinant (hence, from an *Aa* plant, half the gametes contain *A* and the other half contain *a*).
5. The union of male and female reproductive cells is a random process that reunites the hereditary determinants in pairs.

The essential feature of Mendelian genetics is the separation, technically called **segregation**, in unaltered form, of the two hereditary determinants in a hybrid plant in the formation of its reproductive cells (points 3 and 4 in the foregoing list). The principle of segregation is sometimes called *Mendel's first law*, although Mendel never used this term.

The Principle of Segregation: In the formation of gametes, the paired hereditary determinants separate (segregate) in such a way that each gamete is equally likely to contain either member of the pair.

Apart from the principle of segregation, the other key assumption, implicit in points 1 and 5 in the list, is that the hereditary determinants are present as pairs in both the parental organisms and the progeny organisms but as single copies in the reproductive cells.

Important Genetic Terminology

One of the handicaps under which Mendel wrote was the absence of an established vocabulary of terms suitable for describing his concepts. Hence he made a number of seemingly elementary mistakes, such as occasionally confusing the outward appearance of an organism with its hereditary constitution. The necessary vocabulary was developed only after Mendel's work was rediscovered, and it includes the following essential terms.

1. A hereditary determinant of a trait is called a **gene**.
2. The different forms of a particular gene are called **alleles**. In Figure 2.4, the alleles of the gene for seed shape are *A* for round seeds and *a* for wrinkled seeds. *A* and *a* are alleles because they are alternative forms of the gene for seed shape. Alternative alleles are typically represented by the same letter or combination of letters, distinguished either by uppercase and lowercase or by means of superscripts and subscripts or some other typographic identifier.
3. The **genotype** is the genetic constitution of an organism or cell. With respect to seed shape in peas, *AA*, *Aa*, and *aa* are examples of the possible genotypes for the *A* and *a* alleles. Because gametes contain only one allele of each gene, *A* and *a* are examples of genotypes of gametes.
4. A genotype in which the members of a pair of alleles are different, as in the *Aa* hybrids in Figure 2.4, is said to be **heterozygous**; a genotype in which the two alleles are alike is said to be **homozygous**. A homozygous organism may be homozygous dominant (*AA*) or homozygous recessive (*aa*). The terms *homozygous* and *heterozygous* can-

not apply to gametes, which contain only one allele of each gene.

5. The *observable* properties of an organism constitute its **phenotype**. Round seeds, wrinkled seeds, yellow seeds, and green seeds are all phenotypes. The phenotype of an organism does not necessarily tell you anything about its genotype. For example, a seed with the phenotype "round" could have the genotype AA or Aa .

Verification of Mendelian Segregation by the Testcross

A second way in which Mendel tested the genetic hypothesis in Figure 2.4 was by crossing the F_1 heterozygous genotypes with plants that were homozygous recessive. Such a cross, between an organism that is heterozygous for one or more genes (for example, Aa), and an organism that is homozygous for the recessive alleles (for example, aa), is called a **testcross**. The result of such a testcross is shown in Figure 2.6. Because the heterozygous parent is expected to produce A and a gametes in equal numbers, whereas the homozygous recessive produces only a gametes, the expected progeny are $1/2$ with the genotype Aa and $1/2$ with the genotype aa . The former have the dominant phenotype (because A is dominant to a) and the latter have the recessive phenotype. A testcross is often extremely useful in genetic analysis because

In a testcross, the relative frequencies of the different gametes produced by the heterozygous parent can be observed directly in the phenotypes of the progeny, because the recessive parent contributes only recessive alleles.

Mendel carried out a series of testcrosses with the genes for round versus wrinkled seeds, yellow versus green seeds, purple versus white flowers, and long versus short stems. The results are shown Table 2.2. In all cases, the ratio of phenotypes among the progeny is very close to the 1 : 1 ratio expected from segregation of the alleles in the heterozygous parent.

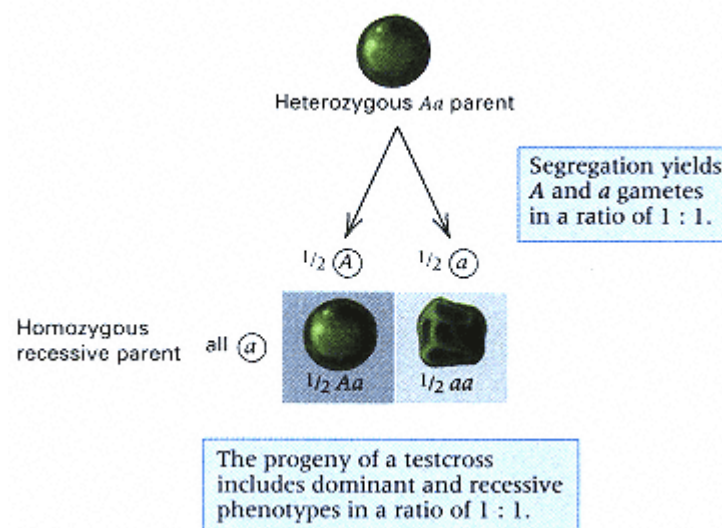


Figure 2.6
In a testcross of an Aa heterozygous parent with an aa homozygous recessive, the progeny are Aa and aa in the ratio of 1 : 1. A testcross shows the result of segregation.

Another valuable type of cross is a **backcross**, in which hybrid organisms are crossed with one of the parental genotypes. Backcrosses are commonly used by geneticists and by plant and animal breeders, as we will see in later chapters. Note that the testcrosses in Table 2.2 are also backcrosses, because in each case, the F_1 heterozygous parent came from a cross between the homozygous dominant and the homozygous recessive.

Table 2.2 Mendel's testcross results

Testcross (F_1 heterozygote $\times$ homozygous recessive)	Progeny from testcross	Ratio
Round $\times$ wrinkled seeds	193 round, 192 wrinkled	1.01 : 1
Yellow $\times$ green seeds	196 yellow,	

	189 green	1.04 : 1
Purple × white flowers	85 purple, 81 white	1.05 : 1
Long × short stems	87 long 79 short	1.10 : 1

2.2—

Segregation of Two or More Genes

Mendel also carried out experiments in which he examined the inheritance of two or more traits simultaneously to determine whether the same pattern of inheritance applied to each pair of alleles separately when more than one allelic pair was segregating in the hybrids. For example, plants from a true-breeding variety with round and yellow seeds were crossed with plants from a variety with wrinkled and green seeds. The F_1 progeny were hybrid for both characteristics, or **dihybrid**, and the phenotype of the seeds was round and yellow. The F_1 phenotype was round and yellow because round is dominant to wrinkled and yellow is dominant to green (Figure 2.2).

Then Mendel self-fertilized the F_1 progeny to obtain seeds in the F_2 generation. He observed four types of seed phenotypes in

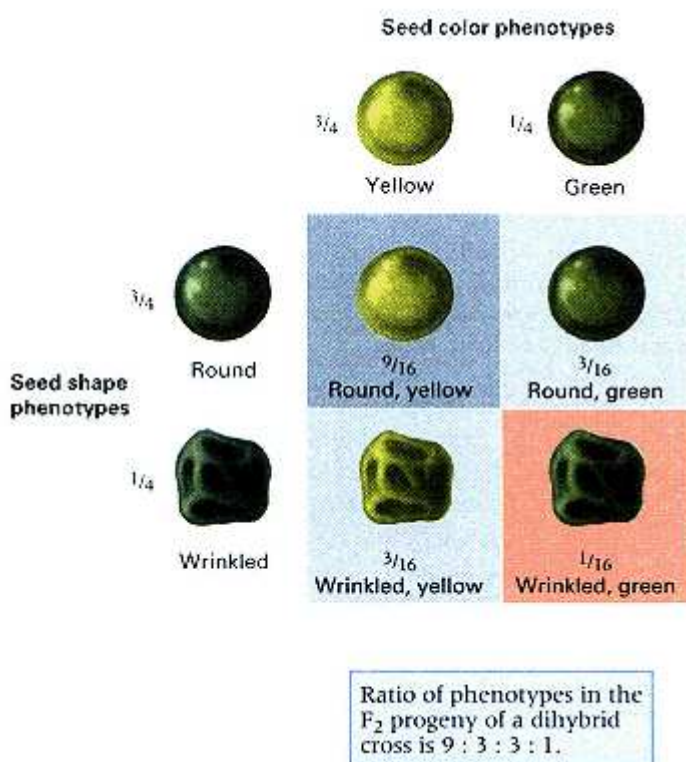


Figure 2.7

The 3 : 1 ratio of round : wrinkled, when combined at random with the 3 : 1 ratio of yellow : green, yields the 9 : 3 : 3 : 1 ratio that Mendel observed in the F_2 progeny of the dihybrid cross.

the progeny and, in counting the seeds, obtained the following numbers:

round, yellow	315
round, green	108
wrinkled, yellow	101
wrinkled, green	32
Total	556

In these data, Mendel noted the presence of the expected monohybrid 3 : 1 ratio for each trait separately. With

respect to each trait, the progeny were

$$\begin{aligned}\text{round : wrinkled} \\ &= (315 + 108) : (101 + 32) \\ &= 423 : 133 \\ &= 3.18 : 1\end{aligned}$$

$$\begin{aligned}\text{yellow : green} \\ &= (315 + 101) : (108 + 32) \\ &= 416 : 140 \\ &= 2.97 : 1\end{aligned}$$

Furthermore, in the F_2 progeny of the dihybrid cross, the separate 3 : 1 ratios for the two traits were combined at random, as shown in Figure 2.7. When the phenotypes of two traits are combined at random, then, among the $3/4$ of the progeny that are round, $3/4$ will be yellow and $1/4$ green; similarly, among the $1/4$ of the progeny that are wrinkled, $3/4$ will be yellow and $1/4$ green. The overall proportions of round yellow to round green to wrinkled yellow to wrinkled green are therefore expected to be $3/4 \times 3/4$ to $3/4 \times 1/4$ to $1/4 \times 3/4$ to $1/4 \times 1/4$ or

$$9/16 : 3/16 : 3/16 : 1/16$$

The observed ratio of 315 : 108 : 101 : 32 equals 9.84 : 3.38 : 3.16 : 1, which is reasonably close to the 9 : 3 : 3 : 1 ratio expected from the cross-multiplication of the separate 3 : 1 ratios in Figure 2.7.

The Principle of Independent Assortment

Mendel carried out similar experiments with other combinations of traits and, for each pair of traits he examined, consistently observed the 9 : 3 : 3 : 1 ratio. He also deduced the biological reason for this observation. To illustrate his explanation using the dihybrid round $\times$ wrinkled cross, we can represent the dominant and reces-

sive alleles of the pair that affect seed shape as W and w , respectively, and the allelic pair that affect seed color as G and g . Mendel proposed that the underlying reason for the 9 : 3 : 3 : 1 ratio in the F_2 generation is that the segregation of the alleles W and w for round or wrinkled seeds has no effect on the segregation of the alleles G and g for yellow or green seeds. Each pair of alleles undergoes segregation into the gametes independently of the segregation of the other pair of alleles. In the P_1 generation, the parental genotypes are $WW\ GG$ (round, yellow seeds) and $ww\ gg$ (wrinkled, green seeds). Then, the genotype of the F_1 is the double heterozygote $Ww\ Gg$. Note that this genotype can also be designated using the symbolism

$W/w; G/g$ or

$$\frac{W}{w} \frac{G}{g}$$

in which the slash (also called a virgule) is replaced with a short horizontal line.

The result of independent assortment in the F_1 plants is that the W allele is just as likely to be included in a gamete with G as with g , and the w allele is just as likely to be included in a gamete with G as with g . The independent segregation is illustrated in Figure 2.8. When two pairs of alleles undergo independent assortment, the gametes produced by the double heterozygote are

$$1/4\ WG\ 1/4\ Wg\ 1/4\ wG\ 1/4\ wg$$

When the four types of gametes combine at random to form the zygotes of the next generation, the result of independent assortment is shown in Figure 2.9. The cross-multiplication-like format, which is used to show how the F_1 female and male gametes may combine to produce the F_2 genotypes, is called a **Punnett square**. In the Punnett square, the possible phenotypes of the F_2 progeny are indicated. Note that the ratio of phenotypes is 9 : 3 : 3 : 1.

The Punnett square in Figure 2.9 also shows that the ratio of *genotypes* in the F_2 generation is not 9 : 3 : 3 : 1. With independent assortment, the ratio of genotypes in the F_2 generation is

$$1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1$$

The reason for this ratio is shown in Figure 2.10. Among seeds with the WW genotype,

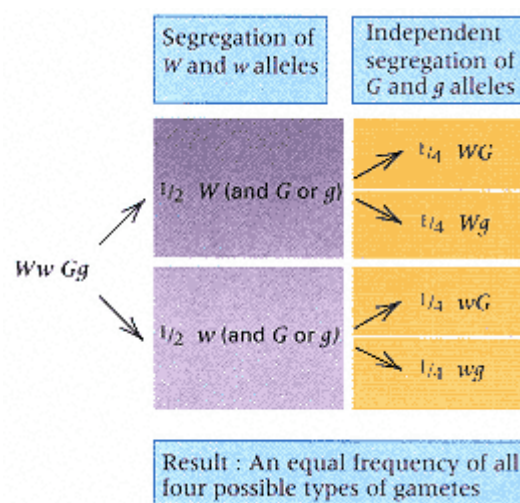


Figure 2.8
Independent segregation of the Ww and Gg allele pairs means that, among each of the W and w classes, the ratio of $G : g$ is 1 : 1. Likewise, among each of the G and g classes, the ratio of $W : w$ is 1 : 1.

the ratio of $GG : Gg : gg$ equals 1 : 2 : 1. Among seeds with the Ww genotype, the ratio is 2 : 4 : 2 (the 1 : 2 : 1 is multiplied by 2 because there are twice as many Ww genotypes as either WW or ww). And among seeds with the ww genotype, the ratio of $GG : Gg : gg$ equals 1 : 2 : 1. The phenotypes of the seeds are shown beneath the

genotypes. The combined ratio of phenotypes is 9 : 3 : 3 : 1. From Figure 2.10, one can also see that among seeds that are GG , the ratio of $WW : Ww : ww$ equals 1 : 2 : 1; among seeds that are Gg , it is 2 : 4 : 2; and among seeds that are gg , it is 1 : 2 : 1. Therefore, the independent segregation means that, among each of the possible genotypes formed by one allele pair, the ratio of homozygous dominant to heterozygous to homozygous recessive for the other allele pair is 1 : 2 : 1.

Mendel tested the hypothesis of independent segregation by ascertaining whether the predicted genotypes were actually present in the expected proportions. He did the tests by growing plants from the F_2 seeds and obtaining F_3 progeny by self-pollination. To illustrate the tests, consider one series of crosses in which he grew plants from F_2 seeds that were round, green. Note in Figures 2.9 and 2.10 that round, green F_2 seeds are expected to have the genotypes $Ww\ gg$ and $WW\ gg$ in the ratio 2 : 1. Mendel grew 102 plants from such seeds and found that 67 of them produced both round, green and wrinkled, green seeds (indicating that the parental plants must have been $Ww\ gg$) and 35 of them produced only round, green seeds (indicating that the parental genotype was

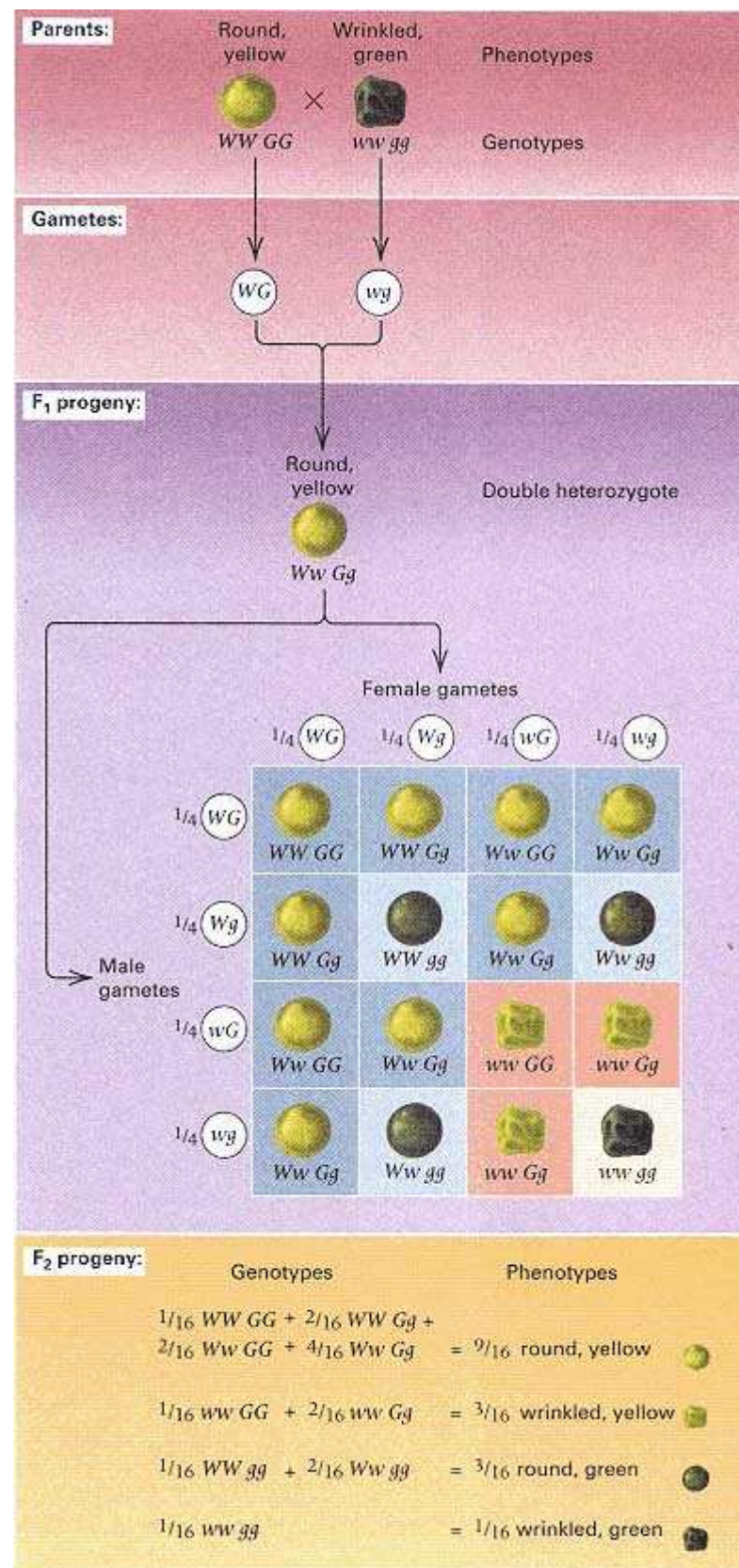


Figure 2.9

Diagram showing the basis for the 9 : 3 : 3 : 1 ratio of F₂ phenotypes resulting from a cross in which the parents differ in two traits determined by genes that undergo independent assortment.

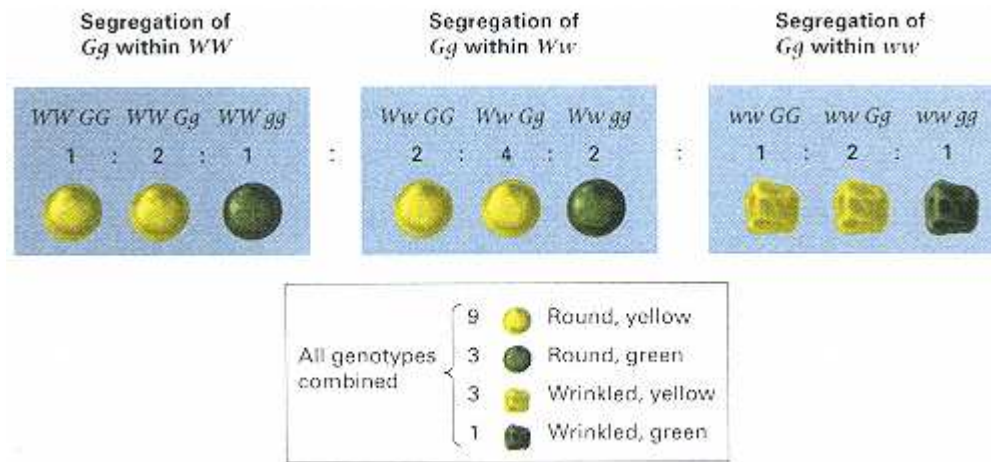


Figure 2.10

The F_2 progeny of the dihybrid cross for seed shape and seed color. In each of the genotypes for one of the allelic pairs, the ratio of homozygous dominant, heterozygous, and homozygous recessive genotypes for the other allelic pair is 1 : 2 : 1.

$WW\ gg$). The ratio 67 : 35 is in good agreement with the expected 2 : 1 ratio of genotypes. Similar good agreement with the predicted relative frequencies of the different genotypes was found when plants were grown from round, yellow or from wrinkled, yellow F_2 seeds. (As expected, plants grown from the wrinkled, green seeds, which have the predicted homozygous recessive genotype $ww\ gg$, produced only wrinkled, green seeds.)

Mendel's observation of independent segregation of two pairs of alleles has come to be known as the principle of independent assortment, or sometimes as *Mendel's second law*:

The Principle of Independent Assortment: Segregation of the members of any pair of alleles is independent of the segregation of other pairs in the formation of reproductive cells.

Although the principle of independent assortment is of fundamental importance in Mendelian genetics, in later chapters we will see that there are important exceptions.

Dihybrid Testcrosses

A second way in which Mendel tested the hypothesis of independent assortment was by carrying out a testcross with the F_1 genotypes that were heterozygous for both genes ($Ww\ Gg$). In a testcross, one parental genotype is always homozygous recessive, in this case $ww\ gg$. As shown in Figure 2.11 the double heterozygotes produce four types of gametes— WG , Wg , wG , and wg —in equal frequencies, whereas the $ww\ gg$ plants produce only wg gametes. Thus the progeny phenotypes are expected to consist of round yellow, round green, wrinkled yellow, and wrinkled green in a ratio of 1 : 1 : 1 : 1; the ratio of phenotypes is a direct demonstration of the ratio of gametes produced by the double heterozygote because no dominant alleles are contributed by the $ww\ gg$ parent to obscure the results. In the actual cross, Mendel obtained 55 round yellow, 51 round green, 49 wrinkled yellow, and 53 wrinkled green, which is in good agreement with the predicted 1 : 1 : 1 : 1 ratio. The results were the same in the reciprocal cross—that is, with the double heterozygote as the female parent and the homozygous recessive as the male parent. This observation confirmed Mendel's assumption that the gametes of both sexes included each possible genotype in approximately equal proportions.

The Big Experiment

Taking his hypothesis a step further, Mendel also carried out crosses between varieties that differed in three traits: seed shape (round or wrinkled, alleles W and w), seed color (yellow or green, alleles G and g), and flower color (purple or white, alleles P and p). The phenotype of the trihybrid F_1 seeds was round and yellow, and the plants grown from these seeds had purple flowers. By analogy with the dihybrid

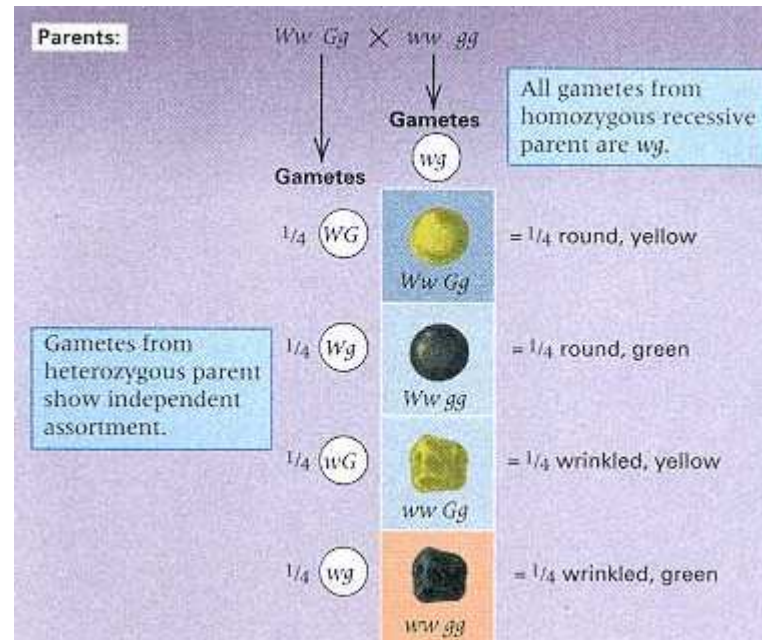


Figure 2.11
Genotypes and phenotypes resulting from a testcross of a $Ww Gg$ double heterozygote.

cross, if the alleles of all three genes undergo independent assortment, then self-fertilization of the F_1 flowers should result in combinations of phenotypes given by successive terms in the multiplication of $[(3/4)D + (1/4)R]^3$, which yields the ratio $27 : 9 : 9 : 9 : 3 : 3 : 3 : 1$. For Mendel's cross, the multiplication is carried out in Figure 2.12. The most frequent phenotype ($27/64$) has the dominant form of all three traits, the next most frequent ($9/64$) has the dominant form of two of the traits, the next most frequent ($3/64$) has the dominant form of only one trait, and the least frequent ($1/64$) is the triple recessive. Observe that if you consider any one of the traits and ignore the other two, then the ratio of phenotypes is $3 : 1$; and if you consider any two of the traits, then the ratio of phenotypes is $9 : 3 : 3 : 1$. This means that all of the possible one- and two-gene independent segregations are present in the overall three-gene segregation.

The observed and expected numbers in Figure 2.12 indicate that agreement with the hypothesis of independent assortment is very good. This, however, did not satisfy Mendel. He realized that there should be 27 different genotypes present in the F_2 progeny, so he self-fertilized each of the 639 plants to determine its genotype for each of the three traits. Mendel alludes to the amount of work this experiment entailed by noting that "of all the experiments, it required the most time and effort."

The result of the experiment is shown in Figure 2.13. From top to bottom, the

$(3/4 W- + 1/4 ww) \times (3/4 G- + 1/4 gg) \times (3/4 P- + 1/4 pp)$				Observed number	Expected number
$27/64$	$W- G- P-$	Round, yellow, purple		269	270
$9/64$	$W- G- pp$	Round, yellow, white		98	90
$9/64$	$W- gg P-$	Round, green, purple		86	90
$9/64$	$ww G- P-$	Wrinkled, yellow, purple		88	90
$3/64$	$W- gg pp$	Round, green, white		27	30
$3/64$	$ww G- pp$	Wrinkled, yellow, white		34	30
$3/64$	$ww gg P-$	Wrinkled, green, purple		30	30
$1/64$	$ww gg pp$	Wrinkled, green, white		7	10

● For any one gene, the ratio of phenotypes is $48 : 16 = 3 : 1$

● For any pair of genes, the ratio of phenotypes is $36 : 12 : 12 : 4 = 9 : 3 : 3 : 1$

Figure 2.12

With independent assortment, the expected ratio of phenotypes in a trihybrid cross is obtained by multiplying the three independent 3 : 1 ratios of the dominant and recessive phenotypes. A dash used in a genotype symbol indicates that either the dominant or the recessive allele is present; for example, $W—$ refers collectively to the genotypes WW and Ww . (The expected numbers total 640 rather than 639 because of round-off error.)

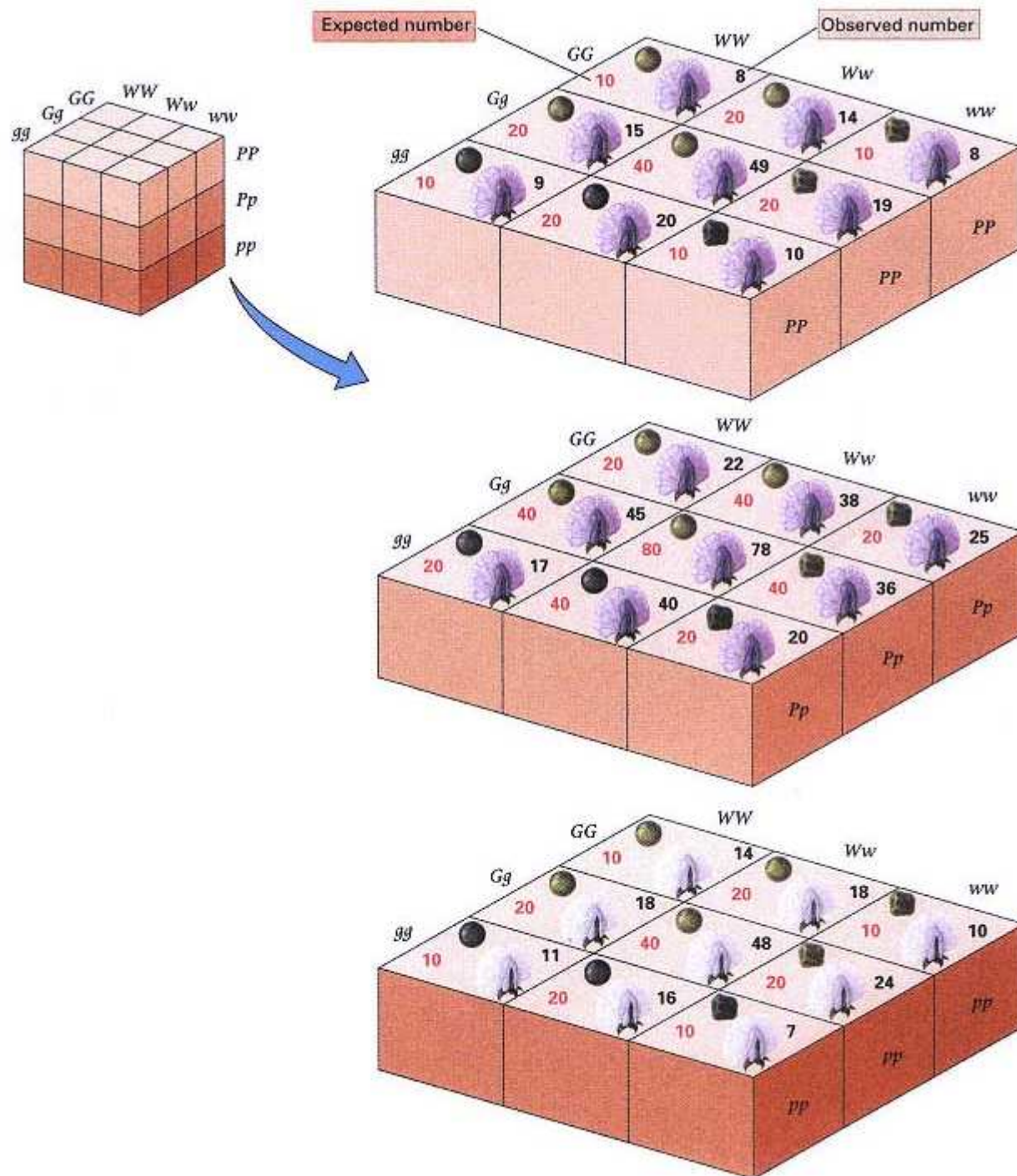


Figure 2.13

Results of Mendel's analysis of the genotypes formed in the F_2 generation of a trihybrid cross with the allelic pairs W, w and G, g and P, p . In each pair of numbers, the red entry is the expected number and the black entry is the observed. Note that each gene, by itself, yields a 1 : 2 : 1 ratio of genotypes and that each pair of genes yields a 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1 ratio of genotypes.

three Punnett squares show the segregation of W and w from G and g in the genotypes PP , Pp , and pp . In each cell, the number in red is the expected number of plants of each genotype, assuming independent assortment, and the number in black is the observed number of each genotype of plant. The excellent agreement

confirmed Mendel in what he regarded as the main conclusion of his experiments:

"Pea hybrids form germinal and pollen cells that in their composition correspond in equal numbers to all the constant forms resulting from the combination of traits united through fertilization."

In this admittedly somewhat turgid sentence, Mendel incorporated both segregation and independent assortment. In modern terms, what he means is that the gametes produced by any hybrid plant consist of equal numbers of all possible combinations of the alleles present in the original true-breeding parents whose cross produced the hybrids. For example, the cross $WW\,gg \times ww\,GG$ produces F_1 progeny of genotype $Ww\,Gg$, which yields the gametes WG , Wg , wG , and wg in equal numbers. Segregation is illustrated by the 1 : 1 ratio of $W : w$ and $G : g$ gametes, and independent assortment is illustrated by the equal numbers of WG , Wg , wG , and wg gametes.

2.3—

Mendelian Inheritance and Probability

A working knowledge of the rules of probability for predicting the outcome of chance events is basic to understanding the transmission of hereditary characteristics. In the first place, the proportions of the different types of offspring obtained from a cross are the cumulative result of numerous independent events of fertilization. Further-more, in each fertilization, the particular combination of dominant and recessive alleles that come together is random and subject to chance variation.

In the analysis of genetic crosses, the probability of an event may be considered as equivalent to the proportion of times that the event is expected to be realized in numerous repeated trials. Likewise, the proportion of times that an event is expected to be realized in numerous repeated trials is equivalent to the probability that it is realized in a single trial. For example, in the F_2 generation of the hybrid between pea varieties with round seeds and those with wrinkled seeds, Mendel observed 5474 round seeds and 1850 wrinkled seeds (Table 2.1). In this case, the proportion of wrinkled seeds was $1850/(1850 + 5474) = 1/3.96$, or very nearly $1/4$. We may therefore regard $1/4$ as the approximate proportion of wrinkled seeds to be expected among a large number of progeny from this cross. Equivalently, we can regard $1/4$ as the probability that any particular seed chosen at random will be wrinkled.

Evaluating the probability of a genetic event usually requires an understanding of the mechanism of inheritance and knowledge of the particular cross. For example, in evaluating the probability of obtaining a round seed from a particular cross, you need to know that there are two alleles, W and w , with W dominant over w ; you also need to know the particular cross, because the probability of round seeds is determined by whether the cross is

$WW \times ww$, in which all seeds are expected to be round,

$Ww \times Ww$, in which $3/4$ are expected to be round, or

$Ww \times ww$, in which $1/2$ are expected to be round.

In many genetic crosses, the possible outcomes of fertilization are equally likely. Suppose that there are n possible outcomes, each as likely as any other, and that in m of these, a particular outcome of interest is realized; then the probability of the outcome of interest is m/n . In the language of probability, an outcome of interest is typically called an *event*. As an example, consider the progeny produced by self-pollination of an Aa plant; four equally likely progeny genotypes (*outcomes*) are possible: namely AA , Aa , aA , and aa . Two of the four possible outcomes are heterozygous, so the probability of a heterozygote is $2/4$, or $1/2$.

Mutually Exclusive Events: The Addition Rule

Sometimes an outcome of interest can be expressed in terms of two or more possibilities. For example, a seed with the pheno-

type of "round" may have either of two genotypes, WW and Ww . A seed that is round cannot have both genotypes at the same time. With events such as the formation of the WW or Ww genotypes, only one event can be realized in any one organism, and the realization of one event in an organism precludes the realization of others in the same organism. In this example, realization of the genotype WW in a plant precludes realization of the genotype Ww in the same plant, and the other way around. Events that exclude each other in this manner are said to be *mutually exclusive*. When events are mutually exclusive, their probabilities are combined according to the addition rule.

Addition Rule: The probability of the realization of one or the other of two mutually exclusive events, A or B, is the sum of their separate probabilities.

In symbols, where *Prob* is used to mean *probability*, the addition rule is written

$$\text{Prob}\{A \text{ or } B\} = \text{Prob}\{A\} + \text{Prob}\{B\}$$

The addition rule can be applied to determine the proportion of round seeds expected from the cross $Ww \times Ww$. The round-seed phenotype results from the expression of either of two genotypes, WW and Ww , which are mutually exclusive. In any particular progeny organism, the probability of genotype WW is $1/4$ and that of Ww is $1/2$. Hence the overall probability of either WW or Ww is

$$\begin{aligned}\text{Prob}\{WW \text{ or } Ww\} \\ &= \text{Prob}\{WW\} + \text{Prob}\{Ww\} \\ &= 1/4 + 1/2 = 3/4\end{aligned}$$

Because $3/4$ is the probability of an individual seed being round, it is also the expected proportion of round seeds among a large number of progeny.

Independent Events: The Multiplication Rule

Events that are not mutually exclusive may be *independent*, which means that the realization of one event has no influence on the possible realization of any others. For example, in Mendel's crosses for seed shape and color, the two traits are independent, and the ratio of phenotypes in the F_2 generation is expected to be $9/16$ round yellow, $3/16$ round green, $3/16$ wrinkled yellow, and $1/16$ wrinkled green. These proportions can be obtained by considering the traits separately, because they are independent. Considering only seed shape, we can expect the F_2 generation to consist of $3/4$ round and $1/4$ wrinkled seeds. Considering only seed color, we can expect the F_2 generation to consist of $3/4$ yellow and $1/4$ green. Because the traits are inherited independently, among the $3/4$ of the seeds that are round, there should be $3/4$ that are yellow, so the overall proportion of round yellow seeds is expected to be $3/4 \times 3/4 = 9/16$. Likewise, among the $3/4$ of the seeds that are round, there should be $1/4$ green, yielding $3/4 \times 1/4 = 3/16$ as the expected proportion of round green seeds. The proportions of the other phenotypic classes can be deduced in a similar way. The principle is that when events are independent, the probability that they are realized together is obtained by multiplication.

Successive offspring from a cross are also independent events, which means that the genotypes of early progeny have no influence on the relative proportions of genotypes in later progeny. The independence of successive offspring contradicts the widespread belief that in each human family, the ratio of girls to boys must "even out" at approximately $1 : 1$, and so, if a family already has, say, four girls, they are somehow more likely to have a boy the next time around. But this belief is not supported by theory, and it is also contradicted by actual data on the sex ratios in human sibships. (The term **sibship** refers to a group of offspring from the same parents.) The data indicate that a human family is no more likely to have a girl on the next birth if it already has five boys than if it already has five girls. The statistical reason is that, though the sex ratios tend to balance out when they are averaged across a large number of sibships, they do not need to balance within individual sibships. Thus, among families in which there are five children, the sibships consisting of five boys balance those consisting of five girls, for an overall sex ratio of $1 : 1$. However, both of these sibships are unusual in their sex distribution.

When events are independent (such as independent traits or successive offspring from a cross), the probabilities are combined by means of the multiplication rule.

Multiplication Rule: The probability of two independent events, A and B, being realized simultaneously is given by the product of their separate probabilities.

In symbols, the multiplication rule is

$$Prob\{A \text{ and } B\} = Prob\{A\} \cdot Prob\{B\}$$

The multiplication rule can be used to answer questions like the following one: Of two offspring from the mating $Aa \times Aa$, what is the probability that both have the dominant phenotype? Because the mating is $Aa \times Aa$, the probability that any particular offspring has the dominant phenotype equals $3/4$. The multiplication rule says that the probability that both of two offspring have the dominant phenotype is $3/4 \times 3/4 = 9/16$.

Here is a typical genetic question that can be answered by using the addition and multiplication rules together: Of two offspring from the mating $Aa \times Aa$, what is the probability of one dominant phenotype and one recessive? Sibships of one dominant phenotype and one recessive can come about in two different ways—with the dominant born first or with the dominant born second—and these outcomes are mutually exclusive. The probability of the first case is $3/4 \times 1/4$ and that of the second is $1/4 \times 3/4$ because the events are mutually exclusive, the probabilities are added. The answer is therefore

$$(3/4 \times 1/4) + (1/4 \times 3/4) = 2(3/4)(1/4)$$

The addition and multiplication rules are very powerful tools for calculating the probabilities of genetic events. Figure 2.14 shows how the rules are applied to determine the expected proportions of the nine different genotypes possible among the F_2 progeny produced by self-pollination of a $Ww Gg$ dihybrid.

In genetics, independence applies not only to the successive offspring formed by a mating, but also to genes that segregate according to the principle of independent assortment (Figure 2.15). The independence means that the multiplication rule can be used to determine the probability of

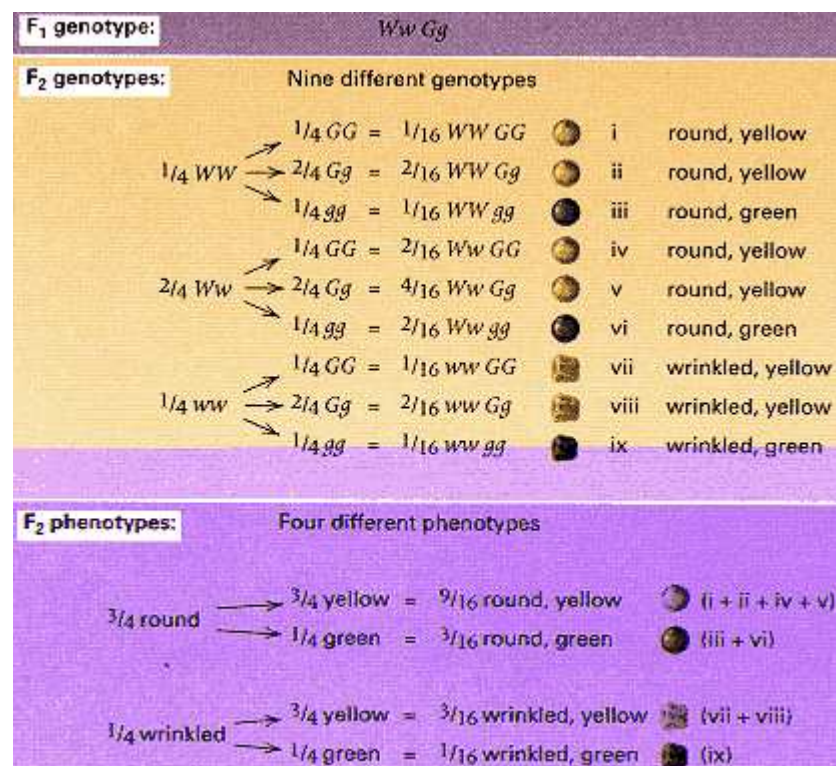


Figure 2.14
Example of the use of the addition and multiplication rules to determine the probabilities of the nine genotypes and four phenotypes in the F_2 progeny

obtained from self-pollination of a dihybrid F_1 . The Roman numerals are arbitrary labels identifying the F_2 genotypes.

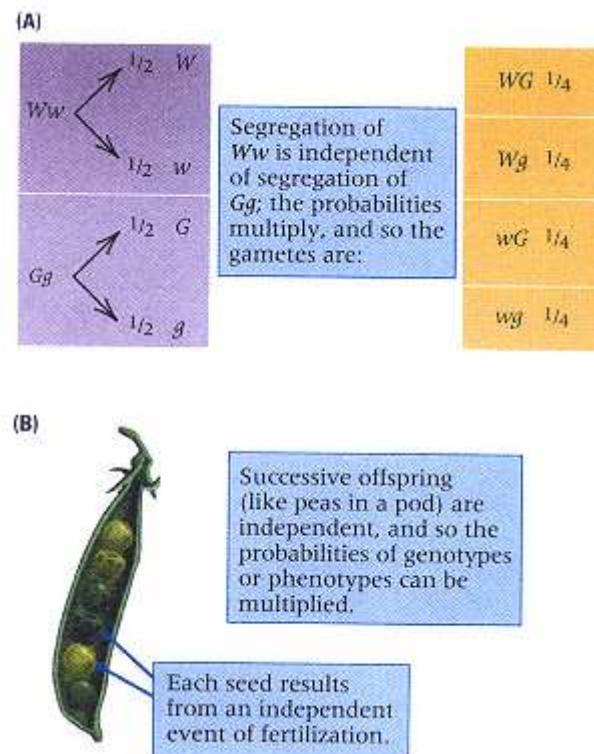


Figure 2.15

In genetics, two important types of independence are (A) independent segregation of alleles that show independent assortment and (B) independent fertilizations resulting in successive offspring. In these cases, the probabilities of the individual outcomes of segregation or fertilization are multiplied to obtain the overall probability.

the various types of progeny from a cross in which there is independent assortment among numerous pairs of alleles. This principle is the theoretical basis for the expected progeny types from a trihybrid cross, shown in Figure 2.12. One can also use the multiplication rule to calculate the probability of a specific genotype among the progeny of a cross. For example, if a quadruple heterozygote of genotype $Aa Bb Cc Dd$ is self-fertilized, the probability of a quadruple heterozygote $Aa Bb Cc Dd$ offspring is $(1/2) (1/2) (1/2) (1/2) = (1/2)^4$, or $1/16$, assuming independent assortment of all four pairs of alleles.

2.4—

Segregation in Human Pedigrees

Determining the genetic basis of a trait from the kinds of crosses that we have considered requires that we control matings between organisms and obtain large numbers of offspring to classify with regard to phenotype. The analysis of segregation by this method is not possible in human beings, and it is not usually feasible for traits in large domestic animals. However, the mode of inheritance of a trait can sometimes be determined by examining the appearance of the phenotypes that reflect the segregation of alleles in several generations of related individuals. This is typically done with a family tree that shows the phenotype of each individual; such a diagram is called a **pedigree**. An important application of probability in genetics is its use in pedigree analysis.

Figure 2.16 depicts most of the standard symbols used in drawing human pedigree. Females are represented by circles and males by squares. (A diamond is used if the sex of an individual is unknown.) Persons with the phenotype of interest are indicated by colored or shaded symbols. For recessive alleles, heterozygous carriers are depicted with half-filled symbols. A mating between a female and a male is indicated by joining their symbols with a horizontal line, which is connected vertically to a second horizontal line below that connects the symbols for their offspring. The offspring within a sibship, called **siblings** or **sibs** regardless of sex, are represented from left to right in order of their birth.

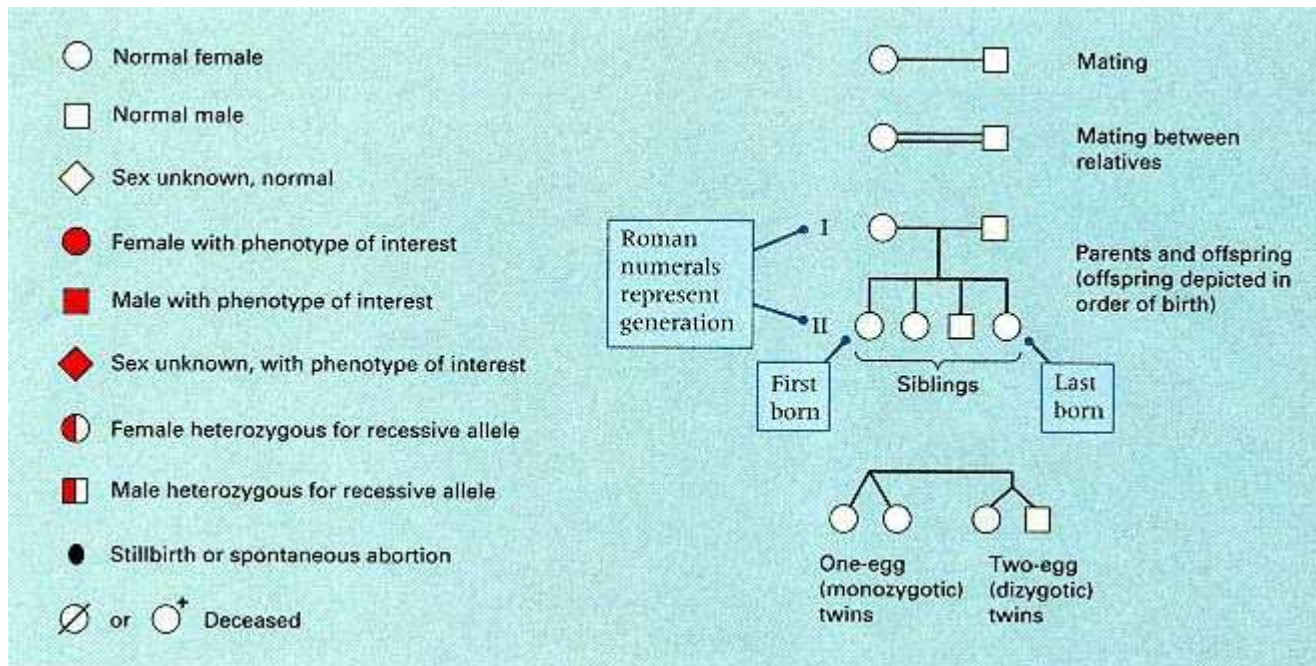


Figure 2.16
Conventional symbols used in depicting human pedigrees.

A pedigree for the trait **Huntington disease**, which is due to a dominant allele, is shown in Figure 2.17. The numbers in the pedigree are for convenience in referring to particular persons. The successive generations are designated by Roman numerals. Within any generation, all of the persons are numbered consecutively from left to right. The pedigree starts with the woman I-1 and the man I-2. He has Huntington disease, which is a progressive nerve degeneration that usually begins about middle age. It results in severe physical and mental disability and then death. The pedigree shows that the trait affects both sexes, that it is transmitted from affected parent to affected offspring, and that about half of all the offspring of an

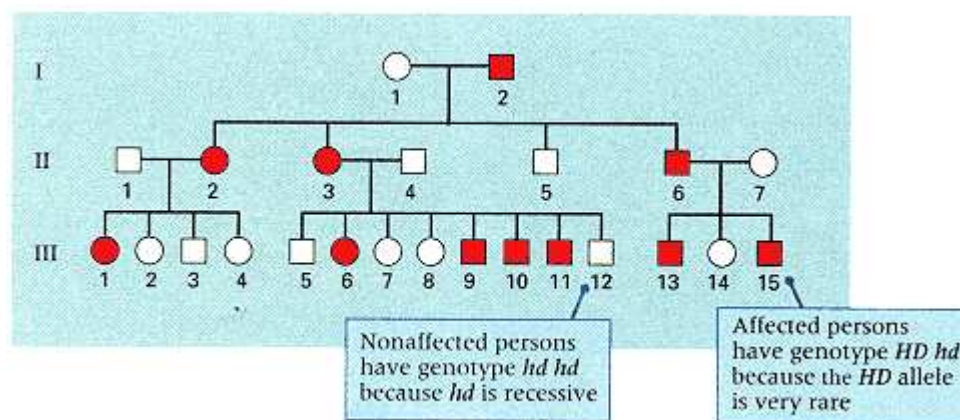


Figure 2.17
Pedigree of a human family showing the inheritance of the dominant gene for Huntington disease. Females and males are represented by circles and squares, respectively. Red symbols indicate persons affected with the disease.

Connection In Memoriam: This Land Is Your Land, This Land Is My Land

The Huntington's Disease Collaborative Research Group 1993

Comprising 58 authors among 9 institutions

A Novel Gene Containing a Trinucleotide Repeat That is Expanded and Unstable on Huntington's Disease Chromosomes

*Modern genetic research is sometimes carried out by large collaborative groups in a number of research institutions scattered across several countries. This approach is exemplified by the search for the gene responsible for Huntington disease. The search was highly publicized because of the severity of the disease, the late age of onset, and the dominant inheritance. Famed folk singer Woody Guthrie, who wrote "This Land Is Your Land" and other well-known tunes, died of the disease in 1967. When the gene was identified, it turned out to encode a protein (now called huntingtin) of unknown function that is expressed in many cell types throughout the body and not, as expected, exclusively in nervous tissue. Within the coding sequence of this gene is a trinucleotide repeat (5'-CAG-3') that is repeated in tandem a number of times according to the general formula (5'-CAG-3')_n. Among normal alleles, the number *n* of repeats ranges from 11 to 34 with an average of 18; among mutant alleles, the number of repeats ranges from 40 to 86. This tandem repeat is genetically unstable in that it can, by some unknown mechanism, increase in copy number ("expand"). In two cases in which a new mutant allele was analyzed, one had increased in repeat number from 36 to 44 and the other from 33 to 49. This is a mutational mechanism that is quite common in some human genetic diseases. The excerpt cites several other examples. The authors also emphasize that their discovery raises important ethical issues, including genetic testing, confidentiality, and informed consent.*

Huntington's disease (HD) is a progressive neurodegenerative disorder characterized by motor disturbance, cognitive loss, and

We consider it of the utmost importance that the current internationally accepted guidelines and counseling protocols for testing people at risk continue to be observed, and that samples from unaffected relatives should not be tested inadvertently or without full consent.

psychiatric manifestations. It is inherited in an autosomal dominant fashion and affects approximately 1 in 10,000 individuals in most populations of European origin. The hallmark of HD is a distinctive choreic [jerky] movement disorder that typically has a subtle, insidious onset in the fourth to fifth decade of life and gradually worsens over a course of 10 to 20 years until death. . . . The genetic defect causing HD was assigned to chromosome 4 in one of the first successful linkage analyses using DNA markers in humans. Since that time, we have pursued an approach to isolating and characterizing the HD gene based on progressively refining its localization. . . . [We have found that a] 500 kb segment is the most likely site of the genetic defect. [The abbreviation kb stands for kilobase pairs; 1 kb equals 1000 base pairs.] Within this region, we have identified a large gene, spanning approximately 210 kb, that encodes a previously undescribed protein. The reading frame contains a polymorphic (CAG)_n trinucleotide repeat with at least 17 alleles in the normal population, varying from 11 to 34 CAG copies. On HD chromosomes, the length of the trinucleotide repeat is substantially increased. . . . Elongation of a trinucleotide repeat sequence has been implicated previously as the cause of three quite different human disorders, the fragile-X syndrome, myotonic dystrophy, and spino-bulbar muscular atrophy. . . . It can be expected that the capacity to monitor directly the size of the trinucleotide repeat in individuals "at risk" for HD will revolutionize testing for the disorder. . . . We consider it of the utmost importance that the current internationally accepted guidelines and counseling protocols for testing people at risk continue to be observed, and that samples from unaffected relatives should not be tested inadvertently or without full consent. . . . With the mystery of the genetic basis of HD apparently solved, [it opens] the next challenges in the effort to understand and to treat this devastating disorder.

Source: *Cell* 72:971–983

affected parent are affected. These are characteristic features of simple Mendelian dominance. The dominant allele, *HD*, that causes Huntington disease is very rare. All affected persons in the pedigree have the heterozygous genotype *HD hd*, whereas nonaffected persons have the homozygous normal genotype *hd hd*.

A pedigree pattern for a trait due to a homozygous recessive allele is shown in Figure 2.18. The trait is **albinism**, absence of pigment in the skin, hair, and iris of the eyes. Both sexes can be affected, but the affected individuals need not have affected parents. The nonaffected parents are called **carriers** because they are heterozygous for the recessive allele; in a mating between carriers ($Aa \times Aa$), each offspring has a 1/4 chance of being affected. The pedigree also illustrates another feature found with

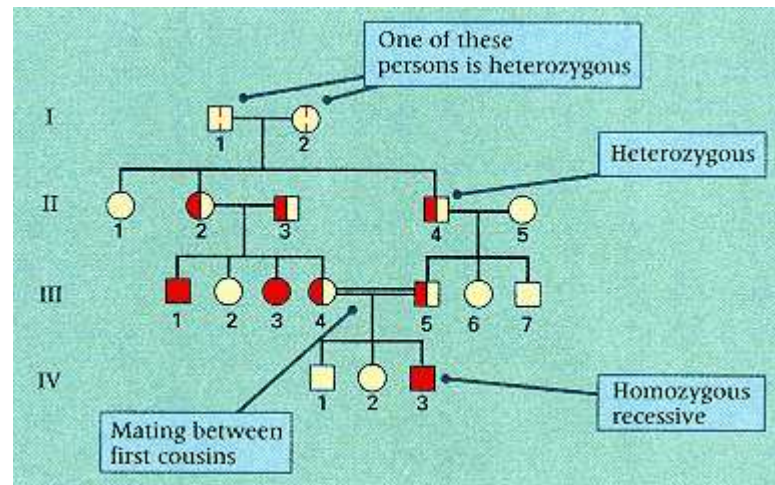


Figure 2.18
Pedigree of albinism. With recessive inheritance, affected persons (filled symbols) often have unaffected parents. The double horizontal line indicates a mating between relatives—in this case, first cousins.

recessive traits, particularly rare traits, which is that the parents of affected individuals are often related. A mating between relatives, in this case first cousins, is indicated with a double line connecting the partners.

Matings between relatives are important for observing rare recessive alleles, because when a recessive allele is rare, it is more likely to become homozygous through inheritance from a common ancestor than from parents who are completely unrelated. The reason is that the carrier of a rare allele may have many descendants who are carriers. If two of these carriers should mate (for example, in a first-cousin mating) the recessive allele can become homozygous with a probability of $1/4$. Mating between relatives constitutes *inbreeding*, and the consequences of inbreeding are discussed further in Chapter 15.

2.5— Genetic Analysis

When a geneticist makes the statement that a single gene with alleles P and p determines whether the color of the flowers on a pea plant will be purple or white, the statement does not imply that this gene is the only one responsible for flower color. The statement means only that this particular gene affecting flower color has been identified owing to the discovery of the recessive p mutation that, when homozygous, changes the color from purple to white. Many genes beside P are also necessary for purple flower coloration. Among these are genes that encode enzymes in the biochemical pathway for the synthesis of the purple pigment, anthocyanin. A geneticist interested in understanding the genetic basis of flower color would rarely be satisfied in having identified only the P gene necessary for purple coloration. The ultimate goal of a genetic analysis of flower color would be to isolate at least one mutation in every gene necessary for purple coloration and then, through further study of the mutant phenotypes, determine the normal function of each of the genes that affect the trait.

The Complementation Test in Gene Identification

In a genetic analysis of flower color, a geneticist would begin by isolating many new mutants with white flowers. Although mutations are usually very rare, their fre-

quency can be increased by treatment with radiation or certain chemicals. The isolation of a set of mutants, all of which show the same type of defect in phenotype, is called a **mutant screen**. Among the mutants that are isolated, some will contain mutations in genes already identified. For example, a genetic analysis of flower color in peas might yield one or more new mutations that changed the wildtype P allele into a defective p allele that prevents the formation of the purple pigment. Each of the $P \rightarrow p$ mutations might differ in DNA sequence, but all of the newly isolated p alleles would be defective forms of P that prevent formation of the purple pigment. On the other hand, a mutant screen should also yield mutations in genes not previously identified. Each of the new genes might be also represented by several recurrences of mutation, analogous to the multiple $P \rightarrow p$ mutations.

In a mutant screen for flower color, all of the new mutations are identified in plants with white flowers. Most of the new mutant alleles will encode an inactive protein needed for the formation of the purple pigment. The mutant alleles will be recessive, because in the homozygous recessive genotype, neither of the mutant alleles can produce the wildtype protein needed for pigmentation, and so the flowers will be white. In the heterozygous genotype, which carries one copy of the mutant allele along with one copy of the wildtype allele, the flowers will be purple because the wildtype allele codes for a functional protein that compensates for the defective protein encoded by the mutant.

Because white flowers may be caused by mutations in any of several genes, any two genotypes with white flowers may be homozygous recessive for alleles of the same gene or for alleles of different genes. After a mutant screen, how can the geneticist determine which pairs of mutations are alleles and which pairs of mutations are not alleles? The issue is illustrated for three particular white-flower mutations in Figure 2.19. Each of the varieties is homozygous for a recessive mutation that causes the flowers to be white. On the one hand, the varieties might carry separate occurrences of a mutation in the same gene; the mutations would be alleles. On the other hand, each mutation might be in a different gene; they would not be alleles.

The issue of possible allelism of the mutations is resolved by observing the phenotype of the progeny produced from a cross between the varieties. As indicated in Figure 2.19, there are two possible outcomes of the cross. The F_1 progeny have either the wildtype phenotype (Figure 2.19A, purple flowers) or the mutant phenotype (Figure 2.19B, white flowers). If the progeny have purple flowers, it means that the mutations in the parental plants are in different genes; this result is called **complementation**. When complementation is observed, it implicates two different genes needed for purple flowers. In this example, mutant strain 1 is homozygous pp for the recessive p allele. The other parent is homozygous for a mutant allele in a different gene, designated cc . The complete genotype of the parental strains should therefore be written $pp\ CC$ for mutant strain 1 and $PP\ cc$ for mutant strain 2. The cross yields the F_1 genotype $Pp\ Cc$, which is heterozygous Pp and heterozygous Cc . Because p and c are both recessive, the phenotype of the F_1 progeny is purple flowers.

The other possible outcome of the cross is shown in Figure 2.19B. In this case, the F_1 progeny have white flowers, which is the mutant phenotype. This outcome is called **noncomplementation**. The lack of complementation indicates that both parental strains have a mutation in the same gene, because neither mutant strain can provide the genetic function missing in the other. In this example, mutant strain 1 is known to have the genotype pp . Mutant strain 3 is homozygous for a different mutation (possibly a recurrence of p). Because the F_1 has white flowers, the genotype of the F_1 must be pp . But the only source of the second p allele must be mutant strain 3, which means that mutant strain 3 is also homozygous pp . In the figure, the particular allele in mutant strain 3 is designated p^* to indicate that this mutation arose independently of the original p allele.

The kind of cross illustrated in Figure 2.19 is a **complementation test**. As we have seen, it is used to determine whether recessive mutations in each of two different strains are alleles of the same gene.

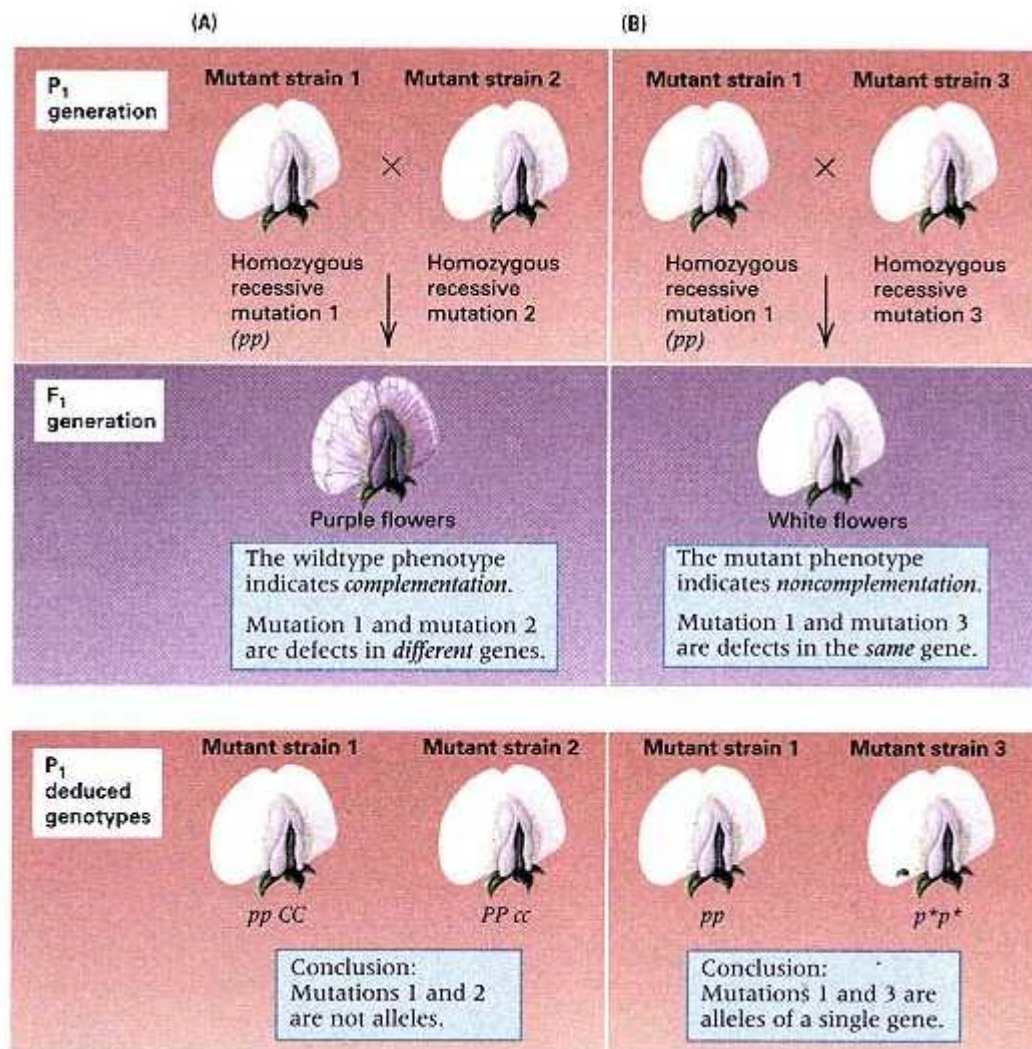


Figure 2.19

The complementation test reveals whether two recessive mutations are alleles of the same gene. In the complementation test, homozygous recessive genotypes are crossed. If the phenotype of the F₁ progeny is nonmutant (A), it means that the mutations in the parental strains are alleles of different genes. If the phenotype of the F₁ progeny is mutant (B), it means that the mutations in the parental strains are alleles of the same gene.

Because the result indicates the presence or absence of allelism, the complementation test is one of the key experimental operations in genetics. To illustrate the application of the test in practice, suppose a mutant screen were carried out to isolate new mutations for white flowers in peas. Starting with a true-breeding strain with purple flowers, we treat pollen with x rays and use the irradiated pollen to fertilize ovules to obtain seeds. The F₁ seeds are grown and the resulting plants allowed to self-fertilize, after which the F₂ plants are grown. A few of the F₁ seeds may contain a new mutation for white flowers, but because the white phenotype is recessive, the flower will be purple. However, the resulting F₁ plant will be heterozygous for the new white mutation, so self-fertilization will result in the formation of F₂ plants with a 3 : 1 ratio of purple : white flowers. Because mutations resulting in a particular phenotype are quite rare, even when induced by radiation, only a few among many thousands of self-fertilized plants, will be found to have a new white-

flower mutation. Let us suppose that we were lucky enough to obtain four new mutations, in addition to the p and c mutations identified by the complementation test in Figure 2.19.

How are we going to name these four new mutations? We can make no assumptions about the number of genes represented. All four could be recurrences of either p or c . On the other hand, each of the four could be a new mutation in a different gene needed for flower color. For the moment, let us call the new mutations $x1$, $x2$, $x3$, and $x4$, where the "x" does not imply a gene but rather that the mutation was obtained with x irradiation. Each mutation is recessive and was identified through the white flowers of the homozygous recessive F_2 seeds (for example, $x1 x1$).

Now the complementation test is used to classify the "x" mutations into groups.

Figure 2.20 shows that the results of a complementation test are typically reported in a triangular array of + and - signs. The crosses that yield F_1 progeny with the wildtype phenotype (in this case, purple flowers) are denoted with a + in the box where imaginary lines from the male parent and the female parent intersect. The crosses that yield F_1 progeny with the mutant phenotype (white flowers) are denoted with a - sign. The + signs indicate complementation between the mutant alleles in the parents; the - signs indicate lack of complementation. The bottom half of the triangle is unnecessary because the reciprocal of each cross produces F_1 progeny with the same genotype and phenotype as the cross that is shown. The diagonal elements are also unnecessary, because a cross between any two organisms carrying the identical mutation, for

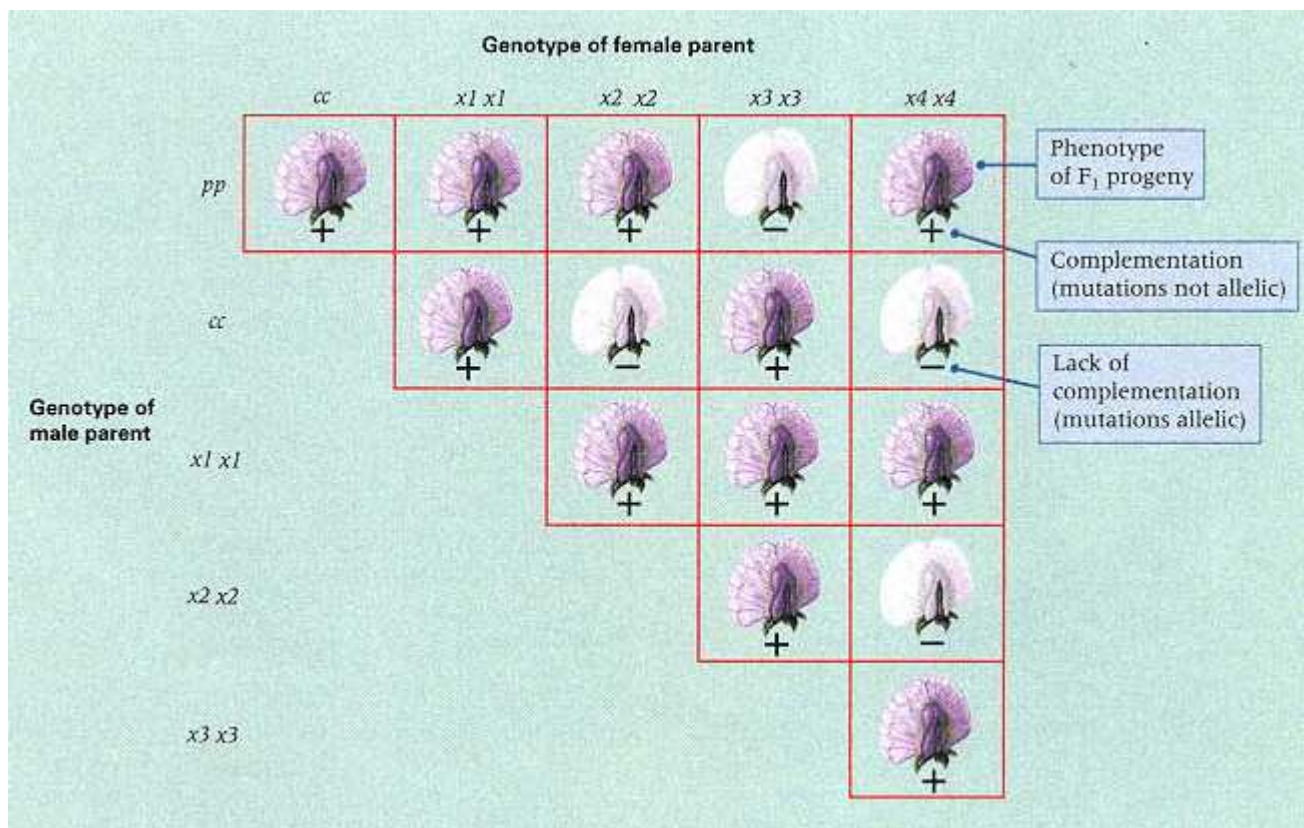


Figure 2.20

Results of complementation tests among six mutant strains of peas, each homozygous for a recessive allele resulting in white flowers. Each box gives the phenotype of the F_1 progeny of a cross between the male parent whose genotype is indicated in the far left column and the female parent whose genotype is indicated in the top row.

example, $x1\ x1 \times x1\ x1$, must yield homozygous recessive $x1\ x1$ progeny, which will be mutant. As we have seen in Figure 2.19, complementation in a cross means that the parental strains have their mutations in different genes. Lack of complementation means that the parental mutations are in the same gene. The principle underlying the complementation test is

The Principle of Complementation: If two recessive mutations are alleles of the same gene, then the phenotype of an organism containing both mutations is mutant; if they are alleles of different genes, then the phenotype of an organism containing both mutations is wildtype (nonmutant).

In interpreting complementation data such as those in Figure 2.20, we actually apply the principle the other way around. Examination of the phenotype of the F_1 progeny of each possible cross reveals which of the mutations are alleles of the same gene:

In a complementation test, if the combination of two recessive mutations results in a mutant phenotype, then the mutations are regarded as alleles of the same gene; if the combination results in a wildtype phenotype, then the mutations are regarded as alleles of different genes.

A convenient way to analyze the data in Figure 2.20 is to arrange the alleles in a circle as shown in Figure 2.21A. Then, for each possible pair of mutations, connect the pair by a straight line if the mutations *fail* to complement (Figure 2.21B). According to the principle of complementation, the lines must connect mutations that are alleles of each other because, in a complementation test, lack of complementation means that the mutations are alleles. In this example, mutation $x3$ is an allele of p , so $x3$ and p are different mutant alleles of the gene P . Similarly, the mutations $x2$, $x4$, and c are different mutant alleles of the gene C . The mutation $x1$ does complement all of the others. It represents a third gene, different from P and C , that affects flower coloration.

In an analysis like that in Figure 2.21, each of the groups of noncomplementing mutations is called a **complementation group**. As we have seen, each complementation group defines a gene.

A gene is defined experimentally as a set of mutations that make up one complementation group. Any pair of mutations in such a group fail to complement one another and result in an organism with an observable mutant phenotype.

The mutations in Figure 2.21 therefore represent three genes, a mutation in any one of which results in white flowers. The gene P is represented by the alleles p and $x3$; the gene C is represented by the alleles c , $x2$, and $x4$; and the allele $x1$ represents a third gene different from either P or C . Each gene coincides with one of the complementation groups.

At this point in a genetic analysis, it is possible to rename the mutations to indicate which ones are true alleles. Because the p allele already had its name before the mutation screen was carried out to obtain more flower-color mutations, the new allele of p , $x3$, should be renamed to reflect its allelism with p . We might rename the $x3$ mutation p_3 , for example, using the subscript to indicate that p_3 arose independently of p . For similar reasons, we might rename the $x2$ and $x4$ mutations c_2 and c_4 to reflect their allelism with the original c mutation and to convey their independent origins. The $x1$ mutation represents an allele of a new gene to which we can assign a name arbitrarily. For example, we might call the mutation *albus* (Latin for white) and assign the $x1$ allele the new name *alb*. The wildtype dominant allele of *alb*, which is necessary for purple coloration, would then be symbolized as *Alb* or as *alb*⁺. The procedure of sorting new mutations into complementation groups and renaming them according to their allelism is an example of how geneticists identify genes and name alleles. Such renaming of alleles is the typical manner in which genetic terminology evolves as knowledge advances.

Why Does the Complementation Test Work?

There is an old Chinese saying that the correct naming of things is the beginning of wisdom, and this is certainly true in the case of genes. The proper renaming of the

p, *c*, and *alb* mutations to indicate which mutations are alleles of the three genes is a wise way to create a terminology that indicates, for each possible genotype, what the phenotype will be with regard to flower color. A purple flower requires the presence of at least one copy of each of the wildtype *P*, *C*, and *Alb* alleles. Any genotype that contains two mutant alleles of *P* will have white flowers. These genotypes are *pp*, *p₃p₃*, and *pp₃*. Likewise, any genotype that contains two mutant alleles of *C* will have white flowers. These genotypes are *cc*, *c₂c₂*, *c₄c₄*, *cc₂*, *cc₄*, and *c₂c₄*. Finally, any genotype that contains two mutant alleles of *Alb* will have white flowers. In this case there is only one such genotype, *alb alb*.

The biological reason why the screen for flower-color mutants yielded mutations in each of three genes is based on the biochemical pathway by which the purple pigment is synthesized in the flowers. Examination of the biochemical pathway also explains why the complementation test works. The pathway is illustrated in Figure 2.22. The purple pigment anthocyanin is produced from a colorless precursor by way of two colorless intermediate compounds denoted X and Y. Each arrow represents a "step" in the pathway, a biochemical conversion from one substance to the next along the way. Each step requires an enzyme encoded by the wildtype allele of the gene indicated at the top. The allele *P*, for example, codes for the enzyme required in the last step in the pathway, which converts intermediate compound Y into anthocyanin. If this enzyme is missing (or is present in an inactive form), the intermediate substance Y cannot be converted into anthocyanin. The pathway is said to be "blocked" at this step. Although the block causes an increase in the concentration of Y inside the cell (because the precursor is still converted into X, and X into Y), no purple pigment is produced and the flower remains white.

Each of the mutant alleles listed across the bottom of the pathway codes for an inactive form of the corresponding enzyme. Any genotype that is homozygous for any of the mutations fails to produce an active form of the enzyme. For example, because the mutant *alb* allele codes for an inactive form of the enzyme for the first

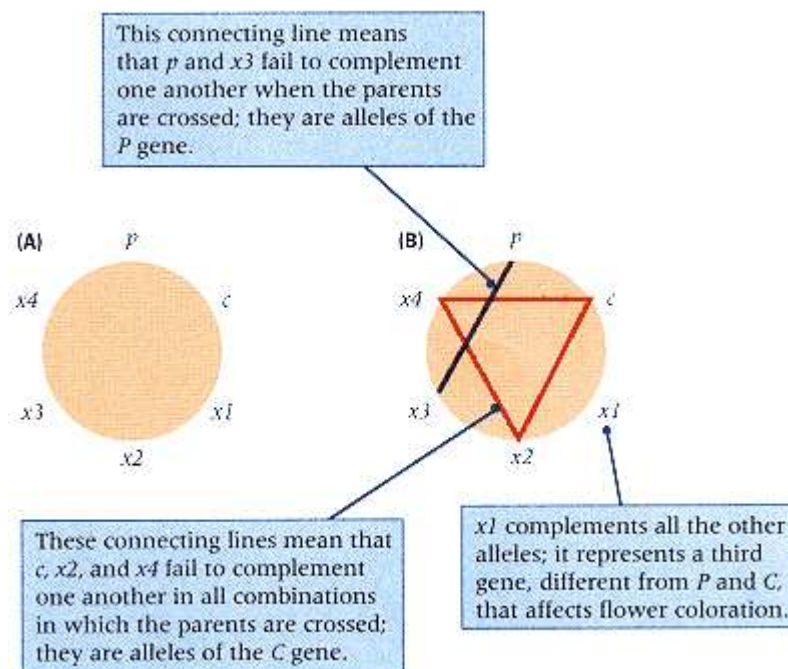


Figure 2.21
A method for interpreting the results of complementation tests. (A) Arrange the mutations in a circle. (B) Connect by a straight line any pair of mutations that fails to complement (that yields a mutant phenotype); any pair of mutations so connected are alleles of the same gene. In this example, there are three complementation groups, each of which represents a single gene needed for purple flower coloration.

step in the pathway, the genotype *alb alb* has no active enzyme for this step. In the homozygous mutant, the pathway is blocked at the first step, so the precursor is not converted to intermediate X. Because no X is produced, there can be no Y, and without Y there can be no anthocyanin, and so homozygous *alb alb* results in white flowers. The *alb* allele is recessive, because in the heterozygous genotype *Alb alb*, the wildtype *Alb* allele codes for a functional enzyme for the first step in the pathway, and so the pathway is not blocked.

Mutant alleles of the *C* gene block the second step in the pathway. In this case, an inactive enzyme is produced not

only in the homozygous genotypes cc , c_2c_2 , and c_4c_4 , but also in the genotypes cc_2 , cc_4 , and c_2c_4 . In the last three genotypes, each mutant allele encodes a *different* (but still inactive) form of the enzyme, so the pathway is blocked at step 2, and the color of the flowers is white. The c , c_2 , and c_4 alleles are all in the same complementation group

(they fail to complement one another) because they all encode inactive forms of the same enzyme.

A similar situation holds for mutations in the *P* gene. The wildtype *P* allele encodes the enzyme for the final step in the pathway to anthocyanin. Any of the genotypes *pp*, *pp₃*, and *p₃p₃* lacks a functional form of the enzyme, which blocks the pathway at the last step and results in white flowers. The alleles *p* and *p₃* are in the same complementation group because they are both mutations in the *P* gene.

Multiple Alleles

The *C* and *P* genes in Figure 2.22 also illustrate the phenomenon of **multiple alleles**, in which there are more than two allelic forms of a given gene. Because the wildtype form of each gene also counts as an allele, there are two alleles of the *Alb* gene (*Alb* and *alb*), four alleles of the *C* gene (*C*, *c*, *c₂*, and *c₄*) and three alleles of the *P* gene (*P*, *p*, and *p₃*).

When a complementation test reveals that two independent mutations are alleles of the same gene, one does not know whether the mutant alleles have identical nucleotide sequences in the DNA. Recall from Chapter 1 that, at the level of DNA, a gene is a sequence of nucleotides that specifies the sequence of amino acids in a protein. Each nucleotide contains a base, either A (adenine), T (thymine), G (guanine), or C (cytosine), so a gene of *n* nucleotides can theoretically mutate at any of the positions to any of the three other nucleotides. The number of possible single-nucleotide differences in a gene of length *n* is therefore $3 \times n$; each of these DNA sequences, if it exists in the population, is an allele. When *n* = 5000, for example, there are potentially 15,000 alleles (not counting any of the possibilities with more than one nucleotide substitution). Most of the potential alleles may not actually exist at any one time, but many of them may be present in any population. The following rules govern the number of alleles.

- A gamete may contain only one allele of each gene.
- Any particular organism or cell may contain up to two different alleles.
- A population of organisms may contain any number of alleles

Many genes have multiple alleles. For example, the human blood groups designated A, B, O, or AB are determined by three types of alleles denoted *I^A*, *I^B*, and *I^O*, and the blood group of any person is determined by the particular pair of alleles present in his or her genotype. (Actually, there are two slightly different variants of the *I^A*

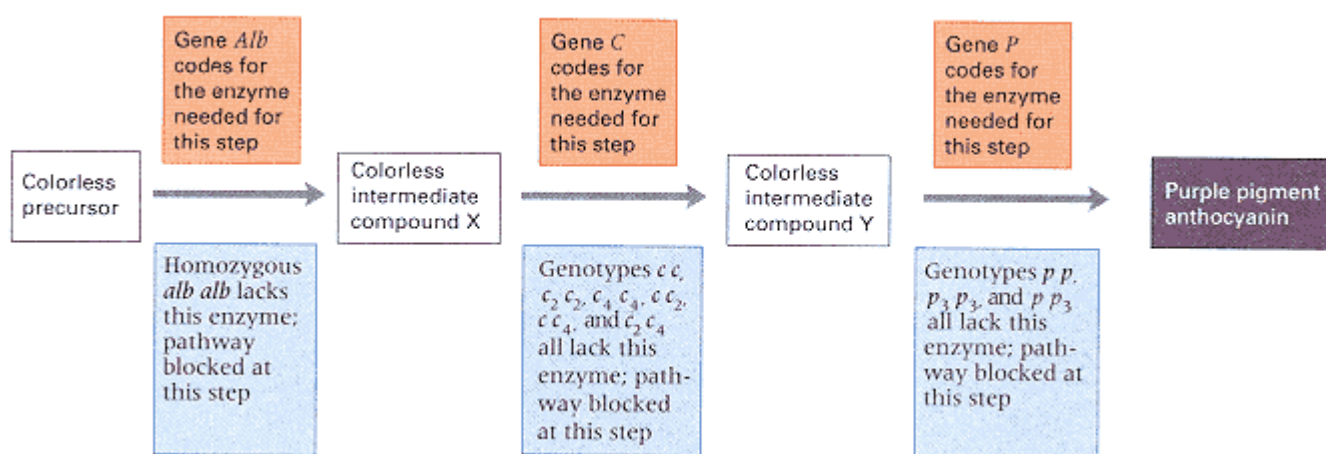


Figure 2.22

Biochemical pathway for the synthesis of the purple pigment anthocyanin from a colorless precursor and colorless intermediates X and Y. Each step (arrow) in the pathway is a biochemical conversion that requires an enzyme encoded in the wildtype allele of the gene indicated.

allele, so four alleles can be distinguished in this case.)

In modern genetics, multiple alleles are encountered in two major settings. One is in genetic analysis when a mutant screen potentially yields two or more mutant alleles of each of a large number of genes. For example, in the early 1980s, mutant screens were carried out in *Drosophila* to obtain new recessive mutations that blocked embryonic development and so led to the death of homozygous recessive embryos. The screens resulted in the identification of approximately 18,000 such mutations, the study of which ultimately earned a 1995 Nobel Prize for Christiane Nüsslein-Volhard and Eric Wieschaus (they shared the prize with Edward B. Lewis, who had already done pioneering work in the genetics of *Drosophila* development).

Geneticists also encounter multiple alleles in studies of natural populations of organisms. In most populations, including the human population, each gene may have many alleles that differ slightly in nucleotide sequence. Most of these alleles, even though they differ in one or more nucleotides in the DNA sequence, are able to carry out the normal function of the gene and produce no observable difference in phenotype.

In human populations, it is not unusual for a gene to have many alleles. Genes used in DNA typing, such as those employed in criminal investigations, usually have multiple alleles in the population. For each of these genes, any person can have no more than two alleles, but often there are 20 or more alleles in the population as a whole. Hence, any two unrelated people are not likely to have the same genotype, especially if several different genes, each with multiple alleles, are examined. Similarly, in the inherited recessive diseases *cystic fibrosis* and *phenylketonuria*, more than 200 different defective alleles of each gene have been identified in studies of affected children throughout the world. The "normal" form of each gene also exists in many alternative forms. Indeed, for most genes in most populations, the "normal" or "wildtype" allele is not a single nucleotide sequence but rather a set of different nucleotide sequences, each capable of carrying out the normal function of the gene.

In some cases, the multiple alleles of a gene exist merely by chance and reflect the history of mutations that have taken place in the population and the dissemination of these mutations among population subgroups by migration and interbreeding. In other cases, there are biological mechanisms that favor the maintenance of a large number of alleles. For example, genes that control self-sterility in certain flowering plants can have large numbers of allelic types. This type of self-sterility is found in species of red clover that grow wild in many pastures. The self-sterility genes prevent self-fertilization because a pollen grain can undergo pollen tube growth and fertilization only if it contains a self-sterility allele different from either of the alleles present in the flower on which it lands. In other words, a pollen grain containing an allele already present in a flower will not function on that flower. Because all pollen grains produced by a plant must contain one of the self-sterility alleles present in the plant, pollen cannot function on the same plant that produced it, and self-fertilization cannot take place. Under these conditions, any plant with a new allele has a selective advantage, because pollen that contains the new allele can fertilize all flowers except those on the same plant. Through evolution, populations of red clover have accumulated hundreds of alleles of the selfsterility gene, many of which have been isolated and their DNA sequences determined. Many of the alleles differ at multiple nucleotide sites, which implies that the alleles in the population are very old.

2.6—

Modified Dihybrid Ratios Caused by Epistasis

In Figure 2.22 we saw how the products of several genes may be necessary to carry out all the steps in a biochemical pathway. In genetic crosses in which two mutations that affect different steps in a single pathway are both segregating, the typical F_2 dihybrid ratio of 9 : 3 : 3 : 1 is not observed. One way in which the ratio may be modified is illustrated by the interaction of the C,

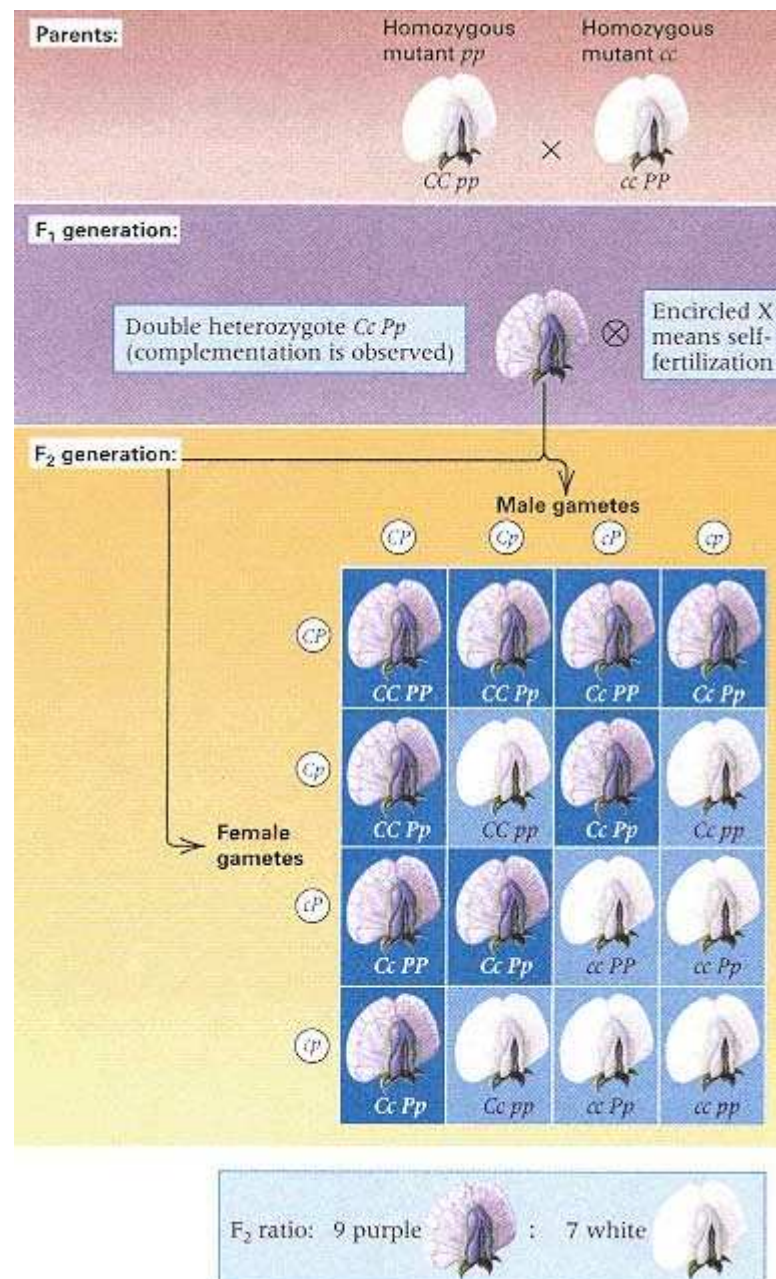


Figure 2.23

A cross showing epistasis in the determination of flower color in sweet peas. Formation of the purple pigment requires the dominant allele of both the C and P genes. With this type of epistasis, the dihybrid F_2 ratio is modified to 9 purple : 7 white.

c and P , p allele pairs affecting flower coloration. Figure 2.23 shows a cross between the homozygous mutants pp and cc . The phenotype of the plants in the F_1 generation is purple flowers; complementation is observed because the p and c mutations are in different genes. Self-fertilization of the F_1 plants (indicated by the encircled cross sign) results in the F_2 progeny genotypes shown in the Punnett square. Because only the progeny with at least one C allele and at least one P allele have purple flowers and all the rest have white flowers, the ratio of purple flowers to white flowers in the F_2 generation is 9 : 7.

Any type of gene interaction that results in the F_2 dihybrid ratio of 9 : 3 : 3 : 1 being modified into some other ratio is called *epistasis*. For a trait determined by the interaction of two genes, each with a dominant allele, there are only a limited number of ways in which the 9:3:3:1 dihybrid ratio can be modified. The possibilities are illustrated in Figure 2.24. In part A are the genotypes produced in the F_2 generation by independent assortment and the ratios in which the genotypes occur. In the absence of epistasis, the F_2 ratio of phenotypes is 9 : 3 : 3 : 1. The possible modified ratios are shown in part B of the figure. In each row, the color coding indicates phenotypes that are indistinguishable because of epistasis, and the resulting modified ratio is given. For example, in the modified ratio at the bottom, the phenotypes of the "3:3:1" classes are indistinguishable, resulting in a 9 : 7 ratio. This is the ratio

observed in the segregation of the *C*, *c* and *P*, *p* alleles in Figure 2.23, and the 9 : 7 ratio is the ratio of purple flowers to white flowers. Taking all the possible modified ratios in Figure 2.24B together, there are nine possible dihybrid ratios when both genes show complete dominance. Examples are known of each of the modified ratios. However, the most frequently encountered modified ratios are 9 : 7, 12 : 3 : 1, 13 : 3, 9:4:3, and 9:6:1. The types of epistasis that result in these modified ratios are illustrated in the following examples, taken from a variety of organisms. Other examples can be found in the problems at the end of the chapter.

9 : 7 This is the ratio observed when a homozygous recessive mutation in either or both of two different genes results in the same mutant phenotype. It is exemplified by the segregation of purple and white flowers in Figure 2.23. Genotypes that are *C*— for the *C* gene

and P — for the P gene have purple flowers; all other genotypes have white flowers. In this notation, the dash in C — means that the unspecified allele could be either C or c , and so C — refers collectively to CC and Cc . Similarly, the dash in P — means that the unspecified allele could be either P or p .

12 : 3 : 1 A modified dihybrid ratio of the 12:3:1 variety results when the presence of a dominant allele of one gene masks the genotype of a different gene. For example, if the A — genotype renders the B — and bb genotypes indistinguishable, then the dihybrid ratio is 12:3:1 because the A — B — and A — bb genotypes are expressed as the same phenotype.

In a genetic study of the color of the hull in oat seeds, a variety having white hulls was crossed with a variety having black hulls. The F_1 hybrid seeds had black hulls. Among 560 progeny in the F_2 generation produced by selffertilization of the F_1 , the following seed phenotypes were observed in the indicated numbers:

418	black hulls
106	gray hulls
36	white hulls

Note that the observed ratio of phenotypes is 11.6 : 2.9 : 1, or very nearly 12:3:1. These results can be explained by a genetic hypothesis in which the black-hull phenotype results from the

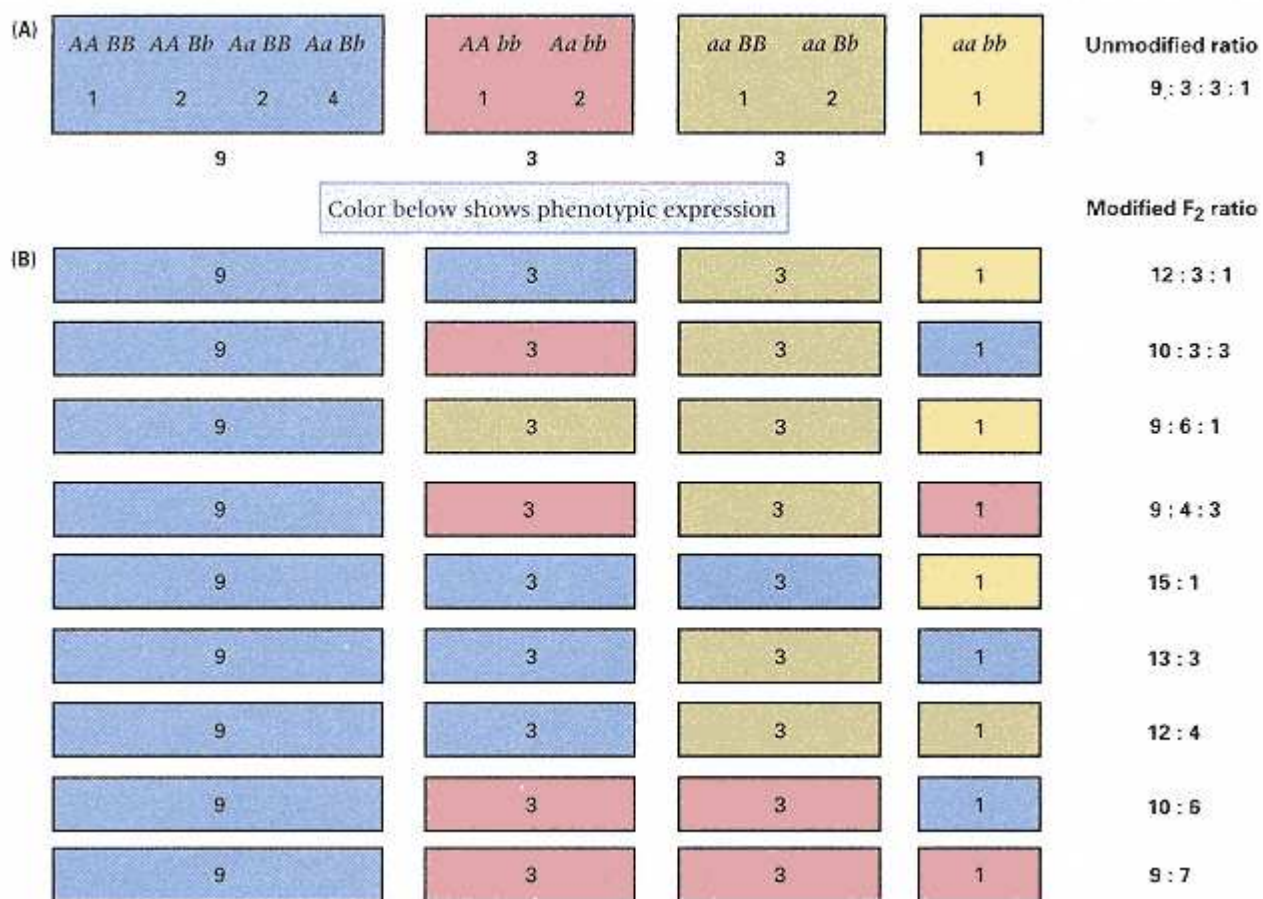


Figure 2.24

Modified F_2 dihybrid ratios. (A) The F_2 genotypes of two independently assorting genes with complete dominance result in a 9 : 3 : 3 : 1 ratio of phenotypes if there is no interaction between the genes (epistasis). (B) If there is epistasis that renders two or more of the phenotypes indistinguishable, indicated by the colors, then the F_2 ratio is modified. The most frequently encountered modified ratios are 9 : 7, 12 : 3 : 1, 13 : 3, 9 : 4 : 3, and 9 : 6 : 1.

presence of a dominant allele (say, A) and the gray-hull phenotype results from another dominant allele (say, B) whose effect is apparent only in the aa homozygotes. On the basis of this hypothesis, the original true-breeding varieties must have had genotypes $aa\ bb$ (white) and $AA\ BB$ (black). The F_1 has genotype $Aa\ Bb$ (black). If the A,a allele pair and the B,b allele pair undergo independent assortment, then the F_2 generation is expected to have the following composition of genotypes:

9/16	$A—$	$B—$	(black hull)
3/16	$A—$	bb	(black hull)
3/16	aa	$B—$	(gray hull)
1/16	aa	bb	(white hull)

This type of epistasis accounts for the 12:3:1 ratio.

13 : 3 This type of epistasis is illustrated by the difference between White Leghorn chickens (genotype $CC\ Ii$) and White Wyandotte chickens (genotype $cc\ ii$). Both breeds have white feathers because the C allele is necessary for colored feathers, but the I allele in White Leghorns is a dominant inhibitor of feather coloration. The F_1 generation of a dihybrid cross between these breeds has the genotype $Cc\ Ii$, which is expressed as white feathers because of the inhibitory effects of the I allele. In the F_2 generation, only the $C-ii$ genotype has colored feathers, so there is a 13:3 ratio of white : colored.

9:4:3 This dihybrid ratio (often stated as 9:3:4) is observed when homozygosity for a recessive allele with respect to one gene masks the expression of the genotype of a different gene. For example, if the aa genotype has the same phenotype regardless of whether the genotype is $B—$ or bb , then the 9:4:3 ratio results.

In the mouse, the grayish coat color called "agouti" is produced by the presence of a horizontal band of yellow pigment just beneath the tip of each hair. The agouti pattern results from the presence of a dominant allele A , and in

aa animals the coat color is black. A second dominant allele, C , is necessary for the formation of hair pigments of any kind, and cc animals are albino (white fur). In a cross of $AA\ CC$ (agouti) $\times$ $aa\ cc$ (albino), the F_1 progeny are $Aa\ Cc$ and agouti. Crosses between F_1 males and females produce F_2 progeny in the following proportions:

9/16	$A—$	$C—$	(agouti)
3/16	$A—$	cc	(albino)
3/16	aa	$C—$	(black)
1/16	aa	cc	(albino)

The dihybrid ratio is therefore 9 agouti: 4 albino : 3 black.

9:6:1 This dihybrid ratio is observed when homozygosity for a recessive allele of either of two genes results in the same phenotype, but the phenotype of the double homozygote is distinct. For example, red coat color in Duroc-Jersey pigs requires the presence of two dominant alleles R and S . Pigs of genotype $R—ss$ and $rr\ S—$ have sandycolor coats, and $rr\ ss$ pigs are white. The F_2 dihybrid ratio is therefore

9/16	$R—$	$S—$	(red)
------	------	------	-------

3/16	R—	ss	(sandy)
3/16	rr	S—	(sandy)
1/16	rr	ss	(white)

The 9:6:1 ratio results from the fact that both single recessives have the same phenotype.

2.7—

Complications in the Concept of Dominance

In Mendel's experiments, all traits had clear dominant-recessive patterns. This was fortunate, because otherwise he might not have made his discoveries. Departures from strict dominance are also frequently observed. In fact, even for such a classical trait as round versus wrinkled seeds in peas, it is an oversimplification to say that round is dominant. At the level of whether a seed is round or wrinkled, round is dominant in the sense that the genotypes WW and Ww cannot be distinguished by the

outward appearance of the seeds. However, as emphasized in Chapter 1, every gene potentially affects many traits. It often happens that the same pair of alleles shows complete dominance for one trait but not complete dominance for another trait. For example, in the case of round versus wrinkled seeds, the genetic defect in wrinkled seeds is the absence of an active form of an enzyme called starch-branching enzyme I (SBEI), which is needed for the synthesis of a branched-chain form of starch known as amylopectin. Compared with homozygous WW , seeds that are heterozygous Ww have only half as much SBEI, and seeds that are homozygous ww have virtually none (Figure 2.25A). Homozygous WW peas contain large, well-rounded starch grains, with the result that the seeds retain water and shrink uniformly as they ripen, so they do not become wrinkled. In homozygous ww seeds, the starch grains lack amylopectin; they are irregular in shape, and when these seeds ripen, they lose water too rapidly and shrink unevenly, resulting in the wrinkled phenotype observed (Figure 2.25B and C).

The w allele also affects the shape of the starch grains in Ww heterozygotes. In heterozygous seeds, the starch grains are intermediate in shape (Figure 2.25B). Nevertheless, their amylopectin content is high enough to result in uniform shrinking of the seeds and no wrinkling (Figure 2.25C). Thus there is an apparent paradox of dominance. If we consider only the overall shape of the seeds, round is dominant over wrinkled. There are only two phenotypes. If we examine the shape of the starch grains with a microscope, all three genotypes can be distinguished from each other: large, rounded starch grains in WW ; large, irregular grains in Ww ; and small, irregular grains in ww . If we consider the amount of the SBEI enzyme, the Ww genotype has an amount about halfway between the amounts in WW and ww .

The round, wrinkled pea example in Figure 2.25 makes it clear that "dominance" is not simply a property of a particular pair of alleles no matter how the resulting phenotypes are observed. When a gene affects multiple traits (as most genes do), then a particular pair of alleles might

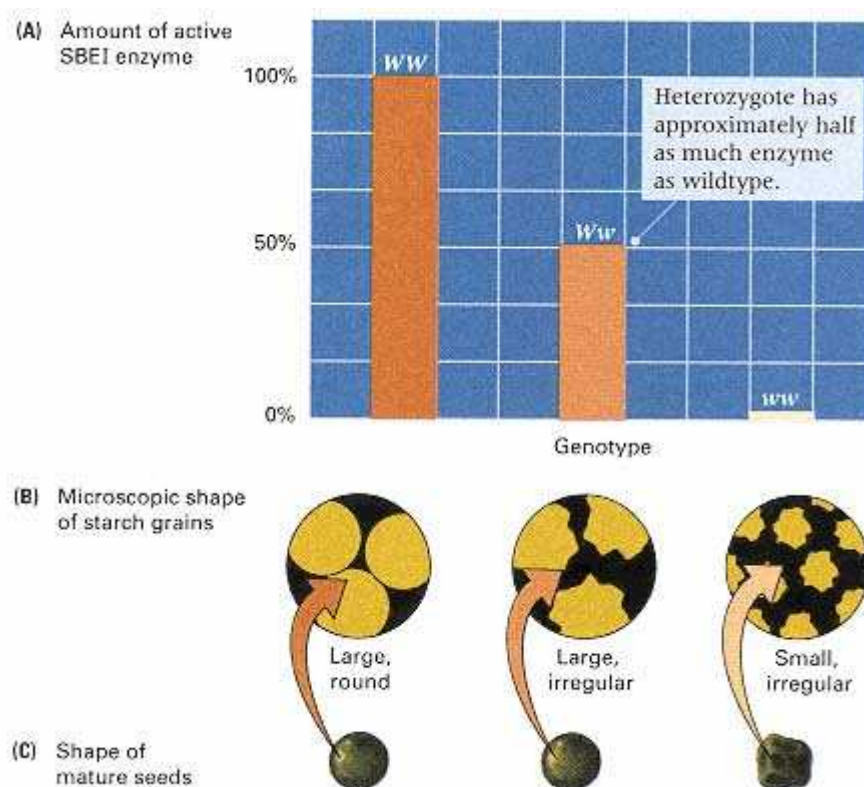


Figure 2.25
Phenotypic expression of three traits affected by Mendel's alleles W and w determining round versus wrinkled seeds. (A) Relative amounts of starch-branching enzyme I (SBEI); the enzyme level in the heterozygous genotype is about halfway between the levels in the homozygous genotypes. (B) Size and shape of the microscopic starch grains; the heterozygote is intermediate. (C) Effect on shape of mature seeds; for seed shape, W is dominant over w .

show simple dominance for some traits but not others. The general principle illustrated in Figure 2.25 is:

The phenotype consists of many different physical and biochemical attributes, and dominance may be observed for some of these attributes and not for others. Thus dominance is a property of a pair of alleles in relation to a particular attribute of phenotype.

Amorphs, Hypomorphs, and Other Types of Mutations

We have seen that a wildtype allele can potentially undergo mutation at any of a large number of nucleotide sites in the DNA, resulting in multiple alleles of a gene. In a series of multiple alleles, some alleles may have a more drastic effect on the phenotype than others. For example, one mutant allele may render the corresponding enzyme completely inactive, whereas another mutant allele may impair the enzyme in such a way as to cause only a partial loss of enzyme activity. Geneticists sometimes classify mutations according to the severity of their effects.

A mutation such as Mendel's *wrinkled* mutation, which encodes an inactive form of the SBEI enzyme, is often called an **amorph**. At the molecular level, an amorphic mutation may result from an amino acid replacement that inactivates the enzyme or even from a deletion of the gene so that no enzyme is produced. A mutation that reduces the enzyme level, but does not eliminate it, is called a **hypomorph**. Hypomorphic mutations typically result from amino acid replacements that impair enzyme activity or that prevent the enzyme from being produced at the normal level. As the prefix *hyper* implies, a **hypermorph** produces a greater-than-normal enzyme level, typically because the mutation changes the regulation of the gene in such a way that the gene product is overproduced.

Relative to their effects on the protein product of the gene they affect, most mutations can be classified as amorphs, hypomorphs, or hypermorphs. They result in none, less, or more of the enzyme activity produced by the wildtype, nonmutant allele. But other types of mutations also arise. A **neomorph** is a type of mutation that qualitatively alters the action of a gene. For example, a neomorph may cause a gene to become active in a type of cell or tissue in which the gene is not normally active. Or a neomorph can result in the expression of a gene in development at a time during which the wildtype gene is not normally expressed. Neomorphic mutations in a *Drosophila* gene called *eyeless*, which cause the wildtype gene product to be expressed in non-eye-forming tissues, can result in the development of parts of compound

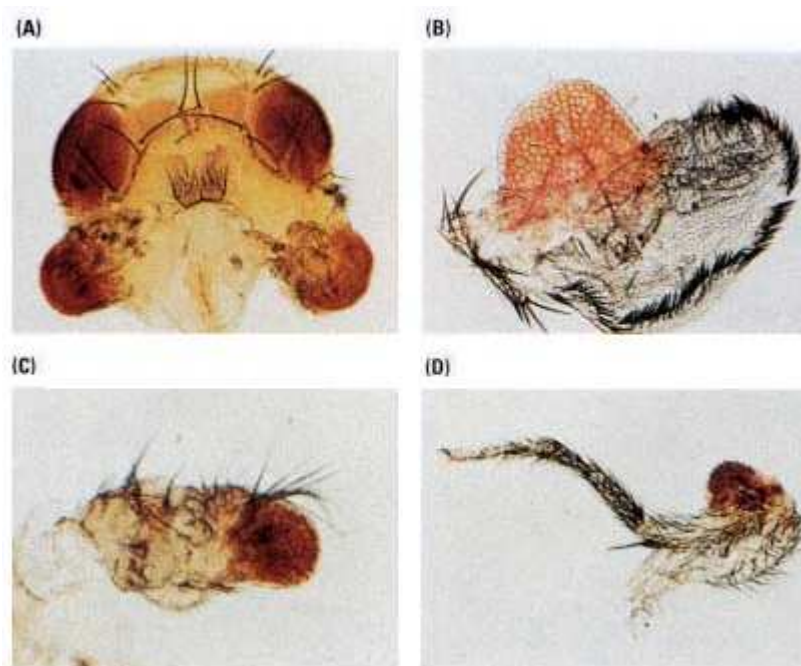


Figure 2.26

Ectopic expression of the wildtype allele of the *eyeless* gene in *Drosophila* results in misplaced eye tissue. (A) An adult head in which both antennae form eye structures. (B) A wing with eye tissue growing out from it. (C) A single antenna in which most of the third segment consists of eye tissue. (D) Middle leg with an eye outgrowth at the base of the tibia.

[Courtesy of G. Halder and W. J. Gehring. From G. Halder, P. Callaerts, and W. J. Gehring, *Science* 1995, 267: 1788.]

eyes, complete with eye pigments, in abnormal locations. The locations can be anywhere the wildtype *eyeless* gene is expressed, including on the legs or mouthparts, in the abdomen, or on the wings (Figure 2.26). Expression of a wildtype gene in an abnormal location is called **ectopic expression**.

Another type of mutation is an **antimorph**, whose mutant product antagonizes the normal product of the gene. In some cases this occurs through an amino acid replacement that causes the mutant protein to combine with the wildtype protein into an inactive complex. These various terms for mutations were coined by Herman J. Muller in 1931. Muller also discovered that mutations can be caused by x rays (Chapter 13). Many x-ray-induced mutations are associated with major disruptions or rearrangements of the DNA sequence, which result in unusual types of patterns of expression of the affected genes. Muller's terms were useful for describing such mutations, and they have come into widespread use for discussing other types of mutations as well.

Incomplete Dominance

When the phenotype of the heterozygous genotype lies in the range between the phenotypes of the homozygous genotypes, there is said to be **incomplete dominance**. Most genes code for enzymes, and each allele in a genotype often makes its own contribution to the total level of the enzyme in the cell or organism. In such cases, the phenotype of the heterozygote falls in the range between the phenotypes of the corresponding homozygotes, as illustrated in Figure 2.27. There is no settled terminology for the situation: the terms *incomplete dominance*, *partial dominance*, and *semidominance* are all in use.

A classical example of incomplete dominance concerns flower color in the snapdragon *Antirrhinum* (Figure 2.28). In wildtype flowers, a red type of anthocyanin pigment is formed by a sequence of enzymatic reactions. A wildtype enzyme, encoded by the *I* allele, is limiting to the rate of the overall reaction, so the amount of red pigment is determined by the amount of enzyme that the *I* allele pro-

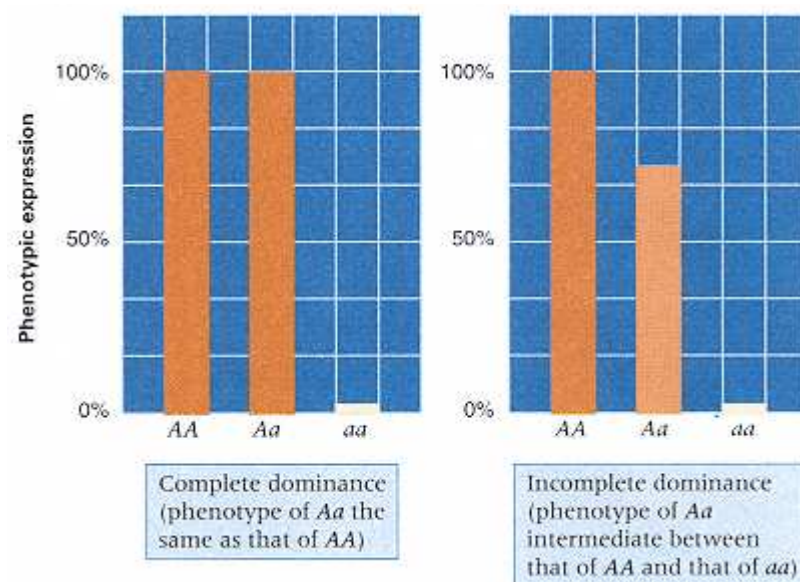


Figure 2.27
Levels of phenotypic expression in heterozygotes with complete dominance and with incomplete dominance.

duces. The alternative *i* allele codes for an inactive enzyme, and *ii* flowers are ivory in color. Because the amount of the critical enzyme is reduced in *Ii* heterozygotes, the amount of red pigment in the flowers is reduced also, and the effect of the dilution is to make the flowers pink.

The result of Mendelian segregation is observed directly when snapdragons that differ in flower color are crossed. For example, a cross between plants from a true-breeding red-flowered variety and a true-breeding ivory-flowered variety results in F_1 plants with pink flowers. In the F_2 progeny obtained by self-pollination of the F_1 hybrids, one experiment resulted in 22 plants with red flowers, 52 with pink flowers, and 23 with ivory flowers. The numbers agree fairly well with the Mendelian ratio of 1 dominant homozygote : 2 heterozygotes : 1 recessive homozygote. In agreement with the predictions from simple Mendelian inheritance, the red-flowered F_2 plants produced only red-flowered progeny, the ivory-flowered plants produced only ivory-flowered progeny, and the pink-flowered plants produced red, pink, and ivory progeny in the proportions 1/4 red: 1/2 pink : 1/4 ivory.

Incomplete dominance is often observed when the phenotype is quantitative

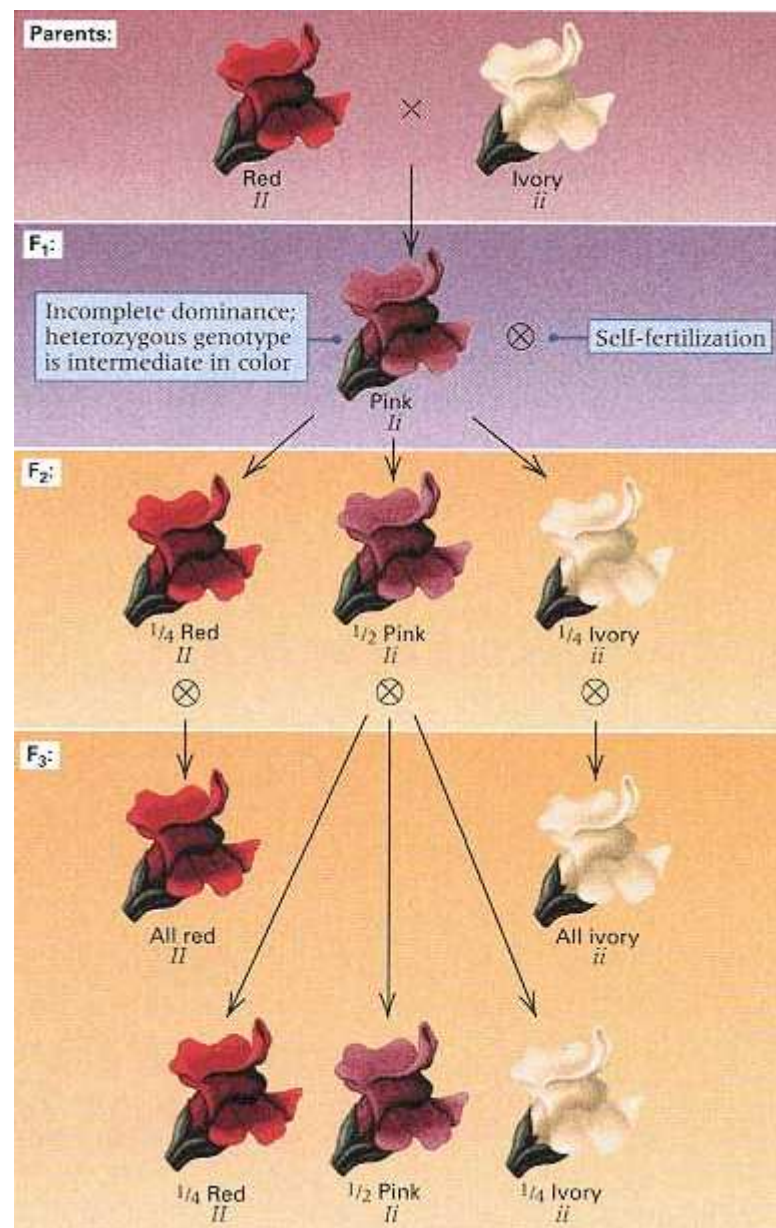


Figure 2.28
Absence of dominance in the inheritance of flower color in snapdragons.

rather than discrete. A trait that is *quantitative* can be measured on a continuous scale; examples include height, weight, number of eggs laid by a hen, time of flowering of a plant, and amount of enzyme in a cell or organism. A trait that is *discrete* is all or nothing; examples include round versus wrinkled seeds, and yellow versus green seeds. With a phenotype that is quantitative, the measured value of a heterozygote usually falls in the range between the homozygotes, and therefore there is incomplete dominance.

Codominance and the Human ABO Blood Groups

A special term, **codominance**, refers to a situation in which the phenotype of a heterozygous genotype is a mixture of the phenotypes of both of the corresponding homozygous genotypes. In such cases, the heterozygous phenotype is not intermediate between the homozygous genotypes (like pink snapdragons) but rather has the characteristics of both homozygous genotypes.

What we mean by "has the characteristics of both homozygous genotypes" is illustrated by one of the classical examples of codominance. These are the alleles that determine the A, B, AB, and O human blood groups, which were discussed earlier in the context of multiple alleles. Blood type is determined by the types of polysaccharides (polymers of sugars) present on the surface of red blood cells. Two different polysaccharides, A and B, can be formed. Both are formed from a precursor substance that is modified by the enzyme product of either the I^A or the I^B allele. The gene products are transferase enzymes that attach either of two types of sugar units to the precursor (Figure 2.29). People of genotype $I^A I^A$ produce red blood cells having only the A polysaccharide and are said to have blood type A. Those of genotype $I^B I^B$ have red blood cells with only the B polysaccharide and have blood

type B. Heterozygous $I^A I^B$ people have red cells with both the A and B polysaccharides and have blood type AB. The $I^A I^B$ genotype illustrates codominance, because the heterozygous genotype has the characteristics of both homozygous genotypes—in this case the presence of both the A and the B carbohydrate on the red blood cells.

The third allele, I^O , does not show codominance. It encodes a defective enzyme that leaves the precursor unchanged; neither the A nor the B type of polysaccharide is produced. Homozygous $I^O I^O$ persons therefore lack both the A and the B polysaccharides; they are said to have blood type O. In $I^A I^O$ heterozygotes, pres-

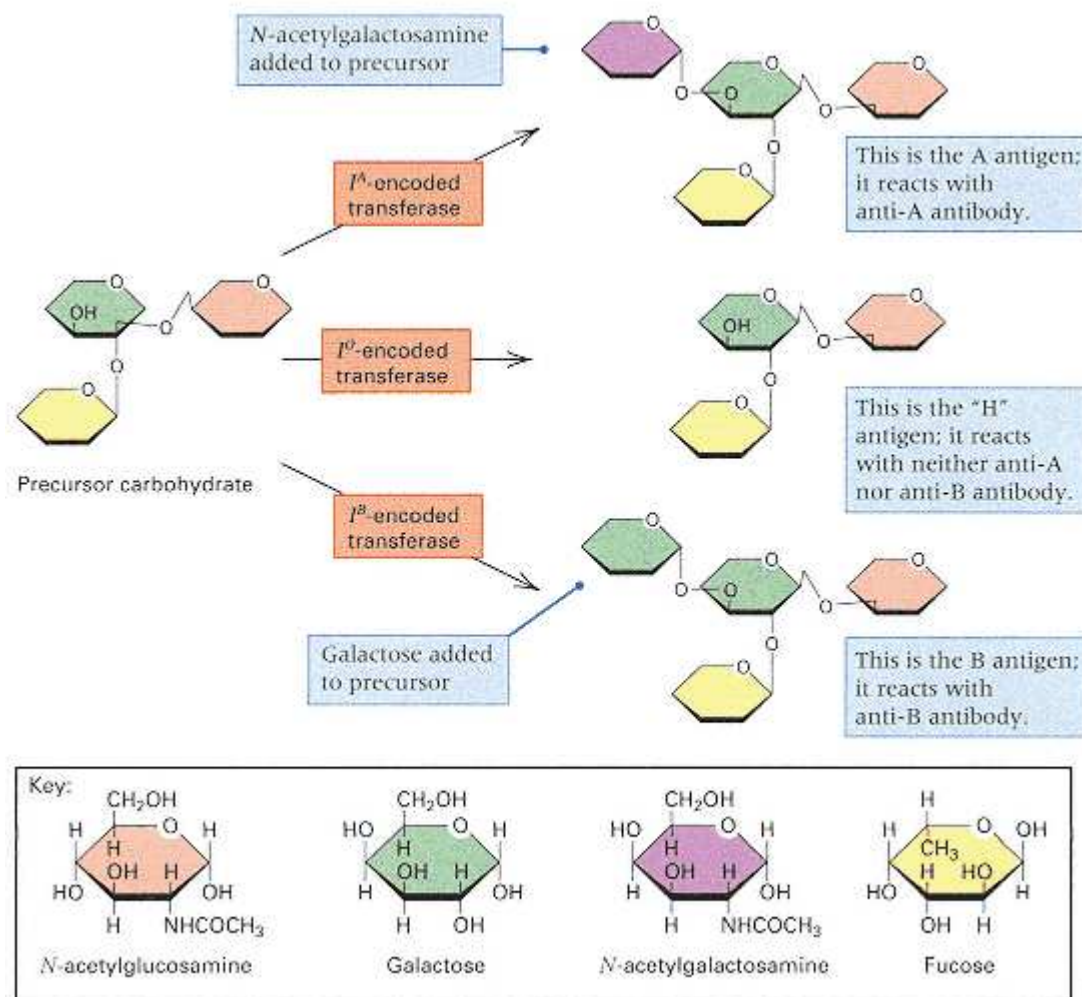


Figure 2.29

The ABO antigens on the surface of human red blood cells are carbohydrates. They are formed from a precursor carbohydrate by the action of transferase enzymes encoded by alleles of the I gene.

Allele I^O codes for an inactive enzyme and leaves the precursor unmodified. The unmodified form is called the H substance. The I^A allele encodes an enzyme that adds N-acetylgalactosamine (purple) to the precursor. The I^B allele encodes an enzyme that adds galactose (green) to the precursor. The other colored sugar units are N-acetylglucosamine (orange), and fucose (yellow). The sugar rings also have side groups attached to one or more of their carbon atoms; these are shown in the detailed structures inside the box.

ence of the I^A allele results in production of the A polysaccharide; and in $I^B I^O$ heterozygotes, presence of the I^B allele results in production of the B polysaccharide. The result is that $I^A I^O$ persons have blood type A and $I^B I^O$ persons have blood type B, so I^O is recessive to both I^A and I^B . The genotypes and phenotypes of the ABO blood group system are summarized in the first three columns of Table 2.3.

The ABO blood groups are important in medicine because of the frequent need for blood transfusions. A crucial feature of the ABO system is that most human blood contains antibodies to either the A or the B polysaccharide. An **antibody** is a protein that is made by the immune system in response to a stimulating molecule called an **antigen** and is capable of binding to the antigen. An antibody is usually specific in that it recognizes only one antigen. Some antibodies combine with antigen and form large molecular aggregates that may precipitate.

Antibodies act in the body's defense against invading viruses and bacteria, as well as other cells, and help in removing such invaders from the body. Although antibodies do not normally form without prior stimulation by the antigen, people capable of producing anti-A and anti-B antibodies do produce them. Production of these antibodies may be stimulated by antigens that are similar to polysaccharides A and B and that are present on the surfaces of many common bacteria. However, a mechanism called *tolerance* prevents an organism from producing antibodies against its own antigens. This mechanism ensures that A antigen or B antigen elicits antibody production only in people whose own red blood cells do not contain A or B, respectively. The end result:

People of blood type O make both anti-A and anti-B antibodies; those of blood type A make anti-B antibodies; those of blood type B make anti-A antibodies; and those of blood type AB make neither type of antibody.

The antibodies found in the blood fluid of people with each of the ABO blood types are shown in the fourth column in Table 2.3. The clinical significance of the ABO blood groups is that transfusion of blood containing A or B red-cell antigens into persons who make antibodies against them results in an agglutination reaction in which the donor red blood cells are clumped. In this reaction, the anti-A anti-body will agglutinate red blood cells of either blood type A or blood type AB, because both carry the A antigen (Figure 2.30). Similarly, anti-B antibody will agglutinate red blood cells of either blood type B or blood type AB. When the blood cells agglutinate, many blood vessels are blocked, and the recipient of the transfusion goes into shock and may die. Incompatibility in the other direction, in which the donor blood contains antibodies against the recipient's red blood cells, is usually acceptable because the donor's antibodies are diluted so rapidly that clumping is avoided. The types of compatible blood transfusions are shown in the last two columns of Table 2.3. Note that a person of blood type AB can receive blood from a person of any other ABO type; type AB is called a *universal recipient*. Conversely, a person of blood type O can donate blood to a person of any ABO type; type O is called a *universal donor*.

Incomplete Penetrance and Variable Expressivity

Monohybrid Mendelian ratios, such as 3 : 1 (or 1 : 2 : 1 when the heterozygote is intermediate), are not always observed even when a trait is determined by the action of a single recessive allele. Regular ratios such as these indicate that organisms with the same genotype also exhibit the same phenotype. Although the phenotypes of organisms with a particular genotype are often very similar, this is not always the case—particularly in natural populations in which

Table 2.3 Genetic control of the human ABO blood group

Genotype	Antigens present on red blood cells	ABO blood group phenotype	Antibodies present in blood fluid	Blood types that can be tolerated in transfusion	Blood types that can accept blood for transfusion
$I^A I^A$	A	Type A	Anti-B	A & O	A & AB
$I^A I^O$	A	Type A	Anti-B	A & O	A & AB
$I^B I^B$	B	Type B	Anti-A	B & O	B & AB
$I^B I^O$	B	Type B	Anti-A	B & O	B & AB
$I^A I^B$	A & B	Type AB	Neither anti-A nor anti-B	A, B, AB & O	AB only
$I^O I^O$	Neither A nor B	Type O	Anti-A & anti-B	O only	A, B, AB & O

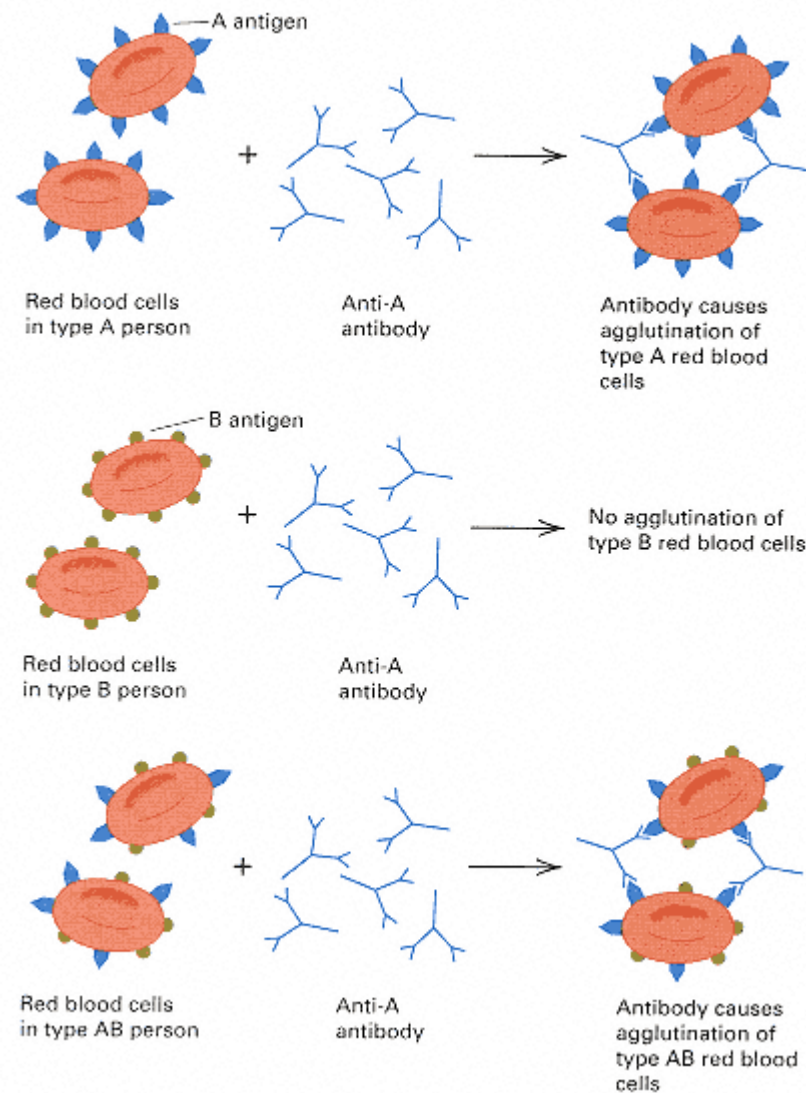


Figure 2.30
Antibody against type-A antigen will agglutinate red blood cells that carry the type A antigen, whether or not they also carry the type-B antigen. Blood fluid containing anti-A antibody will agglutinate red blood cells of type A and type AB, but not red blood cells of type B or type O.

neither the matings nor the environmental conditions are under an experimenter's control. Variation in the phenotypic expression of a particular genotype may happen because other genes modify the phenotype or because the biological processes that produce the phenotype are sensitive to environmental conditions.

The types of variable gene expression are usually grouped into two categories:

- **Variable expressivity** refers to genes that are expressed to different degrees in different organisms. For example, inherited genetic diseases in human beings are often variable in expression from one person to the next. One patient may be very sick, whereas another with the same disease may be less severely affected. Variable expressivity means that the same mutant gene can result in a severe form of the disease in one person but a mild form in another. The different degrees of expression often form a continuous series from full expression to almost no expression of the expected phenotypic characteristics.

- **Incomplete penetrance** means that the phenotype expected from a particular genotype is not always expressed. For example, a person with a genetic predisposition to lung cancer may not get the disease if he or she does not smoke tobacco. A lack of gene expression may result from environmental conditions, such as in the example of

not smoking, or from the effects of other genes. Incomplete penetrance is but an extreme of variable expressivity, in which the expressed phenotype is so mild as to be undetectable. The proportion of organisms whose phenotype matches their genotype for a given character is called the **penetrance** of the genotype. A genotype that is always expressed has a penetrance of 100 percent.

Chapter Summary

Inherited traits are determined by particulate elements called genes. In a higher plant or animal, the genes are present in pairs. One member of each gene pair is inherited from the maternal parent and the other member from the paternal parent. A gene can have different forms owing to differences in DNA sequence. The different forms of a gene are called alleles. The particular combination of alleles present in an organism constitutes its genotype. The observable characteristics of an organism constitute its phenotype. In an organism, if the two alleles of a gene pair are the same (for example, AA or aa), then the genotype is homozygous for the A or a allele; if the alleles are different (Aa), then the genotype is heterozygous. When the phenotype of a heterozygote is the same as that of one of the homozygous genotypes, the allele that is expressed is called dominant and the hidden allele is called recessive.

In genetic studies, the organisms produced by a mating constitute the F_1 generation. Matings between members of the F_1 generation produce the F_2 generation. In a cross such as $AA \times aa$, in which only one gene is considered (a monohybrid cross), the ratio of genotypes in the F_2 generation is 1 dominant homozygote (AA) : 2 heterozygotes (Aa) : 1 recessive homozygote (aa). The phenotypes in the F_2 generation appear in the ratio 3 dominant: 1 recessive. The Mendelian ratios of genotypes and phenotypes result from segregation in gamete formation, when the members of each allelic pair segregate into different gametes, and random union of gametes in fertilization.

The processes of segregation, independent assortment, and random union of gametes follow the rules of probability, which provide the basis for predicting outcomes of genetic crosses. Two basic rules for combining probabilities are the addition rule and the multiplication rule. The addition rule applies to mutually exclusive events; it states that the probability of the realization of either one or the other of two events equals the sum of the respective probabilities. The multiplication rule applies to independent events; it states that the probability of the simultaneous realization of both of two events is equal to the product of the respective probabilities. In some organisms—for example, human beings—it is not possible to perform controlled crosses, and genetic analysis is accomplished through the study of several generations of a family tree, called a pedigree. Pedigree analysis is the determination of the possible genotypes of the family members in a pedigree and of the probability that an individual member has a particular genotype.

The complementation test is the functional definition of a gene. Two recessive mutations are considered alleles of different genes if a cross between the homozygous recessives results in nonmutant progeny. Such alleles are said to complement one another. On the other hand, two recessive mutations are considered alleles of the *same* gene if a cross between the homozygous recessives results in mutant progeny. Such alleles are said to fail to complement. For any group of recessive mutations, a complete complementation test entails crossing the homozygous recessives in all pairwise combinations.

Multiple alleles are often encountered in natural populations or as a result of mutant screens. Multiple alleles means that more than two alternative forms of a gene exist. Examples of large numbers of alleles include the genes used in DNA typing and the self-sterility alleles in some flowering plants. Although there may be multiple alleles in a population, each gamete can carry only one allele of each gene, and each organism can carry at most two different alleles of each gene.

Dihybrid crosses differ in two genes—for example, $AA BB \times aa bb$. The phenotypic ratios in the dihybrid F_2 are 9 : 3 : 3 : 1, provided that both the A and the B alleles are dominant and that the genes undergo independent assortment. The 9 : 3 : 3 : 1 ratio can be modified in various ways by interaction between the genes (epistasis). Different types of epistasis may result in dihybrid ratios such as 9 : 7, 12 : 3 : 1, 13 : 3, 9 : 4 : 3, and 9 : 6 : 1.

In heterozygous genotypes, complete dominance of one allele over the other is not always observed. In most cases, a heterozygote for a wildtype allele and a mutant allele encoding a defective gene product will produce less gene product than in the wildtype homozygote. If the phenotype is determined by the amount of wildtype gene product rather than by its mere presence, the heterozygote will have an intermediate phenotype. This situation is called incomplete dominance. Codominance means that both alleles in a heterozygote are expressed, so the heterozygous genotype exhibits the phenotypic characteristics of both homozygous genotypes. Codominance is exemplified by

the I^A and I^B alleles in persons with blood group AB. Codominance is often observed for proteins when each alternative allele codes for a different amino acid replacement, because it may be possible to distinguish the alternative forms of the protein by chemical or physical means. Genes are not always expressed to the same extent in different organisms; this phenomenon is called variable expressivity. A genotype that is not expressed at all in some organisms is said to have incomplete penetrance.

Key Terms

addition rule	heterozygous	Punnett Square
albinism	homozygous	recessive
allele	Huntington disease	reciprocal cross
amorph	hybrid	segregation
antibody	hypermorph	sib
antigen	hypomorph	sibling
antimorph	incomplete dominance	sibship
backcross	incomplete penetrance	testcross
carrierindependent assortment	true breeding	
codominance	Mendelian genetics	variable expressivity
complementation group	monohybrid	zygote
complementation test	monohybrid	
dihybrid	multiplication rule	
dominant	mutant screen	
ectopic expression	neomorph	
F ¹ generation	P ¹ generation	
F ² generation	partial dominance	
gamete	pedigree	
gene	penetrance	
genotype	phenotype	

Review the Basics

- What is segregation? How would the segregation of a pair of alleles be exhibited in the progeny of a testcross?
- Explain the following statement: "Among the F₂ progeny of a dihybrid cross, the ratio of genotypes is 1: 2 : 1, but among the progeny that express the dominant phenotype, the ratio of genotypes is 1 : 2."

- What is a mutant screen and how is it used in genetic analysis?
- What is a complementation test? How does this test enable a geneticist to determine whether two different mutations are or are not mutations in the same gene?
- What do we mean by a "modified dihybrid F_2 ratio"? Give two examples of a modified dihybrid F_2 ratio and explain the gene interactions that result in the modified ratio.
- What is the distinction between incomplete dominance and codominance? Give an example of each.

Guide to Problem Solving

Problem 1: In tomatoes, the shape of the fruit is inherited, and both round fruit and elongate fruit are true breeding. The cross round $\times$ elongate produces F_1 progeny with round fruit, and the cross $F_1 \times F_1$ produces 3/4 progeny with round fruit and 1/4 progeny with elongate fruit. What kind of genetic hypothesis can explain these data?

Answer: In this kind of problem, a good strategy is to look for some indication of Mendelian segregation. The 3 : 1 ratio in the F_2 generation is characteristic of Mendelian segregation when there is dominance. This observation suggests the genetic hypothesis of a dominant gene R for round fruit and a recessive allele r for elongate fruit. If the hypothesis were correct, then the true-breeding round and elongate genotypes would be RR and rr , respectively. The F_1 progeny of the cross RR (round) $\times$ rr (elongate) would be Rr , which has round fruit, as observed. The $F_1 \times F_1$ cross ($Rr \times Rr$) yields 1/4 RR , 1/2 Rr , and 1/4 rr . Because both RR and Rr have round fruit, the expected F_2 ratio of round : elongate phenotypes is 3 : 1, as expected from the single-gene hypothesis.

Problem 2: In Shorthorn cattle, both red coat color and white coat color are true breeding. Crosses of red $\times$ white produce progeny that are uniformly reddish brown but thickly sprinkled with white hairs; this type of coat color is called *roan*. Crosses of roan $\times$ roan produce 1/4 : 1/2 roan : 1/4 white. What kind of genetic hypothesis can explain these data?

Answer: In this case, the 1:2:1 ratio of phenotypes in the cross roan $\times$ roan suggests Mendelian segregation, because this is the ratio expected from a mating between

Chapter 2 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://www.jbpub.com/genetics>

Select the link to **Genetics: Principles and Analysis** and then choose the link to **GeNETics on the web**. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. Mendel's paper is one of the few nineteenth-century scientific papers that reads almost as clearly as though it had been written today. It is important reading for every aspiring geneticist. You can access a conveniently annotated text by using the keyword **Mendel**. Although modern geneticists make a clear distinction between genotype and phenotype, Mendel made no clear distinction between these concepts. If assigned to do so, make a list of three specific instances in Mendel's paper, each supported by a quotation, in which the concepts of genotype and phenotype are not clearly separated; rewrite each quotation in a way that makes the distinction clear.

2. Although the incidence of **Huntington disease** is only 30 to 70 per million people in most Western countries, it has received great attention in genetics because of its late age of onset and autosomal dominant inheritance. Use the keyword to learn more about this condition. Under the heading *History* there is an *Editor's Note* quoting the blind seer Tiresias confronting Oedipus with the following paradox: "It is sorrow to be wise when wisdom profits not." If assigned to do so, write a 250-word essay explaining what this means in reference to Huntington disease and why DNA-based diagnosis is regarded as an ethical dilemma.

3. The red and purple colors of flowers, as well as of autumn leaves, result from members of a class of pigments called anthocyanins. The biochemical pathway for **anthocyanin** synthesis

(text box continued to next page)

heterozygotes when dominance is incomplete. Supposing that roan is heterozygous (say, Rr). Then the cross roan $\times$ roan ($Rr \times Rr$) is expected to produce $1/4 RR$, $1/2 Rr$, and $1/2 rr$ genotypes. The observed result, that $1/2$ of the progeny are roan (Rr), fits this hypothesis, which implies that the RR and rr genotypes correspond to red and white. The problem states that red and white are true breeding, which is consistent with their being homozygous genotypes. Additional confirmation comes from the cross $RR \times rr$, which yields Rr (roan) progeny, as expected. Note that the g symbols R and r are assigned to red and white arbitrarily, so it does not matter whether RR stands for red and rr for white, or the other way around.

Problem 3: The tailless trait in the mouse results from an allele of a gene in chromosome 17. The cross tailless $\times$ tailless produces tailless and wildtype progeny in a ratio of 2 tailless: 1 wildtype. All tailless progeny from this cross when mated with wildtype, produce a 1 : 1 ratio of tailless to wildtype progeny.

(a) Is the allele for the tailless trait dominant or recessive?

(b) What genetic hypothesis can account for the 2 : 1 ratio of tailless: wildtype and the results of the crosses between the tailless animals?

Answer: (a) If the tailless phenotype were homozygous recessive, then the cross tailless $\times$ tailless should produce only

tailless progeny. This is not the case, so the tailless phenotype must result from a dominant allele, say T . **(b)** Because the cross tailless $\times$ tailless produces both tailless and wildtype progeny, both parents must be heterozygous Tt . The expected ratio of genotypes among the zygotes is $1/4 TT$, $1/2 Tt$, and $1/4 tt$. Because T is dominant, the Tt animals are tailless and the tt animals are wildtype. The 2 : 1 ratio can be explained if the TT zygotes do not survive (that is, the TT genotype is lethal). Because all surviving tailless animals must be Tt , this genetic hypothesis would also explain all of the tailless animals from the cross, when mated with tt , give a 1 : 1 ratio of tailless (Tt) to wildtype (tt). (Developmental studies confirm that about 25 percent of the embryos do not survive.)

Problem 4: The accompanying illustration shows four alternative types of combs in chickens; they are called rose, pea, single, and walnut. The following data summarize the results of crosses. The rose and pea strains used in crosses 1, 2, and 3 are true breeding.

1. rose $\times$ single $\rightarrow$ rose

2. pea $\times$ single $\rightarrow$ pea

3. (rose $\times$ single) $F_1 \times$ (rose $\times$ single) $F_1 \rightarrow$ 3 rose : 1 single

(text box continued from previous page)

in the snapdragon, *Antirrhinum majus*, can be found at this keyword site. The enzyme responsible for the first step in the pathway limits the amount of pigment formed, which explains why red and white flowers in *Antirrhinum* show incomplete dominance. If assigned to do so, identify the enzyme responsible for the first step in the pathway, and give the molecular structures of the substrate (or substrates) and product. Also, examine all of the intermediates in the anthocyanin pathway, and identify which atom that is so prominent in the purine and pyrimidine bases is not found in anthocyanin.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 2, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

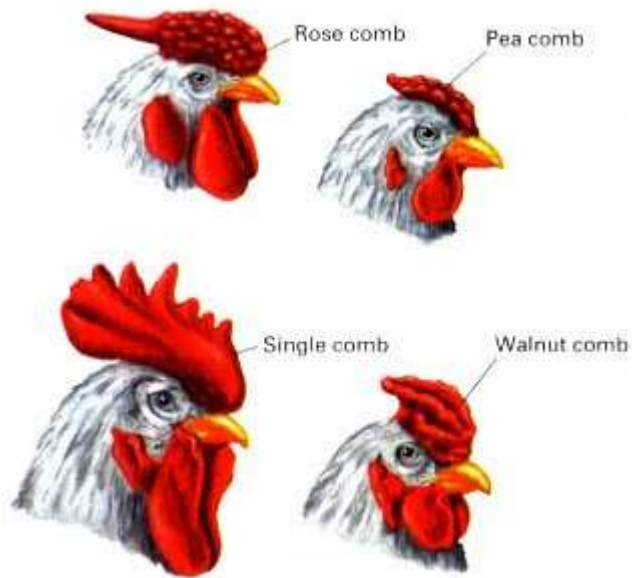
The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 2.



4. (pea $\times$ single) $F_1 \times$ (pea $\times$ single) $F_1 \rightarrow$ 3 pea : 1 single

5. rose $\times$ pea $\rightarrow$ walnut

6. (rose $\times$ pea) $F_1 \times$ (rose $\times$ pea) $F_1 \rightarrow$ 9 walnut : 3 rose : 3 pea : 1 single



- (a) What genetic hypothesis can explain these results?
- (b) What are the genotypes of parents and progeny in each of the crosses?
- (c) What are the genotypes of true-breeding strains of rose, pea, single, and walnut?

Answer:

(a) Cross 6 gives the Mendelian ratios expected when two genes are segregating, so a genetic hypothesis with two genes is necessary. Crosses 1 and 3 give the results expected if rose comb were due to a dominant allele (say, R). Crosses 2 and 4 give the results expected if pea comb were due to a dominant allele (say, P). Cross 5 indicates that walnut comb results from the interaction of R and P . The segregation in cross 6 means that R and P are not alleles of the same gene.

(b) 1. $RRpp \times rrpp \rightarrow Rrpp$.

2. $rrPP \times rrpp \rightarrow rrPp$.

3. $Rrpp \times Rrpp \rightarrow 3/4 R_pp : 1/4 rrpp$.

4. $rrPp \times rrPp \rightarrow 3/4 rrP_ : 1/4 rrpp$.

5. $RRpp \times rrPP \rightarrow RrPp$.

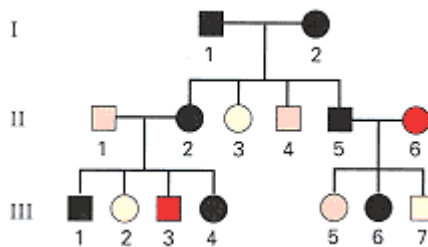
6. $RrPp \times RrPp (\rho)$

$$9/16 R_P_ : 3/16 R_pp : 3/16 rrP_ : 1/16 rrpp.$$

(c) The true-breeding genotypes are $RR\ pp$ (rose), $rr\ PP$ (pea), $rr\ pp$ (single), and $RR\ PP$ (walnut).

Problem 5: The pedigree in the accompanying illustration shows the inheritance of coat color in a group of cocker spaniels. The coat colors and genotypes are as follows:

Black	$A—$	$B—$	(black symbols)
Liver	aa	$B—$	(pink symbols)
Red	$A—$	bb	(red symbols)
Lemon	aa	bb	(yellow symbols)



- (a) Specify in as much detail as possible the genotype of each dog in the pedigree.
- (b) What are the possible genotypes of the animal III-4, and what is the probability of each genotype?
- (c) If a single pup is produced from the mating of III-4 $\times$ III-7, what is the probability that the pup will be red?

Answer:

(a) All three matings (I-1 $\times$ I-2, II-1 $\times$ II-2, and II-5 $\times$ II-6) produce lemon-colored offspring $aa\ bb$, so each parent must carry at least one a allele and at least one b allele. Therefore, in consideration of the phenotypes, the genotypes must be as follows: I-1 $Aa\ Bb$, I-2 $Aa\ Bb$, II-1 $aa\ Bb$, II-2 $Aa\ Bb$, II-5 $Aa\ Bb$, and II-6 $Aa\ bb$. The genotypes of the offspring can be deduced from their own phenotypes and the genotypes of the parents. These are as follows: II-3 $aa\ bb$, II-4 $aa\ B—$, III-1 $Aa\ B—$, III-2 $aa\ bb$, III-3 $Aa\ bb$, III-4 $Aa\ B—$, III-5 $aa\ Bb$, III-6 $A—\ Bb$, III-7 $aa\ bb$.

(b) Animal III-4 is either $Aa\ BB$ or $Aa\ Bb$, and the probabilities of these genotypes are $1/3$ and $2/3$, respectively.

(c) If the animal III-4 is $Aa\ BB$, then the probability of a red pup is 0; and if the animal III-4 is $Aa\ Bb$, then the probability of a red pup is $1/2 \times 1/2 = 1/4$ (that is, the probability of an $A\ b$ gamete from III-4). Overall, the probability of a red pup from the mating is $1/3 \times 0 + 2/3 \times 1/4 = 1/6$.

Problem 6: From the F_2 generation of a cross between mouse genotypes $AA \times aa$, one male progeny of genotype $A—$ was chosen and mated with an aa female. All of the progeny in the resulting litter were $A—$. How large a litter is required for you to be able to assert, with 95 percent confidence, that the father's genotype is AA ? How large a litter is required for 99 percent confidence?

Answer: The a priori ratio of the probabilities that the father is AA versus Aa is $1/3 : 2/3$, because the father was chosen at random from among the $A—$ progeny in the F_2 generation. With one $A—$ progeny in a testcross, the ratio of probabilities drops to $1/3 : (2/3) \times (1/2)$, because $1/2$ of the $A—$ fathers in such a testcross will yield an aa progeny and so identify themselves as Aa . Similarly, with n progeny, the ratio of $AA : Aa$ probabilities is $1/3 : (2/3) \times (1/2)^n$, because the probability that an Aa father has n consecutive $A—$ offspring in a testcross is $(1/2)^n$. For 95 percent confidence we need

$$1/3 : (2/3) \times (1/2)^n \geq 0.95 : 0.05$$

or

$$n \cong \log [(0.95 \times 2)/0.05] / \log (2) = 5.25$$

hence, $n = 6A$ — progeny are necessary for 95 percent confidence that the father is AA . For 99 percent confidence, the corresponding formula is

$$n \cong \log [(0.99 \times 2)/0.01] / \log (2) = 7.6$$

so in this case, $n = 8A$ — progeny are required.

Analysis and Applications

2.1 With respect to homozygosity and heterozygosity, what can be said about the genotype of a strain or variety that breeds true for a particular trait?

2.2 What gametes can be formed by an individual organism of genotype Aa ? Of genotype Bb ? Of genotype $Aa Bb$?

2.3 How many different gametes can be formed by an organism with genotype $AA Bb Cc Dd Ee$ and, in general, by an organism that is heterozygous for m genes and homozygous for n genes?

2.4 Mendel summarized his conclusions about heredity by describing the gametes produced by the F_1 generation in the following manner: "Pea hybrids form germinal and pollen cells that in their composition correspond in equal numbers to all the constant forms resulting from the combination of traits united through fertilization." Explain this statement in terms of the principles of segregation and independent assortment.

2.5 Round pea seeds are planted that were obtained from the F_2 generation of a cross between a true-breeding strain with round seeds and a true-breeding strain with wrinkled seeds. The pollen was collected and used *en masse* to fertilize plants from the true-breeding wrinkled strain. What fraction of the progeny is expected to have wrinkled seeds?

2.6 If an allele R is dominant over r , how many different phenotypes are present in the progeny of a cross between Rr

and Rr , and in what ratio? How many phenotypes are there, and in what ratio, if there is no dominance between R and r ?

2.7 In genetically self-sterile plants like red clover, why are all plants heterozygous for the self-sterility alleles?

2.8 Assuming equal numbers of boys and girls, if a mating has already produced a girl, what is the probability that the next child will be a boy? If a mating has already produced two girls, what is the probability that the next child will be a boy? On what type of probability argument do you base your answers?

2.9 Assuming equal numbers of boys and girls, what is the probability that a family that has two children has two girls? One girl and one boy?

2.10 In the following questions, you are asked to deduce the genotype of certain parents in a pedigree. The phenotypes are determined by dominant and recessive alleles of a single gene.

(a) A homozygous recessive results from the mating of a heterozygote and a parent with the dominant phenotype. What does this tell you about the genotype of the parent with the dominant phenotype?

(b) Two parents with the dominant phenotype produce nine offspring. Two have the recessive phenotype. What does this tell you about the genotype of the parents?

(c) One parent has a dominant phenotype and the other has a recessive phenotype. Two offspring result, and both have the dominant phenotype. What genotypes are possible for the parent with the dominant phenotype?

2.11 Pedigree analysis tells you that a particular parent may have the genotype $AA BB$ or $AA Bb$, each with the same probability. Assuming independent assortment, what is the probability of this parent's producing an Ab gamete? What is the probability of the parent's producing an AB gamete?

2.12 Assume that the trihybrid cross $AA BB rr \times aa bb RR$ is made in a plant species in which A and B are dominant but there is no dominance between R and r . Consider the F_2 progeny from this cross, and assume independent assortment.

(a) How many phenotypic classes are expected?

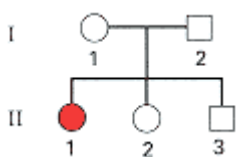
(b) What is the probability of the parental $aa bb RR$ genotype?

(c) What proportion would be expected to be homozygous for all three genes?

2.13 In the cross $Aa Bb Cc Dd \times Aa Bb Cc Dd$, in which all genes undergo independent assortment, what proportion of offspring are expected to be heterozygous for all four genes?

2.14 The pattern of coat coloration in dogs is determined by the alleles of a single gene, with S (solid) being dominant over s (spotted). Black coat color is determined by the dominant allele A of a second gene, tan by homozygosity for the recessive allele a . A female having a solid tan coat is mated with a male having a solid black coat and produces a litter of six pups. The phenotypes of the pups are 2 solid tan, 2 solid black, 1 spotted tan, and 1 spotted black. What are the genotypes of the parents?

2.15 In the human pedigree shown here, the daughter indicated by the red circle (II-1) has a form of deafness determined by a recessive allele. What is the probability that the phenotypically normal son (II-3) is heterozygous for the gene?



2.16 Huntington disease is a rare neurodegenerative human disease determined by a dominant allele, HD . The disorder is usually manifested after the age of forty-five. A young man has learned that his father has developed the disease.

(a) What is the probability that the young man will later develop the disorder?

(b) What is the probability that a child of the young man carries the *HD* allele?

2.17 The Hopi, Zuni, and some other Southwest American Indians have a relatively high frequency of albinism (absence of skin pigment) resulting from homozygosity for a recessive allele, *a*. A normally pigmented man and woman, each of whom has an albino parent, have two children. What is the probability that both children are albino? What is the probability that at least one of the children is albino?

2.18 Which combinations of donor and recipient ABO blood groups are compatible for transfusion? (Consider a combination to be compatible for transfusion if all the antigens in the donor red blood cells are also present in the recipient.)

2.19 Red kernel color in wheat results from the presence of at least one dominant allele of each of two independently segregating genes (in other words, *R— B—* genotypes have red kernels). Kernels on *rr bb* plants are white, and the genotypes *R— bb* and *rr B—* result in brown kernel color. Suppose that plants of a variety that is true breeding for red kernels are crossed with plants true breeding for white kernels.

(a) What is the expected phenotype of the F_1 plants?

(b) What are the expected phenotypic classes in the F_2 progeny and their relative proportions?

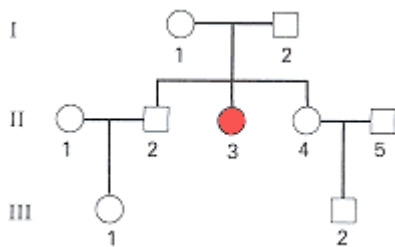
2.20 Heterozygous *Cp cp* chickens express a condition called creeper, in which the leg and wing bones are shorter than normal (*cp cp*). The dominant *Cp* allele is lethal when homozygous. Two alleles of an independently segregating gene determine white (*W—*) versus yellow (*ww*) skin color. From matings between chickens heterozygous for both of

these genes, what phenotypic classes will be represented among the viable progeny, and what are their expected relative frequencies?

2.21 White Leghorn chickens are homozygous for a dominant allele, C , of a gene responsible for colored feathers, and also for a dominant allele, I , of an independently segregating gene that prevents the expression of C . The White Wyandotte breed is homozygous recessive for both genes $cc\ ii$. What proportion of the F_2 progeny obtained from mating White Leghorn $\times$ White Wyandotte F_1 hybrids would be expected to have colored feathers?

2.22 The F_2 progeny from a particular cross exhibit a modified dihybrid ratio of 9 : 7 (instead of 9 : 3 : 3 : 1). What phenotypic ratio would be expected from a testcross of the F_1 ?

2.23 Phenylketonuria is a recessive inborn error of metabolism of the amino acid phenylalanine that results in severe mental retardation of affected children. The female II-3 (red circle) in the pedigree shown here is affected. If persons III-1 and III-2 (they are first cousins) mate, what is the probability that their offspring will be affected? (Assume that persons II-1 and II-5 are homozygous for the normal allele.)



2.24 Black hair in rabbits is determined by a dominant allele, B , and white hair by homozygosity for a recessive allele, b . Two heterozygotes mate and produce a litter of three offspring.

(a) What is the probability that the offspring are born in the order white-black-white? What is the probability that the offspring are born in either the order white-black-white or the order black-white-black?

(b) What is the probability that exactly two of the three offspring will be white?

2.25 Assuming equal sex ratios, what is the probability that a sibship of four children consists entirely of boys? Of all boys or all girls? Of equal numbers of boys and girls?

2.26 Andalusian fowls are colored black, splashed white (resulting from an uneven sprinkling of black pigment through the feathers), or slate blue. Black and splashed white are true breeding, and slate blue is a hybrid that segregates in the ratio 1 black : 2 slate blue : 1 splashed white. If a pair of blue Andalusians is mated and the hen lays three eggs, what is the probability that the chicks hatched from these eggs will be one black, one blue, and one splashed white?

Challenge Problems

2.27 In the mating $Aa \times Aa$, what is the smallest number of offspring, n , for which the probability of at least one aa offspring exceeds 95 percent?

2.28 From the F_2 generation of a cross between mouse genotypes $AA \times aa$, one male progeny of genotype $A-$ was chosen and mated with an aa female. All of the progeny in the resulting litter were $A-$. From this result you would like to conclude that the sire's genotype is AA . How much confidence could you have in this conclusion for each litter size from 1 to 15? (In other words what is the probability that the sire's genotype is AA , given that the a priori probability is $1/3$ and that a litter of n pups resulted in all $A-$ progeny?)

2.29 Meiotic drive is an unusual phenomenon in which two alleles do not show Mendelian segregation from the heterozygous genotype. Examples are known from mammals, insects, fungi, and other organisms. The usual mechanism is one in which both types of gametes are formed, but one of them fails to function normally. The excess of the driving allele over the other can range from a small amount to nearly 100 percent. Suppose that D is an allele showing meiotic drive against its alternative allele d , and suppose that Dd heterozygotes produce functional D -bearing and d -bearing gametes in the proportions $3/4 : 1/4$. In the mating $Dd \times Dd$,

- (a)** What are the expected proportions of DD , Dd , and dd genotypes?
- (b)** If D is dominant, what are the expected proportions of D — and dd phenotypes?
- (c)** Among the D — phenotypes, what is the ratio of $DD : Dd$?
- (d)** Answer parts (a) through (c), assuming that the meiotic drive takes place in only one sex.

Further Reading

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The calico cat illustrates the phenomenon of codominance. This female is heterozygous for an allele for black fur and for an allele for orange (also called "yellow") fur. Some patches of fur are black, whereas other patches are yellow. Because both alleles express their characteristic phenotype when heterozygous, they are considered codominant. Why the black and orange alleles are expressed in alternate patches of cells, rather than in overlapping patches, is explained in Chapter 7. The white spots are caused by an allele of a different gene that prevents any color formation.

Chapter 3— Genes and Chromosomes

CHAPTER OUTLINE

3-1 The Stability of Chromosome Complements

3-2 Mitosis

3-3 Meiosis

The First Meiotic Division: Reduction

The Second Meiotic Division: Equational

3-4 Chromosomes and Heredity

Chromosomal Determination of Sex

X-linked Inheritance

Nondisjunction as Proof of the Chromosome

Theory of Heredity

Sex Determination in *Drosophila*

3-5 Probability in Prediction and Analysis of Genetic Data

Using the Binomial Distribution in Genetics

Evaluating the Fit of Observed Results to

Theoretical Expectations

The Chi-square Method

3-6 Are Mendel's Data Too Good to be True?

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

- Chromosomes in eukaryotic cells are usually present in pairs.
- The chromosomes of each pair separate in meiosis, one going to each gamete.

- In meiosis, the chromosomes of different pairs undergo independent assortment because nonhomologous chromosomes move independently.
- In many animals, sex is determined by a special pair of chromosomes—the X and Y.
- The "criss-cross" pattern of inheritance of X-linked genes is determined by the fact that a male receives his X chromosome only from his mother and transmits it only to his daughters.
- Irregularities in the inheritance of an X-linked gene in *Drosophila* gave experimental proof of the chromosomal theory of heredity.
- The progeny of genetic crosses follow the binomial probability formula.
- The chi-square statistical test is used to determine how well observed genetic data agree with expectations derived from a hypothesis.

CONNECTIONS

CONNECTION: Grasshopper, Grasshopper

E. Eleanor Carothers 1913

The Mendelian ratio in relation to certain Orthopteran chromosomes

CONNECTION: The White-Eyed Male

Thomas Hunt Morgan 1910

Sex limited inheritance in Drosophila

CONNECTION: The Case Against Mendel's Gardener

Ronald Aylmer Fisher 1936

Has Mendel's work been rediscovered?

Mendel's experiments made it clear that in heterozygous genotypes, neither allele is altered by the presence of the other. The hereditary units remain stable and unchanged in passing from one generation to the next. However, at the time, the biological basis of the transmission of genes from one generation to the next was quite mysterious. Neither the role of the nucleus in reproduction nor the details of cell division had been discovered. Once these phenomena were understood, and when microscopy had improved enough that the chromosomes could be observed and were finally recognized as the carriers of the genes, new understanding came at a rapid pace. This chapter examines both the relationship between chromosomes and genes and the mechanism of chromosome segregation in cell division.

3.1—

The Stability of Chromosome Complements

The importance of the cell nucleus and its contents was suggested as early as the 1840s when Carl Nägeli observed that in dividing cells, the nucleus divided first. This was the same Nägeli who would later fail to understand Mendel's discoveries. Nägeli also failed to see the importance of nuclear division when he discovered it. He regarded the cells in which he saw nuclear division as aberrant. Nevertheless, by the 1870s it was realized that nuclear division is a universal attribute of cell division. The importance of the nucleus in inheritance was reinforced by the nearly simultaneous discovery that the nuclei of two gametes fuse in the process of fertilization. The next major advance came a decade later with the discovery of **chromosomes**, which had been made visible by light microscopy when stained with basic dyes. A few years later, chromosomes were found to segregate by an orderly process into the daughter cells formed by cell division as well as into the gametes formed by the division of reproductive cells. Finally, three important regularities were observed about the **chromosome complement** (the complete set of chromosomes) of plants and animals.

1. The nucleus of each **somatic cell** (a cell of the body, in contrast with a **germ cell**, or gamete) contains a fixed number of chromosomes typical of the particular species. However, the numbers vary tremendously among species and bear little relation to the complexity of the organism (Table 3.1).
2. The chromosomes in the nuclei of somatic cells are usually present in pairs. For example, the 46 chromosomes of human beings consist of 23 pairs (Figure 3.1). Similarly, the 14 chromosomes of peas consist of 7 pairs. Cells with nuclei of this sort, containing two similar sets of chromosomes, are called **diploid**. The chromosomes are present in pairs because one chromosome of each pair derives from the maternal parent and the other from the paternal parent of the organism.

Table 3.1 Somatic chromosome numbers of some plant and animal species

Organism	Chromosome number	Organism	Chromosome number
Field horetail	216	Yeast (<i>Saccharomyces cerevisiae</i>)	32
Bracken fern	116	Fruit fly (<i>Drosophila melanogaster</i>)	8
Giant sequoia	22	Nematode (<i>Caenorhabditis elegans</i>)	11 ♂, 12 ♀
Macaroni wheat	28	House fly	12
Bread wheat	42	Scorpion	4
Fava bean	12	Geometrid moth	224
Garden pea	14	Common toad	22
Wall cress (<i>Arabidopsis thaliana</i>)	10	Chicken	78
Corn (<i>Zea mays</i>)	20	Mouse	40
Lily	12	Gibbon	44
Snapdragon	16	Human being	46

3. The germ cells, or gametes, that unite in fertilization to produce the diploid state of somatic cells have nuclei that contain only one set of chromosomes, consisting of one member of each of the pairs. The gamete nuclei are **haploid**.

In multicellular organisms that develop from single cells, the presence of the diploid chromosome number in somatic cells and the haploid chromosome number in germ cells indicates that there are *two* different processes of nuclear division. One of these, mitosis, maintains the chromosome number; the other, meiosis, halves the number. These two processes are examined in the following sections.

3.2— Mitosis

Mitosis is a precise process of nuclear division that ensures that each of two daughter cells receives a diploid complement of chromosomes identical with the diploid complement of the parent cell. Mitosis is usually accompanied by **cytokinesis**, the process in which the cell itself divides to yield two daughter cells. The essential details of mitosis are the same in all organisms, and the basic process is remarkably uniform:

1. Each chromosome is already present as a duplicated structure at the beginning of nuclear division. (The duplication of each chromosome coincides with the replication of the DNA molecule contained within it.)
2. Each chromosome divides longitudinally into identical halves that become separated from each other.
3. The separated chromosome halves move in opposite directions, and each becomes included in one of the two daughter nuclei that are formed.

In a cell not undergoing mitosis, the chromosomes are not visible with a light microscope. This stage of the cell cycle is called **interphase**. In preparation for mitosis, the genetic material (DNA) in the chromosomes is replicated during a period of interphase called **S** (Figure 3.2). (The S stands for *synthesis* of DNA.) DNA replication is accompanied by chromosome dupli-

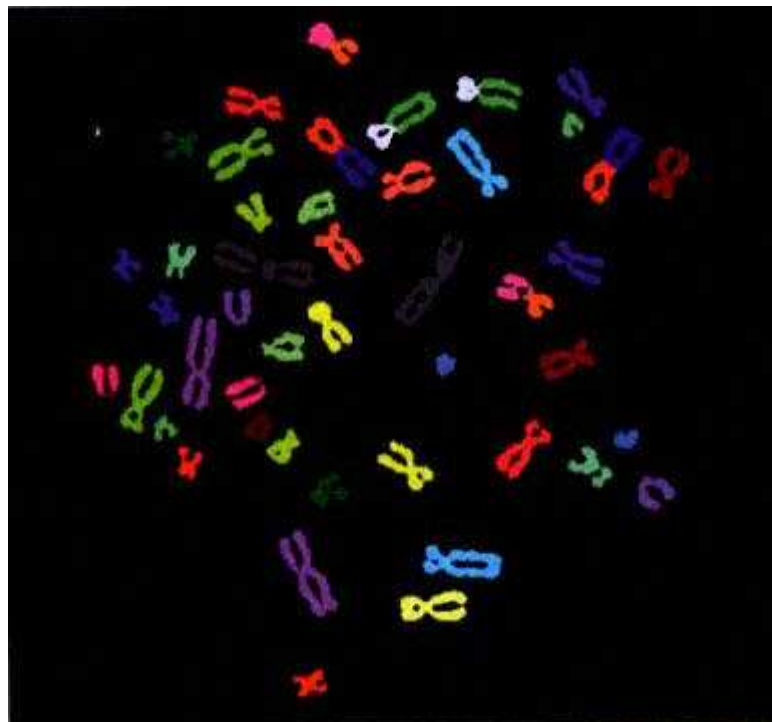


Figure 3.1
Chromosome complement of a human male. There are 46 chromosomes, present in 23 pairs. At the stage of the division cycle in which these chromosomes were observed, each chromosome consists of two identical halves lying side by side longitudinally. Except for the members of one chromosome pair (the pair that determines sex), the members of each of the other chromosome pairs are the same color because they contain DNA molecules that were labeled with the same mixture of fluorescent dyes. The colors differ from one pair to the next because

the dye mixtures for each chromosome differ in color. In some cases, the long and the short arms have been labeled with different colors.

[Courtesy of David C. Ward and Michael R. Speicher.]

cation. Before and after S, there are periods, called G_1 and G_2 , respectively, in which DNA replication does not take place. The **cell cycle**, or the life cycle of a cell, is commonly described in terms of these three interphase periods followed by mitosis, **M**. The order of events is therefore $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$, as shown in Figure 3.2. In this representation, cytokinesis, the division of the cytoplasm into two approximately equal parts containing the daughter nuclei, is included in the M period. The length of time required for a complete life cycle varies with cell type. In higher eukaryotes, the majority of cells require from 18 to 24 hours. The relative duration of the different periods in the cycle also varies considerably with cell type. Mitosis, requiring from 1/2 hour to 2 hours, is usually the shortest period.

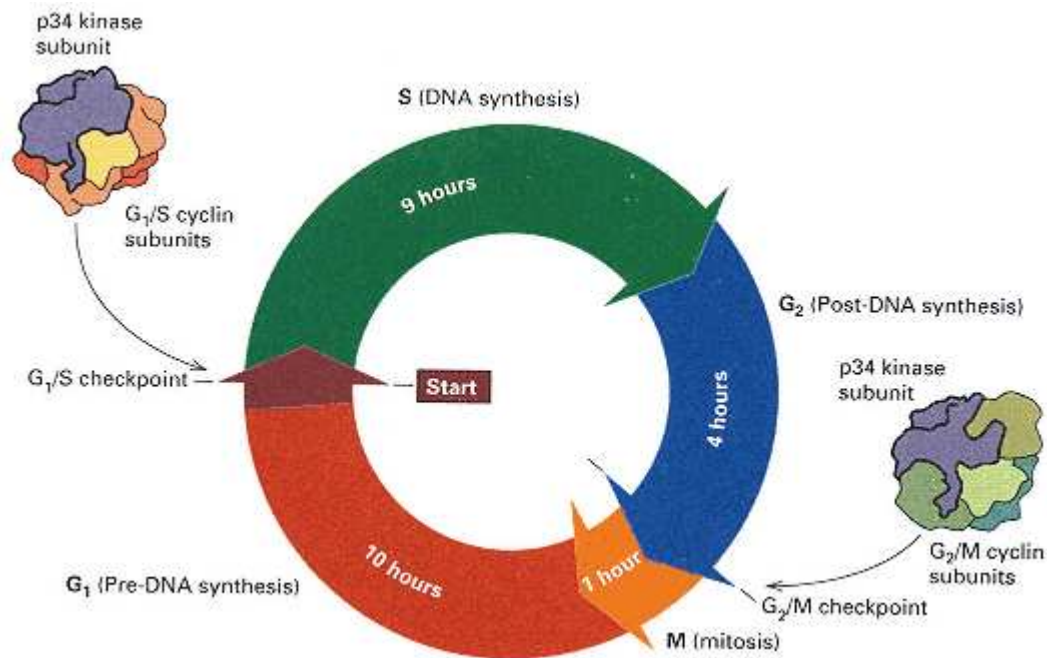


Figure 3.2

The cell cycle of a typical mammalian cell growing in tissue culture with a generation time of 24 hours. The critical control points for the G_1/S and G_2/M transitions are governed by a p34 kinase that is activated by stage-specific cyclins and that regulates the activity of its target proteins through phosphorylation.

The cell cycle itself is under genetic control. The mechanisms of control appear to be essentially identical in all eukaryotes. There are two critical transitions—from G_1 into S and from G_2 into M (Figure 3.2). The G_1/S and G_2/M transitions are called "checkpoints" because the transitions are delayed unless key processes have been completed. For example, at the G_1/S checkpoint, either sufficient time must have elapsed since the preceding mitosis (in some cell types) or the cell must have attained sufficient size (in other cell types) for DNA replication to be initiated. Similarly, the G_2/M checkpoint requires that DNA replication and repair of any DNA damage be completed for the M phase to commence. Both major control points are regulated in a similar manner and make use of a specialized protein kinase (called the p34 kinase subunit in Figure 3.2) that regulates the activity of target proteins by phosphorylation (transfer of phosphate groups). The p34 kinase is one of numerous types of protein kinases that are used to regulate cellular processes. To become activated, the p34 polypeptide subunit must combine with several other polypeptide chains that are known as **cyclins** because their abundance cycles in phase with the cell cycle. At the G_1/S control point, one set of cyclins combines with the p34 subunit to yield the active kinase that triggers DNA replication and other events of the S period. Similarly, at the G_2/M control point, a second set of cyclins combines with the p34 subunit to yield the active kinase that initiates condensation of the chromosomes, breakdown of the nuclear envelope, and reorganization of the cytoskeleton in preparation for cytokinesis.

Illustrated in Figure 3.3 are the essential features of chromosome behavior in

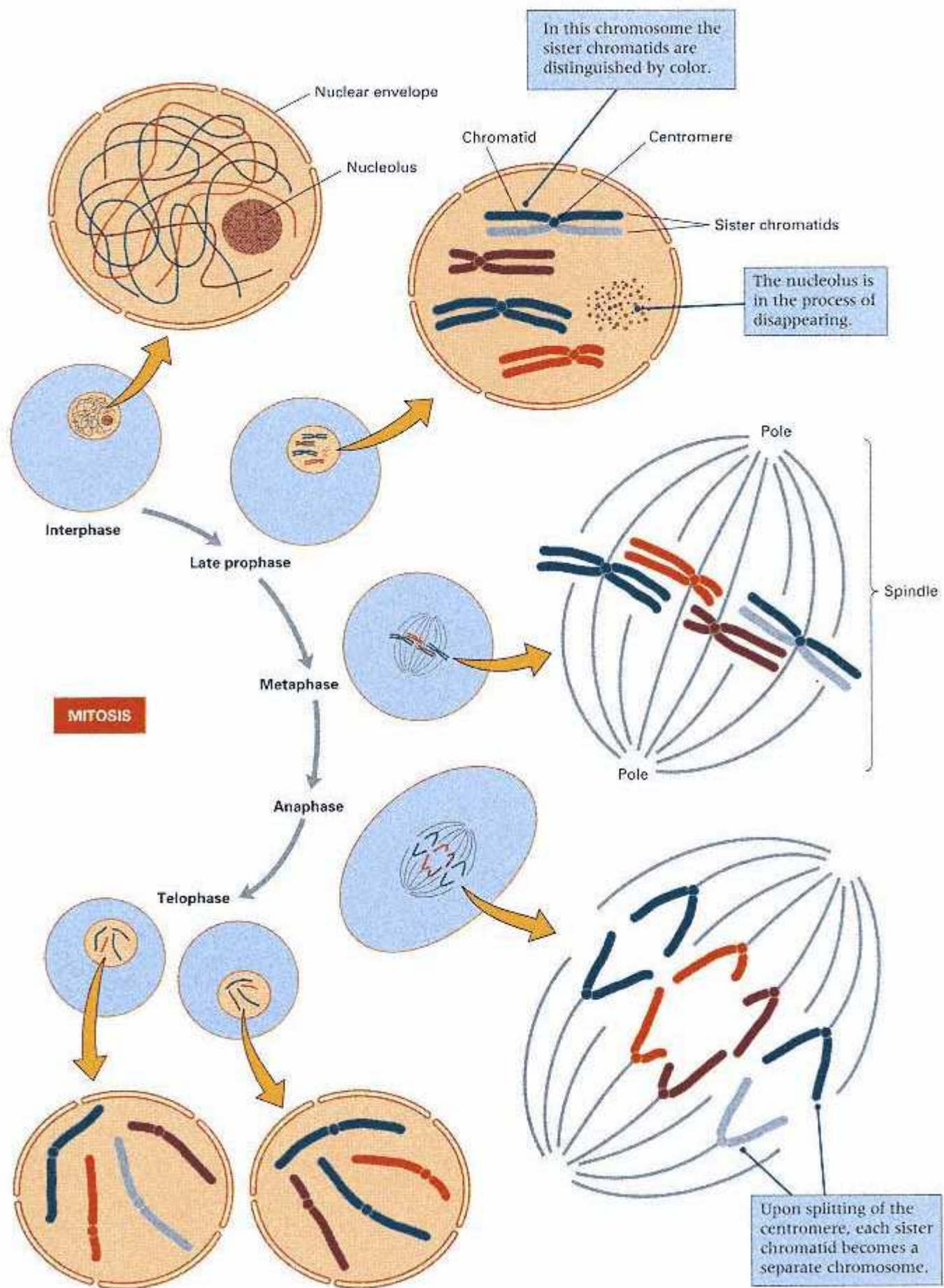


Figure 3.3

Diagram of mitosis in an organism with two pairs of chromosomes (red/rose versus green/blue). At each stage, the smaller inner diagram represents the entire cell, and the larger diagram is an exploded view showing the chromosomes at that stage. Interphase is usually not considered part of mitosis proper; it is typically much longer than the rest of the cell cycle, and the chromosomes are not yet visible. In early prophase, the chromosomes first become visible as fine strands, and the nuclear envelope and one or more nucleoli are intact. As prophase progresses, the chromosomes

condense and each can be seen to consist of two sister chromatids; the nuclear envelope and nucleoli disappear. In metaphase, the chromosomes are highly condensed and aligned on the central plane of the spindle, which forms at the end of prophase. In anaphase, the centromeres split longitudinally, and the sister chromatids of each chromosome move to opposite poles of the spindle. In telophase, the separation of sister chromatids is complete, the spindle breaks down, new nuclear envelopes are formed around each group of chromosomes, the condensation process of prophase is reversed, and the cell cycles back into interphase.

mitosis. Mitosis is conventionally divided into four stages: **prophase**, **metaphase**, **anaphase**, and **telophase**. (If you have trouble remembering the order, you can jog your memory with *peas make awful tarts*.) The stages have the following characteristics:

1. Prophase In interphase, the chromosomes have the form of extended filaments and cannot be seen with a light microscope as discrete bodies. Except for the presence of one or more conspicuous dark bodies (**nucleoli**), the nucleus has a diffuse, granular appearance. The beginning of prophase is marked by the condensation of chromosomes to form visibly distinct, thin threads within the nucleus. Each chromosome is already longitudinally double, consisting of two closely associated subunits called **chromatids**. The longitudinally bipartite nature of each chromosome is readily seen later in prophase. Each pair of chromatids is the product of the duplication of one chromosome in the S period of interphase. The chromatids in a pair are held together at a specific region of the chromosome called the **centromere**. As prophase progresses, the chromosomes become shorter and thicker as a result of intricate coiling. At the end of prophase, the nucleoli disappear and the nuclear envelope, a membrane surrounding the nucleus, abruptly disintegrates.

2. Metaphase At the beginning of metaphase, the **mitotic spindle** forms. The spindle is a bipolar structure consisting of fiber-like bundles of microtubules that extend through the cell between the poles of the spindle. Each chromosome becomes attached to several spindle fibers in the region of the centromere. The structure associated with the centromere to which the spindle fibers attach is technically known as the **kinetochore**. After the chromosomes are attached to spindle fibers, they move toward the center of the cell until all the kinetochores lie on an imaginary plane equidistant from the spindle poles. This imaginary plane is called the **metaphase plate**. Aligned on the metaphase plate, the chromosomes reach their maximum contraction and are easiest to count and examine for differences in morphology.

Proper chromosome alignment is an important cell cycle control checkpoint at metaphase in both mitosis and meiosis. In a cell in which a chromosome is attached to only one pole of the spindle, the completion of metaphase is delayed. By grasping such a chromosome with a micromanipulation needle and pulling, one can mimic the tension that the chromosome would experience were it attached on both sides; the mechanical tension allows the metaphase checkpoint to be passed, and the cell enters the next stage of division. The signal for chromosome alignment comes from the kinetochore, and the chemical nature of the signal seems to be the dephosphorylation of certain kinetochore-associated proteins. The role of the kinetochore is demonstrated by the finding that metaphase is not delayed by an unattached chromosome whose kinetochore has been destroyed by a focused laser beam. The role of dephosphorylation is demonstrated through the use of an antibody that reacts specifically with some kinetochore proteins only when they are phosphorylated. Unattached kinetochores combine strongly with the antibody, but attachment to the spindle weakens the reaction. In chromosomes that have been surgically detached from the spindle, the antibody reaction with the kinetochore reappears. Through the signaling mechanism, when all of the kinetochores are under tension and aligned on the metaphase plate, the metaphase checkpoint is passed and the cell continues the process of division.

3. Anaphase In anaphase, the centromeres divide longitudinally, and the two **sister chromatids** of each chromosome move toward opposite poles of the spindle. Once the centromeres divide, each sister chromatid is regarded as a separate chromosome in its own right. Chromosome movement results in part from progressive shortening of the spindle fibers attached to the centromeres, which pulls the chromosomes in opposite directions toward the poles. At the completion of anaphase, the chromosomes lie in two groups near opposite poles of the spindle. Each group contains the same number of chromosomes that was present in the original interphase nucleus.

4. *Telophase* In telophase, a nuclear envelope forms around each compact group of chromosomes, nucleoli are formed, and the spindle disappears. The chromosomes undergo a reversal of condensation until they are no longer visible as discrete entities. The two daughter nuclei slowly assume a typical interphase appearance as the cytoplasm of the cell divides into two by means of a gradually deepening furrow around the periphery. (In plants, a new cell wall is synthesized between the daughter cells and separates them.)

3.3— **Meiosis**

Meiosis is a mode of cell division in which cells are created that contain only one member of each pair of chromosomes present in the premeiotic cell. When a diploid cell with two sets of chromosomes undergoes meiosis, the result is four daughter cells, each genetically different and each containing one haploid set of chromosomes.

Meiosis consists of two successive nuclear divisions. The essentials of chromosome behavior during meiosis are outlined in Figure 3.4. This outline affords an overview of meiosis as well as an introduction to

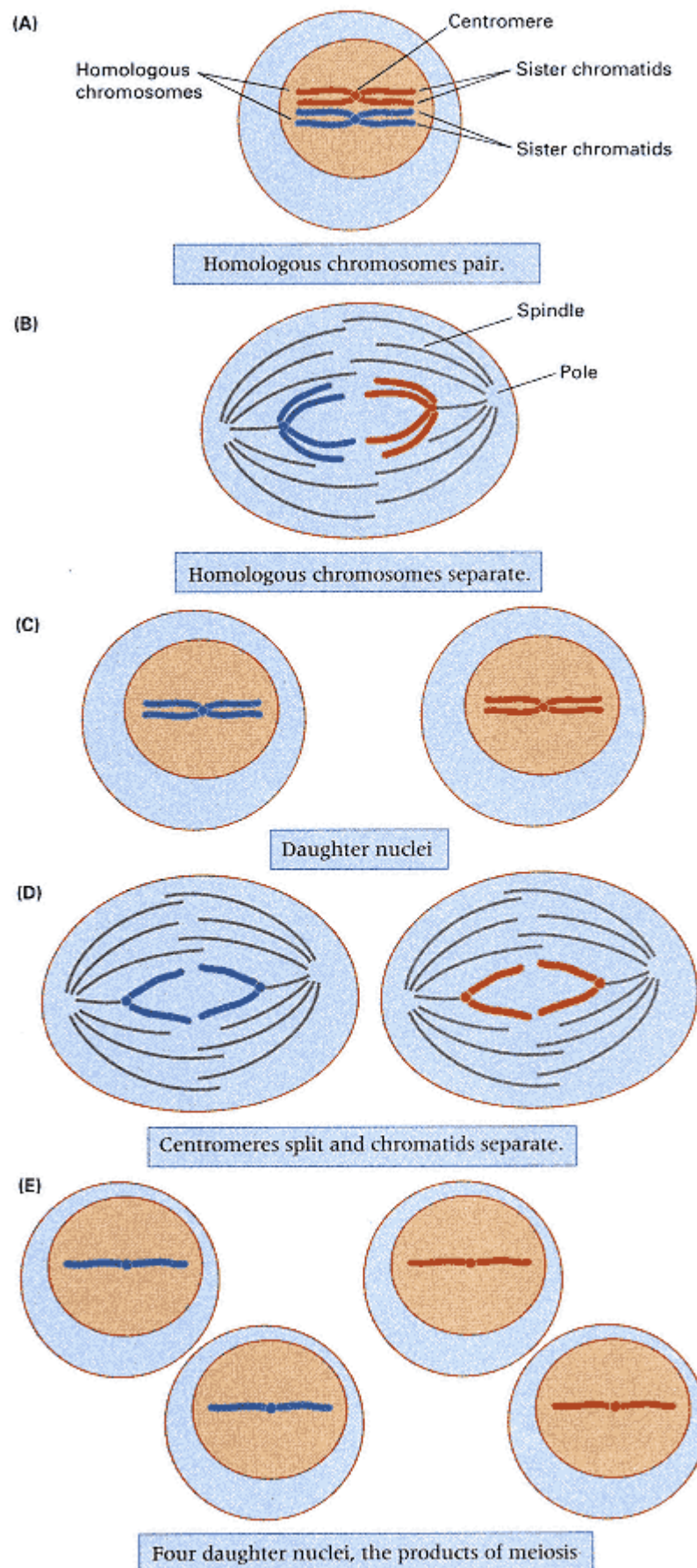


Figure 3.4

Overview of the behavior of a single pair of homologous chromosomes in meiosis. (A) The homologous chromosomes form a pair by coming together; each chromosome consists of two chromatids joined at a single centromere.

(B) The members of each homologous pair separate. (C) At the end of the first meiotic division, each daughter nucleus carries one or the other of the homologous chromosomes. (D) In the second meiotic division, in each of the daughter nuclei formed in meiosis I, the sister chromatids separate. (E) The end result is four products of meiosis, each containing one of each pair of

homologous chromosomes. For clarity, this diagram does not incorporate crossing-over, an interchange of chromosome segments that takes place at the stage depicted in part A. If crossing-over were included, each chromatid would consist of one or more segments of red and one or more segments of blue. (Crossing-over is depicted in Figure 3.7.)

the process as it takes place in a cellular context.

1. Prior to the first nuclear division, the members of each pair of chromosomes become closely associated along their length (Figure 3.4). The chromosomes that pair with each other are said to be **homologous** chromosomes. Because each member of a pair of homologous chromosomes is already replicated, each member consists of two sister chromatids joined at the centromere. The pairing of the homologous chromosomes therefore produces a four-stranded structure.
2. In the first nuclear division, the homologous chromosomes are separated from each other, one member of each pair going to opposite poles of the spindle (Figure 3.4B). Two nuclei are formed, each containing a haploid set of duplex chromosomes (Figure 3.4C) with two chromatids.
3. The second nuclear division loosely resembles a mitotic division, *but there is no chromosome replication*. At metaphase, the chromosomes align on the metaphase plate; and at anaphase, the chromatids of each chromosome are separated into opposite daughter nuclei (Figure 3.4D). The net effect of the two divisions in meiosis is the creation of four haploid daughter nuclei, each containing the equivalent of a single sister chromatid from each pair of homologous chromosomes (Figure 3.4E).

Figure 3.4 does not show that at the time of chromosome pairing, the homologous chromosomes can exchange genes. The exchanges result in the formation of chromosomes that consist of segments from one homologous chromosome intermixed with segments from the other. In Figure 3.4, the exchanged chromosomes would be depicted as segments of alternating color. The exchange process is one of the critical features of meiosis, and it will be examined in the next section.

In animals, meiosis takes place in specific cells called **meiocytes**, a general term for the primary oocytes and spermatocytes in the gamete-forming tissues (Figure 3.5). The oocytes form egg cells, and the spermatocytes form sperm cells. Although the process of meiosis is similar in all sexually reproducing organisms, in the female of both animals and plants, only one of the four products develops into a functional cell (the other three disintegrate). In animals, the products of meiosis form gametes (sperm or eggs).

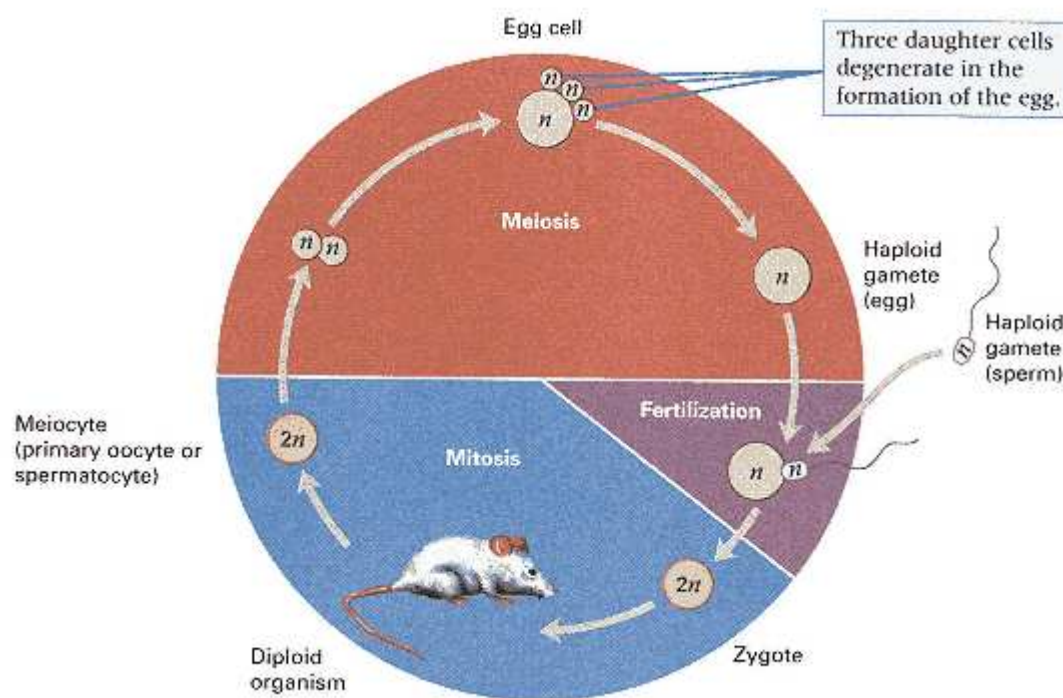


Figure 3.5

The life cycle of a typical animal. The number n is the number of chromosomes in the haploid chromosome complement. In males, the four products of meiosis develop into functional sperm; in females, only one of the four products develops into an egg.

In plants, the situation is slightly more complicated:

1. The products of meiosis typically form **spores**, which undergo one or more mitotic divisions to produce a haploid **gametophyte** organism. The gametophyte produces gametes by mitotic division of a haploid nucleus (Figure 3.6).

2. Fusion of haploid gametes creates a diploid zygote that develops into the **sporophyte** plant, which undergoes meiosis to produce spores and so restarts the cycle.

Meiosis is a more complex and considerably longer process than mitosis and usually requires days or even weeks. The entire process of meiosis is illustrated in its cellular context in Figure 3.7. The essence is that *meiosis consists of two divisions of the nucleus but only one duplication of the chromosomes*. The nuclear divisions—called the **first meiotic division** and the **second meiotic division**—can be separated into a sequence of stages similar to those used to describe mitosis. The distinctive events of this important process occur during the first division of the nucleus; these events are described in the following section.

The First Meiotic Division: Reduction

The first meiotic division (meiosis I) is sometimes called the **reductional division** because it divides the chromosome number in half. By analogy with mitosis, the first meiotic division can be split into the four stages of **prophase I**, **metaphase I**, **anaphase I**, and **telophase I**. These stages are generally more complex than their counterparts in mitosis. The stages

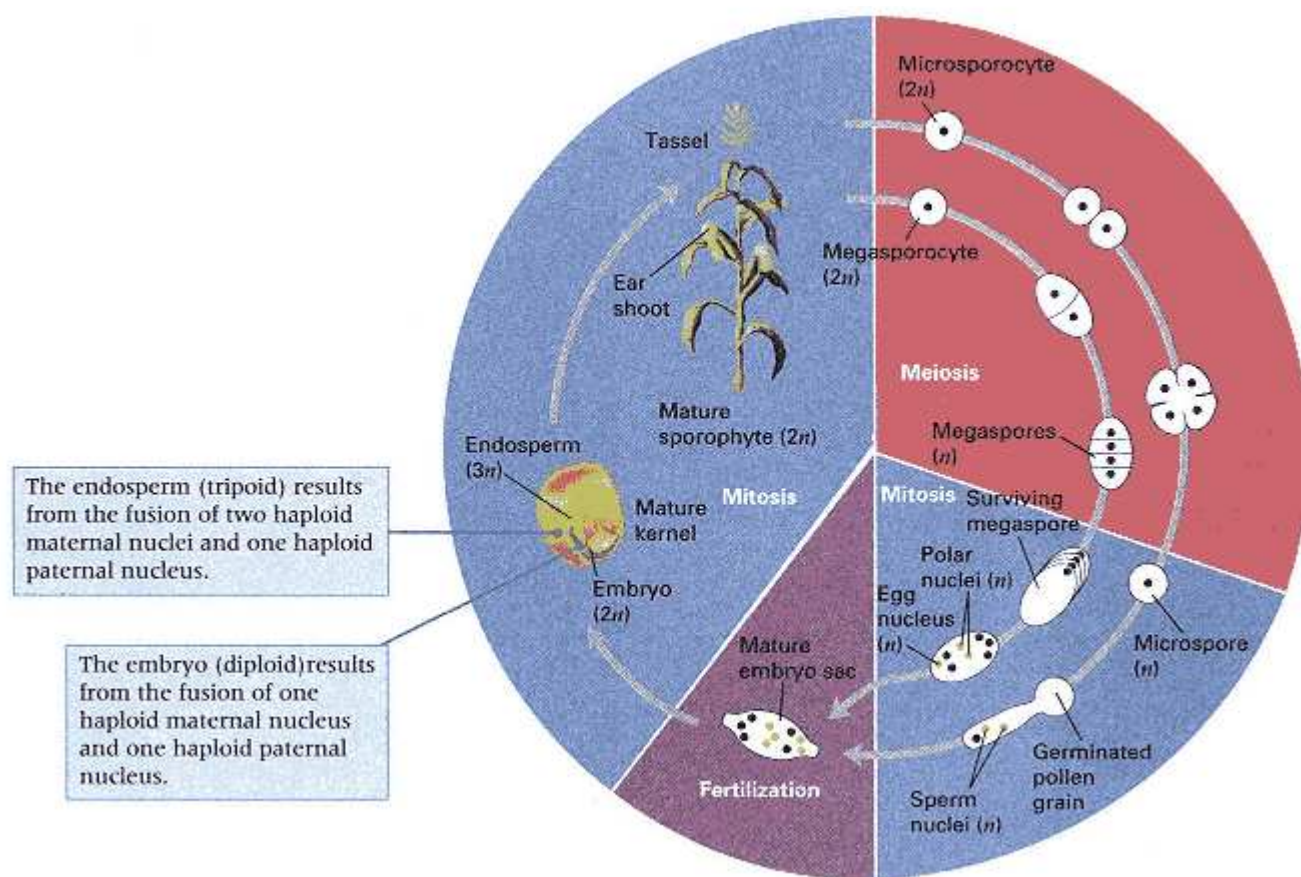


Figure 3.6

The life cycle of corn, *Zea mays*. As is typical in higher plants, the diploid spore-producing (sporophyte) generation is conspicuous, whereas the gamete-producing (gametophyte) generation is microscopic. The egg-producing spore is the *megaspore*, and the sperm-producing spore is the *microspore*. Nuclei participating in meiosis and fertilization are shown in yellow and green.

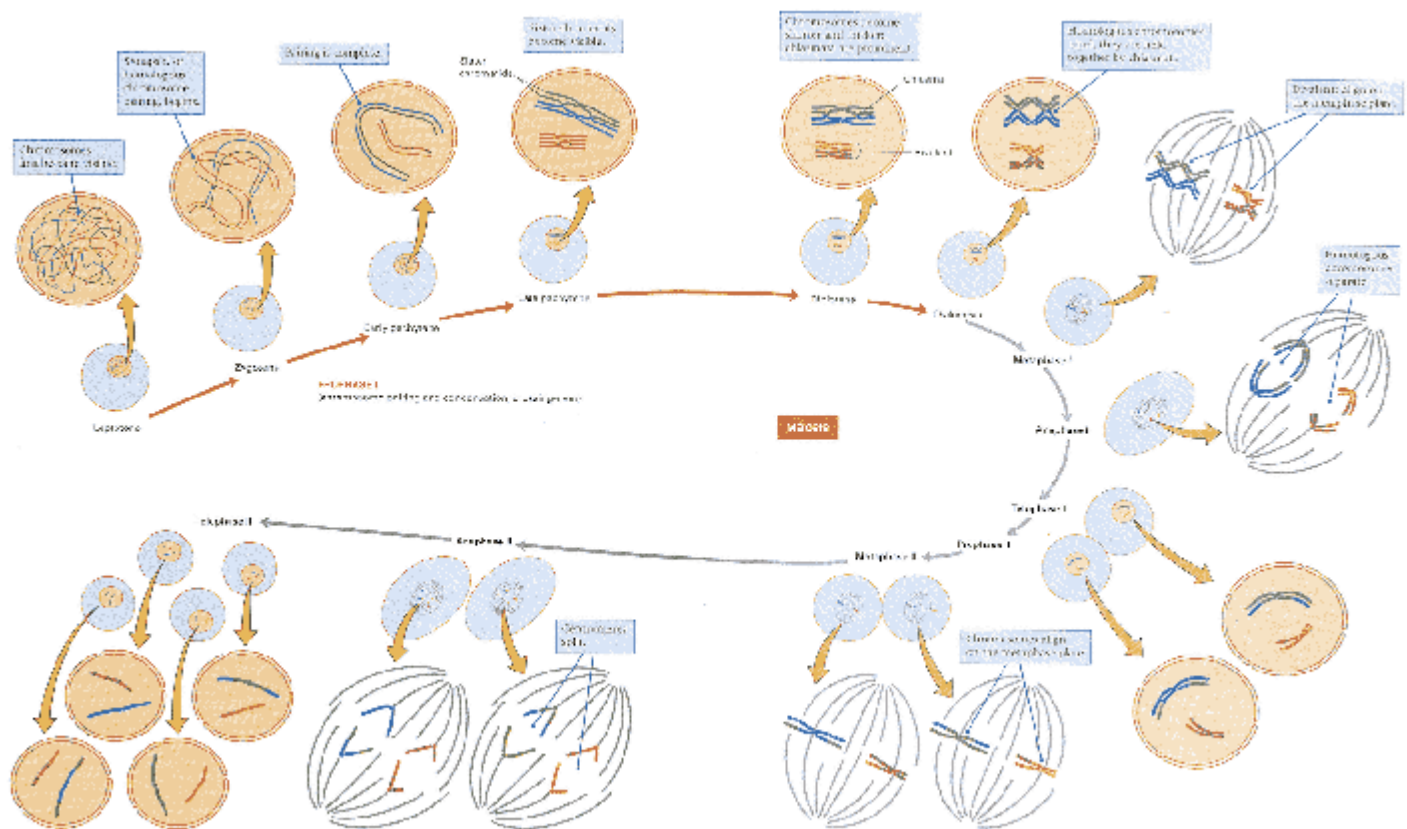


Figure 3.7

Diagram illustrating the major features of meiosis in an organism with two pairs of homologous chromosomes. At each stage, the small diagram represents the entire cell and the larger diagram is an expanded view of the chromosomes at that stage.

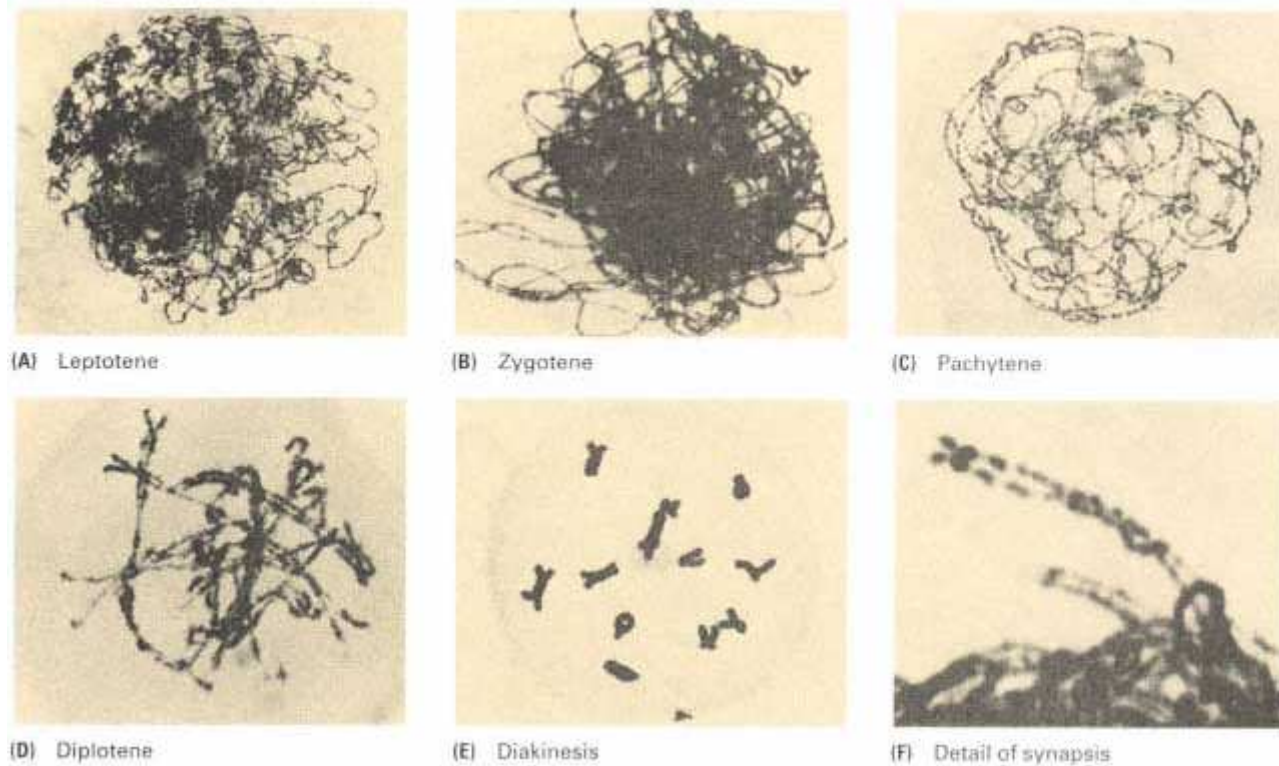


Figure 3.8

Substages of prophase of the first meiotic division in microsporocytes of a lily (*Lilium longiflorum*): (A) leptotene, in which condensation of the chromosomes is initiated and bead-like chromosomes are visible along the length of the chromosomes; (B) zygotene, in which pairing (synopsis) of homologous chromosomes occurs (paired and unpaired regions can be seen particularly at the lower left in this photograph); (C) pachytene, in which crossing-over between homologous chromosomes occurs; (D) diplotene, characterized by mutual repulsion of the paired homologous chromosomes, which remain held together at one or more cross points (chiasmata) along their length; (E) diakinesis, in which the chromosomes reach their maximum contraction; (F) zygotene (at higher magnification in another cell) showing paired homologs and matching of chromomeres during synapsis.

[Courtesy of Marta Walters (parts A, B, C, E, and F) and Herbert Stern (part D).]

and substages can be visualized with reference to Figures 3.7 and 3.8.

1. Prophase I This long stage lasts several days in most higher organisms and is commonly divided into five substages: **leptotene**, **zygotene**, **pachytene**, **diplotene**, and **diakinesis**. These terms describe the appearance of the chromosomes at each substage.

In **leptotene**, which literally means "thin thread," the chromosomes first become visible as long, thread-like structures. The pairs of sister chromatids can be distinguished by electron microscopy. In this initial phase of condensation of the chromosomes, numerous dense granules appear at irregular intervals along their length. These localized contractions, called **chromomeres**, have a characteristic number, size, and position in a given chromosome (Figure 3.8A).

The **zygotene** period is marked by the lateral pairing, or **synapsis**, of homologous chromosomes, beginning at the chromosome tips. (The term *zygotene* means "paired threads.") As the pairing process proceeds along the length of the chromosomes, it results in a precise chromomere-by-chromomere association (Figure 3.8B and F). Each pair of synapsed homologous chromosomes is referred to as a **bivalent**.

During **pachytene** (Figure 3.8C), condensation of the chromosomes continues.

Pachytene literally means "thick thread" and, throughout this period, the chromosomes continue to shorten and thicken (Figure 3.7). By late pachytene, it can sometimes be seen that each bivalent (that is, each set of paired chromosomes) actually consists of a **tetrad** of four chromatids, but the two sister chromatids of each chromosome are usually juxtaposed very tightly. The important event of genetic exchange, which is called **crossing-over**, takes place during pachytene, but crossing-over does not become apparent until the transition to diplotene. In Figure 3.7, the sites of exchange are indicated by the points where chromatids of different colors cross over each other.

At the onset of **diplotene**, the synapsed chromosomes begin to separate. Diplotene means "double thread," and the diplotene chromosomes are clearly double (Figure 3.8D and F). However, the homologous chromosomes remain held together at intervals along their length by cross-connections resulting from crossing-over. Each cross-connection, called a **chiasma** (plural, chiasmata), is formed by a breakage and rejoining between nonsister chromatids. As shown in the chromosome and diagram in Figure 3.9, *a chiasma results from physical exchange between chromatids of homologous chromosomes*. In normal meiosis, each bivalent usually has at least one chiasma, and bivalents of long chromosomes often have three or more.

The final period of prophase I is **diakinesis**, in which the homologous chromosomes seem to repel each other and the segments not connected by chiasmata move apart. Diakinesis means "moving apart." It is at this substage that the chromosomes attain their maximum condensation (Figure 3.8E). The homologous chromosomes in a bivalent remain connected by at least one chiasma, which persists until the first meiotic anaphase. Near the end of diakinesis, the formation of a spindle is initiated, and the nuclear envelope breaks down.

2. Metaphase I The bivalents become positioned with the centromeres of the two homologous chromosomes on opposite sides of the metaphase plate (Figure 3.10A). As each bivalent moves onto the metaphase plate, its centromeres are oriented at random with respect to the poles of the spindle. As shown in Figure 3.11, the bivalents formed from nonhomologous pairs of chromosomes can be oriented on the metaphase plate in either of two ways. The orientation of the centromeres determines which member of each bivalent will subsequently move to each pole. If each of the nonhomologous chromosomes is heterozygous for a pair of alleles, then one type of alignment results in *AB* and *ab* gametes and the other type results in *Ab* and *aB* gametes (Figure 3.11). Because the metaphase alignment takes place at random, the two types of alignment—and

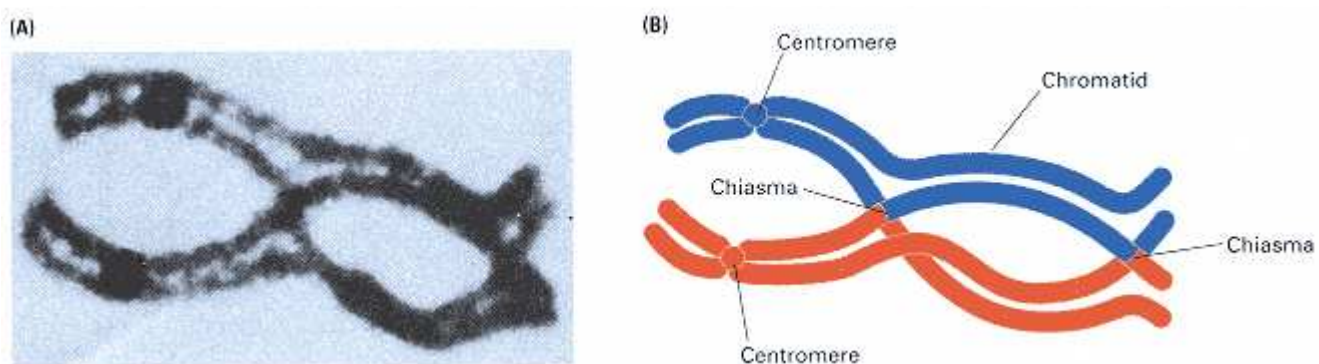


Figure 3.9

Light micrograph (A) and interpretative drawing (B) of a bivalent consisting of a pair of homologous chromosomes. This bivalent was photographed at late diplotene in a spermatocyte of the salamander *Oedipina poelzi*. It shows two chiasmata where the chromatids of the homologous chromosomes appear to exchange pairing partners.

[From F. W. Stahl. 1964. *The Mechanics of Inheritance*. Prentice-Hall, Inc.; courtesy of James Kezer.]

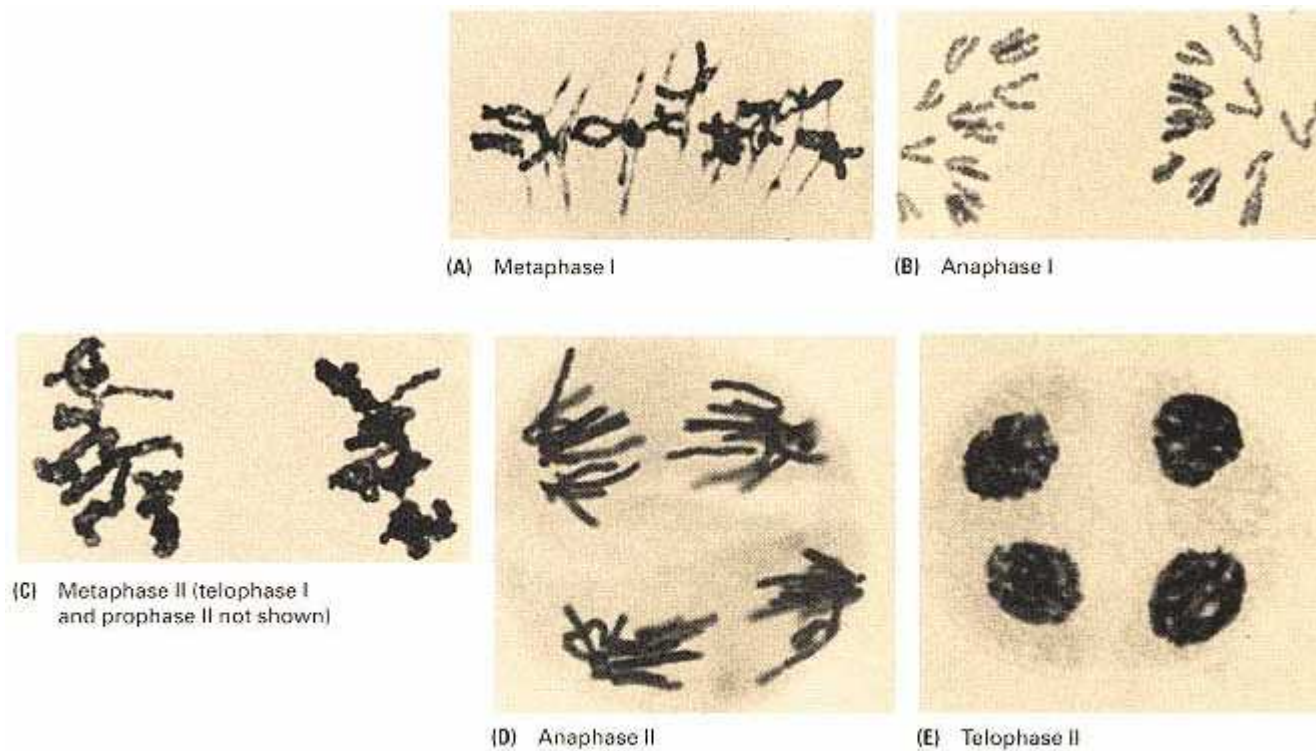


Figure 3.10

Later meiotic stages in microsporocytes of the lily *Lilium longiflorum*: (A) metaphase I; (B) anaphase I; (C) metaphase II; (D) anaphase II; (E) telophase II. Cell walls have begun to form in telophase, which will lead to the formation of four pollen grains.

[Courtesy of Herbert Stern.]

therefore the four types of gametes—are equally frequent. The ratio of the four types of gametes is 1:1:1:1, which means that the *A*, *a* and *B*, *b* pairs of alleles undergo independent assortment. In other words,

Genes on different chromosomes undergo independent assortment because nonhomologous chromosomes align at random on the metaphase plate in meiosis I.

3. Anaphase I In this stage, homologous chromosomes, each composed of two chromatids joined at an undivided centromere, separate from one another and move to opposite poles of the spindle (Figure 3. 10B). Chromosome separation at anaphase is the cellular basis of the segregation of alleles:

The physical separation of homologous chromosomes in anaphase is the physical basis of Mendel's principle of segregation.

4. Telophase I At the completion of anaphase I, a haploid set of chromosomes consisting of one homolog from each bivalent is located near each pole of the spindle (Figure 3.6). In telophase, the spindle breaks down and, depending on the species, either a nuclear envelope briefly forms around each group of chromosomes or the chromosomes enter the second meiotic division after only a limited uncoiling.

The Second Meiotic Division: Equation

The second meiotic division (meiosis II) is sometimes called the **equational division** because the chromosome number remains the same in each cell before and after the second division. In some species, the chromosomes pass directly from telophase I to **prophase II** without loss of condensation; in others, there is a brief pause between the two meiotic divisions and the chromo-

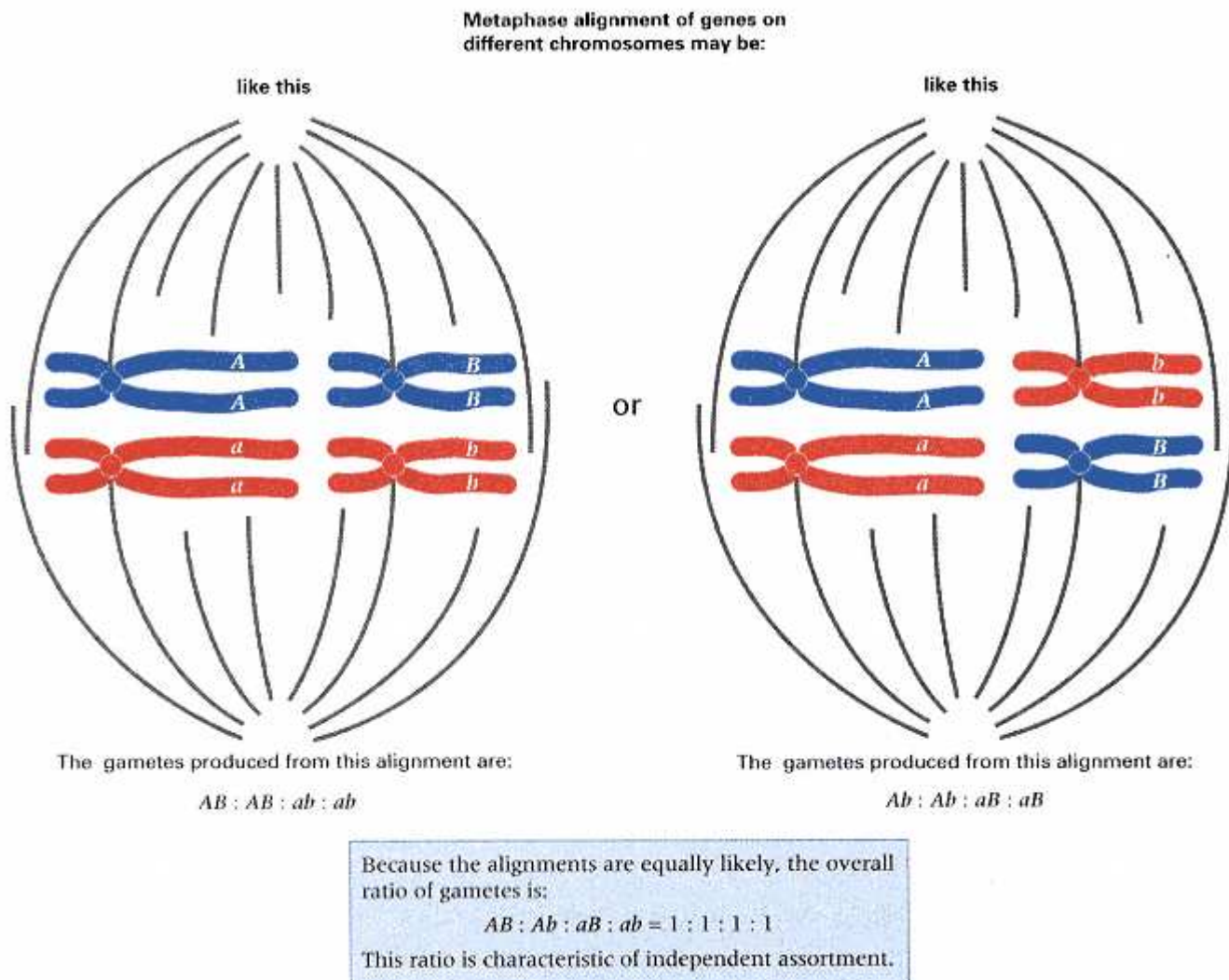


Figure 3.11
Random alignment of nonhomologous chromosomes at metaphase I results in the independent assortment of genes on nonhomologous chromosomes.

somes may "decondense" (uncoil) somewhat. *Chromosome replication never takes place between the two divisions*; the chromosomes present at the beginning of the second division are identical to those present at the end of the first division.

After a short prophase (prophase II) and the formation of second-division spindles, the centromeres of the chromosomes in each nucleus become aligned on the central plane of the spindle at **metaphase II** (Figure 3.10C). In **anaphase II**, the centromeres divide longitudinally and the chromatids of each chromosome move to opposite poles of the spindle (Figure 3.10D). Once the centromere has split at anaphase II, each chromatid is considered to be a separate chromosome.

Telophase II (Figure 3.10E) is marked by a transition to the interphase condition of the chromosomes in the four haploid nuclei, accompanied by division of the cytoplasm. Thus the second meiotic division superficially resembles a mitotic division. However, there is an important difference: *The chromatids of a chromosome are usually not genetically identical sisters along their entire length because of crossing-over associated with the formation of chiasmata during prophase of the first division.*

Connection Grasshopper, Grasshopper

E. Eleanor Carothers 1913

University of Kansas,
Lawrence, Kansas

The Mendelian Ratio in Relation to Certain Orthopteran Chromosomes

As an undergraduate researcher, Carothers showed that nonhomologous chromosomes undergo independent assortment in meiosis. For this purpose she studied a grasshopper in which one pair of homologous chromosomes had members of unequal length. At the first anaphase of meiosis in males, she could determine by observation whether the longer or the shorter chromosome went in the same direction as the X chromosome. As detailed in this paper, she found 154 of the former and 146 of the latter, a result in very close agreement with the 1:1 ratio expected from independent assortment. There is no mention of the Y chromosome because in the grasshopper she studied, the females have the sex chromosome constitution XX, whereas the males have the sex chromosome constitution X. In the males she examined, therefore, the X chromosome did not have a pairing partner. The instrument referred to as a camera lucida was at that time in widespread use for studying chromosomes and other microscopic objects. It is an optical instrument containing a prism or an arrangement of mirrors that, when mounted on a microscope, reflects an image of the microscopic object onto a piece of paper where it may be traced.

The aim of this paper is to describe the behavior of an unequal bivalent in the primary spermatocytes of certain grasshoppers. The distribution of the chromosomes of this bivalent, in relation to the X chromosome, follows the laws of chance; and, therefore, affords direct cytological support of Mendel's laws. This distribution is easily traced on account of a very distinct difference in size of the homologous chromosomes. Thus another link is added to the already long chain of evidence that the chromosomes are distinct morphological individuals continuous from generation to generation, and, as

Another link is added to the already long chain of evidence that the chromosomes are distinct morphological individuals continuous from generation to generation, and, as such, are the bearers of the hereditary qualities.

such, are the bearers of the hereditary qualities. . . . This work is based chiefly on *Brachystola magna* [a short-horned grasshopper]. . . . The entire complex of chromosomes can be separated into two groups, one containing six small chromosomes and the other seventeen larger ones. [One of the larger ones is the X chromosome.] Examination shows that this group of six small chromosomes is composed of five of about equal size and one decidedly larger. [One of the small ones is the homolog of the decidedly larger one, making this pair of chromosomes unequal in size.] . . . In early metaphases the chromosomes appear as twelve separate individuals [the bivalents]. Side views show the X chromosome in its characteristic position near one pole. . . . Three hundred cells were drawn under the camera lucida to determine the distribution of the chromosomes in the asymmetrical bivalent in relation to the X chromosome. . . . In 228 cells the bivalent and the X chromosome were in the same section [the cells had been embedded in wax and thinly sliced]. In 107 cells the smaller chromosome was going to the same pole as the X chromosome, and in the remaining 121 the larger chromosome occupied this position. In the other 72 cells the X chromosome and the bivalent were in different sections, but great care was used to make sure that there was no mistake in identifying the cell or in labeling the drawings. The smaller chromosome is accompanying the X chromosome in 39 of the cells, and the larger in 33. As a net result, then, in the 300 cells drawn, the smaller chromosome would have gone to the same nucleus as the X chromosome 146 times, or in 48.7 percent of the cases; and the larger one, 154 times, or in 51.3 percent of the cases. . . . A consideration of the limited number of chromosomes and the large number of characters in any animal or plant will make it evident that each chromosome must control numerous different characters. . . . Since the rediscovery of Mendel's laws, increased knowledge has been constantly bringing into line facts that at first seemed utterly incompatible with them. There is no cytological explanation of any other form of inheritance. . . . It seems to me probable that all inheritance is, in reality, Mendelian.

3.4—

Chromosomes and Heredity

Shortly after the rediscovery of Mendel's paper, it became widely assumed that genes were physically located in the chromosomes. The strongest evidence was that Mendel's principles of segregation and independent assortment paralleled the behavior of chromosomes in meiosis. But the first undisputable proof that genes are parts of chromosomes was obtained in experiments concerned with the pattern of transmission of the **sex chromosomes**, the chromosomes responsible for the determination of the separate sexes in some plants

and in almost all animals. We will examine these results in this section.

Chromosomal Determination of Sex

The sex chromosomes are an exception to the rule that all chromosomes of diploid organisms are present in pairs of morphologically similar homologs. As early as 1891, microscopic analysis had shown that one of the chromosomes in males of some insect species does not have a homolog. This unpaired chromosome was called the **X chromosome**, and it was present in all somatic cells of the males but in only half the sperm cells. The biological significance of these observations became clear when females of the same species were shown to have two X chromosomes.

In other species in which the females have two X chromosomes, the male has one X chromosome along with a morphologically different chromosome. This different chromosome is referred to as the **Y chromosome**, and it pairs with the X chromosome during meiosis in males because the X and Y share a small region of homology. The difference in chromosomal constitution between males and females is a chromosomal mechanism for determining sex at the time of fertilization. Whereas every egg cell contains an X chromosome, half the sperm cells contain an X chromosome and the rest contain a Y chromosome. Fertilization of an X-bearing egg by an X-bearing sperm results in an XX zygote, which normally develops into a female; and fertilization by a Y-bearing sperm results in an XY zygote, which normally develops into a male (Figure 3.12). The result is a criss-cross pattern of inheritance of the X chromosome in which a male receives his X chromosome from his mother and transmits it only to his daughters.

The XX-XY type of chromosomal sex determination is found in mammals, including human beings, many insects, and other animals, as well as in some flowering plants. The female is called the **homogametic** sex because only one type of gamete (X-bearing) is produced, and the male is called the **heterogametic** sex because two different types of gametes (X-bearing and Y-bearing) are produced. When the union of gametes in fertilization is random, a sex ratio at fertilization of 1:1 is expected because males produce equal numbers of X-bearing and Y-bearing sperm.

The X and Y chromosomes together constitute the sex chromosomes; this term distinguishes them from other pairs of chromosomes, which are called **autosomes**. Although the sex chromosomes control the developmental switch that determines the earliest stages of female or male development, the developmental process itself requires many genes scattered throughout the chromosome complement, including genes on the autosomes. The X chromosome also contains many genes with functions unrelated to sexual differentiation, as will be seen in the next section. In most organisms, including human beings, the Y chromosome carries few genes other than those related to male determination.

X-linked Inheritance

The compelling evidence that genes are in chromosomes came from the study of a *Drosophila* gene for white eyes, which proved to be present in the X chromosome. Recall that in Mendel's crosses, it did not matter which trait was present in the male parent and which in the female parent. Reciprocal crosses gave the same result. One of the earliest exceptions to this rule was found by Thomas Hunt Morgan in 1910, in an early study of a mutant in the fruit fly *Drosophila melanogaster* that had white eyes. The wildtype eye color is a brick-red combination of red and brown pigments (Figure 3.13). Although white eyes can result from certain combinations of autosomal genes that eliminate the pigments individually, the white-eye mutation that Morgan studied results in a metabolic block that knocks out both pigments simultaneously.

Morgan's study started with a single male with white eyes that appeared in a wildtype laboratory population that had been maintained for many generations. In a mating of this male with wildtype females, all of the F₁ progeny of both sexes had red eyes, which showed that the allele for white eyes is recessive. In the F₂ progeny from the mating of F₁ males and females, Morgan observed 2459 red-eyed females, 1011 red-eyed males, and 782 white-eyed males. The white-eyed phenotype was somehow connected with sex because all of the white-eyed flies were males.

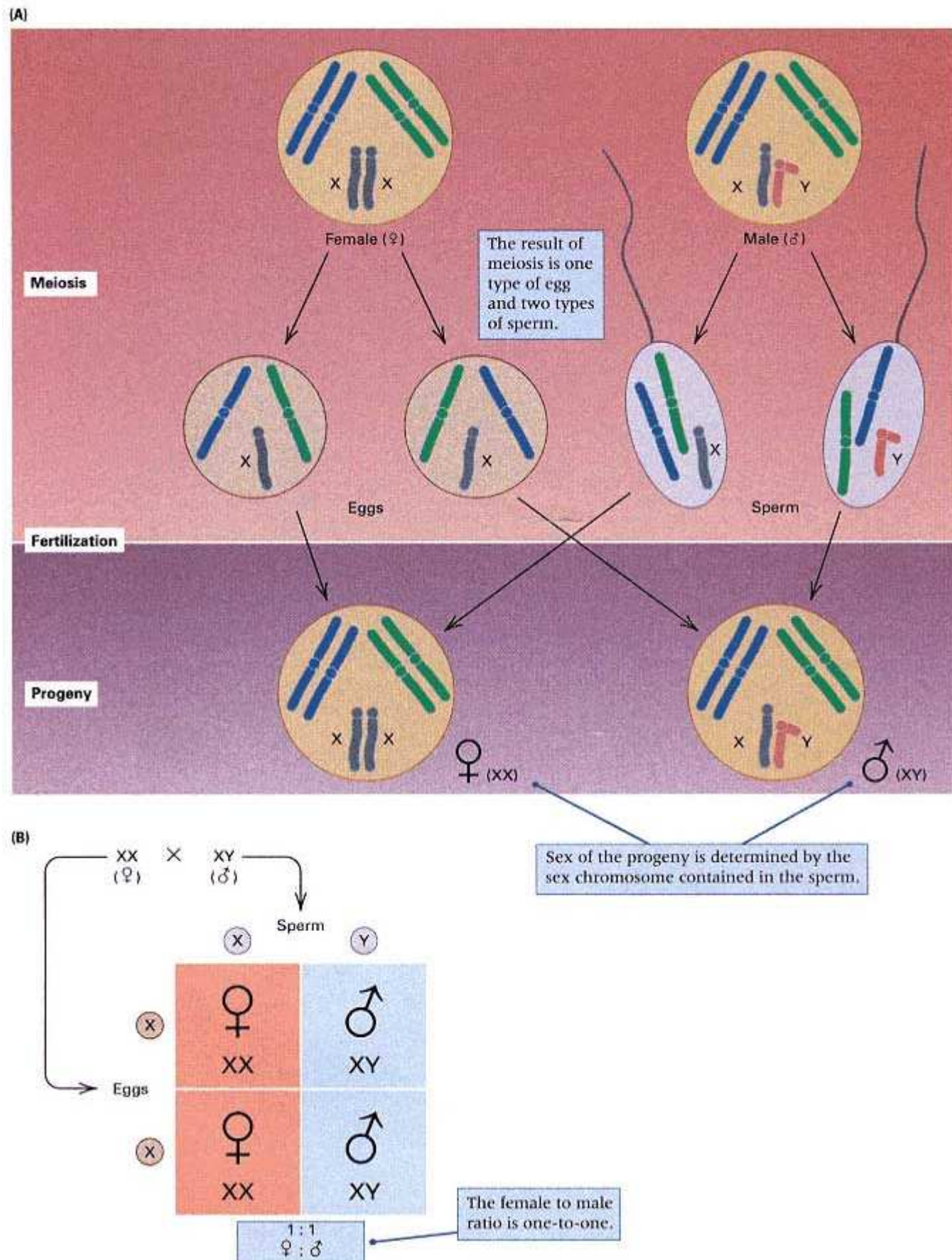


Figure 3.12

The chromosomal basis of sex determination in mammals, many insects, and other animals. (A) In females the X chromosomes segregate from each other; in males the X and Y segregate. In both sexes, each pair of autosomes segregates as well, so a female's gamete contains one X chromosome and a complete set of autosomes, whereas a male gamete carries either an X chromosome or a Y chromosome along with a complete set of autosomes. (B) Punnett square format showing only the sex chromosomes. Note that each son gets his X chromosome from his mother and his Y chromosome from his father.

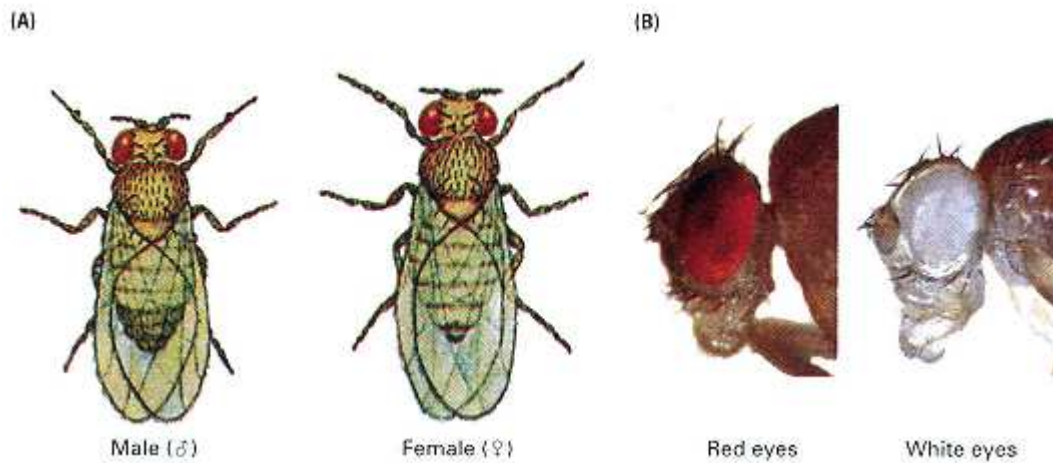


Figure 3.13

Drawings of a male and a female fruit fly, *Drosophila melanogaster*. The photographs show the eyes of a wildtype red-eyed male and a mutant white-eyed male.

[Drawings courtesy of Carolina Biological Supply Company; photographs courtesy of E. R. Lozovskaya.]

On the other hand, white eyes were not restricted to males. For example, when red-eyed F_1 females from the cross of wildtype ♀♀ × white ♂♂; were backcrossed with their white-eyed fathers, the progeny consisted of both red-eyed and white-eyed females and red-eyed and white-eyed males in approximately equal numbers.

A key observation came from the mating of white-eyed females with wildtype males. All the female progeny had wildtype eyes, but all the male progeny had white eyes. This is the reciprocal of the original cross of wildtype ♀♀ × white ♂♂ which had given only wildtype females and wildtype males, so the reciprocal crosses gave different results.

Morgan realized that reciprocal crosses would yield different results if the allele for white eyes were present in the X chromosome. The reason is that the X chromosome is transmitted in a different pattern by males and females. Figure 3.12B shows that a male transmits his X chromosome only to his daughters, whereas a female transmits one of her X chromosomes to the offspring of both sexes. Figure 3.14 shows the normal chromosome complement of *Drosophila melanogaster*. Females have an XX chromosome complement; the males are XY, and the Y chromosome does not contain a counterpart of the *white* gene. A gene on the X chromosome is said to be **X-linked**.

Figure 3.15 illustrates the chromosomal interpretation of the reciprocal crosses wildtype Sbi: male × white Sbi: female (Cross A) and white Sbi: male × wildtype Sbi: female (Cross B). The symbols w and w^+ denote the mutant and wildtype forms of the *white* gene present in the X chromosome. The genotype of a white-eyed male is wY , and that of a wildtype male is w^+Y . Because the w allele is recessive, white-eyed females are of genotype ww and wildtype females are either

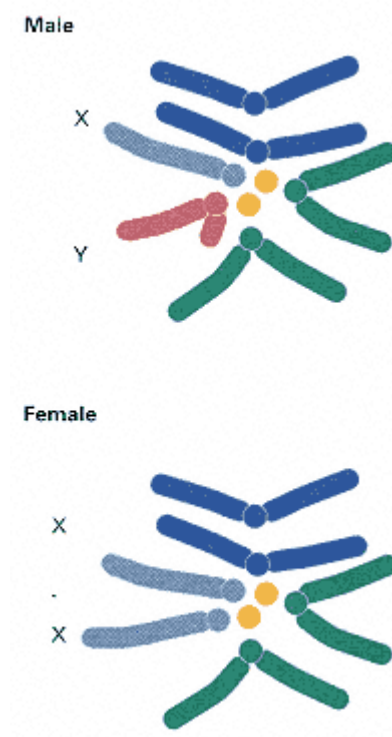


Figure 3.14
The diploid chromosome complements of a male and a female *Drosophila melanogaster*. The centromere of the X chromosome is nearly terminal, but that of the Y chromosome divides the chromosome into two unequal arms. The large autosomes (chromosomes 2 and 3, shown in blue and green) are not easily distinguishable in these types of cells. The tiny autosome (chromosome 4, shown in yellow) appears as a dot.

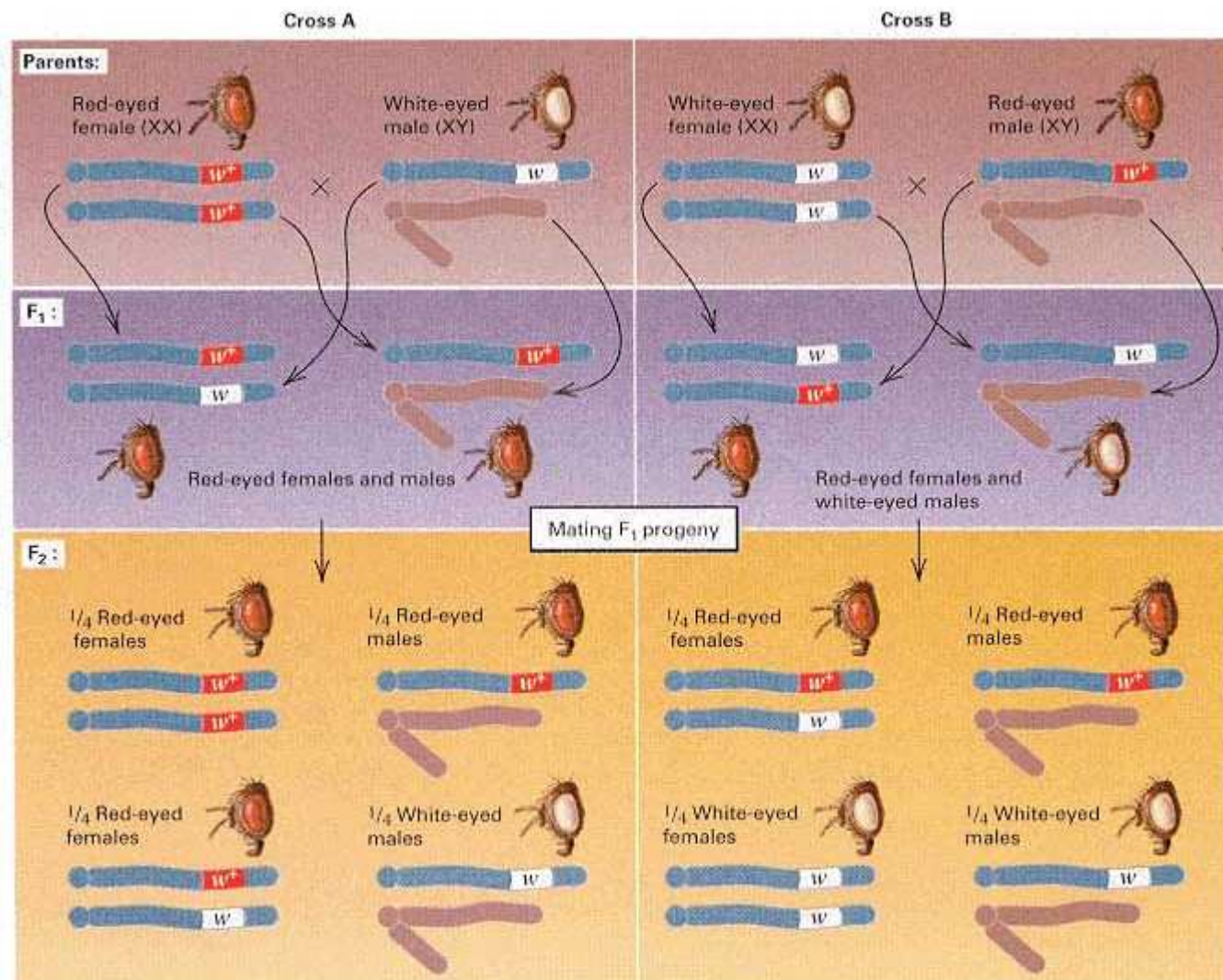


Figure 3.15

A chromosomal interpretation of the results obtained in F₁ and F₂ progenies in crosses of *Drosophila*. Cross A is a mating of a wildtype (red-eyed) female with a white-eyed male. Cross B is the reciprocal mating of a white-eyed female with a red-eyed male. In the X chromosome, the wildtype w^+ allele is shown in red and the mutant w allele in white. The Y chromosome does not carry either allele of the w gene.

heterozygous w^+w or homozygous w^+w^+ . The diagrams in Figure 3.15 account for the different phenotypic ratios observed in the F₁ and F₂ progeny from the crosses. Many other genes were later found in *Drosophila* that also follow the X-linked pattern of inheritance.

The characteristics of X-linked inheritance can be summarized as follows:

1. Reciprocal crosses resulting in different phenotypic ratios in the sexes often indicate X linkage; in the case of white eyes in *Drosophila*, the cross of a red-eyed female with a white-eyed male yields all red-eyed progeny (Figure 3.15, Cross A), whereas the cross of a white-eyed female with a red-eyed male yields red-eyed female progeny and white-eyed male progeny (Figure 3.15, Cross B).
2. Heterozygous females transmit each X-linked allele to approximately half their daughters and half their sons this is illustrated in the F₂ generation of Cross B in Figure 3.15.
3. Males that inherit an X-linked recessive allele exhibit the recessive trait because the Y chromosome does not contain a

Connection The White-Eyed Male

Thomas Hunt Morgan 1910

Columbia University,
New York, New York
Sex Limited Inheritance in Drosophila

Morgan's genetic analysis of the white-eye mutation marks the beginning of Drosophila genetics. It is in the nature of science that as knowledge increases, the terms used to describe things change also. This paper affords an example, because the term sex limited inheritance is used today to mean something completely different from Morgan's usage. What Morgan was referring to is now called X-linked inheritance or sex-linked inheritance. To avoid confusion, we have taken the liberty of substituting the modern equivalent wherever appropriate. Morgan was also unaware that Drosophila males had a Y chromosome. He thought that females were XX and males X, as in grasshoppers (see the Carothers paper). We have also supplied the missing Y chromosome. On the other hand, Morgan's gene symbols have been retained as in the original. He uses R for the wildtype allele for red eyes and W for the recessive allele for white eyes. This is a curious departure from the convention, already introduced by Mendel, that dominant and recessive alleles should be represented by the same symbol. Today we use w for the recessive allele and w⁺ for the dominant allele.

In a pedigree culture of *Drosophila* which had been running for nearly a year through a considerable number of generations, a male appeared with white eyes. The normal flies have brilliant red eyes. The white-eyed male, bred to his red-eyed sisters, produced 1,237 red-eyed offspring . . . The F₁ hybrids, inbred, produced

2,459	red-eyed females
1,011	red-eyed males
782	white-eyed males

No white-eyed females appeared.

No white-eyed females appeared. The new character showed itself to be sex-linked in the sense that it was transmitted only to the grandsons. But that the character is not incompatible with femaleness is shown by the following experiment. The white-eyed male (mutant) was later crossed with some of his daughters (F₁), and produced

129	red-eyed females
132	red-eyed males
88	white-eyed females
86	white-eyed males

The results show that the new character, white eyes, can be carried over to the females by a suitable cross, and is in consequence in this sense not limited to one sex. It will be noted that the four classes of individuals occur in approximately equal numbers (25 percent) . . . The results just described can be accounted for by the following hypothesis. Assume that all of the spermatozoa of the white-eyed male carry the "factor" for white eyes "W"; that half of the spermatozoa carry a sex factor "X," the other half lack it, i. e., the male is heterozygous for sex. [The male is actually XY.] Thus, the symbol for the male is "WXY", and for his two kinds of spermatozoa WX—Y. Assume that all of the eggs of the red-eyed female carry the red-eyed "factor" R; and that all of the eggs (after meiosis) carry one X each, the symbol for the red-eyed female will be therefore RRXX and that for her eggs will be RX. . . . The hypothesis just utilized to explain these results first obtained can be tested in several ways. [There follow four types of

crosses, each yielding the expected result.] . . . In order to obtain these results it is necessary to assume that, when the two classes of spermatozoa are formed in the RXY male, R and X go together. . . . The fact is that this R and X are combined and have never existed apart.

Source: Science 32: 120–122

wildtype counterpart of the gene. Affected males transmit the recessive allele to all of their daughters but none of their sons; this principle is illustrated in the F_1 generation of Cross A in Figure 3.15. Any male that is not affected carries the wildtype allele in his X chromosome.

An example of a human trait with an X-linked pattern of inheritance is **hemophilia A**, a severe disorder of blood clotting determined by a recessive allele. Affected persons lack a blood-clotting protein called factor VIII needed for normal clotting, and they suffer excessive, often life-threatening bleeding after injury. A famous pedigree of hemophilia starts with Queen Victoria of England (Figure 3.16). One of her sons, Leopold, was hemophilic, and two of her daughters were heterozygous carriers of the gene. Two of Victoria's granddaughters were also carriers, and by marriage they introduced the gene into the royal families of Russia and Spain. The heir to the Russian throne of the Romanoffs, Tsarevich Alexis, was afflicted with the condition. He inherited the gene from his mother, the Tsarina Alexandra, one of Victoria's granddaughters. The Tsar, the Tsarina, Alexis, and his four sisters were all executed by the Bolsheviks in the 1918 Russian revolution. Ironically, the present royal family of England is descended from a normal son of Victoria and is free of the disease.

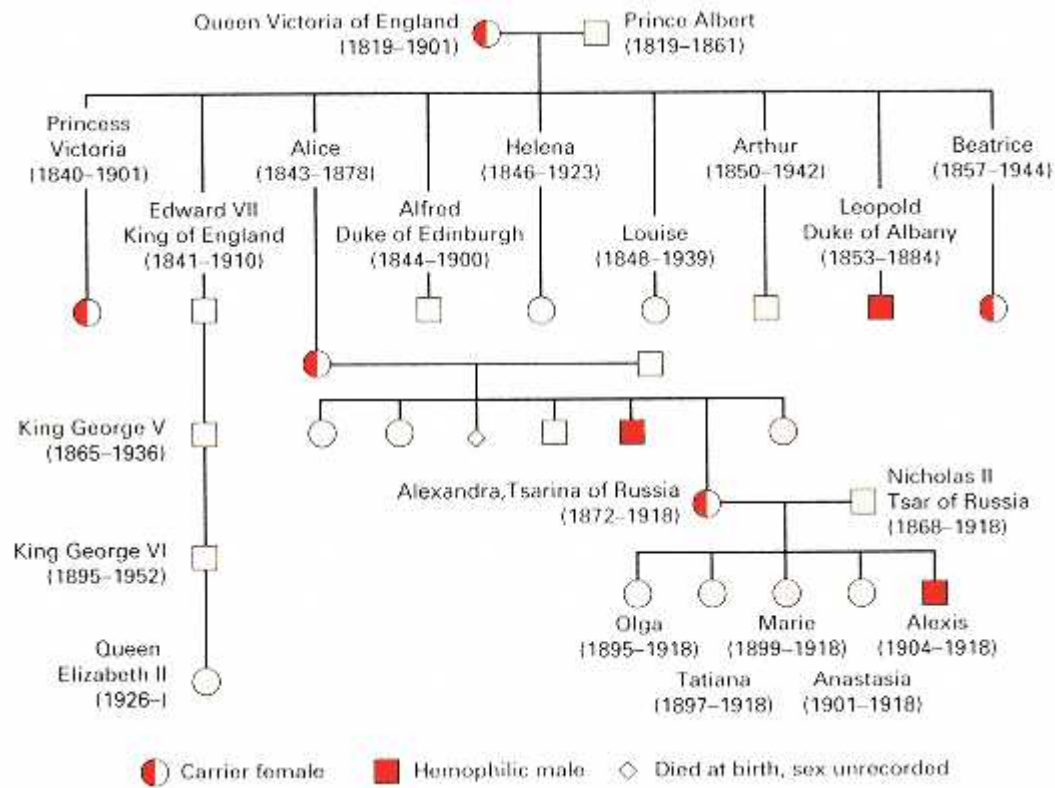


Figure 3.16
Genetic transmission of hemophilia A among the descendants of Queen Victoria of England, including her granddaughter, Tsarina Alexandra of Russia, and Alexandra's five children. The photograph is that of Tsar Nicholas II, Tsarina Alexandra, and the Tsarevich Alexis, who was afflicted with hemophilia.
[Source: Culver Pictures.]

In some organisms, the homogametic and heterogametic sexes are reversed; that is, the males are XX and the females are XY. This type of sex determination is found in birds, in some reptiles and fish, and in moths and butterflies. The reversal of XX and XY in the sexes results in an opposite pattern of nonreciprocal inheritance of X-linked genes. For example, some breeds of chickens have feathers with alternating transverse bands of light and dark color, resulting in a phenotype referred to as barred. The feathers are uniformly colored in the nonbarred phenotypes of other breeds. Reciprocal crosses between truebreeding barred and nonbarred types give the following outcomes:



These results indicate that the gene that determines barring is on the chicken X chromosome and is dominant. To distinguish sex determination in birds, butterflies, and moths from the usual XX-XY mechanism, in these organisms the sex chromosome constitution in the homogametic sex is sometimes designated WW and that in the heterogametic sex as WZ. Hence in birds, butterflies, and moths, males are chromosomally WW and females are chromosomally WZ.

Nondisjunction As Proof of the Chromosome Theory of Heredity

The parallelism between the inheritance of the *Drosophila white* mutation and the genetic transmission of the X chromosome supported the chromosome theory of heredity that genes are parts of chromosomes. Other experiments with *Drosophila* provided the definitive proof.

One of Morgan's students, Calvin Bridges, discovered rare exceptions to the expected pattern of inheritance in crosses with several X-linked genes. For example, when white-eyed *Drosophila* females were mated with red-eyed males, most of the progeny consisted of the expected red-eyed females and white-eyed males. However, about one in every 2000 F₁ flies was an exception, either a white-eyed female or a red-eyed male. Bridges showed that these rare exceptional offspring resulted from occasional failure of the two X chromosomes in the mother to separate

from each other during meiosis—a phenomenon called **nondisjunction**. The consequence of nondisjunction of the X chromosome is the formation of some eggs with two X chromosomes and others with none. Four classes of zygotes are expected from the fertilization of these abnormal eggs (Figure 3.17). Animals with no X chromosome are not detected because embryos that lack an X are not viable; likewise, most progeny with three X chromosomes die early in development. Microscopic examination of the chromosomes of the exceptional progeny from the cross white ♀♀ × wildtype ♂♂ showed that the exceptional white-eyed females had two X chromosomes *plus* a Y chromosome, and the

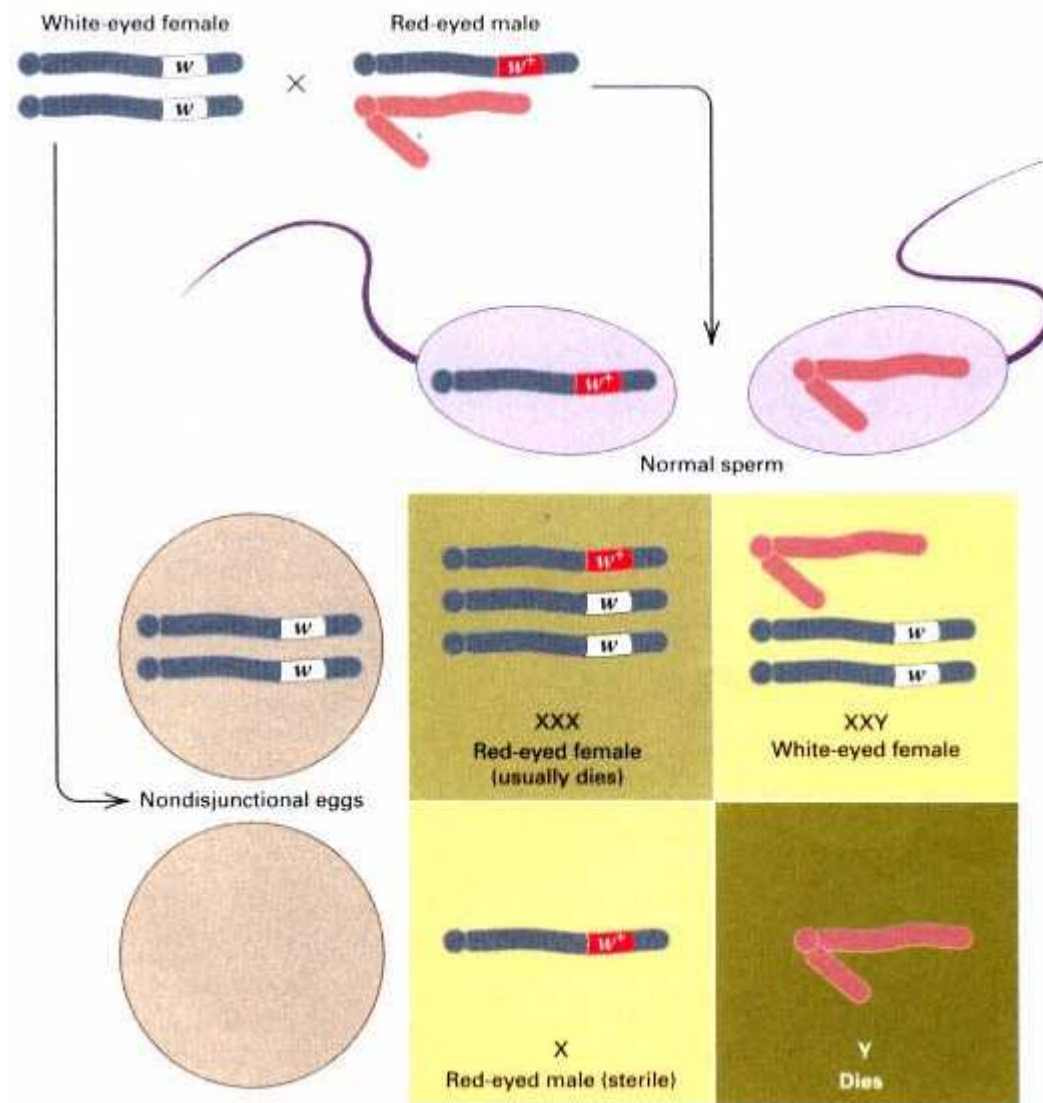


Figure 3.17
The results of meiotic nondisjunction of the X chromosomes in a female *Drosophila*.

exceptional red-eyed males had a single X but were *lacking* a Y. The latter, with a sex-chromosome constitution denoted XO, were sterile males.

These and related experiments demonstrated conclusively the validity of the chromosome theory of heredity.

Chromosome theory of heredity: Genes are contained in the chromosomes.

Bridges's evidence for the chromosome theory was that exceptional behavior on the part of chromosomes is precisely paralleled by exceptional inheritance of their genes. This proof of the chromosome theory ranks among the most important and elegant experiments in genetics.

Sex Determination in Drosophila

In the XX-XY mechanism of sex determination, the Y chromosome is associated with the male. In some organisms, including human beings, this association occurs because the presence of the Y chromosome triggers events in embryonic development

that result in the male sexual characteristics. *Drosophila* is unusual among organisms with an XX-XY type of sex determination because the Y chromosome, although associated with maleness, is not maledetermining. This is demonstrated by the finding, shown in Figure 3.17, that in *Drosophila*, XXY embryos develop into morphologically normal, fertile females, whereas XO embryos develop into morphologically normal, but sterile, males. (The "O" is written in the formula XO to emphasize that a sex chromosome is missing.) The sterility of XO males shows that the Y chromosome, though not necessary for male development, is essential for male fertility; in fact, the *Drosophila* Y chromosome contains six genes required for the formation of normal sperm.

The genetic determination of sex in *Drosophila* depends on the number of X chromosomes present in an individual fly compared with the number of sets of autosomes. In *Drosophila*, a haploid set of autosomes consists of one copy each of chromosomes 2, 3, and 4 (the autosomes). Normal diploid flies have two haploid sets of autosomes (a homologous pair each of chromosomes 2, 3, and 4) plus either two X chromosomes (in a female) or one X and one Y chromosome (in a male). We will use A to represent a complete haploid complement of autosomes; hence

A = Chromosome 2
+ Chromosome 3
+ Chromosome 4

In these terms, a normal male has the chromosomal complement XYAA, and the ratio of X chromosomes to sets of autosomes (the X/A ratio) equals 1 X : 2 A, or 1 : 2. Normal females have the chromosomal complement XXAA, and in this sex the X/A ratio is 2 X : 2 A, or 1 : 1. Flies with X/A ratios smaller than 1 : 2 (for example, XAAA—one X chromosome and three sets of autosomes) are male; those with X/A ratios greater than 1 : 1 (for example, XXXAA—three X chromosomes and two sets of autosomes) are female. Intermediate X/A ratios such as 2 : 3 (for example, XXAAA—two X chromosomes and three sets of autosomes) develop as intersexes with some characteristics of each sex.

Sexual differentiation in *Drosophila* is controlled by a gene called *Sex-lethal* (*Sxl*). The *Sxl* gene codes for two somewhat different proteins, depending on whether a male-specific coding region is included in the messenger RNA. Furthermore, the amount of *Sxl* protein present in the early embryo regulates the expression of the *Sxl* gene by a feedback mechanism. At low levels of *Sxl* protein, the male-specific form of the protein is made and shuts off further expression of the gene. At higher levels of the *Sxl* protein, the female-specific form of the protein is made and the gene continues to be expressed. In some unknown manner, the products of certain genes are sensitive to the X/A ratio and determine the amount of *Sxl* protein available to regulate the *Sxl* gene. The genes for sensing the number of X chromosomes are called *numerator genes* because they determine the "numerator" of the ratio X/A, and the genes for sensing the number of sets of autosomes are known as *denominator genes* because they determine the "denominator" in the ratio X/A. In normal males (X/A = 1 : 2), there is too little *Sxl* protein and the *Sxl* gene shuts down; in the absence of *Sxl* expression, sexual differentiation follows the male pathway, which is the "default" pathway. In normal females (X/A = 1 : 1), there is enough *Sxl* protein that the *Sxl* gene continues to be expressed. Continued expression of the *Sxl* gene initiates a cascade of genetic events, each gene in the cascade controlling one or more other genes downstream, and results in the expression of female-specific gene products and the repression of male-specific gene products. In intermediate situations when the X/A ratio is between 1 : 2 and 1 : 1, some genes specific to each sex are expressed, and the resulting sexual phenotype is ambiguous—an intersex. The *Sxl* protein is an RNA-binding protein that determines the type of mRNA produced by some of the sexdetermining genes. An outline of the genetic control of sex determination in *Drosophila* is shown in Figure 3.18.

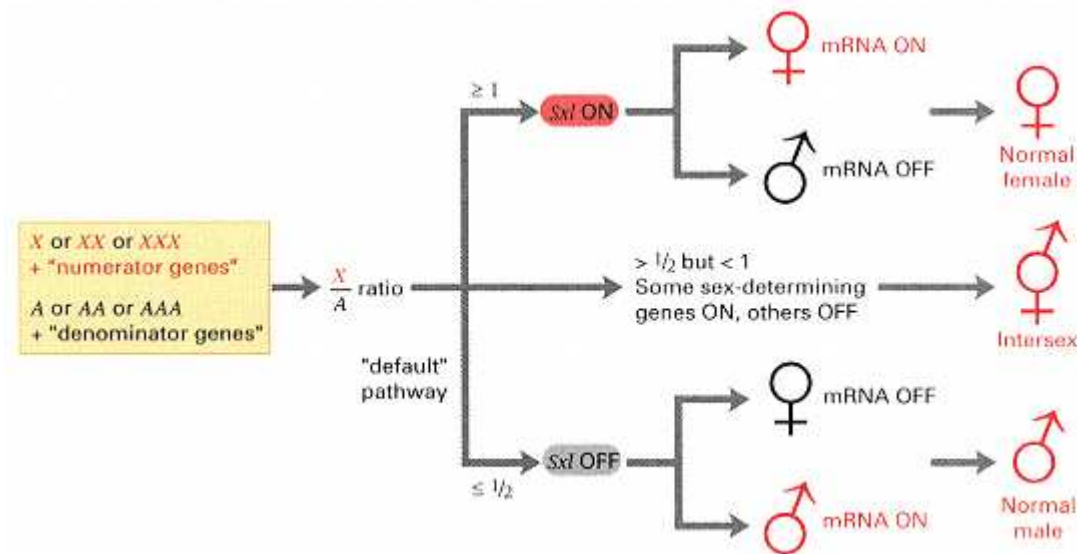


Figure 3.18

Early steps in the genetic control of sex determination in *Drosophila* through the activity of the Sexlethal protein and ultimately through the "numerator" and "denominator" genes that signal the ratio of X chromosomes to sets of autosomes (the X/A ratio).

3.5—

Probability in Prediction and Analysis of Genetic Data

Genetic transmission includes a large component of chance. A particular gamete from an *Aa* organism might or might not include the *A* allele, depending on chance. A particular gamete from an *Aa Bb* organism might or might not include both the *A* and *B* alleles, depending on the chance orientation of the chromosomes on the metaphase I plate. Genetic ratios result not only from the chance assortment of genes into gametes, but also from the chance combination of gametes into zygotes. Although exact predictions are not possible for any particular event it is possible to determine the probability that a particular event might be realized, as we have seen in Chapter 2. In this section, we consider some of the probability methods used in interpreting genetic data.

Using the Binomial Distribution in Genetics

The addition rule of probability deals with outcomes of a genetic cross that are mutually exclusive. Outcomes are "mutually exclusive" if they are incompatible in the sense that they cannot occur at the same time. For example, there are four mutually exclusive outcomes of the sex distribution of sibships with three children—namely, the inclusion of 0, 1, 2, or 3 girls. These have probability 1/8, 3/8, 3/8, and 1/8, respectively. The addition rule states that the overall probability of any combination of mutually exclusive events is equal to the sum of the probabilities of the events taken separately. For example, the probability that a sibship of size 3 contains *at least one girl* includes the outcomes 1, 2, and 3 girls, so the overall probability of at least one girl equals $3/8 + 3/8 + 1/8 = 7/8$.

The multiplication rule of probability deals with outcomes of a genetic cross that are independent. Any two outcomes are independent if the knowledge that one outcome is actually realized provides no information about whether the other is realized also. For example, in a sequence of births, the sex of any one child is not affected by the sex distribution of any children born earlier and has no influence whatsoever on the sex distribution of any siblings born later. Each successive birth is independent of all the others. When possible outcomes are independent, the multi-

plication rule states that the probability of any combination of outcomes being realized equals the product of the probabilities of all of the individual outcomes taken separately. For example, the probability that a sibship of three children will consist of three girls equals $1/2 \times 1/2 \times 1/2$, because the probability of each birth resulting in a girl is $1/2$, and the successive births are independent.

Probability calculations in genetics frequently use the addition and multiplication rules together. For example, the probability that all three children in a family will be of the same sex uses both the addition and the multiplication rules. The probability that all three will be girls is $(1/2)(1/2)(1/2) = 1/8$, and the probability that all three will be boys is also $1/8$. Because these outcomes are mutually exclusive (a sibship of size three cannot include three boys *and* three girls), the probability of either three girls or three boys is the sum of the two probabilities, or $1/8 + 1/8 = 1/4$. The other possible outcomes for sibships of size three are that two of the children will be girls and the other a boy, and that two will be boys and the other a girl. For each of these outcomes, three different orders of birth are possible—for example, GGB, GBG, and BGG—each having a probability of $1/2 \times 1/2 \times 1/2 = 1/8$. The probability of two girls and a boy, disregarding birth order, is the sum of the probabilities for the three possible orders, or $3/8$; likewise, the probability of two boys and a girl is also $3/8$. Therefore, the distribution of probabilities for the sex ratio in families with three children is

GGG	GGB	GBB	BBB
	GBG	BGB	
	BGG	BBG	

$$\begin{array}{ccccccc} (1/2)^3 & + & 3(1/2)^2(1/2) & + & 3(1/2)(1/2)^2 & + & (1/2)^3 = \\ 1/8 & + & 3/8 & + & 3/8 & + & 1/8 = 1 \end{array}$$

The sex ratio information in this display can be obtained more directly by expanding the binomial expression $(p + q)^n$, in which p is the probability of the birth of a girl ($1/2$), q is the probability of the birth of a boy ($1/2$), and n is the number of children. In the present example,

$$(p + q)^3 = 1p^3 + 3p^2q + 3pq^2 + 1q^3$$

in which the red numerals are the possible number of birth orders for each sex distribution. Similarly, the binomial distribution of probabilities for the sex ratios in families of five children is

$$(p + q)^5 = 1p^5 + 5p^4q + 10p^3q^2 + 10p^2q^3 + 5pq^4 + 1q^5$$

Each term tells us the probability of a particular combination. For example, the third term is the probability of three girls (p^3) and two boys (q^2) in a family that has five children—namely,

$$10(1/2)^3(1/2)^2 = 10/32 = 5/16$$

There are $n + 1$ terms in a binomial expansion. The exponents of p decrease by one from n in the first term to 0 in the last term, and the exponents of q increase by one from 0 in the first term to n in the last term. The coefficients generated by successive values of n can be arranged in a regular triangle known as **Pascal's triangle** (Figure 3.19). Note that the horizontal rows of the triangle are symmetrical, and that each number is the sum of the two numbers on either side of it in the row above.

In general, if the probability of event A is p and that of event B is q , and the two events are independent and mutually exclusive (see Chapter 2), the probability that A will be realized four times and B two times—in a specific order—is p^4q^2 , by the multiplication rule. However, suppose that we were interested in the combination of events "four of A and two of B," regardless of order. In that case, we multiply the

n	Coefficients					
0	1					
1	1 1					
2	1 2 1					
3	1 3 3 1					
4	1 4 6 4 1					
5	1 5 10 10 5 1					
6	1 6 15 20 15 6 1					

Figure 3.19
Pascal's triangle. The numbers are the coefficients of each term in the expansion of the polynomial $(p + q)^n$ for successive values of n from 0 through 6.

probability that the combination 4A: 2B will be realized in any one specific order by the number of possible orders. The number of different combinations of six events, four of one kind and two of another, is

$$\frac{6!}{4!2!} = \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6}{(1 \times 2 \times 3 \times 4) \times (1 \times 2)} = 15$$

The symbol ! stands for *factorial*, or the product of all positive integers from 1 through a given number. Except for $n = 0$, the formula for factorial is

$$n! = 1 \times 2 \times 3 \times 4 \times \cdots \times (n-1) \times n$$

The case $n = 0$ is an exception because $0!$ is defined as equal to 1. The first few factorials are

$$0! = 1$$

$$1! = 1$$

$$2! = 1 \times 2 = 2$$

$$3! = 1 \times 2 \times 3 = 6$$

$$4! = 1 \times 2 \times 3 \times 4 = 24$$

$$5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$$

$$6! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$$

The factorial formula

$$6!/(4! \times 2!) = 720/(24 \times 2) = 15$$

is the coefficient of the term p^4q^2 in the expansion of the binomial $(p + q)^6$. Therefore, the probability that event A will be realized four times and event B two times is $15p^4q^2$.

The general rule for repeated trials of event with constant probabilities is as follows:

If the probability of event A is p and the probability of the alternative event B is q , the probability that, in n trials, event A is realized s times and event B is realized t times is

$$\frac{n!}{s!t!} p^s q^t \quad (1)$$

in which $s + t = n$ and $p + q = 1$. Equation (1) applies even when either s or t equals 0 because $0!$ is defined to equal 1. (Remember also that any number raised to the zero power equals 1; for example, $(1/2)^0 = 1$.) Any individual term in the expansion of the binomial $(p + q)^n$ is given by Equation (1) for the appropriate values of s and t .

It is worth taking a few minutes to consider the meaning of the factorial part of the binomial expansion in Equation (1), which equals $n!/(s!t!)$. This ratio enumerates all possible ways in which s elements of one kind and t elements of another kind can be arranged in order, provided that the s elements and the t elements are not distinguished among themselves. A specific example might include s yellow peas and t green peas. Although the yellow peas and the green peas can be distinguished from each other because they have different colors, the yellow peas are not distinguishable from one another (because they are all yellow) and the green peas are not distinguishable from one another (because they are all green).

The reasoning behind the factorial formula begins with the observation that the total number of elements is $s + t = n$. Given n elements, each distinct from the next, the number of different ways in which they can be arranged is

$$n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$$

Why? Because the first element can be chosen in n ways, and once this is chosen, the next can be chosen in $n-1$ ways (because only $n-1$ are left to choose from), and once the first two are chosen, the third can be chosen in $n-2$ ways, and so forth. Finally, once $n-1$ elements have been chosen, there is only 1 way to choose the last element. The $s + t$ elements can be arranged in $n!$ ways, provided that the elements are all distinguished among themselves. However, applying again the argument we just used, each of the $n!$ particular arrangements must include $s!$ different arrangements of the s elements and $t!$ different arrangements of the t elements, or $s! \times t!$

altogether. Dividing $n!$ by $s! \times t!$ therefore yields the exact number of ways in which the s elements and the t elements can be arranged when the elements of each type are not distinguished among themselves.

Let us consider a specific application of Equation (1). in which we calculate the probability that a mating between two heterozygous parents yields exactly the

expected 3 : 1 ratio of the dominant and recessive traits among sibships of a particular size. The probability p of a child showing the dominant trait is $3/4$, and the probability q of a child showing the recessive trait is $1/4$. Suppose we wanted to know how often families with eight children would contain exactly six children with the dominant phenotype and two with the recessive phenotype. This is the "expected" Mendelian ratio. In this case, $n = 8$, $s = 6$, $t = 2$, and the probability of this combination of events is

$$\frac{8!}{6!2!} p^6 q^2 = \frac{6! \times 7 \times 8}{6! \times 2!} (3/4)^6 (1/4)^2 = 0.31$$

That is, in only 31 percent of the families with eight children would the offspring exhibit the expected 3 : 1 phenotypic ratio; the other sibships would deviate in one direction or the other because of chance variation. The importance of this example is in demonstrating that, although a 3 : 1 ratio is the "expected" outcome (and is also the single most probable outcome), the majority of the families (69 percent) actually have a distribution of offspring different from 3 : 1.

Evaluating the Fit of Observed Results to Theoretical Expectations

Geneticists often need to decide whether an observed ratio is in satisfactory agreement with a theoretical prediction. Mere inspection of the data is unsatisfactory because different investigators may disagree. Suppose, for example, that we crossed a plant having purple flowers with a plant having white flowers and, among the progeny, observed 14 plants with purple flowers and 6 with white flowers. Is this result close enough to be accepted as a 1 : 1 ratio? What if we observed 15 plants with purple flowers and 5 with white flowers? Is this result consistent with a 1 : 1 ratio? There is bound to be statistical variation in the observed results from one experiment to the next. Who is to say what results are consistent with a particular genetic hypothesis? In this section, we describe a test of whether observed results deviate too far from a theoretical expectation. The test is

called a test for **goodness of fit**, where the word *fit* means how closely the observed results "fit," or agree with, the expected results.

The Chi-Square Method

A conventional measure of goodness of fit is a value called **chi-square** (symbol, X^2), which is calculated from the number of progeny observed in each of various classes, compared with the number expected in each of the classes on the basis of some genetic hypothesis. For example, in a cross between plants with purple flowers and those with white flowers, we may be interested in testing the hypothesis that the parent with purple flowers is heterozygous for one pair of alleles determining flower color and that the parent with white flowers is homozygous recessive. Suppose further that we examine 20 progeny plants from the mating and find that 14 are purple and 6 are white. The procedure for testing this genetic hypothesis (or any other genetic hypothesis) by means of the chisquare method is as follows:

1. ***State the genetic hypothesis in detail, specifying the genotypes and phenotypes of the parents and the possible progeny.*** In the example using flower color, the genetic hypothesis implies that the genotypes in the cross *purple* $\times$ *white* could be symbolized as $Pp \times pp$. The possible progeny genotypes are either Pp or pp .
2. ***Use the rules of probability to make explicit predictions of the types and proportions of progeny that should be observed if the genetic hypothesis is true. Convert the proportions to numbers of progeny (percentages are not allowed in a X^2 test).*** If the hypothesis about the flower-color cross is true, then we should expect the progeny genotypes Pp and pp to occur in a ratio of 1 : 1. Because the hypothesis is that Pp flowers are purple and pp flowers are white, we expect the phenotypes of the progeny to be purple or white in the ratio 1 : 1. Among 20 progeny, the expected numbers are 10 purple and 10 white.

3. **For each class of progeny in turn, subtract the expected number from the observed number. Square this difference and divide the result by the expected number.** In our example, the calculation for the purple progeny is $(14-10)^2/10 = 1.6$, and that for the white progeny is $(6-10)^2/10 = 1.6$.

4. **Sum the result of the numbers calculated in step 3 for all classes of progeny. The summation is the value of X^2 for these data.** The sum for the purple and white classes of progeny is $1.6 + 1.6 = 3.2$, and this is the value of X^2 for the experiment, calculated on the assumption that our genetic hypothesis is correct.

In symbols, the calculation of X^2 can be represented by the expression

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

in which Σ means the summation over all the classes of progeny. Note that X^2 is calculated using the observed and expected *numbers*, not the proportions, ratios, or percentages. Using something other than the actual numbers is the most common beginner's mistake in applying the X^2 method. The X^2 value is reasonable as a measure of goodness of fit, because the closer the observed numbers are to the expected numbers, the smaller the value of X^2 . A value of $X^2 = 0$ means that the observed numbers fit the expected numbers perfectly.

As another example of the calculation of X^2 , suppose that the progeny of an $F_1 \times F_1$ cross includes two contrasting phenotypes observed in the numbers 99 and

Table 3.2 Calculation of X^2 for a monohybrid ratio

Phenotype (class)	Observed number	Expected number	Deviation from expected	$\frac{(\text{Deviation})^2}{\text{expected number}}$
Wildtype	99	108	-9	0.75
Mutant	45	36	+9	2.25
Total	144	144		$\chi^2 = 3.00$

45. In this case the genetic hypothesis might be that the trait is determined by a pair of alleles of a single gene, in which case the expected ratio of dominant: recessive phenotypes among the F_2 progeny is 3 : 1. Considering the data, the question is whether the observed ratio of 99 : 45 is in satisfactory agreement with the expected 3 : 1. Calculation of the value of X^2 is illustrated in Table 3.2. The total number of progeny is $99 + 45 = 144$. The *expected* numbers in the two classes, on the basis of the genetic hypothesis that the true ratio is 3 : 1, are calculated as $(3/4) \times 144 = 108$ and $(1/4) \times 144 = 36$. Because there are two classes of data, there are two terms in the X^2 calculation:

$$\begin{aligned}\chi^2 &= \frac{(99 - 108)^2}{108} + \frac{(45 - 36)^2}{36} \\ &= 0.75 + 2.25 \\ &= 3.00\end{aligned}$$

Once the X^2 value has been calculated, the next step is to interpret whether this value represents a good fit or a bad fit to the expected numbers. This assessment is done with the aid of the graphs in Figure 3.20. The x-axis gives the X^2 values that reflect goodness of fit, and the y-axis gives the probability P that a worse fit (or one equally bad) would be obtained by chance, assuming that the genetic hypothesis is true. If the genetic hypothesis is true, then the observed numbers should be reasonably close to the expected numbers. Suppose that the observed X^2 is so large that the probability of a fit as bad or worse is very small. Then the observed results do *not* fit the theoretical expectations. This means that the genetic hypothesis used to calculate the expected numbers of progeny must be rejected, because the observed numbers of progeny deviate too much from the expected numbers.

In practice, the critical values of P are conventionally chosen as 0.05 (the 5 percent level) and 0.01 (the 1 percent level). For P values ranging from 0.01 to 0.05, the probability that chance alone would lead to a fit as bad or worse is between 1 in 20 experiments and between 1 in 100, respectively. This is the purple region in Figure 3.20; if the P

value falls in this range, the

correctness of the genetic hypothesis is considered very doubtful. The result is said to be **significant** at the 5 percent level. For P values smaller than 0.01, the probability that chance alone would lead to a fit as bad or worse is less than 1 in 100 experiments. This is the green region in Figure 3.20; in this case, the result is said to be **highly significant** at the 1 percent level, and the genetic hypothesis is rejected outright. If the terminology of statistical significance seems backward, it is because the term "significant" refers to the magnitude of the deviation between the observed and the expected numbers; in a result that is statistically significant, there is a large ("significant") difference between what is observed and what is expected.

To use Figure 3.20 to determine the P value corresponding to a calculated X^2 , we

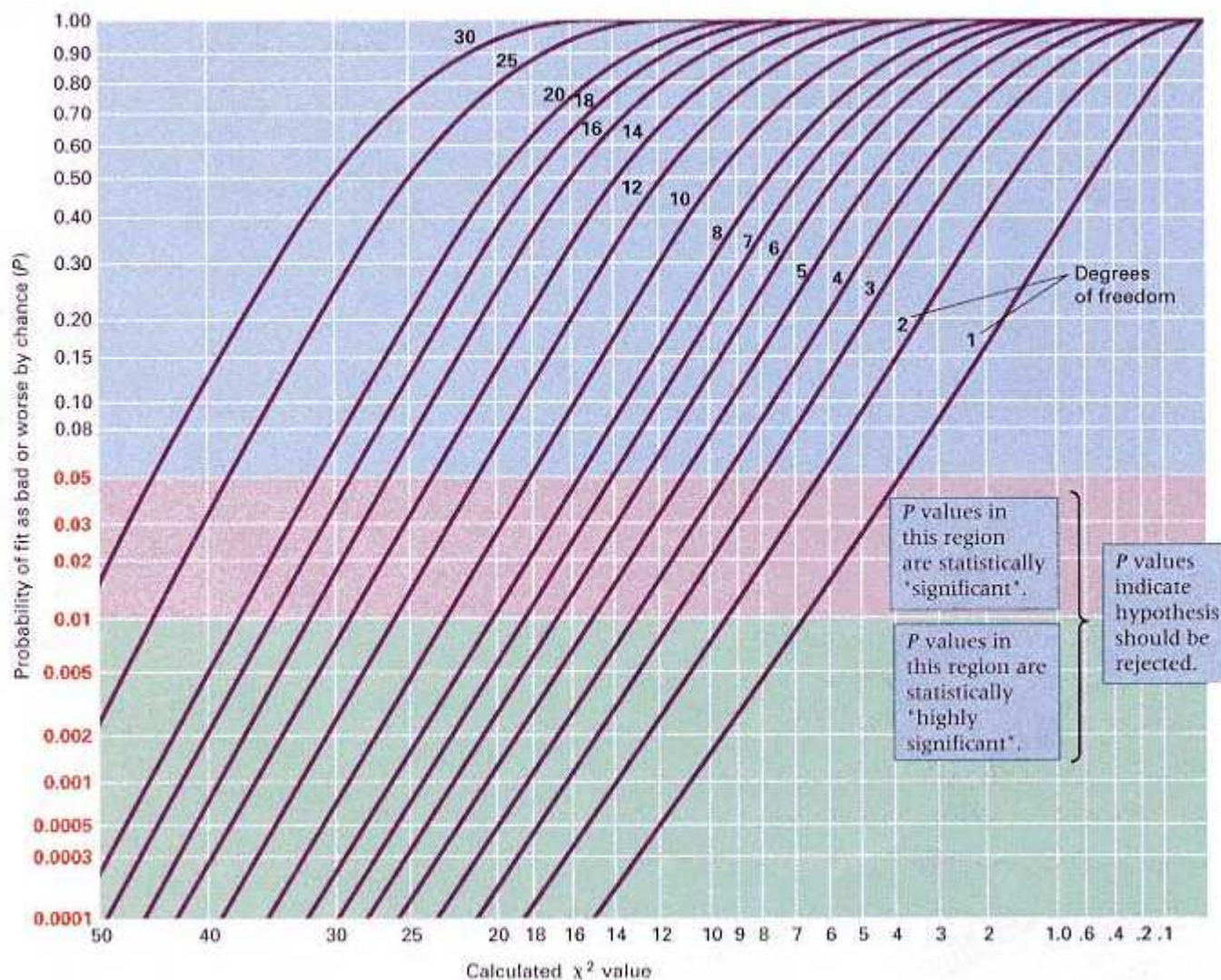


Figure 3.20

Graphs for interpreting goodness of fit to genetic predictions using the chi-square test. For any calculated value of X^2 along the x -axis, the y -axis gives the probability P that chance alone would produce a fit as bad as or worse than that actually observed, when the genetic predictions are correct. Tests with P in the purple region (less than 5 percent) or in the green region (less than 1 percent) are regarded as statistically significant and normally require rejection of the genetic hypothesis leading to the prediction. Each X^2 test has a number of degrees of freedom associated with it. In the tests illustrated in this chapter, the number of degrees of freedom equals the number of classes in the data minus 1.

Connection The Case Against Mendel's Gardener

Ronald Aylmer Fisher 1936

University College, London, England
Has Mendel's Work Been Rediscovered?

R. A. Fisher, one of the founders of modern statistics, was also interested in genetics. He gave Mendel's data a thorough going over and made an "abominable discovery." Fisher's unpleasant discovery was that some of Mendel's experiments yielded a better fit to the wrong expected values than they did to the right expected values. At issue are two series of experiments consisting of progeny tests in which F_2 plants with the dominant phenotype were self-fertilized and their progeny examined for segregation to ascertain whether each parent was heterozygous or homozygous. In the first series of experiments, Mendel explicitly states that the cultivated 10 seeds from each plant. What Mendel did not realize, apparently, is that inferring the genotype of the parent on the basis of the phenotypes of 10 progeny introduces a slight bias. The reason is shown in the accompanying illustration. Because a fraction $(3/4)^{10}$ of all progenies from a heterozygous parent will not exhibit segregation, purely as a result of chance, this proportion of Aa parents gets misclassified as AA. The expected proportion of "apparent" AA plants is $(1/3) + (2/3)(3/4)^{10}$ and that of Aa plants is $(2/3)[1-(3/4)^{10}]$, for a ratio of 0.37: 0.63. In the first series of experiments, among 600 plants tested, Mendel reports a ratio of 0.335: 0.665, which is in better agreement with the incorrect expectation of 0.33 : 0.67 than with 0.37 : 0.63. In the second series of experiments, among 473 progeny, Mendel reports a ratio of 0.32 : 0.68, which is again in better agreement with 0.33 : 0.67 than

The reconstruction [of Mendel's experiments] gives no doubt whatever that his report is to be taken entirely literally, and that his experiments were carried out in just the way and much in the order that they are recounted.

with 0.37 : 0.63. This is the "abominable discovery." The reported data differ highly significantly from the true expectation. How could this be? Fisher suggested that Mendel may have been deceived by an overzealous assistant. Mendel did have a gardener who tended the fruit orchards, a man described as untrustworthy and excessively fond of alcohol, and Mendel was also assisted in his pea experiments by two fellow monks. Another possibility, also suggested by Fisher, is that in the second series of experiments, Mendel cultivated more than 10 seeds from each plant. (Mendel does not specify how many seeds were tested from each plant in the second series.) If he cultivated 15 seeds per plant, rather than 10, then the data are no longer statistically significant and the insinuation of data tampering evaporates.

In connection with these tests of homozygosity by examining ten offspring formed by self-fertilization, it is disconcerting to find that the proportion of plants misclassified by this test is not inappreciable.

Between 5 and 6 percent of the heterozygous plants will be classified as homozygous. . . . Now among 600 plants tested by Mendel 201 were classified as homozygous and 399 as heterozygous. . . . The deviation [from the true expected values of 222 and 378] is one to be taken seriously. . . . A deviation as fortunate as Mendel's is to be expected once in twenty-nine trials. . . . [In the second series of experiments], a total deviation of the magnitude observed, and in the right direction, is only to be expected once in 444 trials; there is therefore a serious discrepancy. . . . If we could suppose that larger progenies, say fifteen plants, were grown on this occasion, the greater part of the discrepancy would be removed. . . . Such an explanation, however, could not explain the discrepancy observed in the first group of experiments, in which the procedure is specified, without the occurrence of a coincidence of considerable

(text box continued on next page)

need the number of **degrees of freedom** of the particular X^2 test. For the type of X^2 test illustrated in Table 3.2, the number of degrees of freedom equals the number of classes of data minus 1. Table 3.2 contains two classes of data

(wildtype and mutant), so the number of degrees of freedom is $2 - 1 = 1$. The reason for subtracting 1 is that, in calculating the expected numbers of progeny, we make sure that the total number of progeny is the same as that actually observed. For this reason, one of the classes of data is not really "free" to contain any number we might specify; because the expected number in one class must be adjusted to make the total come out correctly, one "degree of freedom" is lost. Analogous X^2 tests with three classes of data have 2 degrees of freedom, and those with four classes of data have 3 degrees of freedom.

Once we have decided the appropriate number of degrees of freedom, we can interpret the X^2 value in Table 3.2. Refer to

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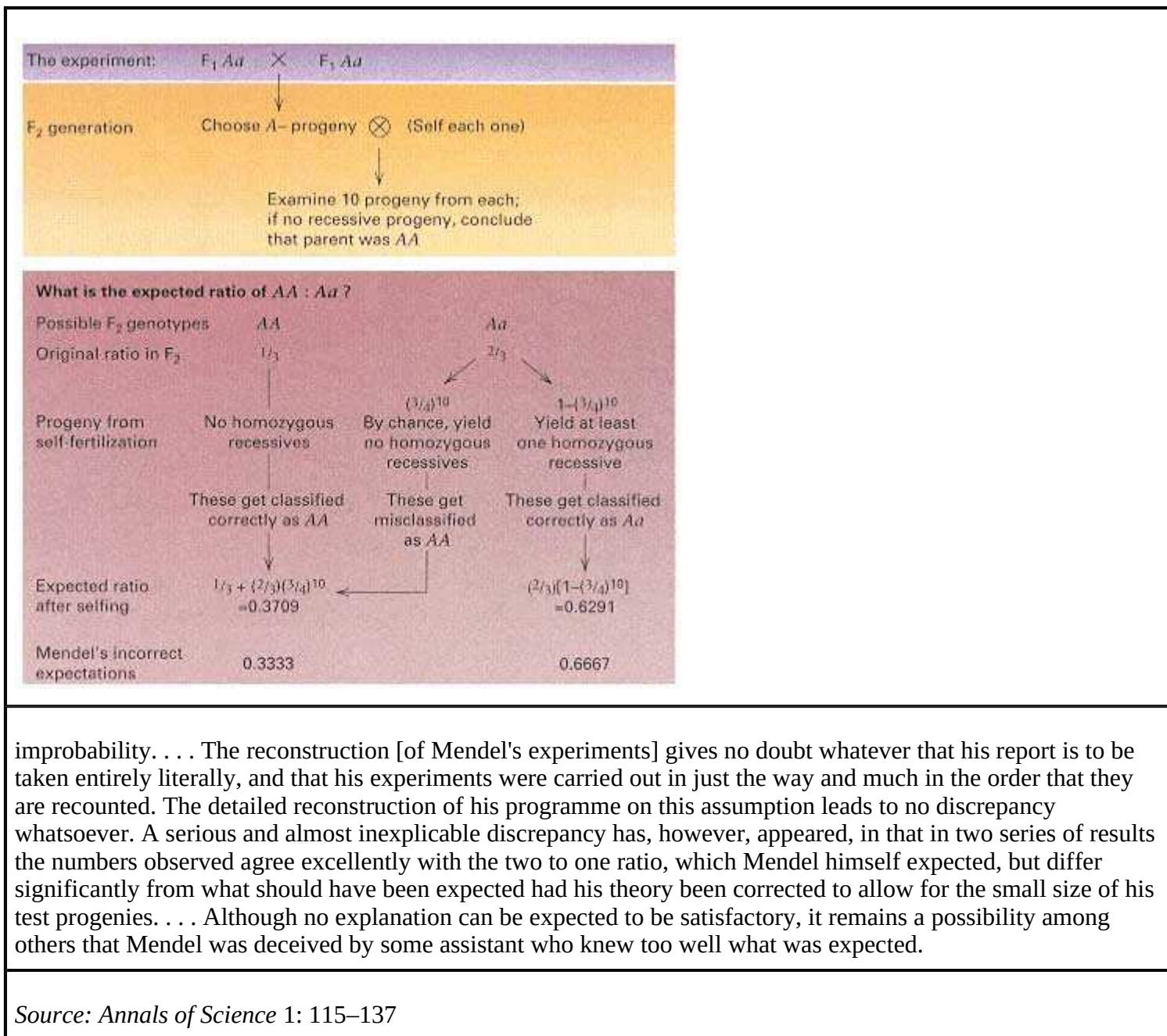


Figure 3.20, and observe that each curve is labeled with its degrees of freedom. To determine the P value for the data in Table 3.2, in which the X^2 value is 3 (3.00), first find the location of $X^2 = 3$ along the x-axis in Figure 3.20. Trace vertically from 3 until you intersect the curve with 1 degree of freedom. Then trace horizontally to the left until you intersect the y-axis, and read the P value; in this case, $P = 0.08$. This means that chance alone would produce a X^2 value as great as or greater than 3 in about 8 percent of experiments of the type in Table 3.2; and, because the P value is within the blue region, the goodness of fit to the hypothesis of a 3 : 1 ratio of wildtype: mutant is judged to be satisfactory.

As a second illustration of the X^2 test, we will determine the goodness of fit of Mendel's round versus wrinkled data to the expected 3 : 1 ratio. Among the 7324 seeds that he observed, 5474 were round and 1850 were wrinkled. The expected numbers are $(3/4) \times 7324 = 5493$ round and

$(1/4) \times 7324 = 1831$ wrinkled. The X^2 value is calculated as

$$\chi^2 = \frac{(5474 - 5493)^2}{5493} + \frac{(1850 - 1831)^2}{1831} \\ = 0.26$$

The fact that the X^2 is less than 1 already implies that the fit is very good. To find out how good, note that the number of degrees of freedom equals $2 - 1 = 1$ because there are two classes of data (round and wrinkled). From Figure 3.20, the P value for $X^2 = 0.26$ with 1 degree of freedom is approximately 0.65. This means that in about 65 percent of all experiments of this type, a fit as bad or worse would be expected simply because of chance; only about 35 percent of all experiments would yield a better fit.

3.6—

Are Mendel's Data Too Good to Be True?

Many of Mendel's experimental results are very close to the expected values. For the ratios listed in Table 2.1 in Chapter 2, the X^2 values are 0.26 (round versus wrinkled seeds), 0.01 (yellow versus green seeds), 0.39 (purple versus white flowers), 0.06 (inflated versus constricted pods), 0.45 (green versus yellow pods), 0.35 (axial versus terminal flowers), and 0.61 (long versus short stems). (As an exercise in X^2 , you should confirm these calculations for yourself.) All of the X^2 tests have P values of 0.45 or greater (Figure 3.20), which means that the reported results are in excellent agreement with the theoretical expectations.

The statistician Ronald Fisher pointed out in 1936 that Mendel's results are *suspiciously* close to the theoretical expectations. In a large number of experiments, some experiments can be expected to yield fits that appear doubtful simply because of chance variation from one experiment to the next. In Mendel's data, the doubtful values that are to be expected appear to be missing. Figure 3.21 shows the observed deviations in Mendel's experiments compared with the deviations expected by chance. (The measure of deviation is the square root of the X^2 value, assigned either a plus or a minus sign according to whether the dominant or the recessive phenotypic class was in excess of the expected number.) For each magnitude of deviation, the height of the yellow bar gives the number of experiments that Mendel observed with such a magnitude of deviation, and the orange bar gives the number of experiments expected to deviate by this amount as a result of chance alone. There are clearly too few experiments with deviations smaller than -1 or larger than +1. This type of discrepancy could be explained if Mendel discarded or repeated a few experiments with large deviations that made him suspect that the results were not to be trusted.

Did Mendel cheat? Did he deliberately falsify his data to make them appear better? Mendel's paper reports extremely deviant ratios from individual plants, as well as experiments repeated a second time when the first results were doubtful. These are not the kinds of things that a dishonest person would admit. Only a small bias is necessary to explain the excessive goodness of fit in Figure 3.21. In a count of seeds or individual plants, only about 2 phenotypes per 1000 would need to be assigned to the wrong category to account for the bias in the 91 percent of the data generated by the testing of monohybrid ratios. The excessive fit could also be explained if three or four entire experiments were discarded or repeated because deviant results were attributed to pollen contamination or other accident. After careful reexamination of Mendel's data in 1966, the evolutionary geneticist Sewall Wright concluded,

Mendel was the first to count segregants at all. It is rather too much to expect that he would be aware of the precautions now known to be necessary for completely objective data. . . . Checking of counts that one does not like, but not of others, can lead to systematic bias toward agreement. I doubt whether there are many geneticists even now whose data, if extensive, would stand up wholly satisfactorily under the X^2 test. . . . Taking everything into account, I am confident that there was no deliberate effort at falsification.

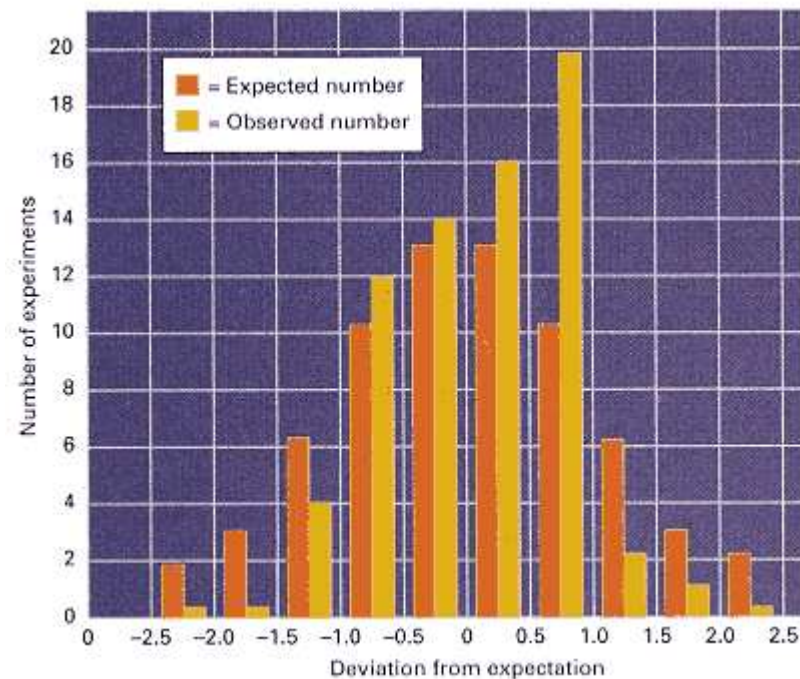


Figure 3.21

Distribution of deviations observed in 69 of Mendel's experiments (yellow bars) compared with expected values (orange bars). There is no suggestion that the data in the middle have been adjusted to improve the fit. However, several experiments with large deviations may have been discarded or repeated, because there are not so many experiments with large deviations as might be expected.

Mendel's data are some of the most extensive and complete "raw data" ever published in genetics. Additional examinations of the data will surely be carried out as new statistical approaches are developed. However, the principal point to be emphasized is that up to the present time, no reputable statistician has alleged that Mendel knowingly and deliberately adjusted his data in favor of the theoretical expectation.

Chapter Summary

The chromosomes in somatic cells of higher plants and animals are present in pairs. The members of each pair are homologous chromosomes, and each member is a homolog. Pairs of homologs are usually identical in appearance, whereas nonhomologous chromosomes often show differences in size and structural detail that make them visibly distinct from each other. A cell whose nucleus contains two sets of homologous chromosomes is diploid. One set of chromosomes comes from the maternal parent and the other from the paternal parent. Gametes are haploid. A gamete contains only one set of chromosomes, consisting of one member of each pair of homologs.

Mitosis is the process of nuclear division that maintains the chromosome number when a somatic cell divides. Before mitosis, each chromosome replicates, forming a two-part structure consisting of two sister chromatids joined at the centromere (kinetochore). At the onset of mitosis, the chromosomes become visible and, at metaphase, become aligned on the metaphase plate perpendicular to the spindle. At anaphase, the centromere of each chromosome divides, and the sister chromatids are pulled by spindle fibers to opposite poles of the cell. The separated sets of chromosomes present in telophase nuclei are genetically identical.

Meiosis is the type of nuclear division that takes place in germ cells, and it reduces the diploid number of chromosomes to the haploid number. The genetic material is replicated before the onset of meiosis, so each chromosome consists of two sister chromatids. The first meiotic division is the reduction division, which reduces the chromosome number by half. The homologous chromosomes first pair (synapsis) and then, at anaphase I, separate. The resulting products contain chromosomes that consist of two chromatids attached to a common centromere. However, as a result of crossing-over, which takes place in prophase I, the chromatids may not be genetically identical along their entire length. In the second meiotic division, the centromeres

divide and the homologous chromatids separate. The end result of meiosis is the formation of four genetically different haploid nuclei.

A distinctive feature of meiosis is the synapsis, or side-by-side pairing, of homologous chromosomes in the zygotene substage of prophase I. During the pachytene substage, the paired chromosomes become connected by chiasmata (the physical manifestations of crossing-over) and do not separate until anaphase I. This separation is called disjunction (unjoining), and failure of chromosomes to separate is called nondisjunction. Nondisjunction results in a gamete that contains either two copies or no copies of a particular chromosome. Meiosis is the physical basis of the segregation and independent assortment of genes. In *Drosophila*, an unexpected pattern of inheritance of the X-linked *white* gene was shown to be accompanied by nondisjunction of the X chromosome; these observations gave experimental proof of the chromosome theory of heredity.

Unlike other chromosome pairs, the X and Y sex chromosomes are visibly different and contain different genes. In mammals and in many insects and other animals, as well as in some flowering plants, the female contains two X chromosomes (XX) and hence is homogametic, and the male contains one X chromosome and one Y chromosome (XY) and hence is heterogametic. In birds, moths, butterflies, and some reptiles, the situation is the reverse: Females are the heterogametic sex (WZ) and males the homogametic sex (WW). The Y chromosome in many species contains only a few genes. In human beings and other mammals, the Y chromosome includes a male-determining factor. In *Drosophila*, sex is determined by a male-specific or female-specific pattern of gene expression that is regulated by the ratio of the number of X chromosomes to the number of sets of autosomes. In most organisms, the X chromosome contains many genes unrelated to sexual differentiation. These X-linked genes show a characteristic pattern of inheritance that is due to their location in the X chromosome.

The progeny of genetic crosses often conform to the theoretical predictions of the binomial probability formula. The degree to which the observed numbers of different genetic classes of progeny fit theoretically expected numbers is usually found with a chi-square (χ^2) test. On the basis of the criterion of the χ^2 test, Mendel's data fit the expectations somewhat more closely than chance would dictate. However, the bias in the data is relatively small and is unlikely to be due to anything more than recounting or repeating certain experiments whose results were regarded as unsatisfactory.

Key Terms

anaphase	heterogametic	spore
anaphase I	highly significant	spore
anaphase II	homogametic	statistical significance
autosomes	homologous	synapsis
bivalent	interphase	telophase
cell cycle	kinetochore	telophase I
centromere	leptotene	telophase II
chiasma	M period	tetrad
chi-square	meiocyte	X chromosome
chromatid	meiosis	X-linked
chromomere	metaphase	Y chromosome

chromosome	metaphase plate	zygotene
chromosome complement	metaphase I	
crossing-over	metaphase II	
cyclins	mitosis	
cytokinesis	mitotic spindle	
degrees of freedom	nondisjunction	
diakinesis	nucleolus	
diploid	pachytene	
diplotene	Pascal's triangle	
equational division	prophase	
first meiotic division	prophase I	
G ₁ period	prophase II	
G ₂ period	reductional division	
gametophyte	S phase	
germ cell	second meiotic division	
goodness of fit	sister chromatids	
haploid	sex chromosome	
hemophilia A	somatic cell	

Review the Basics

- Explain the following statement: "Independent alignment of nonhomologous chromosomes at metaphase I of meiosis is the physical basis of independent assortment of genes on different chromosomes."
- Draw a diagram of a bivalent, and label the following parts: centromere, sister chromatids, nonsister chromatids, homologous chromosomes, chiasma.
- What is the genetic consequence of the formation of a chiasma in a bivalent?
- T. H. Morgan discovered X-linkage by following up his observation that reciprocal crosses in which one parent was wildtype for eye color and the other had white eyes yielded different types of progeny. Diagram the reciprocal crosses, indicating the X and Y chromosomal genotypes of each parent and each class of offspring.
- What does it mean to say that two outcomes of a cross are mutually exclusive? What does it mean to say that two outcomes of a cross are independent?
- In what way does the chi-square value indicate "goodness of fit"?
- What are the conventional P values for "significant" and "highly significant" and what do these numbers mean?
- If Mendel did discard the results of some experiments because he considered them excessively deviant as a result of to pollen contamination or some other factor, do you consider this a form of "cheating"? Why or why not?

Guide to Problem Solving

Problem 1: The black and yellow pigments in the fur of cats are determined by an X-linked pair of alleles, c^b (black) and c^y (yellow). Males are black (c^b) or yellow (c^y), and females are either homozygous black $c^b c^b$, homozygous yellow ($c^y c^y$), or heterozygous ($c^b c^y$). The phenotype of the heterozygous female has patches of black and patches of yellow, a pattern known as *calico*. (The white spotting usually also present in domestic short-hair cats is caused by a separate gene.)

- (a) What genotypes and phenotypes would be expected among the offspring of a cross between a black female and a yellow male?
- (b) In a litter of eight kittens, there are two calico females, one yellow female, two black males, and three yellow males. What are the genotypes and phenotypes of the parents?
- (c) Rare calico males are the result of nondisjunction. What are their sex-chromosome constitution and their genotype?

Answer: (a) The black female has genotype $c^b c^b$ and the yellow male has genotype c^y . The female offspring receive an X chromosome from each parent, so their genotype is $c^b c^y$ and their phenotype is calico. The male offspring receive an X chromosome from their mother, so their genotype is c^b and their phenotype is black. (b) The male offspring provide information about the X chromosomes in the mother. Because some males are black (c^b) and some yellow (c^y), the mother must have the genotype $c^b c^y$ and have a calico coat. The fact that one of the offspring is a yellow female means that both parents carry an X chromosome with the c^y allele, so the father must have the genotype c^y and have a yellow coat. The occurrence of the calico female offspring is also consistent with these parental genotypes. (c) Nondisjunction in a female of genotype $c^b c^y$ can produce an XX egg with the alleles c^b and c^y . When the XX egg is fertilized by a Y-bearing sperm, the result is an XXY male of genotype $c^b c^y$, which is the genotype of a male calico cat. (The XXY males are sterile but otherwise are similar to normal males.)

Problem 2: Certain breeds of chickens have reddish gold feathers because of a recessive allele, g , on the W chromosome (the avian equivalent of the X chromosome); presence of the dominant allele results in silver plumage. An autosomal recessive gene, s , results in feathers called silkies that remain soft like chick down. What genotypes and phenotypes of each sex would be expected from a cross of a red rooster heterozygous for silkies and a silkie hen with silver plumage? (Remember that, in birds, females are the heterogametic sex, WZ, and males the homogametic sex, WW.)

Answer: In this problem, you need to keep track of both the W-linked and the autosomal inheritance, and the situation has the additional complication that males are WW and females WZ for the sex chromosomes. The

parental red rooster that is heterozygous for silkie has the genotype gg for the W chromosome and Ss for the relevant autosome, and the silver, silkie hen has the genotype G for the W chromosome and ss for the autosome. The female (WZ) offspring from the cross receive their W chromosome from their father (the reverse of the situation in most animals), so they have genotype g (reddish gold feathers) and half of them are Ss (wildtype) and half ss (silkie). The male (WW) offspring have genotype Gg (silver feathers), and again, half are Ss (wildtype) and half ss (silkie).

Chapter 3 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

<http://www.jbpub.com/genetics>

Select the link to **Genetics: Principles and Analysis** and then choose the link to **GeNETics on the web**. You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. See **meiosis** in action by using this keyword. Then locate the still photographs of chromosomes in various stages of prophase I. How many bivalents are formed in this organism? What is the diploid chromosome number of the organism? If assigned to do so, identify each of the prophase I stages depicted as leptotene, zygotene, pachytene, diplotene, or diakinesis.

2. Some of the main characteristics of **X-linked inheritance** can be examined at this site. Mate wildtype females with white males and observe the results. Then cross the F_1 females with their white-eye fathers and observe the results. Make a diagram of the crosses, giving the genotypes and phenotypes of all the flies and the numbers observed in each class of offspring. If assigned to do so, prepare a similar report of an initial mating of white-eyed females with wildtype males, followed by mating of the F_1 female progeny with their wildtype fathers.

3. *FlyBase* is the main internet repository of information about *Drosophila* genetics. Using the keyword **white**, you can learn about the metabolic defect in the mutation that Thomas Hunt Morgan originally discovered. Enter the keyword into the search engine, and

(text box continued on next page)

Problem 3: Ranch mink with the dark gray coat color known as aleutian are homozygous recessive, *aa*. Genotypes *AA* and *Aa* have the standard deep brown color. A mating of $Aa \times Aa$ produces eight pups.

(a) What is the probability that none of them has the aleutian coat color?

(b) What is the probability of a perfect 3 : 1 distribution of standard to aleutian?

(c) What is the probability of a 1 : 1 distribution in this particular litter?

Answer: This kind of problem demonstrates the effect of chance variation in segregation ratios in small sibships. In the mating $Aa \times Aa$, the probability that any particular pup has the standard coat color is $3/4$, and that of aleutian is $1/4$. Therefore, in a litter of size 8, the probability of 0, 1, 2, 3, . . . pups with the standard coat color is given by successive terms in the binomial expansion of $[(3/4) + (1/4)]^8$. The specific probability of r standard pups and $8 - r$ aleutian pups is given by $8!/[r!(8 - r)!] \times (3/4)^r \times (1/4)^{(8-r)}$.

(a) The probability that none is aleutian means that $r = 8$, so the probability is

$$(3/4)^8 = 6561/65,536 = 0.10$$

(b) A perfect 3 : 1 ratio in the litter means that $r = 6$, and the probability is

$$[8!/(6!2!)](3/4)^6(1/4)^2 = 28(729/4096)(1/16) = 0.31$$

(That is, a little less than a third of the litters would have the "expected" Mendelian ratio.)

(c) A 1 : 1 distribution in the litter means that $r = 4$, and the probability is

$$[8!/(4!4!)](3/4)^4(1/4)^4 = 70(81/256)(1/256) \\ = 0.09$$

Problem 4: Certain varieties of maize are true-breeding either for colored aleurone (the outer layer of the seed) or for colorless aleurone. A cross of a colored variety with a colorless variety gave an F_1 with colored seeds. Among 1000 seeds produced by the cross $F_1 \times$ colored, all were colored; and, among 1000 seeds produced by the cross $F_1 \times$ colorless, 525 were colored, and 475 were colorless.

(a) What genetic hypothesis can explain these data?

(b) Using the criterion of a X^2 test, evaluate whether the data are in satisfactory agreement with the model.

Answer: (a) Mendelian segregation in a heterozygote is suggested by the 525 : 475 ratio, and dominance is implied by

(text box continued from previous page)

then select *w*. Near the bottom of the report, select *Full*, and then read the section on phenotypic information. If assigned to do so, select the allele number 1 (not +1) and write a 100-word report on the molecular basis of the *w*¹ mutation. In preparing this report, you will find it helpful to return to the search engine and enter the keyword *Doc*.

4. Using the keyword **controversies**, you can learn about at least nine controversial issues concerning Mendel's motivation for doing his work or about aspects of the work itself. If assigned to do so, pick any three of these controversial issues and write a paragraph about each, describing the issue and summarizing the opinions of those on opposite sides of the matter.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 3, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 3.



the crosses of colored $\times$ colorless and $F_1 \times$ colored. Therefore, a hypothesis that seems to fit the data is that there is a dominant allele for colored aleurone (say, *C*) and the true-breeding colored and colorless varieties are *CC* and *cc*, respectively. The F_1 has the genotype *Cc*. The cross $F_1 \times$ colored (*CC*) yields 1 *CC* : 1 *Cc* progeny (all colored), and the cross $F_1 \times$ colorless (*cc*) is expected to produce a 1 : 1 ratio of colored (*Cc*) to colorless (*cc*). **(b)** The expected numbers are 500 colored and 500 colorless, and therefore the X^2 equals $(525 - 500)^2/500 + (475 - 500)^2/500 = 2.5$. Because there are two classes of data, there is 1 degree of freedom, and the *P* value is approximately 0.12 (see Figure 3.20). The *P* value is greater than 0.05, so the goodness of fit is regarded as satisfactory.

Analysis and Applications

3.1 At what stage in mitosis and meiosis are the chromosomes replicated? When do the chromosomes first become visible in the light microscope?

3.2 If a cell contains 23 pairs of chromosomes immediately after completion of mitotic telophase, how many chromatids were present in metaphase?

3.3 The Greek roots of the terms *leptotene*, *zygotene*, *pachytene*, *diplotene*, and *diakinesis* literally mean *thin thread*,

paired thread, *thick thread*, *doubled thread* and *moving apart*, respectively. How are these terms appropriate in describing the appearance and behavior of the chromosomes during prophase I? (Incidentally, the ending *-tene* denotes the adjective; the nouns are formed by dropping *-tene* and adding *-nema*, yielding the alternative terms *leptonema*, *zygonema*, *pachynema*, *diplonema*, and *diakinesis*, which are preferred by some authors.)

3.4 The first meiotic division is often called the *reductional* division and the second meiotic division the *equational* division. Which feature of the chromosome complement is reduced in the first meiotic division and which is kept equal in the second?

3.5 Maize is a diploid organism with 10 pairs of chromosomes. How many chromatids and chromosomes are present in the following stages of cell division:

- (a) Metaphase of mitosis?
- (b) Metaphase I of meiosis?
- (c) Metaphase II of meiosis?

3.6 Sweet peas have a somatic chromosome number of 14. If the centromeres of the 7 homologous pairs are designated as Aa, Bb, Cc, Dd, Ee, Ff, and Gg.

- (a) how many different combinations of centromeres can be produced during meiosis?
- (b) what is the probability that a gamete will contain only centromeres designated by capital letters?

3.7 Emmer wheat (*Triticum dicoccum*) has a somatic chromosome number of 28, and rye (*Secale cereale*) has a somatic chromosome number of 14. Hybrids produced by crossing these cereal grasses are highly sterile and have many characteristics intermediate between the parental species. How many chromosomes do the hybrids possess?

3.8 X-linked inheritance is occasionally called *crisscross* inheritance. In what sense does the X chromosome move back and forth between the sexes every generation? In what sense is the expression misleading?

3.9 The most common form of color blindness in human beings results from an X-linked recessive gene. A phenotypically normal couple have a normal daughter and a son who is color-blind. What is the probability that the daughter is heterozygous?

3.10 The mutation for Bar-shaped eyes in *Drosophila* has the following characteristics of inheritance:

- (a) Bar males X wildtype females produce wildtype sons and Bar daughters,
- (b) The Bar females from the mating in part a, when mated with wildtype males, yield a 1 : 1 ratio of Bar : wildtype sons and Bar : wildtype daughters.

What mode of inheritance do these characteristics suggest?

3.11 Vermilion eye color in *Drosophila* is determined by the recessive allele, v , of an X-linked gene, and the wildtype eye color determined by the v^+ allele is brick red. What genotype and phenotype ratios would be expected from the following crosses:

- (a) vermilion male $\times$ wildtype female
- (b) vermilion female $\times$ wildtype male
- (c) daughter from mating in part a $\times$ wildtype male
- (d) daughter from mating in part a $\times$ vermilion male

3.12 The autosomal recessive allele, bw for brown eyes in *Drosophila*, interacts with the X-linked recessive allele, v , for vermilion, to produce white eyes. What eye-color phenotypes, and in what proportions, would be expected from a cross of a white-eyed female (genotype $v v:bw bw$) with a brown-eyed male (genotype $v^+ Y:bw bw$)?

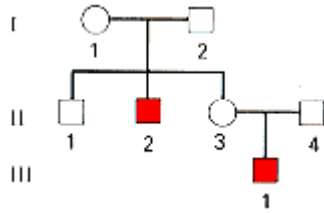
3.13 It is often advantageous to be able to determine the sex of newborn chickens from their plumage. How could this be done by using the W-linked dominant allele S for silver plumage and the recessive allele s for gold plumage? (Remember that, in chickens, the homogametic and heterogametic sexes are the reverse of those in mammals.)

3.14 A recessive mutation of an X-linked gene in human beings results in hemophilia, marked by a prolonged increase in the time needed for blood clotting. Suppose that phenotypically normal parents produce two normal daughters and a son affected with hemophilia.

(a) What is the probability that both of the daughters are heterozygous carriers?

(b) If one of the daughters mates with a normal man and produces a son, what is the probability that the son will be affected?

3.15 In the pedigree illustrated, the shaded symbols represent persons affected with an X-linked recessive form of mental retardation. What are the genotypes of all the persons in this pedigree?



3.16 For an autosomal gene with five alleles, there are fifteen possible genotypes (five homozygotes and ten heterozygotes). How many genotypes are possible with five alleles of an X-linked gene?

3.17 Attached-X chromosomes in *Drosophila* are formed from two X chromosomes attached to a common centromere. Females of genotype C(1)RM Y, in which C(1)RM denotes the attached-X chromosomes, produce C(1)RM-bearing and Y-bearing gametes in equal proportions. What progeny are expected to result from the mating between a male carrying the X-linked allele, *w*, for white eyes and an attached-X female with wildtype eyes? How does this result differ from the typical pattern of X-linked inheritance? (Note: *Drosophila* zygotes containing three X chromosomes or no X chromosomes do not survive.)

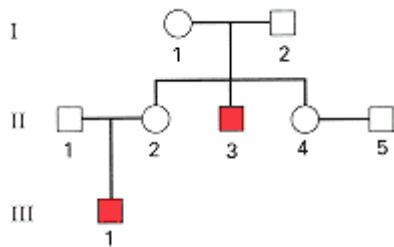
3.18 People who have the sex chromosome constitution XXY are phenotypically male. A woman heterozygous for an X-linked mutation for color blindness mates with a normal man and produces an XXY son, who is color-blind. What kind of nondisjunction can explain this result?

3.19 Mice with a single X chromosome and no Y chromosome (an XO sex-chromosome constitution) are fertile females. Assuming that at least one X chromosome is required for viability, what sex ratio is expected among surviving progeny from the mating $XO \text{ ♀} \times XY \text{ ♂}$?

3.20 In the accompanying pedigree, the shaded symbols represent persons affected with X-linked hemophilia, a blood-clotting disorder.

(a) If the woman identified as II-2 has two more children, what is the probability that neither will be affected?

(b) What is the probability that the first child of the mating II-4 $\times$ II-5 will be affected?



3.21 Assume a sex ratio at birth of 1 : 1 and consider two sibships, A and B, each with three children.

(a) What is the probability that A consists only of girls and B only of boys?

(b) What is the probability that one sibship consists only of girls and the other only of boys?

Challenge Problems

3.22 A hybrid corn plant with green leaves was testcrossed with a plant having yellow-striped leaves. When 250 seedlings were grown, 140 of the seedlings had green leaves and 110 had yellow-striped leaves. Using the chi-squared method, test this result for agreement with the expected 1 : 1 ratio.

3.23 A cross was made to produce *D. melanogaster* flies heterozygous for two pairs of alleles: dp^+ and dp , which determine long versus short wings, and e^+ and e , which determine gray versus ebony body color. The following F_2 data were obtained:

Long wing, gray body	462
Long wing, ebony body	167
Short wing, gray body	127
Short wing, ebony body	44

Test these data for agreement with the 9 : 3 : 3 : 1 ratio expected if the two pairs of alleles segregate independently.

Further Reading

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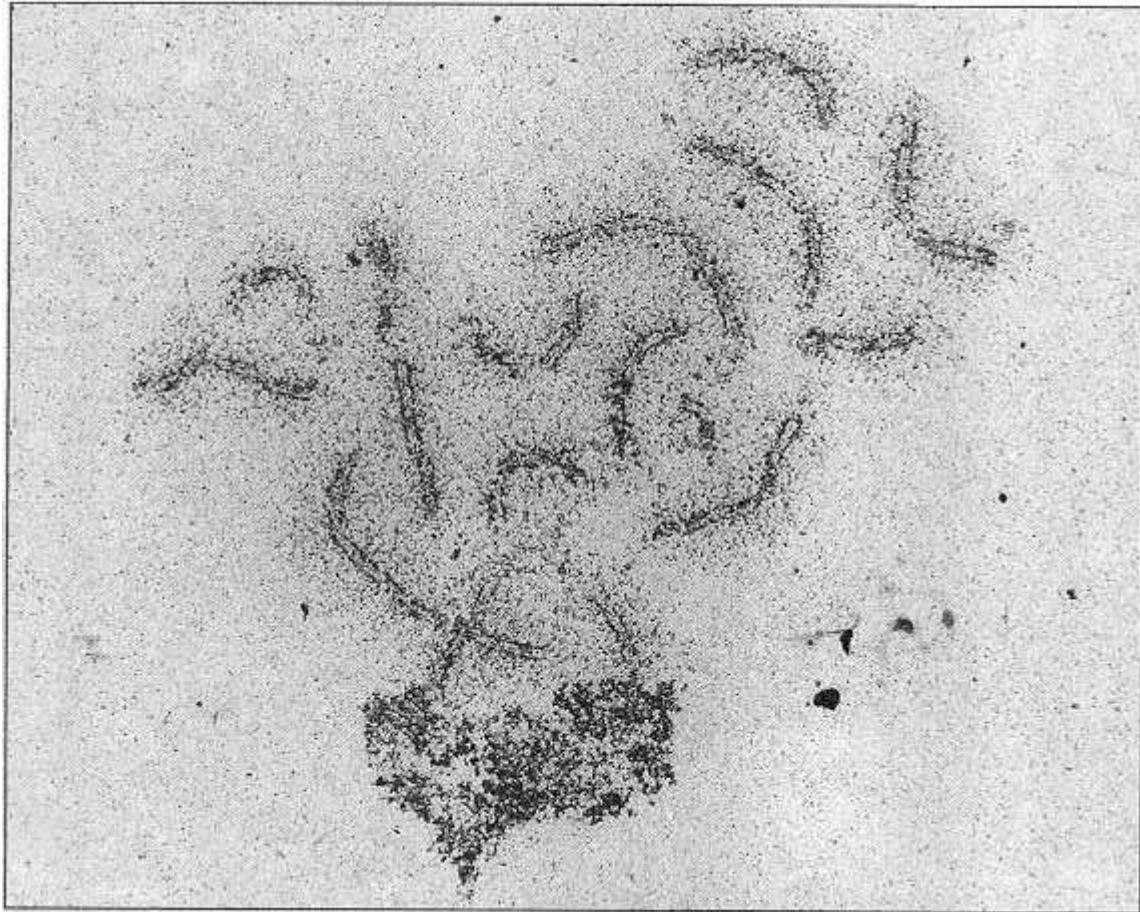
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Electron micrograph of the chromosomes in a haploid yeast cell (*Saccharomyces cerevisiae*) in prophase of mitosis, showing the full complement of 16 chromosomes. The darkly stained mass at the bottom is the nucleus, which is associated with chromosome XII.
[Courtesy of Kuei-Shu Tung and Shirleen Roeder.]

Chapter 4— Genetic Linkage and Chromosome Mapping

CHAPTER OUTLINE

4-1 Linkage and Recombination of Genes in a Chromosome

4-2 Genetic Mapping

Crossing-over

Crossing-over Takes Place at the Four-Strand Stage of Meiosis

Molecular Basis of Crossing-over

Multiple Crossing-over

4-3 Gene Mapping from Three-Point Testcrosses

Chromosome Interference in Double Crossing-over

Genetic Mapping Functions

Genetic Distance and Physical Distance

4-4 Genetic Mapping in Human Pedigrees

4-5 Mapping by Tetrad Analysis

of Unordered Tetrads

The Analysis of Ordered Tetrads

4-6 Mitotic Recombination

4-7 Recombination Within Genes

4-8 A Closer Look at Complementation

Chapter Summary

Key Terms

Review the Basics

Guide to Problem Solving

Analysis and Applications

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

- Genes that are located in the same chromosome and that do not show independent assortment are said to be linked.

- The alleles of linked genes that are present together in the same chromosome tend to be inherited as a group.
- Crossing-over between homologous chromosomes results in recombination that breaks up combinations of linked alleles.
- A genetic map depicts the relative positions of genes along a chromosome.
- The map distance between genes in a genetic map is related to the rate of recombination between the genes.
- Physical distance along a chromosome is often—but not always—correlated with map distance.
- Tetrads are sensitive indicators of linkage because each contains all the products of a single meiosis.
- Recombination can also take place between nucleotides within a gene.
- The complementation test is the experimental determination of whether two mutations are, or are not, alleles of the same gene.

CONNECTIONS

CONNECTION: Genes All in a Row

Alfred H. Sturtevant 1913

The linear arrangement of six sex-linked factors in Drosophila, as shown by their mode of association

CONNECTION: DosXX

Lilian V. Morgan 1922

Non-crisscross inheritance in Drosophila melanogaster

In meiosis, homologous chromosomes form pairs, and the individual members of each pair separate from one another. The observation that homologous chromosomes behave as complete units when they separate led to the expectation that genes located in the same chromosome would not undergo independent assortment but rather would be transmitted together with complete **linkage**. As we shall see, Thomas Hunt Morgan examined this issue using two genes that he knew were both present in the X chromosome of *Drosophila*. One was a mutation for white eyes, the other a mutation for miniature wings. Morgan did observe linkage, but it was incomplete. Morgan found that the *white* and *miniature* alleles present in each X chromosome of a female tended to remain together in inheritance, but he also observed that some X chromosomes were produced that had new combinations of the *white* and *miniature* alleles.

In this chapter, we will see that Morgan's observation of incomplete linkage is the rule for genes present in the same chromosome. The reason why linkage is incomplete is that the homologous chromosomes, when they are paired, can undergo an exchange of segments. An exchange event between homologous chromosomes, crossing-over, results in the **recombination** of genes in the homologous chromosomes. The probability of crossing-over between any two genes serves as a measure of genetic distance between the genes and makes possible the construction of a **genetic map**, a diagram of a chromosome showing the relative positions of the genes. The genetic mapping of linked genes is an important research tool in genetics because it enables a new gene to be assigned to a chromosome and often to a precise position relative to other genes within the same chromosome. Genetic mapping is usually a first step in the identification and isolation of a new gene and the determination of its DNA sequence. Genetic mapping is essential in human genetics for the identification of genes associated with hereditary diseases, such as the genes whose presence predisposes women carriers to the development of breast cancer.

4.1—

Linkage and Recombination of Genes in a Chromosome

As we saw in Chapter 3, a direct test of independent assortment is to carry out a testcross between an F_1 double heterozygote ($Aa Bb$) and the double recessive homozygote ($aa bb$). When the genes are on different chromosomes, the expected gametes from the $Aa Bb$ parent are as shown in Figure 4.1. Because the pairs of homologous chromosomes segregate independently of each other in meiosis, the double heterozygote produces all four possible types of gametes— AB , Ab , aB , and ab —in equal proportions.

Independent assortment takes place in the $Aa Bb$ genotype whether the *parents* were genotypically $AA bb$ and $aa BB$. The four products of meiosis are still expected in equal proportions. An expected 50 percent of the testcross progeny result from gametes with the same combination of alleles present in the parents of the double heterozygote (**parental combinations**), and 50 percent result from gametes with new combinations of the alleles (**recombinants**). For example, if the double heterozygote came from the mating $AA BB \times aa bb$, then the AB and ab gametes would be parental and the Ab and aB gametes recombinant. On the other hand, if the double heterozygote came from the mating $AA bb \times aa BB$, then the Ab and aB gametes would be parental and the ab and AB gametes recombinant. In either case, with independent assortment, the genotypes of the testcross progeny are expected in the ratio of 1 : 1 : 1 : 1. In this chapter, we will examine phenomena that cause deviations from this expected ratio.

In his early experiments with *Drosophila*, Morgan found mutations in each of several X-linked genes that provided ideal materials for studying the inheritance of genes in the same chromosome. One of these genes, with alleles w^+ and w , determined normal red eye color versus white eyes, as discussed in Chapter 3; another such gene, with the alleles m^+ and m , determined whether the size of the wings was normal

or miniature. The initial cross was between females with white eyes and normal wings and males with red eyes and miniature wings. We will use the slash in this instance to help us follow these X-linked traits:

$$w m^+ / w m^+ \times w^+ m / Y$$

The resulting F_1 progeny consisted of wildtype females and white-eyed, nonminiature males. When these were crossed,

$$w m^+ / w^+ m \times w m^+ / Y$$

the female progeny consisted of a 1 : 1 ratio of red : white eyes (all were nonminiature), and the male progeny were as follows:

white eye, normal wing ($w m^+ / Y$)	226	} 66.5 percent are parental phenotypes
red eye, miniature wing ($w^+ m / Y$)	202	
red eye, normal wing ($w^+ m^+ / Y$)	114	} 33.5 percent are recombinant phenotypes
white eye, miniature wing ($w m / Y$)	102	
	<u>644</u>	

Because each male receives his X chromosome from his mother, the phenotype reveals the genotype of the X chromosome that he inherited. The results of the experiment show a great departure from the 1 : 1 : 1 : 1 ratio of the four male phenotypes expected with independent assortment. If genes in the same chromosome tended to remain together in inheritance but were not completely linked, this pattern of deviation might be observed. In this case, the combinations of phenotypic traits in the parents of the original cross (parental phenotypes) were present in 428/644 (66.5 percent) of the F_2 males, and nonparental combinations (recombinant phenotypes) of the traits were present in 216/644 (33.5 percent). The 33.5 percent recombinant X chromosomes is called the **frequency of recombination**, and it should be contrasted with the 50 percent recombination expected with independent assortment.

The recombinant X chromosomes $w^+ m^+$ and $w m$ result from crossing-over in meiosis in F_1 females. In this example, the frequency of recombination between the linked w and m genes was 33.5 percent, but

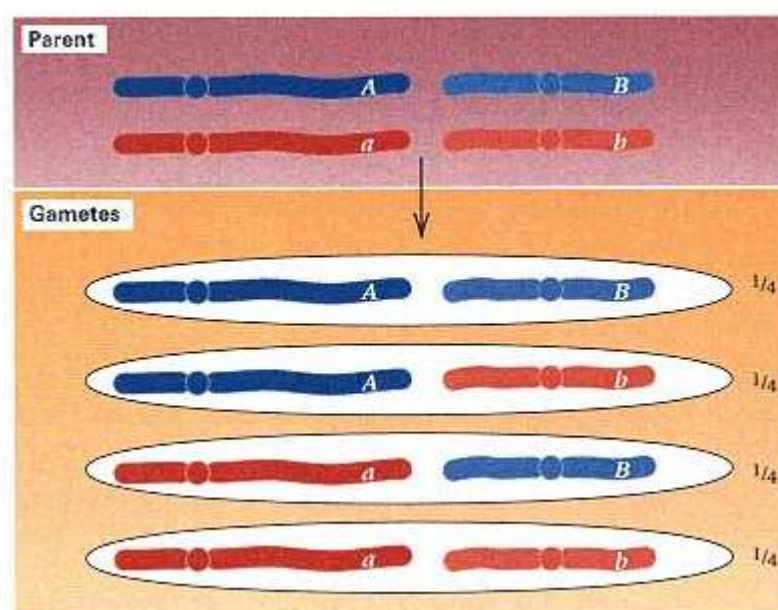


Figure 4.1

Alleles of genes in different chromosomes undergo independent assortment. The pairs of homologous chromosomes segregate at random with respect to one another, so an A -bearing chromosome is as likely to go to the same anaphase pole with a B -bearing chromosome as with a b -bearing chromosome. The result is that each possible combination

of chromatids is equally likely among the gametes: 1/4 each for AB , Ab , aB , and ab .

with other pairs of linked genes it ranges from near 0 to 50 percent. Even genes in the same chromosome can undergo independent assortment (frequency of recombination equal to 50 percent) if they are sufficiently far apart. This implies the following principle:

Genes with recombination frequencies smaller than 50 percent are present in the same chromosome (linked). Two genes that undergo independent assortment, indicated by a recombination frequency equal to 50 percent, either are in nonhomologous chromosomes or are located far apart in a single chromosome.

Geneticists use a notation for linked genes that has the general form $w^+ m/w m^+$. This notation is a simplification of a more descriptive but cumbersome form:

$$\begin{array}{cc} w^+ & m \\ \hline w & m^+ \end{array}$$

In this form of notation, the horizontal line separates the two homologous chromosomes in which the alleles of the genes are

located. The linked genes in a chromosome are always written in the same order for consistency. In the system of gene notation used for *Drosophila*, and in a similar system used for other organisms, this convention makes it possible to indicate the wildtype allele of a gene with a plus sign in the appropriate position. For example, the genotype $w m^+/w^+ m$ can be written without ambiguity as $w^+ / w m$.

"A genotype that is heterozygous for each of two linked genes can have the alleles in either of two possible configurations, as shown in Figure 4.2 for the w and m genes. In one configuration, called the **trans**, or **repulsion**, configuration, the mutant alleles are in opposite chromosomes, and the genotype is written as $w^+ / w m$ (Figure 4.2A). In the alternative configuration, called the **cis**, or **coupling**, configuration, the mutant alleles are present in the same chromosomes, and the genotype is written as $w m / +$.

Morgan's study of linkage between the *white* and *miniature* alleles began with the *trans* configuration. He also studied progeny from the *cis* configuration of the w and m alleles, which results from the cross of white, miniature females with red, non-miniature males:

$$\frac{w m}{w m} \times \frac{+ +}{Y}$$

In this case, the F_1 females were phenotypically wildtype double heterozygotes, and the males had white eyes and miniature wings. When these F_1 progeny were crossed,

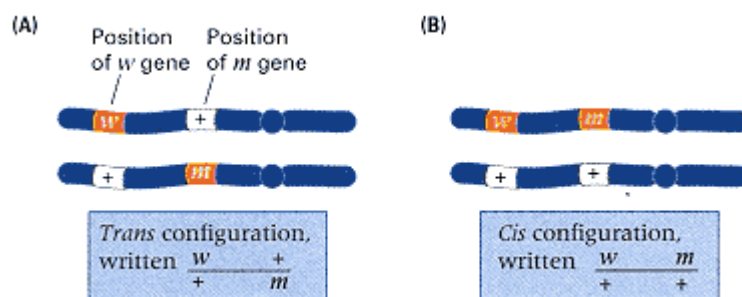


Figure 4.2

There are two possible configurations of the mutant alleles in a genotype that is heterozygous for both mutations. (A) The *trans*, or repulsion, configuration has the mutant alleles on opposite chromosomes. (B) The *cis*, or coupling, configuration has the mutant alleles on the same chromosome.

$$\frac{w m}{+ +} \times \frac{w m}{Y}$$

they produced the following progeny:

red eye, normal wing ($+ + / w m$ ♀♀ and $+ + / Y$ ♂♂)	395	62.3 percent are parental phenotypes
white eye, miniature wing ($w m / w m$ ♀♀ and $w m / Y$ ♂♂)	382	
white eye, normal wing ($w + / w m$ ♀♀ and $w + / Y$ ♂♂)	223	37.7 percent are recombinant phenotypes
red eye, miniature wing ($+ m / w m$ ♀♀ and $+ m / Y$ ♂♂)	247	
	1247	

Compared to the preceding experiment with w and m , the frequency of recombination between the genes is approximately the same: 37.7 percent versus 33.5 percent. The difference is within the range expected from random variation from experiment to experiment. However, in this case, the phenotypes constituting the parental and recombinant classes of offspring are reversed. They are reversed because the original parents of the F_1 female were different. In the first cross, the F_1 female was the *trans* double heterozygote ($w^+ / w m$); in the second cross,

the F_1 female had the *cis* configuration ($w^+ m^+ / +$). The repeated finding of equal recombination frequencies in experiments of this kind leads to the following conclusion:

Recombination between linked genes takes place with the same frequency whether the alleles of the genes are in the *trans* configuration or in the *cis* configuration; it is the same no matter how the alleles are arranged.

The recessive allele y of another X-linked gene in *Drosophila* results in yellow body color instead of the usual gray color determined by the y^+ allele. When white-eyed females were mated with males having yellow bodies, and the wildtype F_1 females were testcrossed with yellow-bodied, white-eyed males,

$$\begin{array}{l} + w^+ / + w \times y^+ / Y \\ \downarrow \\ + w / y^+ \times y w / Y \end{array}$$

the progeny were

wildtype body, white eye (maternal gamete, + w)	4292	} 98.6 percent are parental phenotypes
yellow body, red eye (maternal gamete, y +)	4605	
wildtype body, red eye (maternal gamete, + +)	86	} 1.4 per- cent are recombinant phenotypes
yellow body, white eye (maternal gamete, y w)	44	
	9027	

In a second experiment, yellow-bodied, white-eyed females were crossed with wildtype males, and the F_1 wildtype females and F_1 yellow-bodied, white-eyed males were intercrossed:

$$\begin{array}{c} yw/yw \times ++/Y \\ \downarrow \\ ++/yw \times yw/Y \end{array}$$

In this case, 98.6 percent of the F_2 progeny had parental phenotypes and 1.3 percent had recombinant phenotypes. The parental and recombinant phenotypes were reversed in the reciprocal crosses, but the recombination frequency was virtually the same. Females with the *trans* genotype $y +/+ w$ produced about 1.4 percent recombinant progeny, carrying either of the recombinant chromosomes $y w$ or $+ +$; similarly, females with the *cis* genotype $y w/+ +$ produced about 1.4 percent recombinant progeny, carrying either of the recombinant chromosomes $y +$ or $+ w$. However, the recombination frequency was much lower between the genes for yellow body and white eyes than between the genes for white eyes and miniature wings (1.4 percent versus about 35 percent). These and other experiments have led to the following conclusions:

- The recombination frequency is a characteristic of a particular pair of genes.
- Recombination frequencies are the same in *cis* (coupling) and *trans* (repulsion) heterozygotes.

In experiments with other genes, Morgan also discovered that *Drosophila* is unusual in that recombination does not take place in males. Although it is not known how (or why) crossing-over is prevented in males, the result of the absence of recombination in *Drosophila* males is that all alleles located in a particular chromosome show complete linkage in the male. For example, the genes *cn* (cinnabar eyes) and *bw* (brown eyes) are both in chromosome 2 but are so far apart that, in females, there is 50 percent recombination. Thus the cross

$$\frac{cn\ bw}{++} \text{♀} \times \frac{cn\ bw}{cn\ bw} \text{♂}$$

yields progeny of genotype $++/cn\ bw$ and $cn\ bw/cn\ bw$ (the nonrecombinant types) as well as $cn +/cn\ bw$ and $bw/cn\ bw$ (the recombinant types) in the proportions 1 : 1 : 1 : 1. However, because there is no crossing-over in males, the reciprocal cross

$$\frac{cn\ bw}{cn\ bw} \text{♀} \times \frac{cn\ bw}{++} \text{♂}$$

yields progeny only of the nonrecombinant genotypes $++/cn\ bw$ and $cn\ bw/cn\ bw$ in equal proportions. The absence of recombination in *Drosophila* males is a convenience often made use of in experimental design; as shown in the case of *cn* and *bw*, all the alleles present in any chromosome in a male must be transmitted as a group, without being recombined with alleles present in the homologous chromosome. The absence of crossing-over in *Drosophila* males is atypical; in most other animals and plants, recombination takes place in both sexes.

4.2— Genetic Mapping

The linkage of the genes in a chromosome can be represented in the form of a **genetic map**, which shows the linear order of the genes along the chromosome with the distances between adjacent genes proportional to the frequency of recombination between them. A genetic map is also called a **linkage map** or a **chromosome map**. The concept

of genetic mapping was first developed by Morgan's student, Alfred H. Sturtevant, in 1913. The early geneticists understood that recombination between genes takes place by an exchange of

segments between homologous chromosomes in the process now called crossing-over. Each crossing-over is manifested physically as a chiasma, or cross-shaped configuration, between homologous chromosomes; chiasmata are observed in prophase I of meiosis (Chapter 3). Each chiasma results from the breaking and rejoining of chromatids during synapsis, with the result that there is an exchange of corresponding segments between them. The theory of crossing-over is that each chiasma results in a new association of genetic markers. This process is illustrated in Figure 4.3. When there is no crossing-over (Figure 4.3A), the alleles present in each homologous chromosome remain in the same combination. When crossing-over does take place (Figure 4.3B), the outermost alleles in two of the chromatids are interchanged (recombined).

The unit of distance in a genetic map is called a **map unit**; 1 map unit is equal to 1 percent recombination. For example, two genes that recombine with a frequency of 3.5 percent are said to be located 3.5 map units apart. One map unit is also called a **centimorgan**, abbreviated cM, in honor of T. H. Morgan. A distance of 3.5 map units therefore equals 3.5 centimorgans and indicates 3.5 percent recombination between the genes. For ease of reference, we list the four completely equivalent ways in which a genetic distance between two genes may be represented.

- As a *frequency of recombination* (in the foregoing example, 0.035)
- As a *percent recombination* (here 3.5 percent)
- As a *map distance* in map units (in this case, 3.5 map units)
- As a map distance in *centimorgans* (here 3.5 centimorgans, abbreviated 3.5 cM)

Physically, 1 map unit corresponds to a length of the chromosome in which, on the

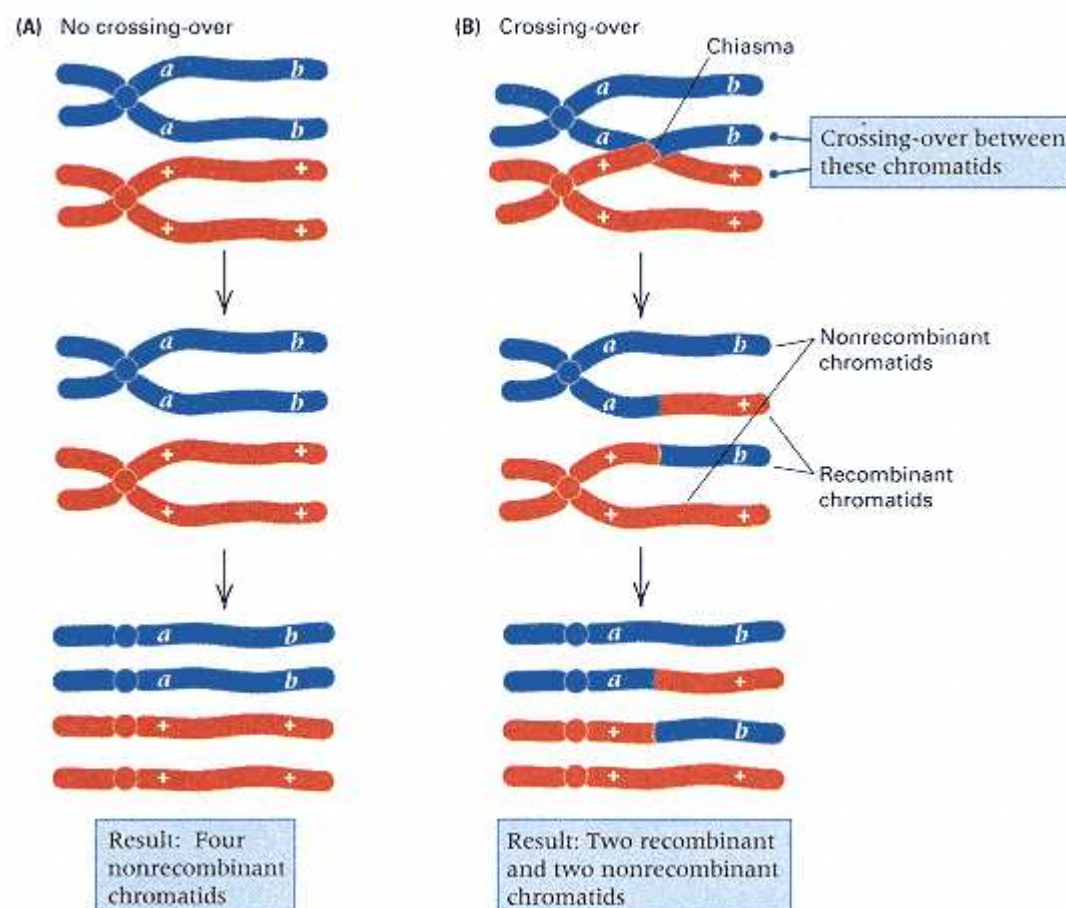


Figure 4.3

Diagram illustrating crossing-over between two genes. (A) When there is no crossing-over between two genes, the alleles are not recombined. (B) When there is crossing-over between them, the result of the crossover is two recombinant and two nonrecombinant products, because the exchange is between only two of the four chromatids.

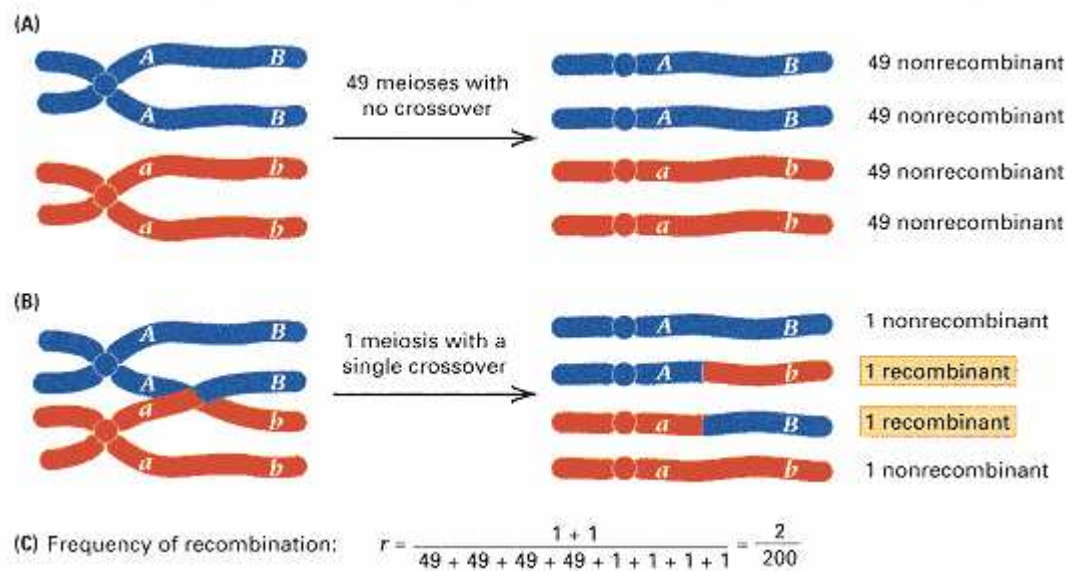


Figure 4.4

Diagram of chromosomal configurations in 50 meiotic cells, in which one has a crossover between two genes. (A) The 49 cells without a crossover result in 98 AB and 98 ab chromosomes; these are all nonrecombinant. (B) The cell with a crossover yields chromosomes that are AB , Ab , aB , and ab , of which the middle two types are recombinant chromosomes. (C) The recombination frequency equals 2/200, or 1 percent, which is also called 1 map unit or 1 cM. Hence 1 percent recombination means that 1 meiotic cell in 50 has a crossover in the region between the genes.

average, one crossover is formed in every 50 cells undergoing meiosis. This principle is illustrated in Figure 4.4. If one meiotic cell in 50 has a crossing-over, the frequency of *crossing-over* equals 1/50, or 2 percent. Yet the frequency of *recombination* between the genes is 1 percent. The correspondence of 1 percent recombination with 2 percent crossing-over is a little confusing until you consider that a crossover results in two recombinant chromatids and two nonrecombinant chromatids (Figure 4.4). A frequency of crossing-over of 2 percent means that of the 200 chromosomes that result from meiosis in 50 cells, exactly 2 chromosomes (the two involved in the exchange) are recombinant for genetic markers spanning the particular chromosome segment. To put the matter in another way, 2 percent crossing-over corresponds to 1 percent recombination because only half the chromatids in each cell with an exchange are actually recombinant.

In situations in which there are **genetic markers** along the chromosome, such as the A , a and B , b pairs of alleles in Figure 4.4, recombination between the marker genes takes place only when crossing-over occurs *between* the genes. Figure 4.5 illustrates a case in which crossing-over takes place between the gene A and the centromere, rather than between the genes A and B . The crossing-over does result in the physical exchange of segments between the innermost chromatids. However, because it is located outside the region between A and B , all of the resulting gametes must carry either the AB or ab allele combinations. These are nonrecombinant chromosomes. The presence of the crossing-over is undetected because it is not in the region between the genetic markers.

In some cases, the region between genetic markers is large enough that two (or even more) crossovers can be formed in a single meiotic cell. One possible configuration for two crossovers is shown in Figure 4.6. In this example, both crossovers are between the same pair of chromatids. The result is that there is a physical exchange of a segment of chromosome between the marker genes, but the double crossover remains undetected because the markers themselves are not recombined. The absence of recombination results from the fact that the second crossover reverses the

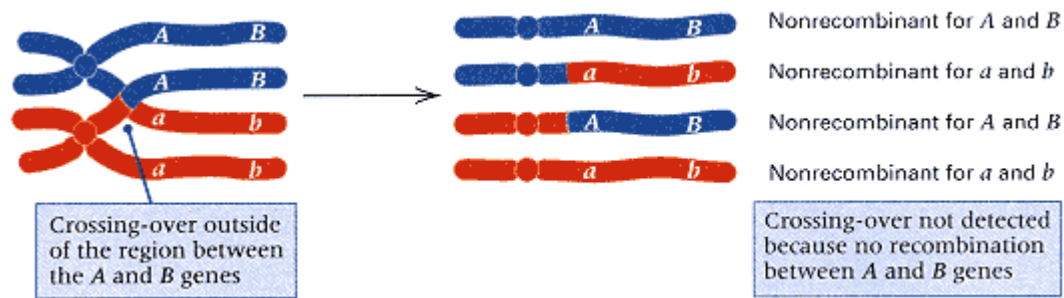


Figure 4.5

Crossing-over outside the region between two genes is not detectable through recombination. Although a segment of chromosome is exchanged, the genetic markers stay in the nonrecombinant configurations, in this case *A B* and *a b*.

effect of the first, insofar as recombination between *A* and *B* is concerned. The resulting chromosomes are either *A B* or *a b*,] both of which are nonrecombinant.

Given that double crossing-over in a region between two genes can remain undetected because it does not result in recombinant chromosomes, there is an important distinction between the distance between two genes as measured by the recombination frequency and as measured in map units. Map units measure how much crossing-over takes place between the genes. For any two genes, the map distance between them equals one-half times the average number of crossover events that take place in the region per meiotic cell. The recombination frequency, on the other hand, reflects how much recombination is actually *observed* in a particular experiment. Double crossovers that do not yield recombinant gametes, such as the one in Figure 4.6, *do* contribute to the map distance but *do not* contribute to the recombination frequency. The distinction is important only when the region in question is large enough that double crossingover can occur. If the region between the genes is so short that no more than one crossover can be formed in the region in any one meiosis, then map units and recombination frequencies are the same (because there are no multiple crossovers that can undo each other). This is the basis for defining a map unit as being equal to 1 percent recombination. Over an interval so short as to yield 1 percent observed recombination, multiple crossovers are usually precluded, so the map distance equals the recombination frequency in this case.

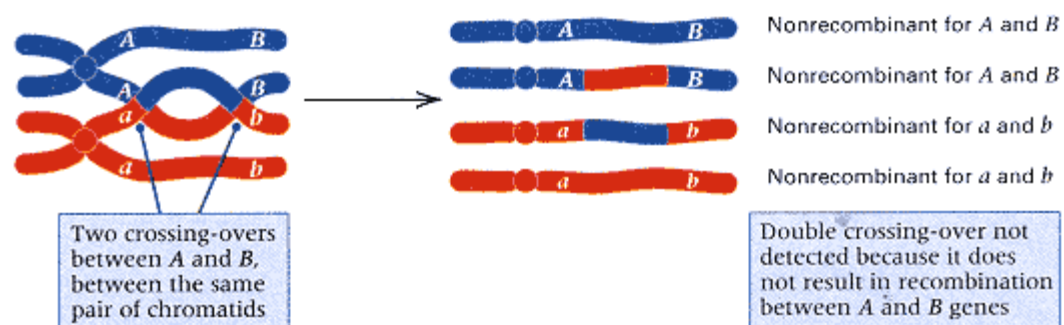


Figure 4.6

If two crossovers take place between marker genes, and both involve the same pair of chromatids, then neither crossover is detected because all of the resulting chromosomes are nonrecombinant *A B* or *a b*.

Connection Genes All in a Row

Alfred H. Sturtevant 1913

Columbia University,
New York, New York

The Linear Arrangement of Six Sex-Linked Factors in Drosophila, As Shown by Their Mode of Association

Genetic mapping remains the cornerstone of genetic analysis. It is the principal technique used in modern human genetics to identify the chromosomal location of mutant genes associated with inherited diseases, as we saw with Huntington disease in Chapter 2. The genetic markers used in human genetics are homologous DNA fragments that differ in length from one person to the next, but the basic principles of genetic mapping are the same as those originally enunciated by Sturtevant. In this excerpt, we have substituted the symbols presently in use for the genes, y (yellow body), w (white eyes), v (vermillion eyes), m (miniature wings), and r (rudimentary wings). (The sixth gene mentioned is another mutant allele of white, now called white-eosin.) In this paper, Sturtevant uses the term crossing-over instead of recombination and crossovers instead of recombinant chromosomes. We have retained his original terms but, in a few cases, have put the modern equivalent in brackets.

Morgan, by crossing white eyed, long winged flies to those with red eyes and rudimentary wings (the new sex-linked character), obtained, in F_2 , white eyed rudimentary winged flies. This could happen only if "crossing-over" [recombination] is possible; which means, on the assumption that both of these factors are in the X chromosome, that an interchange of materials between homologous chromosomes occurs (in the female only, since the male has only one X chromosome). A point not noticed at this time came out later in connection with other sex-linked factors in *Drosophila*. It became evident that some of the sex-linked factors are associated, i. e., that crossing-over does not occur freely between some factors, as shown by the fact that the combinations present in the F_1 flies are much more frequent in the F_2 than

These results form a new argument in favor of the chromosome view of inheritance, since they strongly indicate that the factors investigated are arranged in a linear series.

are new combinations of the same characters. This means, on the chromosome view, that the chromosomes, or at least certain segments of them, are much more likely to remain intact during meiosis than they are to interchange materials. . . . It would seem, if this hypothesis be correct, that the proportion of "crossovers" [recombinant chromosomes] could be used as an index of the distance between any two factors. Then by determining the distances (in the above sense) between A and B and between B and C, one should be able to predict AC. . . . Just how far our theory stands the test is shown by the data below, giving observed percent of crossovers [recombinant chromosomes] and the distances calculated [from the summation of shorter intervals].

	Calculated	Observed
Factors	distance	percentage of crossovers
y-v	30.7	32.2
y-m	33.7	35.2
y-r	57.6	37.6
w-m	32.7	33.7
w-r	56.6	45.2

It will be noticed at once that the longer distances, y—r and w—r, give smaller per cent of crossovers, than the calculation calls for. This is a point which was to be expected and is probably due to the occurrence of two breaks in the same chromosome, or "double crossing-over." But in the case of the shorter distances the correspondence with expectation is perhaps as close as was to be expected with the small numbers that are available. . . . It has been found possible to arrange six sex-linked factors in *Drosophila* in a linear series, using the number of

crossovers per 100 cases [the frequency of recombination] as an index of the distance between any two factors. A source of error in predicting the strength of association between untried factors is found in double crossing-over. The occurrence of this phenomenon is demonstrated. . . . These results form a new argument in favor of the chromosome view of inheritance, since they strongly indicate that the factors investigated are arranged in a linear series.

Source: *Journal of Experimental Zoology* 14: 43–59

When adjacent chromosome regions separating linked genes are sufficiently short that multiple crossovers are not formed, the recombination frequencies (and hence the map distances) between the genes are additive. This important feature of recombination, and also the logic used in genetic mapping, is illustrated by the example in Figure 4.7. The genes are all in the X chromosome of *Drosophila*: *y* (yellow body), *rb* (ruby eye color), and *cv* (shortened wing crossvein). The recombination frequency between genes *y* and *rb* is 7.5 percent, and that between *rb* and *cv* is 6.2 percent. The genetic map might be any one of three possibilities, depending on which gene is in the middle (*y*, *cv*, or *rb*). Map A, which has *y* in the middle, can be excluded

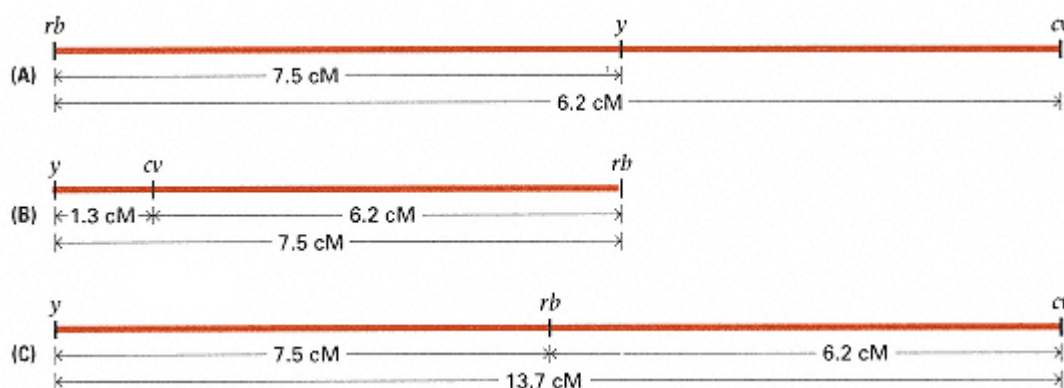


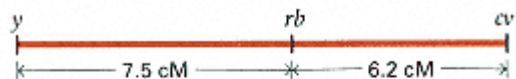
Figure 4.7

In *Drosophila*, the genes *y* (yellow body) and *rb* (ruby eyes) have a recombination frequency of 7.5 percent, and *rb* and *cv* (shortened wing crossvein) have a recombination frequency of 6.2 percent. There are three possible genetic maps, depending on whether *y* is in the middle (A), *cv* is in the middle (B), or *rb* is in the middle (C). Map A can be excluded because it implies that *rb* and *y* are closer than *rb* and *cv*, whereas the observed recombination frequency between *rb* and *y* is larger than that between *rb* and *cv*. Maps B and C are compatible with the data given.

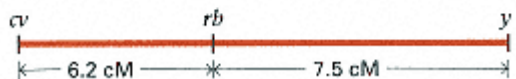
because it implies that the recombination frequency between *rb* and *cv* should be greater than that between *rb* and *y*, and this contradicts the observed data.

Maps B and C are both consistent with the recombination frequencies. They differ in their predictions regarding the recombination frequency between *y* and *cv*. In map B the predicted distance is 1.3 map units, whereas in map C the predicted distance is 13.3 map units. In reality, the observed recombination frequency between *y* and *cv* is 13.3 percent. Map C is therefore correct.

There are actually two genetic maps corresponding to map C. They differ only in whether *y* is placed at the left or the right. One map is



The other map is



These two ways of depicting the genetic map are completely equivalent.

A genetic map can be expanded by this type of reasoning to include all the known genes in a chromosome; these genes constitute a **linkage group**. The number of linkage groups is the same as the haploid number of chromosomes of the species. For example, cultivated corn (*Zea mays*) has ten pairs of chromosomes and ten linkage groups. A partial genetic map of chromosome 10 is shown in Figure 4.8, along with the dramatic phenotypes shown by some of the mutants. The ears of corn in the two photographs (Figure 4.8C and 4.8F) demonstrate the result of Mendelian segregation. The photograph in Figure 4.8C shows a 3 : 1 segregation of yellow : orange kernels produced by the recessive *orange pericarp-2* (*orp-2*) allele in a cross between two heterozygous genotypes. The ear in the photograph in Figure 4.8F shows a 1 : 1 segregation of marbled : white kernels produced by the dominant allele *R1-mb* in a cross between a heterozygous genotype and a homozygous normal.

Crossing-over

The orderly arrangement of genes represented by a genetic map is consistent with the conclusion that each gene occupies a well-defined site, or **locus**, in the chromosome, with the alleles of a gene in a heterozygote occupying corresponding locations in the pair of homologous chromosomes. Crossing-over, which is brought about by a physical exchange of segments

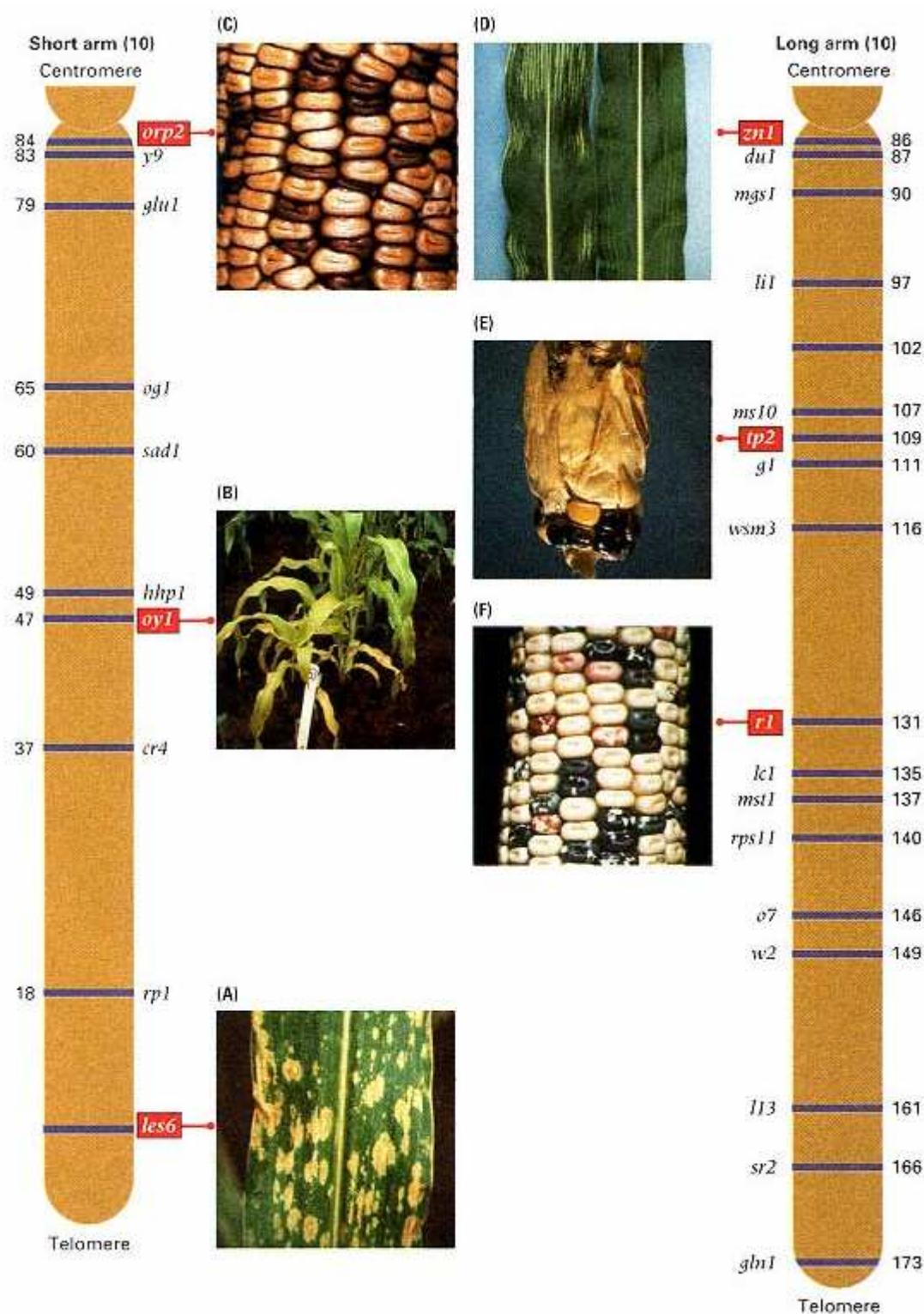


Figure 4.8

Genetic map of chromosome 10 of corn, *Zea mays*. The map distance to each gene is given in standard map units (centimorgans) relative to a position 0 for the telomere of the short arm (lower left). Mutations in the gene *lesion-6* (*les6*), result in many small to medium-sized, irregularly spaced, discolored spots on the leaf blade and sheath; (A) shows the phenotype of a heterozygote for *Les6*, a dominant allele.

Mutations in the gene *oil yellow-1* (*oy1*) result in a yellow-green plant. In (B), the plant in front is heterozygous for the dominant allele *Oy1*; behind is a normal plant. The *orp2* allele is a recessive expressed as orange pericarp, a maternal tissue that surrounds the kernels; (C) shows the segregation of *orp2* in a cross between two heterozygous genotypes, yielding a 3 : 1 ratio of yellow : orange seeds. The gene *zn1* is *zebra necrotic-1*, in which dying tissue appears in transverse leaf bands; in (D), the left leaf is homozygous *zn1*, the right leaf wildtype. Mutations in the gene *teopod-2* (*tp2*) result in many small, partially podded ears and a simple tassel; one of the ears in a plant heterozygous for the dominant allele *Tp2* is shown (E). The mutation *R1-mb* is an allele of the *r1* gene resulting in red or purple color in the aleurone layer of the seed; (F) shows the marbled color in kernels of an ear segregating for *R1-mb*.

[Photographs courtesy of M. G. Neuffer; genetic map courtesy of E. H. Coe.]

that results in a new association of genes in the same chromosome, has the following features:

1. The exchange of segments between parental chromatids takes place in the first meiotic prophase, *after the chromosomes have duplicated*. The four chromatids (strands) of a pair of homologous chromosomes are closely synapsed at this stage. Crossing-over is a physical exchange between chromatids in a pair of homologous chromosomes.
2. The exchange process consists of the breaking and rejoining of the two chromatids, resulting in the *reciprocal* exchange of equal and corresponding segments between them (see Figure 4.3).
3. The sites of crossing-over are more or less random along the length of a chromosome pair. Hence the probability of crossing-over between two genes increases as the physical distance between the genes along the chromosome becomes larger. This principle is the basis of genetic mapping.

The demonstration that crossing-over (as detected by the recombination of two heterozygous markers) is associated with a physical exchange of segments between homologous chromosomes was made possible by the discovery of two structurally altered chromosomes that permitted the microscopic recognition of parental and recombinant chromosomes. In 1931, Curt Stern discovered two X chromosomes of *Drosophila* that had undergone structural changes that made them distinguishable from each other and from a normal X chromosome. He used these structurally altered X chromosomes in an experiment that provided one of the classical proofs of the physical basis of crossing-over (Figure 4.9). One of the altered X chromosomes was missing a segment that had become attached to chromosome 4. This altered X chromosome could be identified by its missing terminal segment. The second aberrant X chromosome had a small piece of a Y chromosome attached as a second arm. The mutant alleles *car* (abbreviated *c*, a recessive allele resulting in carnation eye color instead of wildtype red) and *B* (a dominant allele resulting in bar-shaped eyes instead of round) were present in the first altered X chromosome, and the wild-type alleles of these genes were in the second altered X. Females with the two structurally and genetically marked X chromosomes were mated with males having a normal X that carried the recessive alleles of the genes (Figure 4.9). In the progeny from this cross, flies with parental or recombinant combinations of the phenotypic traits were recognized by their eye color and shape, and their chromosomal makeup could be determined by microscopic examination of the offspring they produced in a testcross. In the genetically recombinant progeny from the cross, the X chromosome had the morphology that would be expected if recombination of the genes were accompanied by an exchange that recombined the chromosome markers; that is, the progeny with wildtype (red), bar-shaped eyes had an X chromosome with a missing terminal segment and the attached Y arm; similarly, progeny with carnation-colored, round eyes had a structurally normal X chromosome with no missing terminus and no Y arm. As expected, the nonrecombinant progeny were found to have an X chromosome morphologically identical with one in their mothers.

Crossing-over Takes Place at the Four-Strand Stage of Meiosis

So far we have asserted, without citing experimental evidence, that crossing-over takes place in meiosis after the chromosomes have duplicated, at the stage when each bivalent has four chromatid strands. One experimental proof that crossing-over takes place after the chromosomes have duplicated came from a study of laboratory stocks of *D. melanogaster* in which the two X chromosomes in a female are joined to a common centromere to form an aberrant chromosomes called an **attached-X**, or **compound-X**, chromosome. The normal X chromosome in *Drosophila* has a centromere almost at the end of the chromosome, and the attachment of two of these chromosomes to a single centromere results in a chromosome with two equal arms, each consisting of a virtually complete X. Females with a compound-X chro-

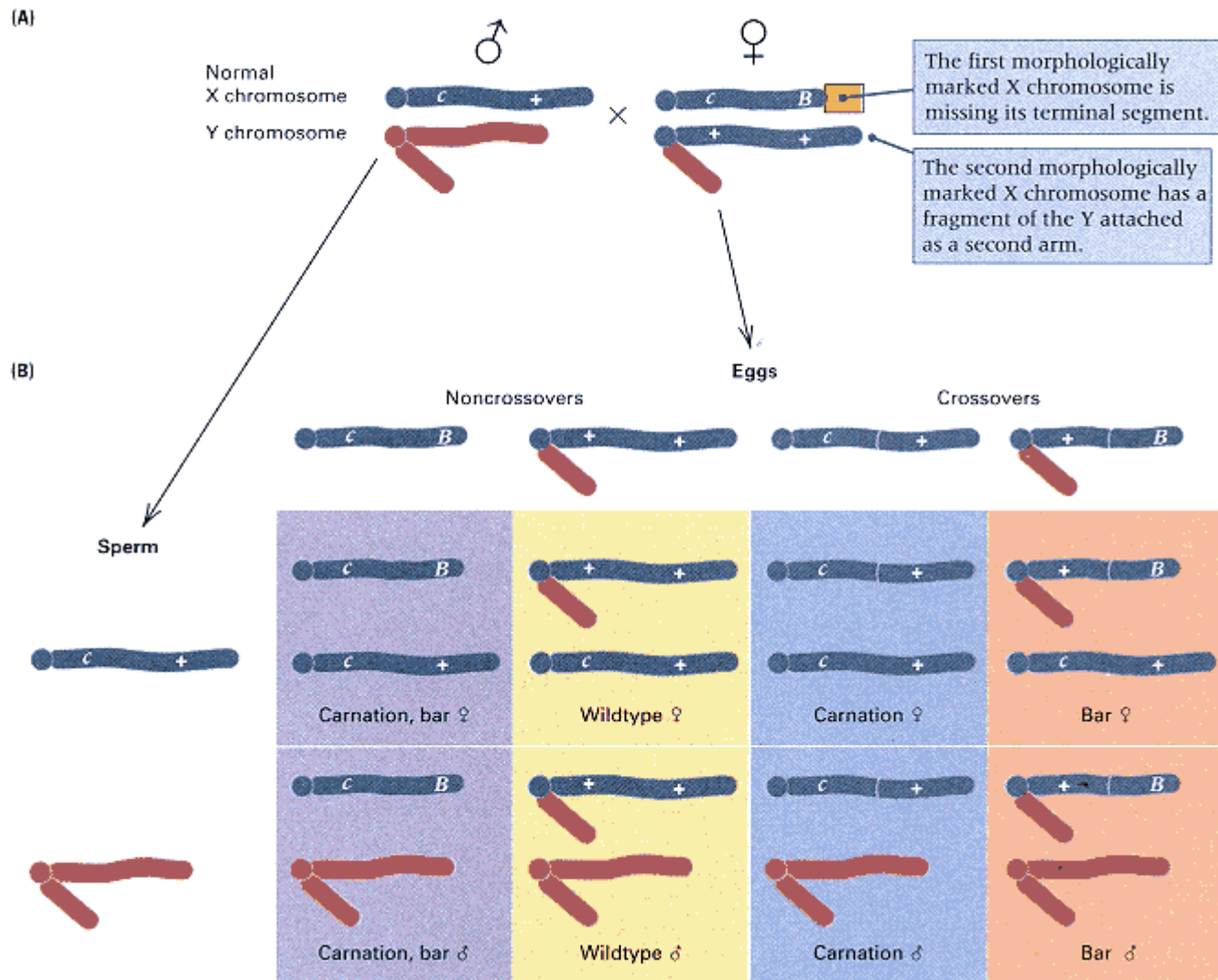


Figure 4.9

(A) Diagram of a cross in which the two X chromosomes in a *Drosophila* female are morphologically distinguishable from each other and from a normal X chromosome. One X chromosome has a missing terminal segment, and the other has a second arm consisting of a fragment of the Y chromosome. (B) Result of the cross. The carnation offspring contain a structurally normal X chromosome, and the bar offspring contain an X chromosome with both morphological markers. The result demonstrates that genetic recombination between marker genes is associated with physical exchange between homologous chromosomes. Segregation of the missing terminal segment of the X chromosome, which is attached to chromosome 4, is not shown.

mosome usually contain a Y chromosome as well, and they produce two classes of viable offspring: females who have the maternal compound-X chromosome along with a paternal Y chromosome, and males with the maternal Y chromosome along with a paternal X chromosome (Figure 4.10). Attached-X chromosomes are frequently used to study X-linked genes in *Drosophila* because a male carrying any X-linked mutation, when crossed with an attached-X female, produces sons who also carry the mutation and daughters who carry the attached-X chromosome. In matings with attached-X females, therefore, the inheritance of an X-linked gene in the male passes from father to son to grandson, and so forth, which is the opposite of usual X-linked inheritance.

In an attached-X chromosome in which one X carries a recessive allele and the other carries the wildtype nonmutant allele,

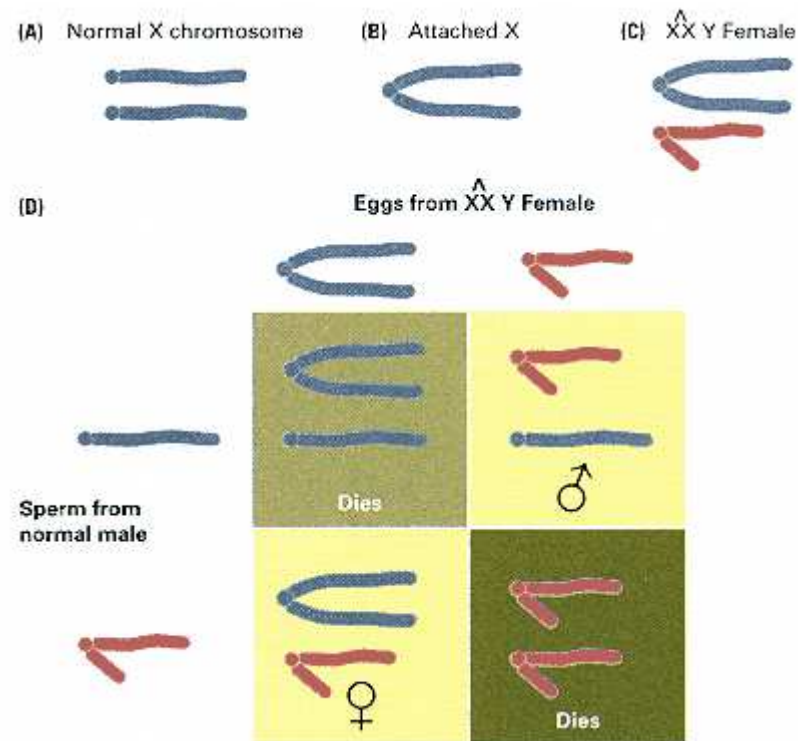


Figure 4.10

Attached-X (compound-X) chromosomes in *Drosophila*. (A) A structurally normal X chromosome in a female. (B) An attached-X chromosome, with the long arms of two normal X chromosomes attached to a common centromere. (C) Typical attached-X females also contain a Y chromosome. (D) Outcome of a cross between an attached-X female and a normal male. The eggs contain either the attached-X or the Y chromosome, which combine at random with X-bearing or Y-bearing sperm. Genotypes with either three X chromosomes or no X chromosomes are lethal. Note that a male fly receives its X chromosome from its father and its Y chromosome from its mother—the opposite of the usual situation in *Drosophila*.

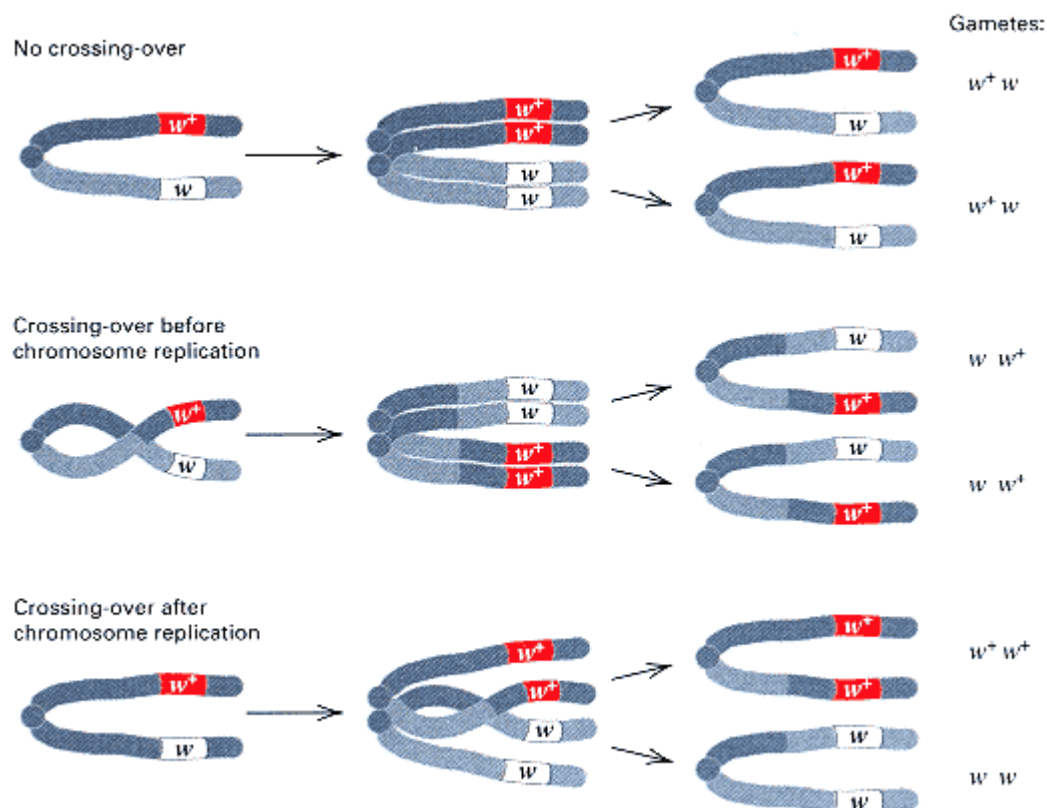


Figure 4.11

Diagram showing that crossing-over must take place at the four-strand stage in meiosis to produce a homozygous attached-X chromosome from one that is heterozygous for an allele. To yield homozygosity, the exchange must take place between the centromere and the gene.

Connection Dos XX

Lilian V. Morgan 1922
Columbia University, New York, New York <i>Non-crisscross Inheritance in Drosophila melanogaster</i>
<i>Lilian V. Morgan was a first-rate geneticist long associated with T. H. Morgan as his collaborator and wife. She discovered the first attached-X chromosome as a single exceptional female in a routine mapping cross. She realized instinctively that this female was extremely important. There is an old story, of uncertain validity, that the female temporarily escaped, causing consternation and a mad search by everyone in the laboratory, until finally it was found resting on a window pane. The attached-X chromosome is still one of the most important genetic tools available to Drosophila geneticists.</i>
A complete reversal of the ordinary crisscross inheritance of recessive X-linked characters occurs in a line of <i>Drosophila</i> recently obtained. In ordinary X-linked inheritance, the recessive X-linked characters of the mother are transmitted to the sons, while the daughters show the dominant allele of the father. In the present case, the daughters show a recessive X-linked character of the mother and the sons show the dominant allele of the father. The reversal is explicable on the assumption that the two X-chromosomes of the mother are united and behave in meiosis as a single body. The cytological evidence verifies the genetic deduction. The eggs of these
<i>The reversal is explicable on the assumption that the two X-chromosomes of the mother are united and behave in meiosis as a single body.</i>
females do have two united X-chromosomes. . . . [A single female fly with a yellow abdomen was found in a cross between a homozygous nonyellow (gray) female and a yellow male.] She was mated to a gray male and produced 43 daughters and 59 sons. The daughters were, without exception, all yellow and the sons were all gray. The conclusion was at once evident that the [mother] had received from its father two yellow-bearing chromosomes, inseparable from one another, and that these inseparable chromosomes were transmitted together to the next generation producing (wherever they occurred) females, because there were always two of them. No male offspring could be yellow, because no single yellow chromosome was transmitted. . . . The F₁ females were fertile. . . . The daughters were all yellow, but differed from their yellow mother in having, besides the "yellow-bearing" double chromosome [the attached-X], a Y-bearing chromosome from their father. . . . The genetic behavior of the line of flies having the two inseparable X chromosomes is in entire accord with the condition of the chromosomes as seen in cytological preparations. . . . The origin of the [attached-X] can be explained if at some division in spermatogenesis of the father (perhaps at the equational division) the two halves of the X chromosome failed to become completely detached, but remained fastened together at one of their ends, producing the V-shaped chromosome found in the germ cells of the female descendants.
Source: <i>Biological Bulletin</i> 42: 267-274

crossing-over between the X-chromosome arms can yield attached-X products in which the recessive allele is present in both arms of the attached-X chromosome (Figure 4.11). Hence, attached-X females that are heterozygous can produce some female progeny that are homozygous for the recessive allele. The frequency with which homozygosity is observed increases with increasing map distance of the gene from the centromere. From the diagrams in Figure 4.11, it is clear that homozygosity can result only if the crossover between the gene and the centromere takes place after the chromosome has duplicated. The implication of finding homozygous attached-X female progeny is therefore that crossing-over takes place at the *four-strand* stage of meiosis. If this were not the case, and crossing-over happened before duplication of the chromosome (at the *two-strand* stage), it would result only in a swap of the allele between the chromosome arms and would never yield the homozygous products that are actually observed.

The Molecular Basis of Crossing-over

As we will see in Chapter 6, each chromosome in a eukaryote contains a single, long molecule of duplex DNA complexed with proteins that undergoes a process of **condensation**, forming a hierarchy of coils upon coils that becomes progressively tighter as the chromosome progresses through nuclear division and reaches a state of maximum condensation at metaphase. Crossing-over along a chromosome must therefore correspond to some

type of exchange of genetic information between DNA molecules.

The first widely accepted model of recombination between DNA molecules was proposed by Robin Holliday in 1964. Although it is overly simplistic in some of the details, the model has formed the basis of more realistic models favored today that account for most observations related to recombination. These models, and the evidence on which they are based, are discussed in detail in Chapter 13 in the context of DNA breakage and repair. It is, however, appropriate to introduce the Holliday model at this point to connect crossing-over observed in chromosomes to exchange between DNA molecules as envisaged in the Holliday model.

An outline of the Holliday model is illustrated in Figure 4.12. The DNA molecules depicted are those present in the chromatids that participate in the recombination event. The DNA duplexes in the other two chromatids, which are also present at the time of recombination, are not shown. The exchange is initiated by a single-stranded break in each molecule (Figure 4.12A), the ends of which are joined crosswise (Figure 4.12B). DNA is a dynamic molecule that "breathes" in the sense that local regions of paired bases frequently come apart and form again. Such "breathing" in the region of the exchange allows the molecules to exchange pairing partners along a region near the point of exchange (Figure 4.12C); the exchange of pairing partners is called *branch migration*. At any time, breaks at the positions of the arrows in part C, followed by crosswise rejoining, result in separate DNA molecules (Figure 4.12D) that are recombinant for the outside genetic markers—namely, *Ab* and *aB*. In part E, the second pair of breaks rejoin to resolve the interconnected Holliday structure in part C.

We need to make one additional comment relative to scale. The molecular events in Figure 4.12 are submicroscopic, and the Holliday structure can be observed only under favorable conditions through an electron microscope. Therefore, the cross-shaped exchange between the DNA strands indicated in part C is invisible through the light microscope. What, then, is a chiasma, the cross-shaped structure that connects nonsister chromatids in a bivalent? In pachytene, at the time of crossing-over, the chromatids are already condensed enough to be visible through the light microscope. In Figure 4.12, the DNA is shown in an elongated form rather than in the highly convoluted form actually present in condensed chromatin. The events in Figure 4.12 take place in a local region of DNA where the molecules are able to undergo the exchange. The events themselves are invisible. However, the resulting connection between the chromatids forms a visible chiasma between nonsister chromatids. Like a loose knot sliding along a rope, a chiasma can also slide along a chromosome, so the physical position of a chiasma may not necessarily represent the physical location of the DNA exchange that led to its formation.

Multiple Crossing-over

When two genes are located far apart along a chromosome, more than one crossover can be formed between them in a single meiosis, and this complicates the interpretation of recombination data. The probability of multiple crossovers increases with the distance between the genes. Multiple crossing-over complicates genetic mapping because map distance is based on the number of physical exchanges that are formed, and some of the multiple exchanges between two genes do not result in recombination of the genes and hence are not detected. As we saw in Figure 4.6, the effect of one crossover can be canceled by another crossover farther along the way. If two exchanges between the same two chromatids take place between the genes *A* and *B*, then their net effect will be that all chromosomes are nonrecombinant, either *AB* or *ab*. Two of the products of this meiosis have an interchange of their middle segments, but the chromosomes are not recombinant for the genetic markers, and so are genetically indistinguishable from noncrossover chromosomes. The possibility of such canceling events means that the observed recombination value is an *underestimate* of the true exchange frequency and the map distance between the genes. In higher organisms, double crossing-over is effectively precluded in chromosome segments that are sufficiently short. Therefore,

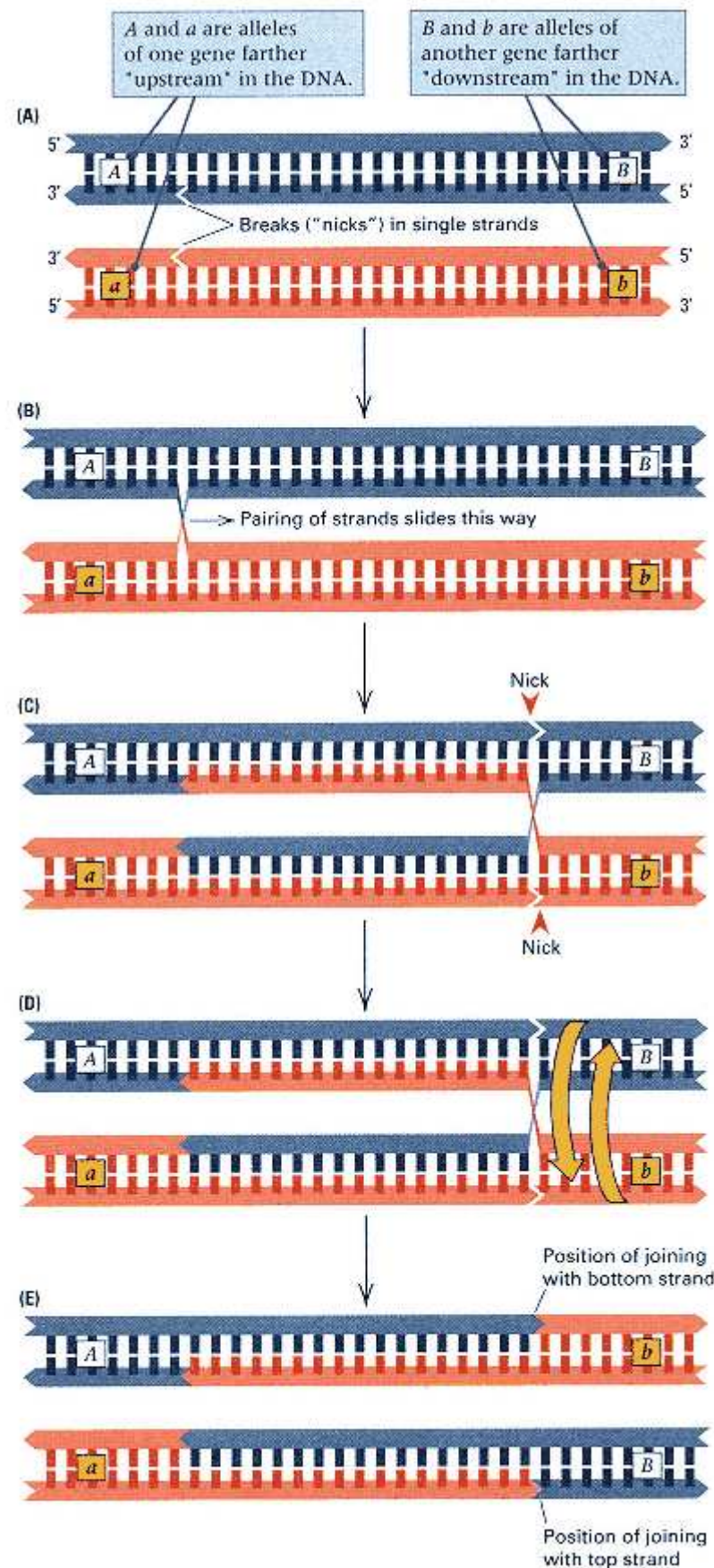


Figure 4.12

The Holliday model of recombination. (A) In the participating DNA molecules, the exchange process is initiated by a single-stranded break in one strand of each duplex. (B) The ends of the broken strands are joined crosswise, resulting in a connection between the molecules. (C) The newly joined strands "unzip" a little and exchange pairing partners (the exchange of pairing partners is called *branch migration*). The exchange can be resolved by the breaking and rejoining of the

outer strands. (D) In resolving the structure, the nicked outer strands exchange places. (E) Sealing of the gaps results in molecules that are recombinant for the outside genetic markers (*A b* and *a B*).

by using recombination data for closely linked genes to build up genetic linkage maps, we can avoid multiple crossovers that cancel each other's effects.

The *minimum* recombination frequency between two genes is 0. The recombination frequency also has a maximum:

No matter how far apart two genes may be, the maximum frequency of recombination between any two genes is 50 percent.

Fifty percent recombination is the same value that would be observed if the genes were on nonhomologous chromosomes and assorted independently. The maximum frequency of recombination is observed when the genes are so far apart in the chromosome that at least one crossover is almost always formed between them.

Figure 4.3B, shows that a single exchange in every meiosis would result in half of the products having parental combinations and the other half having recombinant combinations of the genes. Two exchanges between two genes have the same effect, as shown in Figure 4.13. Figure 4.13A shows a two-strand double crossover, in which the same chromatids participate in both exchanges; no recombination of the marker genes is detectable. When the two exchanges have one chromatid in common (three-strand double crossover, Figure 4.13B and C), the result is indistinguishable from that of a single exchange; two products with parental combinations and two with recombinant combinations are produced. Note that there are two types of three-strand doubles, depending on which three chromatids participate. The final pos-

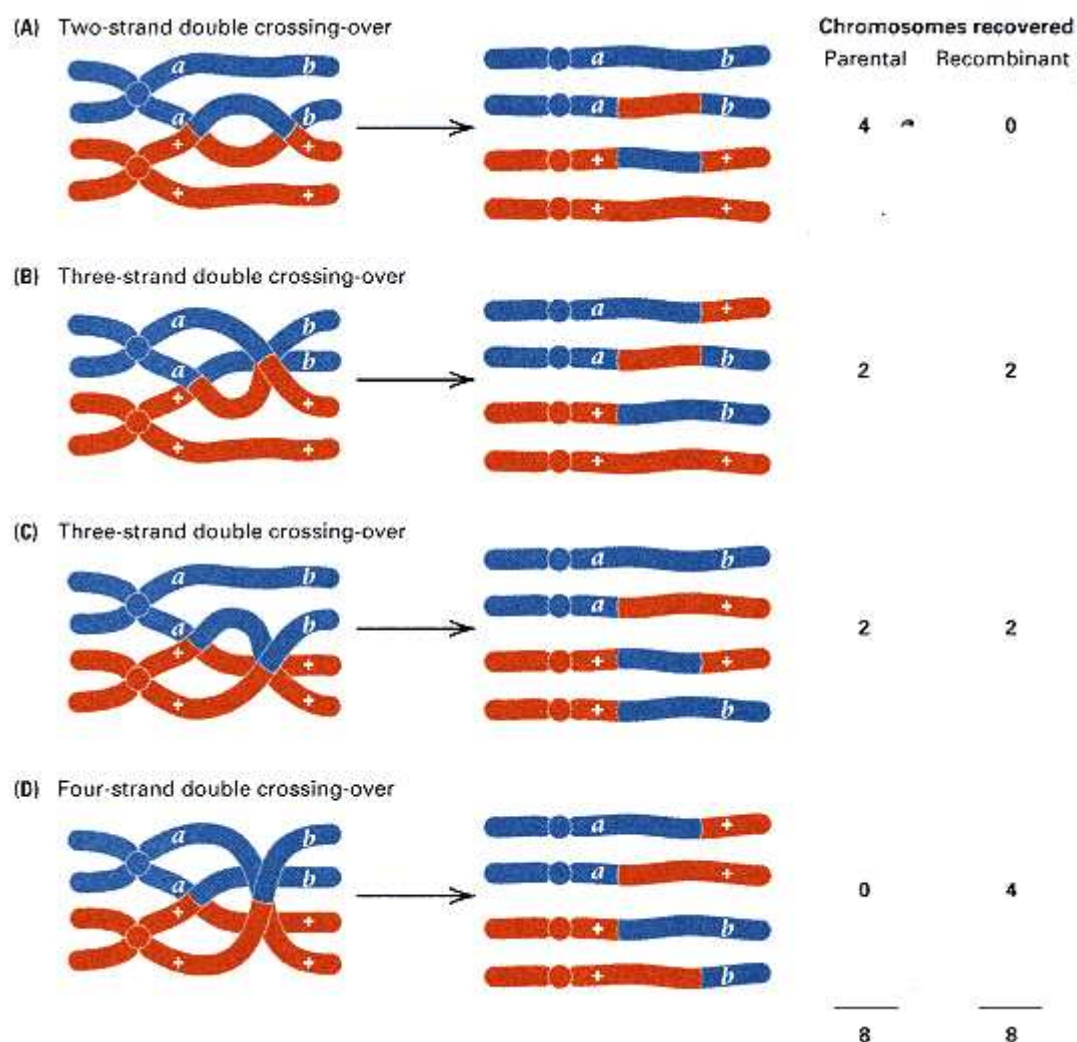


Figure 4.13

Diagram showing that the result of two exchanges in the interval between two genes is indistinguishable from independent assortment of the genes, provided that the chromatids participate at random in the exchanges. (A) A two-strand double crossing-over. (B and C) The two types of three-strand double crossing-overs. (D) A four-strand double crossing-over.

sibility is that the second exchange connects the chromatids that did not participate in the first exchange (four-strand double crossover, Figure 4.13D), in which case all four products are recombinant.

In most organisms, when double crossovers are formed, the chromatids that take part in the two exchange events are selected at random. In this case, the expected proportions of the three types of double exchanges are 1/4 four-strand doubles, 1/2 three-strand doubles, and 1/4 two-strand doubles. This means that, on the average,

$$(1/4)(0) + (1/2)(2) + (1/4)(4) = 2$$

recombinant chromatids will be found among the 4 chromatids produced from meioses with two exchanges between a pair of genes. This is the same proportion obtained with a single exchange between the genes. Moreover, a maximum of 50 percent recombination is obtained for any number of exchanges.

In the discussion of Figure 4.13, we emphasized that, in most organisms, the chromatids taking part in double-exchange events are selected at random. Then the maximum frequency of recombination is 50 percent. When there is a *nonrandom* choice of chromatids in successive crossovers, the phenomenon is called **chromatid interference**. It can be seen in Figure 4.13 that, relative to a random choice of chromatids, an excess of four-strand double crossing-over (*positive chromatid interference*) results in a maximum frequency of recombination greater than 50 percent; likewise, an excess of two-strand double crossing-over (*negative chromatid interference*) results in a maximum frequency of recombination smaller than 50 percent. Therefore, the finding that the maximum frequency of recombination between two genes in the same chromosome is not 50 percent can be regarded as evidence for chromatid interference. Positive chromatid interference has not yet been observed in any organism; negative chromatid interference has been reported in some fungi.

Double crossing-over is detectable in recombination experiments that employ **three-point crosses**, which include three pairs of alleles. If a third pair of alleles, c^+ and c , is located between the two with

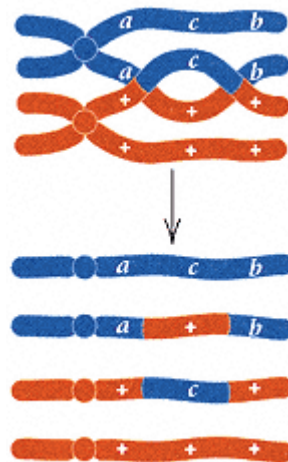


Figure 4.14
Diagram showing that two exchanges between the same chromatids and spanning the middle pair of alleles in a triple heterozygote will result in a reciprocal exchange of that pair of alleles between the two chromatids.

which we have been concerned (the outermost genetic markers), then double exchanges in the region can be detected when the crossovers flank the c gene (Figure 4.14). The two crossovers, which in this example take place between the same pair of chromatids, would result in a reciprocal exchange of the c^+ and c alleles between the chromatids. A three-point cross is an efficient way to obtain recombination data; it is also a simple method for determining the order of the three genes, as we will see in the next section.

Gene Mapping from Three-Point Testcrosses

The data in Table 4.1, which result from a testcross in corn with three genes in a single chromosome, illustrates the analysis of a three-point cross. The recessive alleles of the genes in this cross are *lz* (for lazy or prostrate growth habit), *gl* (for glossy leaf), and *su* (for sugary endosperm), and the multiply heterozygous parent in the cross has the genotype

$$\frac{Lz\ Gl\ Su}{lz\ gl\ su}$$

Therefore, the two classes of progeny that inherit noncrossover (parental-type) gametes are the normal plants and those with the lazy-glossy-sugary phenotype. These classes are far larger than any of the crossover classes. If the combination of dominant and recessive alleles in the chromosomes of the heterozygous parent were

Table 4.1 Progeny from a three-point testcross in corn

Phenotype of testcross progeny	Genotype of gamete from hybrid parent	Number
Normal (wildtype)	<i>Lz Gl Su</i>	286
Lazy	<i>lz Gl Su</i>	33
Glossy	<i>Lz gl Su</i>	59
Sugary	<i>Lz Gl su</i>	4
Lazy, glossy	<i>lz gl Su</i>	2
Lazy, sugary	<i>lz Gl su</i>	44
Glossy, sugary	<i>Lz gl su</i>	40
Lazy, glossy, sugary	<i>lz gl su</i>	272

unknown, then we could deduce from their relative frequency in the progeny that the noncrossover gametes were *Lz Gl Su* and *lz gl su*. This is a point important enough to state as a general principle:

In any genetic cross involving linked genes, no matter how complex, the two most frequent types of gametes with respect to any pair of genes are *nonrecombinant*; these provide the linkage phase (*cis* versus *trans*) of the alleles of the genes in the multiply heterozygous parent.

In mapping experiments, the gene sequence is usually not known. In this example, the order in which the three genes are shown is entirely arbitrary. However, there is an easy way to determine the correct order from three-point data. Simply identify the genotypes of the double-crossover gametes produced by the heterozygous parent and compare them with the nonrecombinant gametes. Because the probability of two simultaneous exchanges is considerably smaller than of either single exchange, the double-crossover gametes will be the least frequent types. Table 4.1 shows that the classes composed of four plants with the sugary phenotype and two plants with the lazy-glossy phenotype (products of the *Lz Gl su* and *Lz gl Su* gametes, respectively) are the least frequent and therefore constitute the double-crossover progeny.

The effect of double crossing-over, as Figure 4.14 shows, is to interchange the members of the *middle* pair of alleles between the chromosomes.

This means that if the parental chromosomes are

Lz Gl Su and *lz gl su*

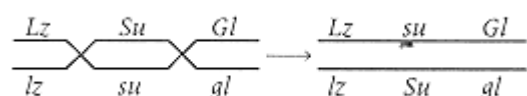
and the double-crossover chromosomes are

Lz Gl su and *lz gl Su*

then *Su* and *su* are interchanged by the double crossing-over and must be the middle pair of alleles. Therefore, the genotype of the heterozygous parent in the cross should be written as

$$\begin{array}{ccc} \underline{Lz} & \underline{Su} & \underline{Gl} \\ \hline lz & su & gl \end{array}$$

which is now diagrammed correctly with respect to both the order of the genes and the array of alleles in the homologous chromosomes. A two-strand double crossover between chromatids of these parental types is diagrammed below, and the products can be seen to correspond to the two types of gametes identified in the data as the double crossovers.



From this diagram, it can also be seen that the reciprocal products of a single crossover between *lz* and *su* would be *Lz su gl* and *Lz Su Gl* and that the products of a single exchange between *su* and *gl* would be *Lz Su gl* and *lz su Gl*.

We can now summarize the data in a more informative way, writing the genes in correct order and identifying the numbers of the different chromosome types produced by the heterozygous parent that are present in the progeny.

<i>Lz</i>	<i>Su</i>	<i>Gl</i>	286	} Parental types
<i>lz</i>	<i>su</i>	<i>gl</i>	272	
<i>Lz</i>	<i>su</i>	<i>gl</i>	40	} Single crossovers between <i>lz</i> and <i>su</i>
<i>lz</i>	<i>Su</i>	<i>Gl</i>	33	
<i>Lz</i>	<i>Su</i>	<i>gl</i>	59	} Single crossovers between <i>su</i> and <i>gl</i>
<i>lz</i>	<i>su</i>	<i>Gl</i>	44	
<i>Lz</i>	<i>su</i>	<i>Gl</i>	4	} Double-crossover types
<i>lz</i>	<i>Su</i>	<i>gl</i>	2	
			740	

Note that each class of single recombinants consists of two reciprocal products

and that these are found in approximately equal frequencies (40 versus 33 and 59 versus 44). This observation illustrates an important principle:

The two reciprocal products that result from any crossover, or any combination of crossovers, are expected to appear in approximately equal frequencies among the progeny.

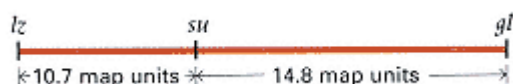
In calculating the frequency of recombination from the data, remember that the double-recombinant chromosomes result from *two* exchanges, one in each of the chromosome regions defined by the three genes. Therefore, chromosomes that are recombinant between *lz* and *su* are represented by the following chromosome types:

<i>Lz</i>	<i>su</i>	<i>gl</i>	40
<i>lz</i>	<i>Su</i>	<i>Gl</i>	33
<i>Lz</i>	<i>su</i>	<i>Gl</i>	4
<i>lz</i>	<i>Su</i>	<i>gl</i>	2
			79

That is, 79/740, or 10.7 percent, of the chromosomes recovered in the progeny are recombinant between the *lz* and *su* genes, so the map distance between these genes is 10.7 map units or 10.7 centimorgans. Similarly, the chromosomes that are recombinant between *su* and *gl* are represented by

<i>Lz</i>	<i>Su</i>	<i>gl</i>	59
<i>lz</i>	<i>su</i>	<i>Gl</i>	44
<i>Lz</i>	<i>su</i>	<i>Gl</i>	4
<i>lz</i>	<i>Su</i>	<i>gl</i>	2
			109

The recombination frequency between this second pair of genes is 109/740, or 14.8 percent, so the map distance between them indicated by these data is 14.8 map units or 14.8 centimorgans. The genetic map of the chromosome segment in which the three genes are located is therefore



The error that students most commonly make as they are learning how to interpret three-point crosses is to forget to include the double recombinants when calculating the recombination frequency between adjacent genes. You can keep from falling into this trap by remembering that the double recombinant chromosomes have single recombination in *both* regions.

Chromosome Interference in Double Crossing-over

The detection of double crossing-over makes it possible to determine whether exchanges in two different regions of a pair of chromosomes are formed independently of each other. Using the information from the example with corn, we know from the recombination frequencies that the probability of recombination is 0.107 between *lz* and *su* and

0.148 between *su* and *gl*. If crossing-over is independent in the two regions (which means that the formation of one exchange does not alter the probability of the second exchange), then the probability of an exchange in both regions is the product of these separate probabilities, or $0.107 \times 0.148 = 0.0158$ (1.58 percent). This implies that in a sample of 740 gametes, the expected number of double crossovers would be 740×0.0158 , or 12, whereas the number actually observed was only 6. Such deficiencies in the observed number of double crossovers are common and identify a phenomenon called **chromosome interference**, in which crossing-over in one region of a chromosome reduces the probability of a second crossover in a nearby region. Because chromosome interference is nearly universal, and chromatid interference is virtually unknown, the term *interference*, when used without qualification, almost always refers to chromosome interference.

The **coefficient of coincidence** is the observed number of double recombinant chromosomes divided by the expected number. Its value provides a quantitative measure of the degree of interference, defined as

$$\begin{aligned} i &= \text{interference} \\ &= 1 - \text{coefficient of coincidence} \end{aligned}$$

From the data in our example, the coefficient of coincidence is $6/12 = 0.50$, which